ICHNAEA-MST

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What is Ichnaea-MST and what is it used for?

Ichnaea-MST is a tool designed by environmental microbiologists for other environmental microbiologists to facilitate the traceability of the origin of microbial contamination in a desired matrix. Ichnaea-MST is a novel and autonomous iteration of Ichnaea 2.0 and 1.0. It comprises a comprehensive suite of scripts written in the R language, which have been developed for training and classification tasks using the H2O machine learning framework.

The tool has been developed for novice users with little to no experience in machine learning. The objective is to provide the microbiology community with a straightforward method for exploring machine learning in Microbial Source Tracking (MST) scenarios.

For local use on personal computers, we recommend the utilisation of the R Notebooks under RStudio, which is an integrated development environment (IDE) for the R language.

Individuals with no prior experience of R or RStudio are welcome to access the software remotely via the provided web address (http://biost3.bio.ub.edu:3838/apps/Ichnaea_MST/), which can be used as a graphical user interface (GUI) under a Shiny application.

Previous versions of Ichnaea

The European project entitled "Tracking the origin of faecal pollution in surface water (TOFPSW) " (EVK1-CT-2000-00080), which was executed between 2001 and 2004, was one of the first interdisciplinary studies in the European region to evaluate potential MST markers that would allow differentiation between human and non-human contamination. The publications by Blanch et al. (2004 and 2006) collates and summarises the results of this European project.

First attempt to develop a classification system based on machine learning dated from 2008 (Belanche-Muñoz & Blanch (2008)) a pioneering innovation that utilises a combination of MST markers, in conjunction with machine learning methodologies, to predict the origin of contamination. But the first version of Ichnaea (version 1.0) was developed in 2011 Belanche & Blanch (2011), and a refined version (version 2.0) was developed in 2015 (Casanovas-Massana et al., 2015).

This software was developed by a multidisciplinary team of microbiologists and computer scientists, led by Anicet R. Blanch from the University of Barcelona and Lluis Belanche from the Polytechnic University of Catalonia. Ichnaea eliminates subjectivity in the selection of relevant indicators, thereby providing reliable predictions even in complex cases where contamination is highly diluted or degraded.

Nonetheless, the utilisation of machine learning in the context of Microbial Source Tracking (MST) can be traced back to earlier periods. In a study conducted by different authors (Blanch 2006 and 2011, Casanovas-Massana et al. 2015), initial classifications were commenced with a curated set of MST parameters.

Why a new Ichnaea?

The motivation behind the development of a new version of Ichnaea (now named Ichnaea-MST) is attributable to the recent advancements in the extended use of machine learning. This movement has resulted in the development of tools that enable the creation of learning models with minimal or no user intervention.

These advancements include low-code and no-code platforms, which allow users to construct machine learning models with negligible programming expertise. H2O is an open-source machine learning and predictive analytics platform that facilitates the construction of machine learning models on substantial datasets and simplifies their deployment within enterprise contexts. A salient feature of the software is H2O AutoML, which automates the process of training and tuning a wide selection of models. H2O AutoML is characterised by its user-friendliness, requiring minimal input from the user. Furthermore, H2O AutoML has been demonstrated to possess the capacity to manage instances of missing data, with the capability to generate predictions even in the presence of missing MST markers. For the aforementioned reasons, the necessity to create new software using the H2O framework as a machine learning tool has been considered.

So, Ichnaea-MST is an innovative software that uses H2O AutoML for the training and classification of a dataset of MST markers. This allows the system to leverage these advancements to provide more efficient and accurate Microbial Source Tracking with minimal user intervention. It is evident that even users with limited technical expertise can reap the benefits of sophisticated machine learning models in identifying sources of contamination in water bodies.

How Ichnaea-MST works?

Ichnaea-MST is a frontend that can be distributed as a Shiny application, which can be used on a remote computer, or as a R Notebook that can be used under RStudio. It is important to note that both of these approaches have their respective advantages and disadvantages. The execution of Ichnaea-MST as a Shiny application is a more straightforward process, as it enables the user to manage the program via a graphical interface. However, it should be noted that the functionality of the program is constrained to the form in which it was designed. Conversely, utilising a R Notebook facilitates the execution of the program locally, thereby enabling users with a certain degree of proficiency in R to undertake modifications to the code.

Very briefly, in both scenarios, the user is prompted to upload the dataset intended for training purposes, the set of samples intended for classification, and following this the users are encouraged to engage in a thoughtful consideration of the integration of environmental decay and even to considerer to reduce the number of MST markers.

Once completed the previous options, the program begins with the augmentation of the training dataset. From a statistical perspective, the determination of the minimum number of samples required for a study is of paramount importance in ensuring the attainment of statistically significant and reliable results. This objective is realised through a power analysis, which calculates the necessary sample size to detect a real effect in the population.

Conversely, it is well-established that machine learning algorithms generally necessitate substantial datasets to ensure the accuracy of predictions, whether for classification or regression. It is common for datasets utilised in MST studies to be constrained to a mere few tens or hundreds of samples. This limitation is attributable to the inherent constraints of such studies, including the variability found in environmental samples, particularly in faecal or wastewater samples, as well as the time-consuming and costly nature of the analyses.

To deal with this fact, a data augmentation has been designed. Briefly, data augmentation involves several techniques to enhance the dataset for machine learning models. One approach is resampling with replacement, where samples are repeatedly drawn from the original dataset, allowing for the creation of a larger and more varied dataset. Additionally, a random dilution factor can be applied within an empirical probability distribution or using a probability density function (pdf) of a continuous uniform distribution. This introduces variability and simulates different conditions within the dataset.

For quantitative samples, each sample value can be used as the lambda parameter of a Poisson distribution. This method generates new samples based on the original values, reflecting the natural variability of the data. After generating these new values, they are compared to a threshold (detection limit) determined for each methodology used for measuring the selected parameters. If the obtained value is below this methodological threshold, it is assigned a value of zero (no information contribution), ensuring that only significant data points are considered.

These techniques collectively enhance the dataset, making it more robust and representative of real-world conditions, which improves the accuracy and reliability of machine learning models.

Machine learning requires a training step for the chosen algorithm. In the case of MST, the machine learning corresponds to a supervised machine learning, where the occurrences of MST markers are accompanied by their respective labels that identify the type of contamination source.

To elaborate further, the process of supervised machine learning involves the training of learning algorithms using labelled data. This means that each sample in the dataset includes both the input features (results for the measured parameters usually named MST markers) and the corresponding output labels (contamination faecal sources). H2O AutoML facilitates the evaluation of disparate learning algorithms, which are designated as 'models'. In the context of H2O, a model signifies a machine learning algorithm that has been trained on a dataset to facilitate predictions.

This approach facilitates the models' learning of the relationship between the markers and their sources, thereby enabling accurate predictions (classification) on new, unseen data. It is imperative to acknowledge the significance of this point: it is of the utmost importance that the models under consideration are never fed with samples that are subsequently to be used for classification.

Consequently, the dataset is segmented into two subsets: a training subset and a validation subset. Alternatively, if a more sophisticated approach is required, the dataset can be divided into three subsets: a training subset, a validation subset, and a test subset. The larger subset, comprising approximately 70% to 75% of the samples, is utilised for the training of the model. In scenarios where three subsets are employed, the training phase may run concurrently with the validation subset. This subset, which typically encompasses approximately 20% of the total sample, serves as a checkpoint to monitor the model's performance during training, allowing for adjustments and improvements in real-time.

Following the training of the model, the final subset (comprising either 30% or 10% of the dataset, depending on the number of subsets utilised) is employed for the evaluation of the model's predictive accuracy once the training is complete. This step is of crucial importance, as it provides an unbiased assessment of the model's performance on unseen data, ensuring its generalizability and effectiveness in real-world applications. In conclusion, the model that has been determined to offer the optimum yield in the validation process is selected.

What type of algorithm should be selected?

H2O AutoML offers 6 model families, which we will briefly discuss. The models are presented in order from least to most computationally intensive. The models are presented in this manner because it is an established fact that an increase in computational resources will result in a corresponding increase in calculation time. It is imperative to comprehend this relationship when utilising H2O AutoML (or any other machine learning framework). The time required to obtain results is contingent upon the available computational resources and capabilities. The duration for processing is contingent upon the available resources and can range from a few minutes to several tens of minutes.

Furthermore, it is important to consider that executing the program multiple times can be advantageous. It is important to note that each execution may yield slightly different results due to variations in data handling, model initialization, and other stochastic (random) processes involved in machine learning. It is imperative to acknowledge the significance of randomness in this context, as it constitutes a fundamental element in the sequence of these steps. Furthermore, the augmentation of data and the partitioning of samples into training and validation subsets are processes that are governed by randomness ensuring that the models are exposed to a diverse set of data points. It can be posited that the repetition of the process may result in outcomes that are more robust and reliable. This is due to the fact that by repeating the training process and classification multiple times, the effects of any single random partitioning or augmentation can be mitigated. Consequently, this results in outcomes that are more robust and reliable, and it is possible to demonstrate which samples after classification can show differences in their origin. This phenomenon is more likely to occur when complex mixtures from multiple distinct animal species are present within a water sample.

In light of these considerations, it is advised to employ the default option, GBM, which is esteemed for its balanced performance. Please consider visiting the H2O website for more information.

In the ensuing lines, the models are presented in accordance with the computational power that is required for their operation.

Computationally light:

- Generalised Linear Models (GLM). GLM models are characterised by their simplicity and ease of comprehension, which renders them highly efficient for the management of large datasets. These options include regularization techniques, which are employed to mitigate the risk of overfitting. Notwithstanding their simplicity, GLM have been observed to be incapable of capturing complex relationships in data, and they generally demonstrate a reduced level of power in comparison to more complex models, such as the GBM or DRF models.

Moderately computationally intensive:

- Distributed Random Forest (DRF) and Extremely Randomized Trees (XRT) models. DRF is a machine learning method that generates a forest of classification or regression trees. This process reduces variance and improves prediction robustness by averaging the results of multiple trees. The system is scalable and efficient for large datasets, making it suitable for big data applications. The employment of Extremely Randomized Trees (XRT) has been demonstrated to introduce an additional element of randomness in the process of split computation. This, in turn, has the potential to further reduce variance, though it may concomitantly result in an increase in bias. It has been demonstrated that XRT is both computationally efficient and capable of providing insights into feature importance. Nevertheless, the incorporation of randomness can complicate the interpretation of the model. It is evident that both the Decision Tree (DT) and the Random Forest (RF) are valuable tools in the H2O AutoML framework for the creation of robust and efficient machine learning models. Nevertheless, the training of numerous trees can be computationally intensive and time-consuming, with the potential for overfitting if not adequately tuned. The fact that DRF is an ensemble model has implications for the complexity of the model and the ease with which it can be interpreted, in comparison to simpler models. Ichnaea 1.0 and 2.0 versions utilise random forest algorithms.

- Gradient Boosted Machines (GBM) are a machine learning algorithm that employs a gradient-based optimization technique to improve the accuracy of predictions by learning from previously observed data. GBM is distinguished by its high predictive performance, particularly in the context of structured data. The system is distinguished by its flexibility, which enables it to accommodate a wide range of data and loss functions, thereby facilitating the extraction of insights pertaining to feature importance. However, GBM necessitates substantial computational resources and time for training, and there is a risk of overfitting if not properly tuned. H2O AutoML's default implementation of the gradient-based boosting (GBM) algorithm has been determined as offering a balanced optimisation of processing speed and accuracy.

Computationally intensive:

- XGBoost has been demonstrated to offer high performance and efficient handling of large datasets, thus rendering it ideal for complex machine learning tasks. The software incorporates a built-in regularisation process to prevent overfitting, in addition to providing feature importance scores to facilitate interpretability. Nonetheless, the process necessitates meticulous calibration of parameters, can prove computationally onerous, and may present interpretative challenges due to its intricacy.

- Stacked Ensembles. The concept of stacked ensembles involves the integration of multiple models, with the objective of enhancing the predictive accuracy and mitigating the risk of overfitting. This is achieved by leveraging the diversity inherent in the models. Whilst the products in question offer improved performance and robustness, they are more complex to implement and understand, and they require greater computational power due to the involvement of multiple models.

- Deep Learning Models demonstrate a high level of proficiency in tasks such as image and speech recognition, a proficiency attributable to two key factors. Firstly, these models possess a high degree of accuracy. Secondly, they are able to automatically learn features from raw data. However, these models require substantial computational power and time, and their complexity makes them difficult to interpret and understand.

What are the capabilities of Ichnaea-MST?

Capabilities:

- Ichnaea-MST can accurately predict the source of faecal pollution in water bodies and other matrices, whether it originates from human or different animal sources.
- Ichnaea-MST automatise the process of selection of the datasets for training and classification, the data augmentation process for the training of the machine learning, and the application of the environmental decay of the MST markers.
- Ichnaea-MST has the capacity to accept data with varying concentration levels, process quantitative and qualitative MST markers, and permit modification of the dataset in several steps prior to the classification process. It is acknowledged that the user has the capacity to consider the environmental persistence, thus rendering it adaptable to a range of geographical and climatic areas.
- Ichnaea-MST utilises in its backend H2O AutoML, a sophisticated tool that empowers users to investigate a diverse range of machine learning algorithms. Benefits of using H2O AutoML:
    - Its capacity to automate the process of model creation and tuning, thereby minimising the necessity for manual intervention.
    - This automation facilitates the training of multiple models concurrently and the selection of the most efficacious model based on various metrics.
    - Users can train the system with their own data or use a pre-trained model, allowing for customised and precise analysis based on specific environmental conditions.
    - H2O AutoML provides robust methods for explaining the generated models. These methods highlight the MST markers that are more important in the classification.

This facilitates the comprehension of the underlying mechanics and decision-making processes of machine learning models by users.

- Furthermore, the user has the option of simplifying the number of MST markers used in the classification process, thereby reducing the redundant parameters.

What are the limitations of Ichnaea-MST?

It is crucial to recognize the inherent limitations of this approach. Some of these are associated with the characteristics of data augmentation and machine learning classification, while others are related to the characteristics of sampling and determining the occurrence of the MST markers, and the MST markers themselves.

Methodological limitations to be considered when determining the occurrence of MST markers

It is imperative to acknowledge that the presence of MST markers, as determined at the sources of contamination, such as faeces or wastewater, is contingent on the methodology employed for their detection and/or quantification. The accuracy and reliability of these determinations can be influenced by various factors inherent to the detection methods.

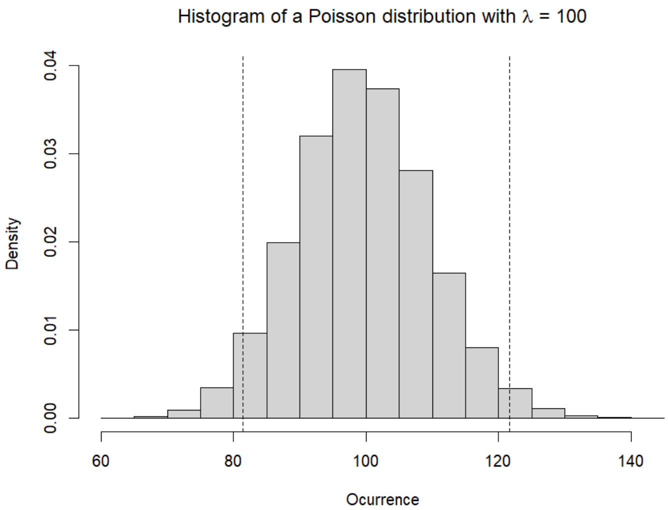
For instance, consider the determination of the occurrence of an MST marker by quantitative PCR (qPCR). qPCR is a powerful and sensitive technique for detecting specific DNA sequences; however, its effectiveness is contingent on the quality of the sample preparation. In order to accurately determine the presence of this marker it is essential to begin with a properly homogenised sample. The process of homogenisation is crucial in ensuring the uniformity and representativeness of the sample, thereby ensuring the reliability of the results obtained.

In the context of samples originating from faeces or wastewater, the process of achieving adequate homogenisation often poses a considerable challenge. Such samples are frequently characterised by heterogeneity, comprising diverse particles, debris, and microorganisms, which can difficult the process of homogenisation. In the event that the sample has not been adequately homogenised, the qPCR results may be skewed, resulting in an inaccurate determination of the MST marker. Moreover, individual faecal samples or those composed of a limited number of individuals can distort the models when used as a training set, as they do not adequately represent the biodiversity of an MST marker in the host population that generates the faecal contamination.

Furthermore, the methodology employed for sample collection, storage, and preparation has the capacity to influence the detection and quantification of MST markers. Factors such as the preservation of sample integrity, the prevention of contamination, and the efficient extraction of DNA are critical for ensuring the validity and reliability of the qPCR results.

Notwithstanding the exclusion of instrumental and methodological uncertainties, and under the assumption of correct homogenisation of the sample, it must be acknowledged that the presence of an MST marker in the analysis of an aliquot is contingent upon stochastic processes. The random nature of the distribution of microorganisms within the sample gives rise to these stochastic processes. Consequently, the occurrence of this MST markers can vary between different aliquots of the same sample. The utilisation of a Poisson distribution is a viable modelling approach for this variability.

For instance, the histogram below illustrates a Poisson distribution with an expected value of 100. The dotted lines represent the 95% confidence intervals, which correspond to the values 81.36 and 121.63, respectively.



Data augmentation process.

In the context of a data augmentation process that involves resampling with the objective of expanding the dataset for training purposes, a number of potential issues may emerge. The following is a summary of the potential issues:

The process of sample poisoning involves the introduction of misleading data into the training set. This phenomenon may occur when the dataset for training uploaded by the user contains a reduced number of samples, and the simulation of dilution considers a very restricted range of possible dilutions. This results in the generation of a dataset comprising highly similar samples, which has the potential to compromise the model's performance or induce unpredictable behaviour. Conversely, excessive augmentation or improper resampling techniques have the potential to introduce artefacts that the model assimilates, thereby diminishing its efficacy on real-world data or even engendering data patterns within the dataset. The presence of these patterns can result in overfitting, a phenomenon in which the model becomes overly focused on the noise and details of the augmented data, rather than the underlying patterns that are of interest.

In the event of the dataset exhibiting an imbalanced number of classes, meaning that one class is clearly predominant over the others, data augmentation has been known to exacerbate class imbalances if not executed with the requisite degree of care. This can result in a model that is biased towards the majority class. Furthermore, in the event that the original dataset contains biases, augmentation has the potential to amplify these biases, resulting in a model that perpetuates or exacerbates the bias. Conversely, aggressive augmentations have the potential to distort the data, resulting in a dataset that no longer accurately represents the real-world scenario. It is important to note that random transformations or resampling techniques have the potential to introduce irrelevant variations to the data. This has the effect of confusing the model and degrading its performance.

It is also important to note that an excessive number of training data can lead to an increase in computational load. The augmentation of data results in an increase in the size of the dataset. It is well established that larger datasets require more memory and processing power, and this has the potential to result in a slowdown in the training process, particularly when using computationally demanding algorithms.

One potential solution to these issues is to limit the number of samples after data augmentation (including the original training dataset uploaded by the user) to five times the minimum number of samples indicated by the power analysis. Furthermore, it is recommended that, if feasible, the user provides a marker that has been quantified in the samples used for training and in the matrix to be classified. This marker does not necessarily have to be an MST marker. A potential marker for the load of faecal contamination could be *Escherichia coli*, somatic coliphages, or any genetic marker that is universally found in the faeces of humans and animals. This marker would serve as a heuristic device, indicating, with a certain degree of approximation, the appropriate dilution factor to be applied in the simulation.

Probabilistic models and matrices containing complex mixing of polluted waters.

In a very short summary, H2O AutoML uses tree-based algorithms (DRF, XRT, GBM, and XGBoost), neural network-based algorithms (Deep Learning), and a stacking of simpler models that are trained to form part of a more complex model (Stacked Ensembles). All of these can be said to be probabilistic models to a greater or lesser extent, which, once correctly trained, provide a probabilistic response when classifying, indicating the probabilities of each of the possible sources of faecal contamination. The class with the highest probability is the one that is chosen in the result of the classification. It is important for the user to understand that these classifications do not show the percentage of each class, but rather the probability that the classified sample belongs to one class or another.

For example, a dataset that has two classes: human and non-human. After the training, the classification of a particular sample showed that it was most likely human. It is not the same if the chances of human and non-human contamination are 90% and 10%, respectively, as opposed to 51% and 49%. The first can be assumed to be a human-polluted sample, while the second is clearly a mix, so we should not be able to say where the pollution came from, even if the classification selects the class human as the most likely. It is important to remember that selection of the origin of the pollution can be tricky. This complexity can become even more challenging when more animal species contribute to the pollution of the sample.

Ichnaea-MST displays the probabilities for each class. However, it is imperative for the user to comprehend the rationale behind the necessity of repeating the training and classification process on multiple occasions. This is particularly the case when the probabilities of being assigned to one class, or another are unclear. In addition, it is crucial to ascertain whether each sample is consistently classified into the same class and with similar probabilities. It is therefore recommended that the training and classification be reiterated, employing either the same model or an alternative one. This is particularly advisable in cases where the samples are ambiguous and uncertain as to which class they belong.

Environmental decay.

The presence of MST markers in sources of contamination is determined through the analysis in samples that have not undergone a significant environmental decay. This is not generally the case with the matrices that are to be classified; many of them are environmental samples in which MST markers may be subjected to decay processes.

The user should consider the environmental conditions to which the sample may be subjected, including but not limited to temperature, solar radiation, altitude, the chemical composition of the water, pH, and the presence of predators.

These decay processes have the capacity to significantly alter the distribution of MST markers in the water matrix, especially if the decay affects the chosen and the most important (which carry more weight in the prediction) MST markers unevenly. In order to address this issue, Ichnaea-MST has introduced a small function that modifies the occurrence of the MST parameters according to a linear model based on t90. It should be noted that if a linear decay kinetics (which may be incorrect) is assumed for a given marker, the t90 value is the residence time for which the MST marker has decayed by 90%.

It is advisable that users employ this function to observe the impact of varying residence times of MST markers in the matrix on the classification process. In addition, it is recommended that the user conducts a search for those values that most closely align with the characteristics of their marker (several t90 values collected from the literature are included).

MST marker occurrences below the limit of detection

In any training and classification process, it is imperative to understand the critical role of the limit of detection (LOD). Ideally, the detection limits of each MST marker used during the training phase should be equivalent to—or at least closely aligned with—the LOD applied when quantifying the same markers in the matrix intended for classification. Discrepancies between these limits can introduce inconsistencies that may compromise the reliability of the classification outcomes.

It is particularly important to consider this during data augmentation, where the presence or absence of a marker in a given sample is determined relative to its LOD. Moreover, it is crucial to recognise the role of LOD, particularly when the LOD is related to the methodologies employed in the determination of the occurrences in the matrix of the origin and the matrix to classify. In order to ascertain the occurrences for a particular MST marker in these two matrices, it is recommended that methodologies with similar LODs be employed. This is especially important when LOD of the training set > LOD of the classification set, because during the augmentation, values below the limit of detection are considered null, this means that values are assigned a value of 0. Consequently, the probability of achieving a reliable and accurate classification outcome is significantly diminished. For instance, the occurrence of the HF183 marker has been quantified via qPCR in both a source matrix and a target matrix. In many cases, the volume of the sample analysed from the source matrix is significantly smaller than that of the target matrix, resulting in a higher detection limit for the former. This difference may directly affects how low-concentration values are interpreted. During data augmentation, any values falling below the LOD are typically treated as non-detects—that is, they are assigned a value of zero. While classification can still proceed under these conditions, it is important to recognize the potential for bias. If the HF183 marker is indeed present in the sample to be classified, but its measured concentration is below the detection threshold used during training, the algorithm may fail to recognise it appropriately. This is because the model has not been exposed to such low values during training and therefore lacks the capacity to interpret them accurately.

In such scenarios, the classification model is deprived of the necessary input signals to make a confident and informed prediction. In order to address this challenge, particular attention should be given to matrices that are inherently prone to low marker concentrations, such as groundwater or other oligotrophic environments. In such cases, the application of sample concentration techniques prior to molecular analysis is strongly recommended. In addition, it is crucial to ensure that a sufficient number of MST markers are detectable in each sample, as this is a prerequisite for robust classification. In circumstances where the majority of markers are undetectable, the model functions in conditions of extreme data sparsity. This phenomenon has the potential to compromise both the accuracy of classification and the risk of misclassification.

Conversely, while it is to be expected that some divergence will occur between the probability distribution of the classification dataset and the dataset generated through data augmentation, it is advisable for this divergence to remain moderate. It is evident that a significant disparity between the two variables is indicative of an inadequate training of the algorithm in regard to the range of data present within the classification set. In order to emphasise these potential disparities between the probability distributions, two metrics have been calculated [6]. The Jensen-Shannon Divergence (JSD) is a statistical measure that quantifies the similarity between two probability distributions. The distribution is characterised by its symmetry and the fact that it always yields a finite value between 0 and 1. A JSD value of 0.0 indicates that the distributions are identical, meaning there is no difference between the datasets. Distributions falling between 0.0 and 0.2 indicate highly similar outcomes, suggesting that the model is likely to generalise effectively. When the JSD falls between 0.2 and 0.4, it reflects moderate differences, which may introduce slight bias in classification. A range of 0.4 to 0.6 indicates significant disparities, suggesting that the model may be influenced by distributional shifts. Finally, a JSD value greater than 0.6 indicates markedly divergent distributions, implying that the model may struggle to generalise due to substantial differences between the training and classification data.

The Total Variation Distance (TVD) is another fundamental metric used to assess the dissimilarity between two probability distributions. The term is defined as the mean of the arithmetic mean of the absolute differences between the probabilities assigned to each outcome. As with JSD, TVD ranges from 0 to 1, where 0 indicates identical distributions and 1 indicates completely disjoint distributions. A TVD value below 0.1 suggests very similar distributions, while values between 0.1

and 0.3 indicate moderate differences. Values between 0.3 and 0.5 indicate significant differences, while values above 0.5 suggest substantial divergence.

Collectively, both constitute a robust and complementary framework for the evaluation of distributional similarity and it is particularly advantageous in machine learning for detecting distributional changes, such as when comparing training and prediction datasets, to evaluate potential model bias or generalisation issues. While the Jensen-Shannon Divergence (JSD) quantifies the similarity between two probability distributions by measuring the average information content shared between them, it does so in a symmetric and smoothed manner, making it particularly useful for comparing distributions in a stable and interpretable way. In contrast, the Total Variation Distance (TVD) captures the maximum possible difference between the two distributions by identifying the largest discrepancy in their assigned probabilities over all possible outcomes. This makes TVD a more sensitive and stringent measure, highlighting the point of greatest divergence rather than an average or smoothed difference.

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R NOTEBOOK

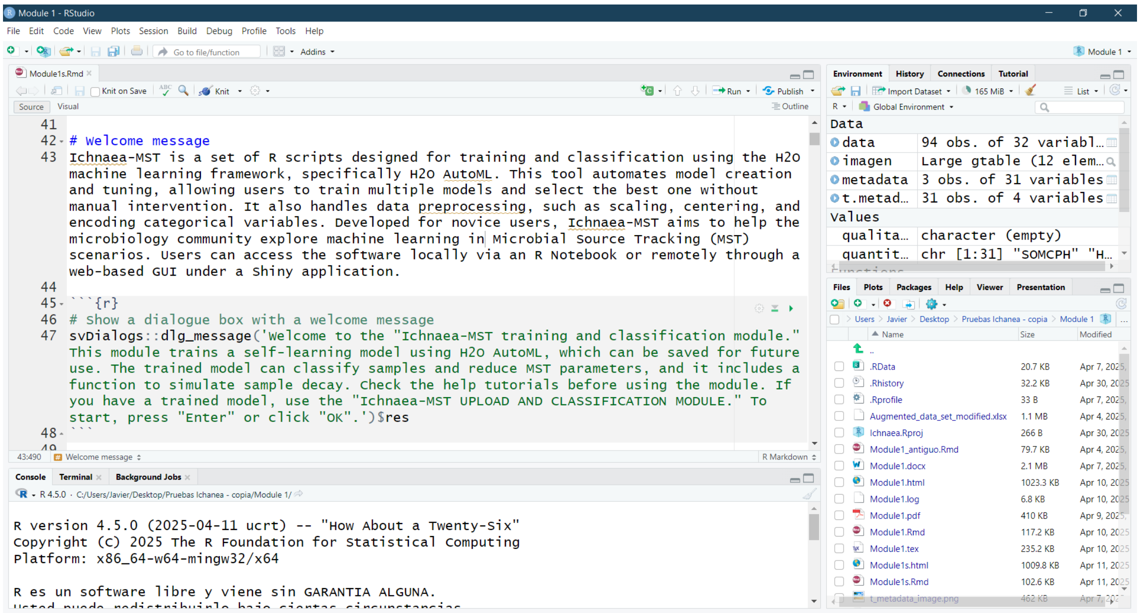
MODULE I: TRAINING, CLASSIFICATION AND REDUCTION OF THE NUMBER OF MST PREDICTORS

R NOTEBOOK FOR TRAINING, CLASSIFICATION AND REDUCTION OF THE NUMBER OF MST PREDICTORS

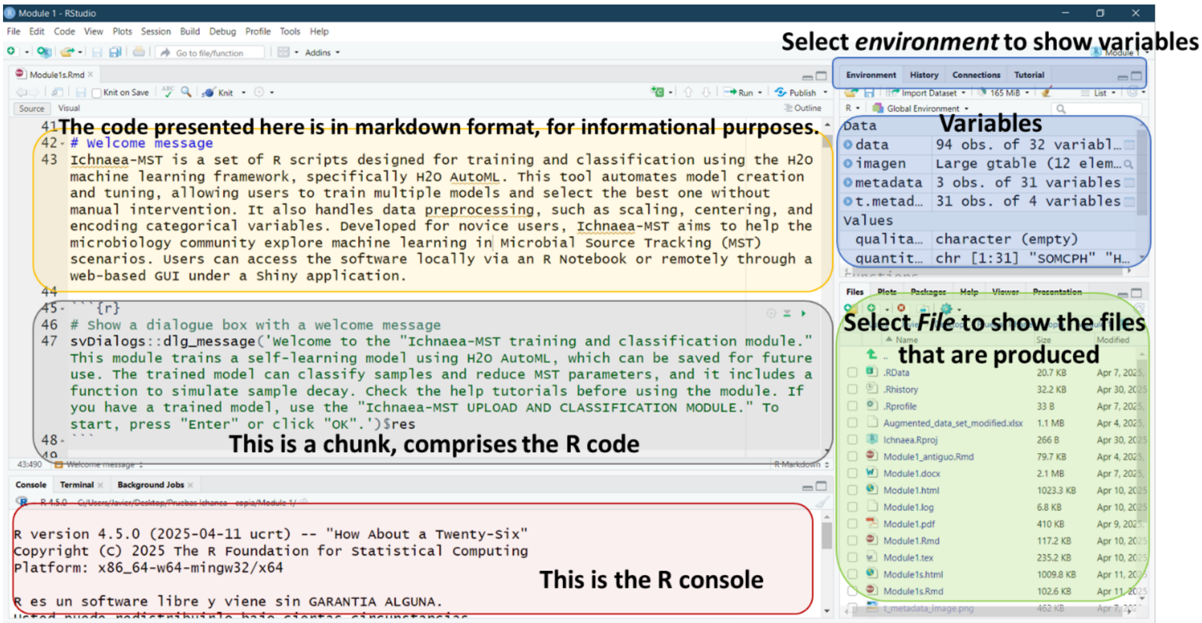
AIM OF THIS MANUAL

The present user manual has been designed for users who possess a fundamental understanding of the R programming language and the RStudio software environment. The following pages will elucidate the functionality of each constituent element (chunk) of the R Notebook, along with the outcomes of each one.

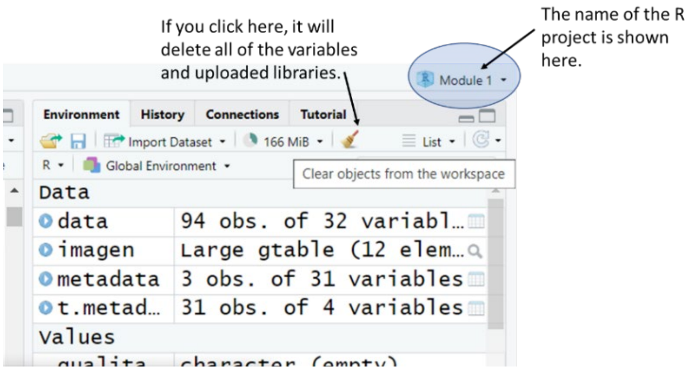
RStudio is organised in a series of panes.



Each pane has a function, below these lines are presented the main function of each pane.

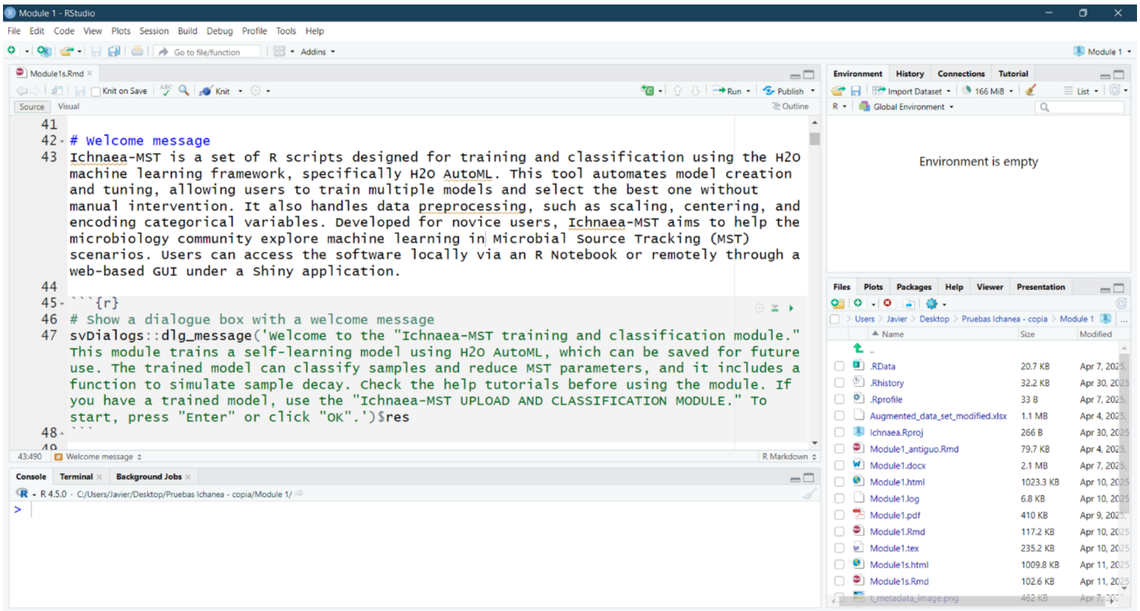


Now, let us now turn our attention to the *Environment* option.

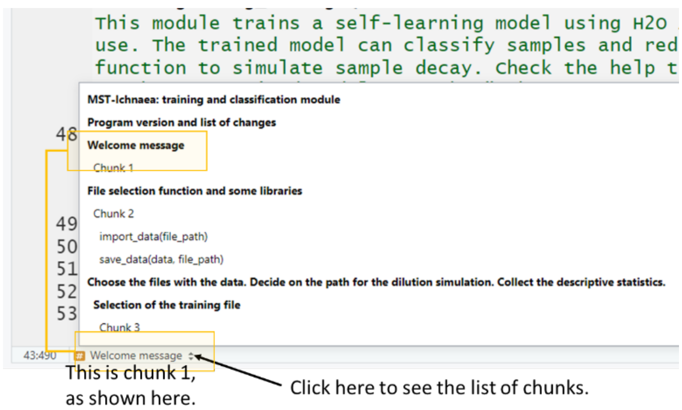


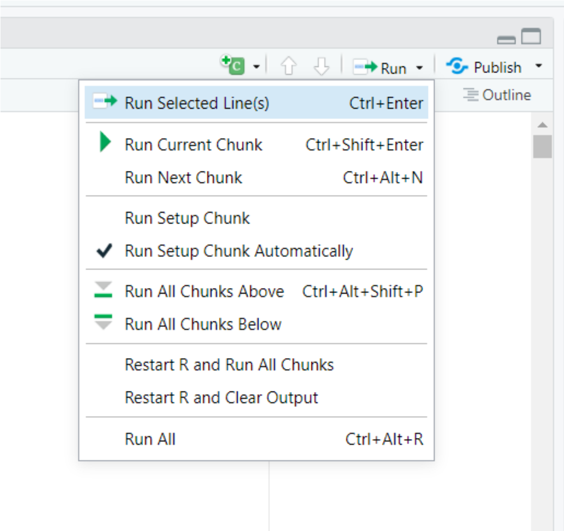
After clicking on the broom icon, the variables and libraries loaded in memory are cleared.

Chunk 1



Now, let us now turn our attention to the pane that contains the R code. Below this line a screen capture show the chunk 1. Henceforth, each chunk will be marked to ensure that it can be easily followed by the user.

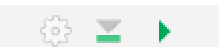




There are a number of ways to execute a chunk. One of the simplest and most convenient methods is to clear the environment, open the Run menu and select 'Restart R and Run All Chunks'. This enables the user to initiate the process with a pristine system, ensuring that the chunks are executed automatically.

However, the downside is that execution continues regardless of whether errors occur, which limits your ability to analyse or repeat a specific chunk.

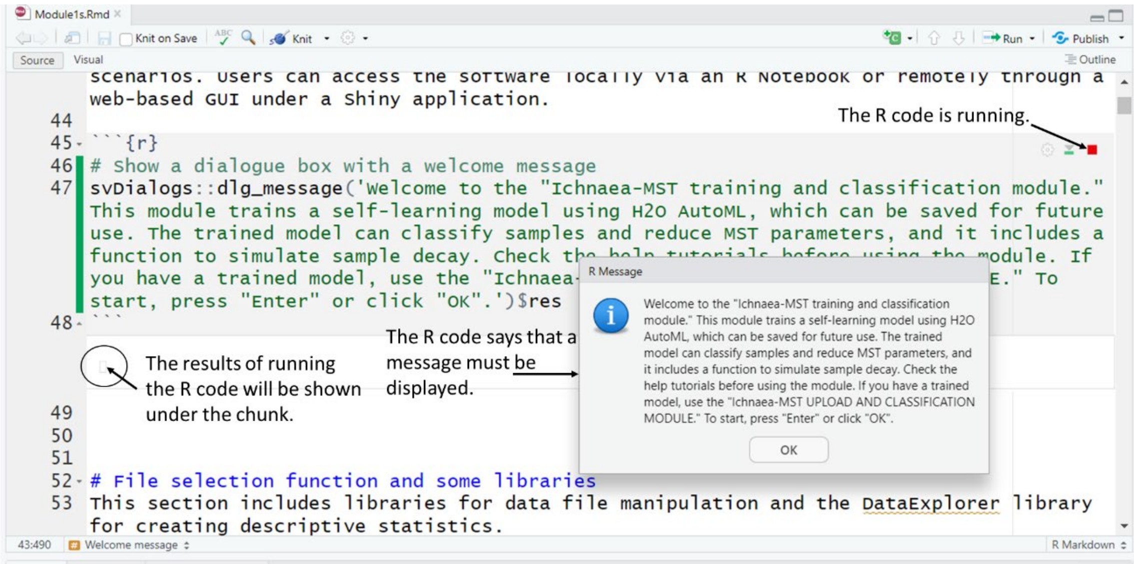
Though this method may not be the most convenient option when running the notebook locally for the first time, it is the recommended option for performing repetitions and variations in training and classification. It is important to note that machine learning is based on stochastic processes, and therefore, the user is encouraged to perform different training and classification processes. Therefore, when using the program for the first time, it is advisable to run chunk by chunk and verify that the entered data and the program's behaviour are correct.

To execute a chunk individually. It is possible to use either the keyboard, by pressing Ctrl + Shift + Enter, or by using the mouse. In every chunk, in its top left corner, there is a series of buttons . The button located at the extreme left of the interface (the 'play button') is the function that initiates the execution of the chunk code. A screen capture showing the first chunk is shown below these lines.

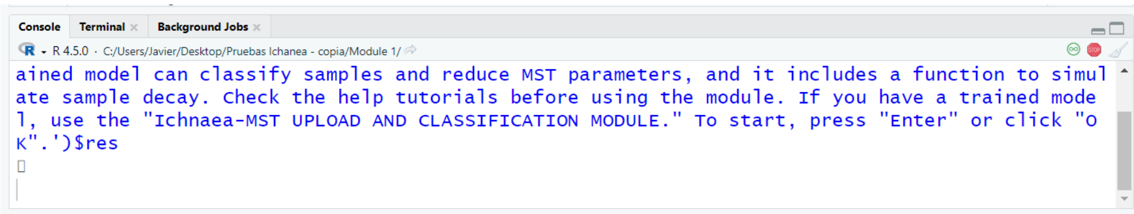
On the following page, an example is presented to show how the code is organised in a R Notebook. The illustration corresponds to the first chunk, which generates a welcome pop-up. In brief, a chunk is defined by a starting flag—in our case, the chunks begin with the flag ```{r} and ends with the flag ```. The R code to be executed is placed between these two flags. In the R notebook the chunk is presented in shaded. Below this shaded section are shown the results of executing the R code.

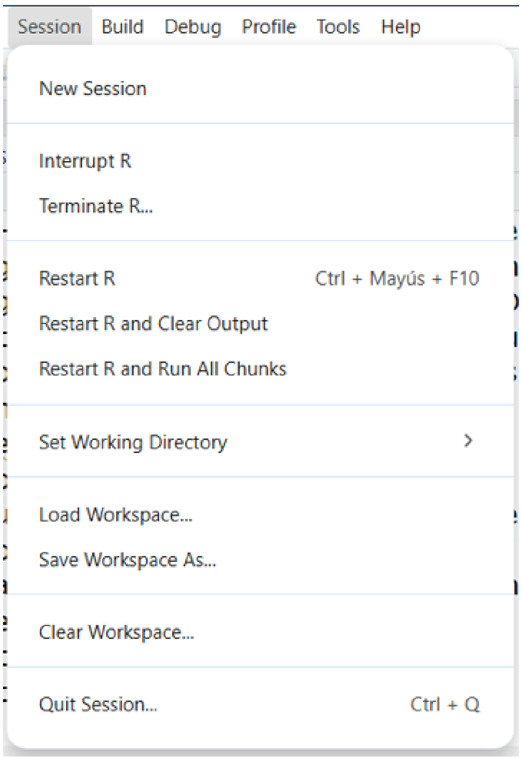
Annotations can be incorporated using the # symbol. Inside a chunk, a line starting with # represents a comment intended solely for informational purposes. Outside the chunk, the text may represent Markdown code, which can serve various functions. In the illustration, # (shown in blue) indicates a heading. In fact, the chunk below the heading inherits its name from this heading. Therefore, an R Notebook is a type of interactive document in RStudio that combines code, output, and narrative text within a single file.

When the ‘play button’ is clicked a green line appears on the left and moves downward, indicating the progression through the different lines of R code.



In the console appears the results of the R code. The console not only reproduces the code from the chunk, but also allows you to enter R code directly and execute it. If you wish to reproduce a chunk in the console, copy only the content found between the start and end flags. Remember that chunks containing R code begin with ```{r. R Notebooks allow you to work with different programming languages, so various starting flags can be used, such as {python} for Python, {bash} for shell scripts, {sql} for SQL queries, {cpp} or {c} for C/C++, {js} for JavaScript, and {julia} for Julia.

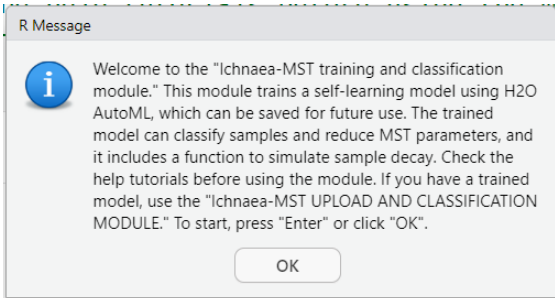




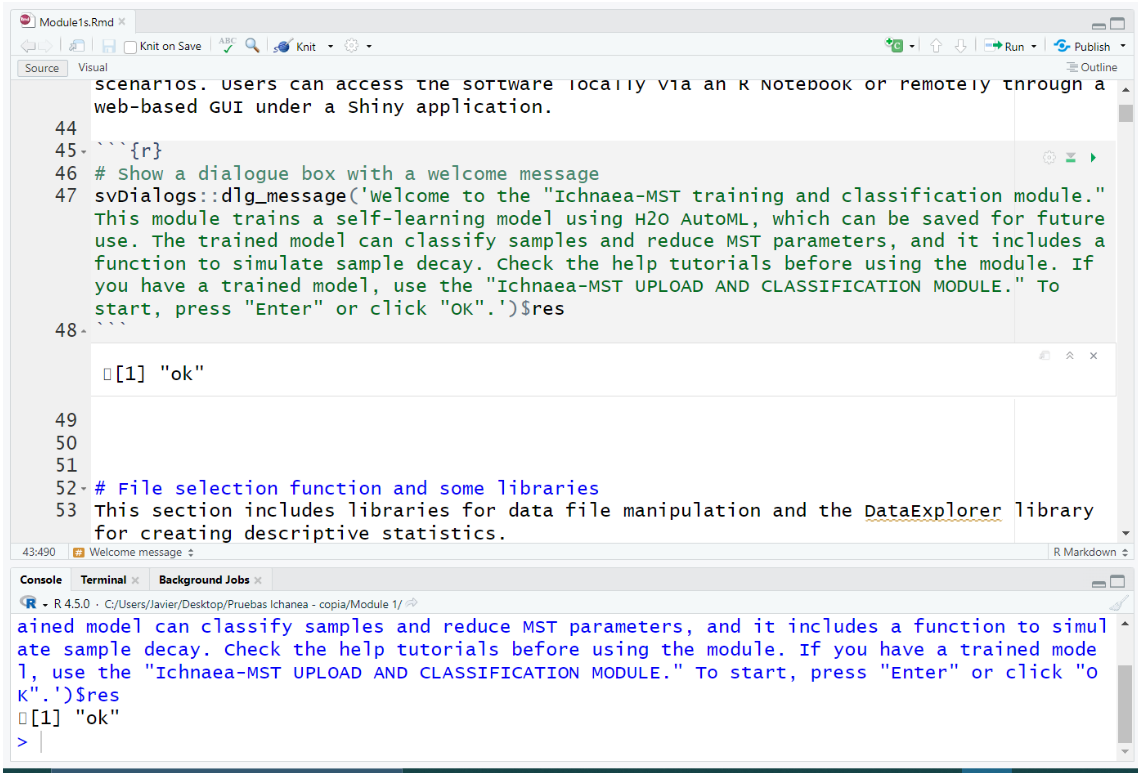
The icon depicting a stop sign 🔴, located on the top right within the console, can be selected to halt the execution of the code. This can be advantageous when entering an infinite loop or when calculations exceed a reasonable execution time. Sometimes this button may be no responsible, when this occurs R can be halted by interrupting R that causes the pause/stop current task, keep session alive. If choose terminate R, the session will be killed entirely and starting over.

Nonetheless, this cessation may, on occasion, result in system instability. Consequently, it may be necessary to initiate a new R session, which may result in the forfeiture of results and data that have been executed during the session.

Let us now direct our attention to the window title R message and click the ‘OK’ button.



Following the selection of the button, a number of changes will be implemented. An "ok" appears below the chunk and below the console. This "ok" is actually the result obtained after executing the R code. Let us proceed with the next chunk.



For now, the environment continues empty. From this point onwards, we will briefly explain the purpose and results of each chunk. Please be aware that when running the example, it is possible that you may obtain slightly different results; this is a standard occurrence. Please use the provided sample data to familiarise yourself with the notebook. If you encounter any difficulties, rest assured that you have access to the web application.

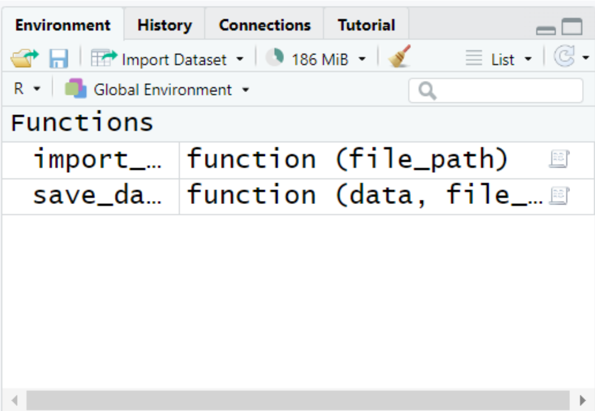
Chunk 2

The second chunk contains several functions. The first one deals with the process of upload files in the form of spreadsheet containing information for the program. The permitted formats encompass CSV files, Excel files (XLSX), Calc files (ODS), and files from SPSS, SATA and SAS (under development).

Notwithstanding the variety of formats, it is recommended that users employ spreadsheets in Excel or LibreOffice Calc.

In addition, this chunk encodes for a function to launch Excel or LibreOffice for editing Excel, Calc and CSV files. It also provides a function to calculate Jensen-Shannon Divergence (JSD) and Total Variation Distance (TVD). These are measures of the similarity between two probability distributions, indicating the extent to which information is shared between them. This is a measure of the maximum possible difference between two distributions, capturing the largest discrepancy in probabilities across all possible outcomes.

Once you have run this chunk (please click on the green play button) the new defined 'variables' (in our case a function for import spreadsheets and other for save) appear in the environment.



And the present chunk concludes at this point.

Chunk 3

We will now proceed to the next chunk, where you will find detailed instructions on preparing the spreadsheet containing the data for the training. Below is shown the description of this chunk.

```
# Choose the files with the data. Decide on the path for the dilution simulation. Collect the descriptive statistics.
## Selection of the training file

The user is prompted to select the training data file, ensuring the dataset is properly structured:

- **Column headings** should indicate variable names, with the class in the first column.
- **Row 1**: Quantitative (1) or qualitative (0) variable.
- **Row 2**: Detection limit (cfus, pfps, gcs, etc.).
- **Row 3**: MST marker (1) or not (0).


| Class     | Parameter 1  | Parameter 2  | Parameter 3  |
|----------|-------------|-------------|-------------|
|           | 1            | 1            | 0            |
|           | 95           | 19           | 0.001        |
|           | 1            | 1            | 0            |
| HUMAN     | 101          | 2523         | 1            |
| POULTRY   | 17090        | 125          | 0            |
| BOVINE    | 254          | 242          | 0            |
| PORCINE   | 327          | 0            | 0            |

For more details, refer to the help section.

```{r message=FALSE, warning=FALSE}
```

Here is a detailed guide on how to prepare the spreadsheet to be loaded into the program:

Column Headers:

Please ensure that the first row of your spreadsheet contains the column headers. These headers should provide clear descriptions of the data in each column. R distinguishes between upper and lowercase letters. It is generally considered poor practice for variable names to start with a number. Please do not use:

- Special characters such as #, @, !, $, %, ^, &, *, (,), -, +, =, {, }, [,], |, \\, :, ;, \", \', <, >, ,, ?, /.
- The use of spaces is discouraged to use in variable names.
- Do not start a variable name with a number.
- Although underscores can be used in variable names do not start a name with an underscore (_).

There are some reserved words that cannot be used as variable names. Some of these words include:

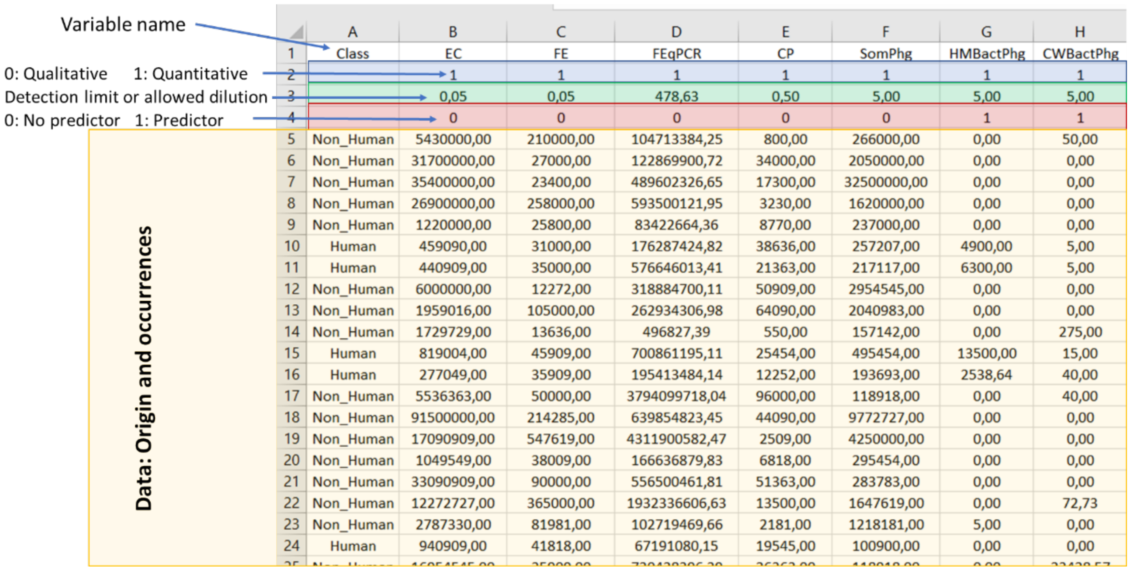
TRUE, FALSE, NULL, if, else, repeat, while, function, for, in, next, break, Inf, NaN, NA, NA_integer_, NA_real_, NA_complex_, NA_character_.

For instance, the Excel file named 'to_train_binary.xlsx' will be utilised as a case study.

The first column should be called "Class", and this column will contain the class of contamination (above these lines, in the example used in the description of the chunk, the classes are: HUMAN, POULTRY, BOVINE, PORCINE). The first three rows of this column must remain empty, as these three rows are reserved for defining the type of variable, the limit of detection and whether it is part of the set of predictors to be used in the training or not. The remaining columns should contain the occurrences of the variables to be used in the study.
```

Second to fourth row: metadata

The definition of each variable is determined by the numbering used in their first four rows. The first row corresponds to the label which identifies the parameters, rows 2 to 4 contains the metadata.



	A	B	C	D	E	F	G	H
1	Class	EC	FE	FEqPCR	CP	SomPhg	HMBactPhg	CWBactPhg
2		1	1	1	1	1	1	1
3		0,05	0,05	478,63	0,50	5,00	5,00	5,00
4		0	0	0	0	0	1	1
5	Non_Human	5430000,00	210000,00	104713384,25	800,00	266000,00	0,00	50,00
6	Non_Human	31700000,00	27000,00	122869900,72	34000,00	2050000,00	0,00	0,00
7	Non_Human	35400000,00	23400,00	489602326,65	17300,00	32500000,00	0,00	0,00
8	Non_Human	26900000,00	258000,00	593500121,95	3230,00	1620000,00	0,00	0,00
9	Non_Human	1220000,00	25800,00	83422664,36	8770,00	237000,00	0,00	0,00
10	Human	459090,00	31000,00	176287424,82	38636,00	257207,00	4900,00	5,00
11	Human	440909,00	35000,00	576646013,41	21363,00	217117,00	6300,00	5,00
12	Non_Human	6000000,00	12272,00	318884700,11	50909,00	2954545,00	0,00	0,00
13	Non_Human	1959016,00	105000,00	262934306,98	64090,00	2040983,00	0,00	0,00
14	Non_Human	1729729,00	13636,00	496827,39	550,00	157142,00	0,00	275,00
15	Human	819004,00	45909,00	700861195,11	25454,00	495454,00	13500,00	15,00
16	Human	277049,00	35909,00	195413484,14	12252,00	193693,00	2538,64	40,00
17	Non_Human	5536363,00	50000,00	3794099718,04	96000,00	118918,00	0,00	40,00
18	Non_Human	91500000,00	214285,00	639854823,45	44090,00	9772727,00	0,00	0,00
19	Non_Human	17090909,00	547619,00	4311900582,47	2509,00	4250000,00	0,00	0,00
20	Non_Human	1049549,00	38009,00	166636879,83	6818,00	295454,00	0,00	0,00
21	Non_Human	33090909,00	90000,00	556500461,81	51363,00	283783,00	0,00	0,00
22	Non_Human	12272727,00	365000,00	1932336606,63	13500,00	1647619,00	0,00	72,73
23	Non_Human	2787330,00	81981,00	102719469,66	2181,00	1218181,00	5,00	0,00
24	Human	940909,00	41818,00	67191080,15	19545,00	100900,00	0,00	0,00

The second row indicated if the variable is qualitative (for example a PCR with only presence or absence) or quantitative (an enumeration or a qPCR). If the variables are qualitative only 0 and 1 are allowed to indicate absence or presence. Other characters are not allowed. If the variables are quantitative only numeric values are allowed, “< values” or “> values” are not permit. When enumerating, the temptation to indicate the detection limit is evident, given that the significance of a "not detected" result differs significantly between highly sensitive and less sensitive methods. However, from a machine learning perspective, it is preferable to indicate that it has not been detected (value of 0), and if it is detected and the occurrence is known, to indicate this. The aforementioned principles also apply in the event of values that exceed the quantification limit. In such cases, the symbol "> 100" is rendered meaningless. It is preferable to indicate a value of 100. Ideally, dilutions should be made.

The third row contains the data related to the detection limit

Data Types

Apart from the Class column, in which text entries (of the character variety) are permitted, the remaining columns must contain numerical values. It is advisable to use complete datasets without missing values, but in any case, H2O AutoML can manage missing rows.

The image below illustrates a table that has been poorly formatted. The table contains both qualitative and quantitative data that has been invented. The table is shown as an example of how it should be formatted for use as a training dataset. It is imperative to refrain from utilising this table in the context of an MST study.

	A	B	C
1	Origin	hCYTB48	HF183
2	human	presence	105325
3	Human	positive	109321
4	Cow	negative	non tested
5	Bovine	absence	105
6	human	positive	72592
7	human	positive	98325
8	Bovine	negative	no data
9	Pig	absence	0

Text are not allowed

Bad name for class

R is case sensitive

One origin, one label

	A	B	C
1	Origin	hCYTB48	HF183
2	human	presence	105325
3	Human	positive	109321
4	Cow	negative	non tested
5	Bovine	absence	105
6	human	positive	72592
7	human	positive	98325
8	Bovine	negative	no data
9	Pig	absence	0

Step 1: Rectify the 'Class' label and the labels concerning to the origin of pollution. Modify the text entries to the appropriate numerical ones.

	A	B	C
1	Class	hCYTB48	HF183
2	Human	1	105325
3	Human	1	109321
4	Cow	0	
5	Cow	0	105
6	Human	1	72592
7	Human	1	98325
8	Cow	0	
9	Pig	0	0

Step 2: Write the metadata.

Qualitative (presence/absence)

Quantitative (numeric)

	A	B	C
1	Class	hCYTB48	HF183
2		0	1
3		0.0005	100
4		1	1
5	Human	1	105325
6	Human	1	109321
7	Cow	0	
8	Cow	0	105
9	Human	1	72592
10	Human	1	98325
11	Cow	0	
12	Pig	0	0

Please ensure that these rows are empty

For hCYTB48 (PCR)
0.0005 = dilution 1/2000

For HF183 (qPCR)
100 = detection limit

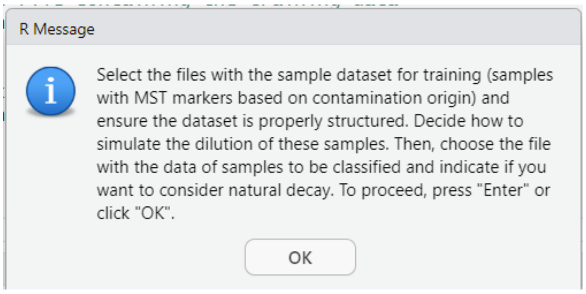
Both will be used in the training

Remove text entries

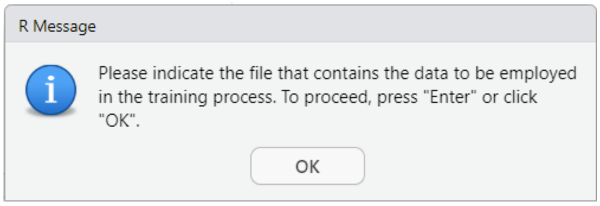
Note that the detection limit for qualitative samples is related to the maximum dilution at which a positive result is obtained using a presence/absence method.

Finally, once your data is properly formatted, save the file in a compatible format such as Excel or LibreOffice Calc.

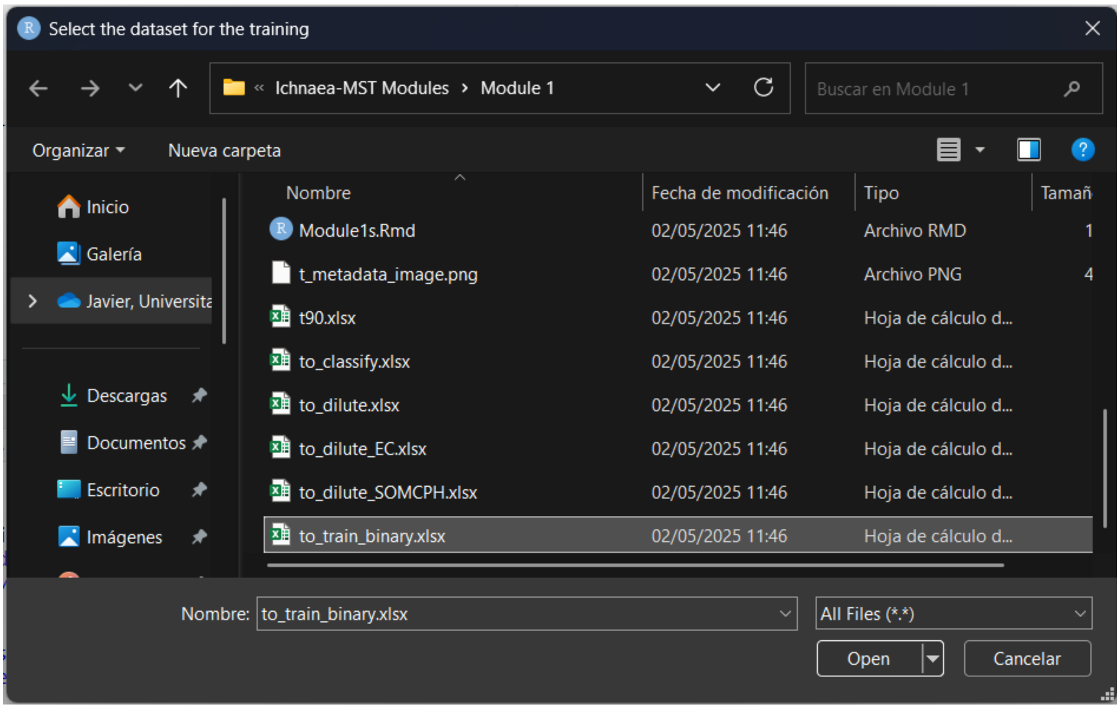
Turn back to the program. Chunk 3 starts with an info pop up indicating what steps are being taken by the program.



Once the ‘Ok’ button is clicked a new information pop up appears:



Now you can select which file contains your dataset for training.



The present chunk concludes at the point at which the file is uploaded.

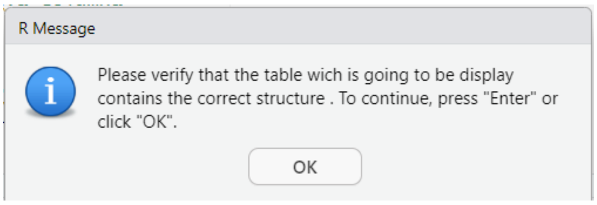
Chunk 4

This chunk contains the R script that retrieves the metadata from the uploaded data.

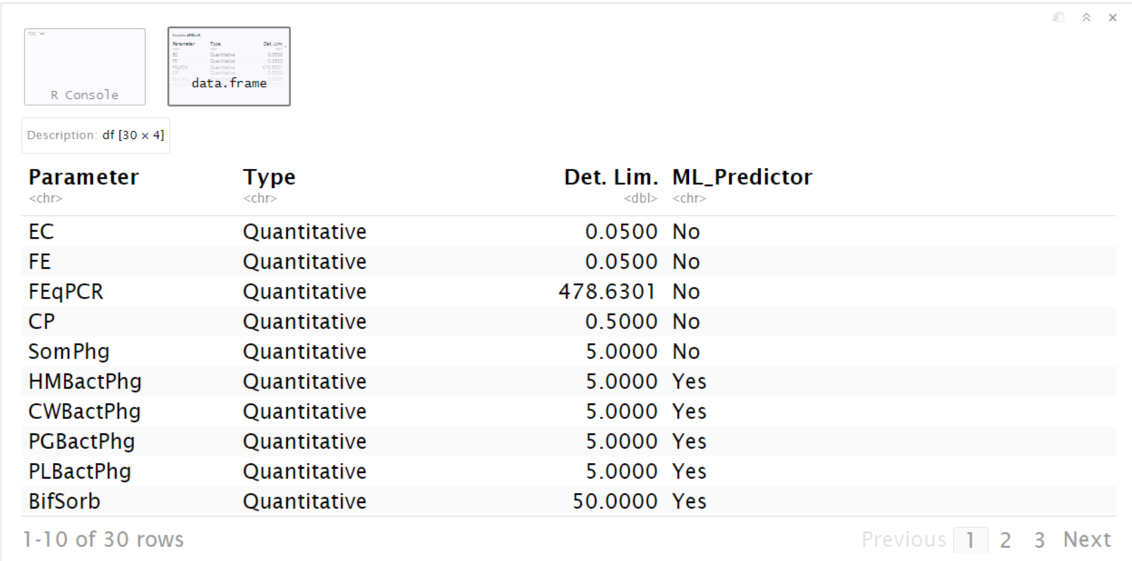
Chunk 5

This chunk contains the R code that displays how the uploaded dataset is organised. Please check that your data is correct. If not, make the necessary amendments, go back to Chunk 3 and try again.

The chunk starts with an info pop-up.



After that, the user can verify the structure of the data in two ways. The first method involves navigating below the chunk, where the user can click on the number in the bottom right-hand corner to switch between parts of the table. Alternatively, they can visualise the PNG file named 't_metadata_image.png', which is displayed on the next page.



Parameter <chr>	Type <chr>	Det. Lim. <dbl>	ML_Predictor <chr>
EC	Quantitative	0.0500	No
FE	Quantitative	0.0500	No
FEqPCR	Quantitative	478.6301	No
CP	Quantitative	0.5000	No
SomPhg	Quantitative	5.0000	No
HMBactPhg	Quantitative	5.0000	Yes
CWBactPhg	Quantitative	5.0000	Yes
PGBactPhg	Quantitative	5.0000	Yes
PLBactPhg	Quantitative	5.0000	Yes
BifSorb	Quantitative	50.0000	Yes

1-10 of 30 rows   Previous 1 2 3 Next

	Parameter	Type	Det. Lim.	ML_Predictor
1	EC	Quantitative	0.0500	No
2	FE	Quantitative	0.0500	No
3	FEqPCR	Quantitative	478.6301	No
4	CP	Quantitative	0.5000	No
5	SomPhg	Quantitative	5.0000	No
6	HMBactPhg	Quantitative	5.0000	Yes
7	CWBactPhg	Quantitative	5.0000	Yes
8	PGBactPhg	Quantitative	5.0000	Yes
9	PLBactPhg	Quantitative	5.0000	Yes
10	BifSorb	Quantitative	50.0000	Yes
11	BifTot	Quantitative	50.0000	No
12	HMBif	Quantitative	64000.0000	Yes
13	CWBif	Quantitative	30000.0000	Yes
14	PGNeo	Quantitative	36000.0000	Yes
15	PLBif	Quantitative	66000.0000	Yes
16	TLBif	Quantitative	64000.0000	No
17	BacR	Quantitative	478.6301	Yes
18	Pig2Bac	Quantitative	478.6301	Yes
19	AllBac	Quantitative	478.6301	No
20	HF183TaqMan	Quantitative	478.6301	Yes
21	HMMit	Quantitative	16000.0000	Yes
22	CWMit	Quantitative	10000.0000	Yes
23	PGMit	Quantitative	16000.0000	Yes
24	PLMit	Quantitative	10000.0000	Yes
25	Acesulfame	Quantitative	50.0000	Yes
26	Cyclamate	Quantitative	50.0000	Yes
27	Saccharain	Quantitative	50.0000	Yes
28	Sucralose	Quantitative	50.0000	Yes
29	Adeno	Quantitative	40000.0000	Yes
30	NoV	Quantitative	800.0000	Yes

In the image above, the markers indicated are:

Marker	Description	Used for	Type	Detected by
EC	*Escherichia coli*	Pollution load	Bacteria	Culture
FE	Faecal (intestinal) enterococci	Pollution load	Bacteria	Culture
CP	Spores of *Clostridium perfringens*	Pollution load	Spore	Culture
SomPhg	Somatic coliphages	Pollution load	Virus	Culture
HMBactPhg	Human phage	Human	Virus	Culture
CWBactPhg	Bovine phage	Bovine	Virus	Culture
PGBactPhg	Porcine phage	Porcine	Virus	Culture
PLBactPhg	Poultry phage	Poultry	Virus	Culture

Marker	Description	Used for	Type	Detected by
BifSorb	Bifidobacteria cultivated onto sorbitol	Human	Bacteria	Culture
BifTot	Total bifidobacteria 1	Pollution load	Bacteria	Culture
HMBif	Human bifidobacteria	Human	Bacteria	Molecular
CWBif	Bovine bifidobacteria	Bovine	Bacteria	Molecular
PGNeo	Porcine marker *Neoscardovia arbecensis*	Porcine	Bacteria	Molecular
PLBif	Poultry bifidobacteria	Poultry	Bacteria	Molecular
TLBif	Total bifidobacteria 2	Pollution load	Bacteria	Molecular
BacR	Marker of bovine contamination	Bovine	Bacteria	Molecular
Pig2Bac	Marker of porcine contamination	Porcine	Bacteria	Molecular
AllBac	Unspecific marker of faecal contamination	Pollution load	Bacteria	Molecular
HF183TaqMan	Marker of human contamination	Human	Bacteria	Molecular
FEqPCR	Faecal enterococci by qPCR	Pollution load	Bacteria	Molecular
HMMit	Human mitochondrial marker	Human	Eukaryote	Molecular
CWMit	Bovine mitochondrial marker	Bovine	Eukaryote	Molecular
PGMit	Porcine mitochondrial marker	Porcine	Eukaryote	Molecular
PLMit	Poultry mitochondrial marker	Poultry	Eukaryote	Molecular
Acesulfame	Artificial sweetener		Chemical	Chemical
Cyclamate	Artificial sweetener		Chemical	Chemical
Saccharain	Artificial sweetener		Chemical	Chemical
Sucralose	Artificial sweetener		Chemical	Chemical
Adeno	Human adenovirus	Human	Virus	Molecular
NoV	Norovirus, marker of human and porcine	Human / Porcine	Virus	Molecular

Chunk 6

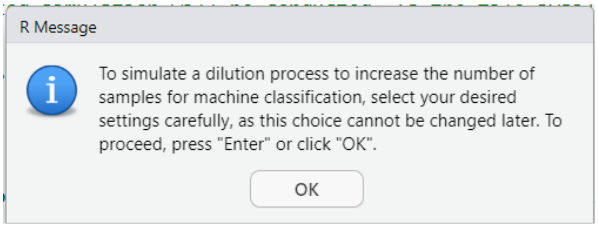
This chunk comprises the code that is responsible for the selection of the dilution process. In this section, the user can choose whether to employ a pair of dilution markers to simulate the dilution process or to carry out the dilution process itself from a state of no dilution to a maximum threshold.

It is imperative for users to comprehend this element in order to utilise H2O AutoML in an effective manner. The H2O AutoML system has been developed to automate data preparation steps such as centring and scaling. This functionality enables users to bypass these steps and instead input raw occurrence data directly. This approach has been shown to simplify the process and ensure that the data is ready for analysis without the need for additional preprocessing.

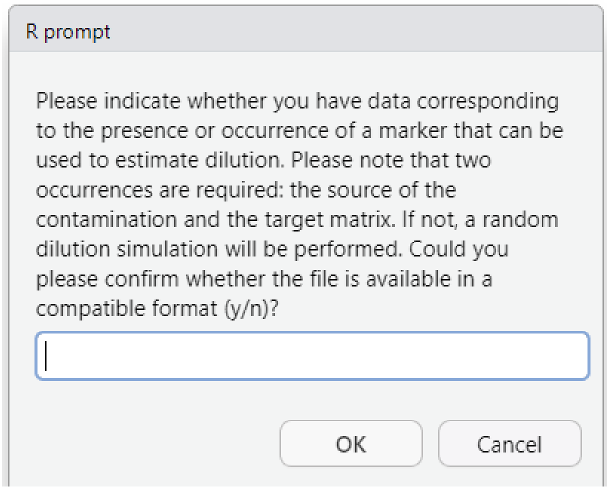
When discussing dilution markers, we are referring to the presence of specific entities such as bacteriophages, bacteria, molecules, or the number of genomic copies. These markers are measured both in the matrix where MST markers are determined (pollution origin) and in a similar matrix intended for classification. From the occurrence in both matrices, it is possible to define an empirical probability distribution for the marker in each matrix. The ratio between these two empirical distributions forms a new probability distribution that represents the dilution factor. Keep in mind that this factor is not a single value, but rather a pool of possible values.

When the MST marker has no numerical values (presence/absence) or there is no information available about the possible dilution factor, the user can decide the maximum dilution value to which the sample can be subjected. Note that we have chosen to set only a maximum value and have fixed the minimum to undiluted, so a pool of possible values will be generated ranged from no dilution until the value indicated by the user. To facilitate the introduction of this factor, it has been defined as (occurrence in the source matrix) / (occurrence in the matrix to be classified). This way, if there is no dilution process, the factor is 1; if it is diluted 10 times, the factor is 10; for 100 times, it would be 100, and so on.

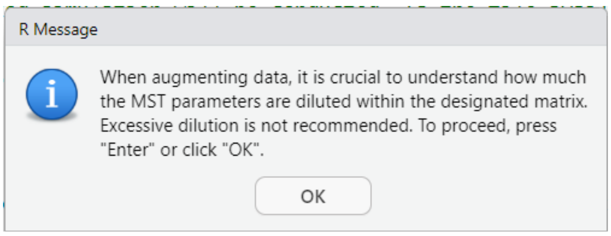
Turn back to the program, when running this chunk, the following pop-up appears.



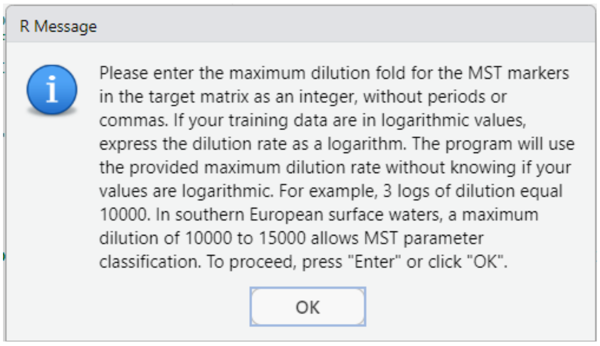
After clicking the 'OK' button, the user is prompted to indicated if a dataset containing the occurrences of a parameter measured in the matrix origin of pollution and the matrix to be classified that can be used as a marker for determine the factor of dilution.



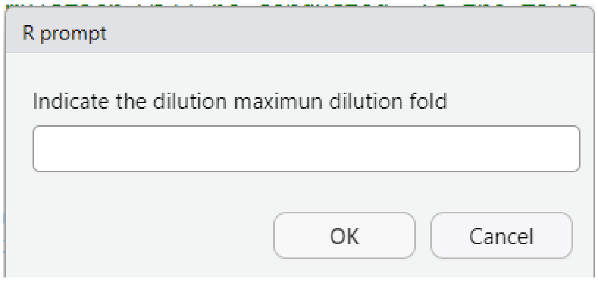
If the answer is negative the program displays a series of pop-up windows indicating that the user should indicate the maximum threshold of the dilution factor, please be careful and indicate a realistic value.



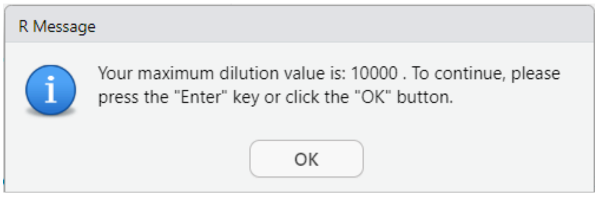
After clicking the 'OK' button, a new pop-up appears. This pop-up provides guidance on how to enter the numerical dilution threshold value.



And it continues asking for the maximum dilution factor.



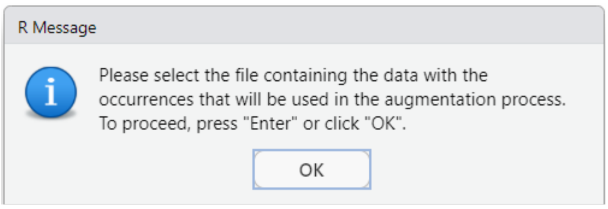
For example, after introducing a dilution factor of 10000, a new pop-up windows appears indicating the value.



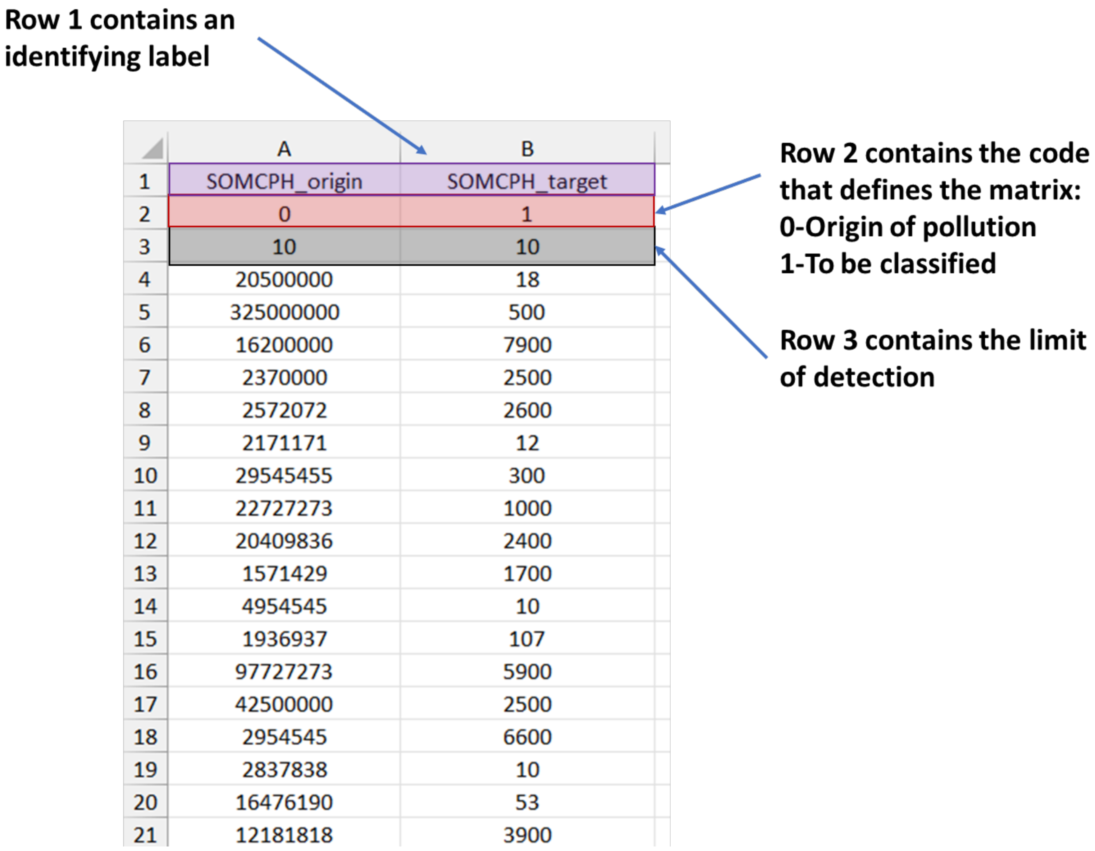
After clicking 'OK' this chunk ends here and the program continues in the chunk 7.

On the contrary, if the user answers affirmatively to the question about the availability of a dataset containing the both occurrences, the program continues from the pop-up with the text: *'Please indicate whether you have data corresponding to the presence or occurrence of a marker that can be used to estimate dilution. Please note that two occurrences are required: the source of the contamination and the target matrix. If not, a random dilution simulation will be performed. Could you please confirm whether the file is available in a compatible format (y/n)?'*

Once the user answer affirmatively, the program continues with the following pop-up window.

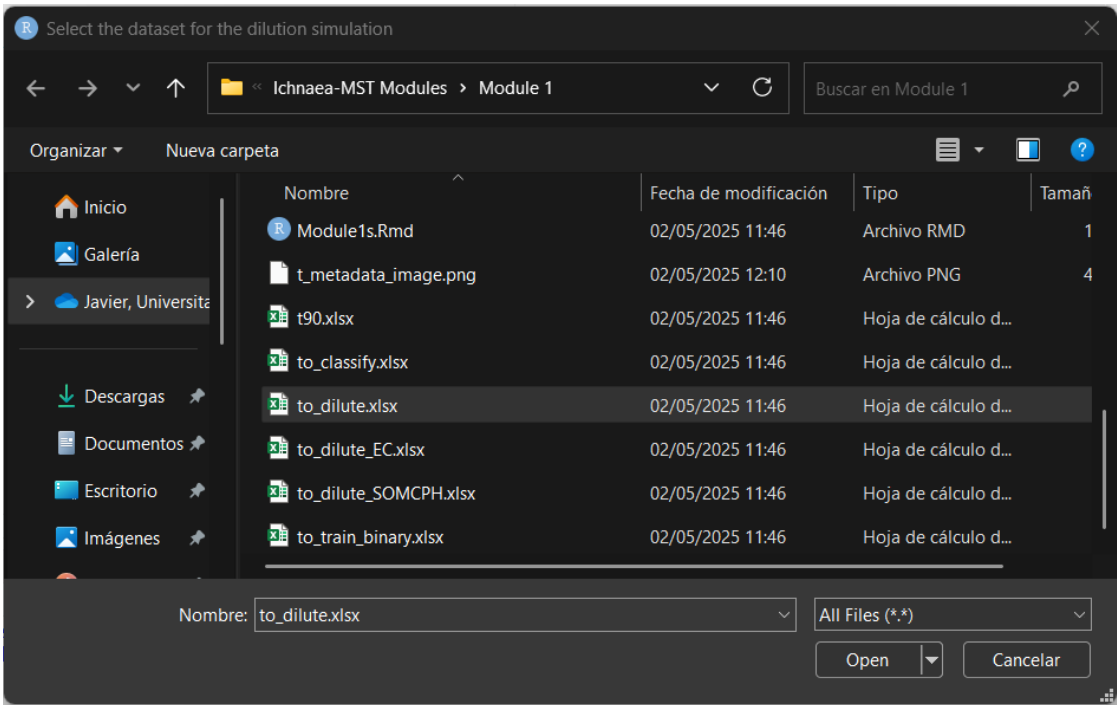


In the following page, is illustrated how the data should be organised to ensure it is in a compatible format.



	A	B
1	SOMCPH_origin	SOMCPH_target
2	0	1
3	10	10
4	20500000	18
5	325000000	500
6	16200000	7900
7	2370000	2500
8	2572072	2600
9	2171171	12
10	29545455	300
11	22727273	1000
12	20409836	2400
13	1571429	1700
14	4954545	10
15	1936937	107
16	97727273	5900
17	42500000	2500
18	2954545	6600
19	2837838	10
20	16476190	53
21	12181818	3900

After clicking the “OK” button, the program proceeds to a window where the user can upload the file containing the dataset to be used in the dilution process.



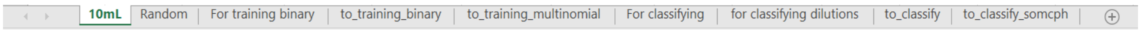
Once introduced the file, the present chunk concludes at this point.

Chunk 7

In this instance, the programme prompts the user to upload the data file that contains the data to be classified. In the web page can be find both an Excel and Calc file containing several sheets.

Both files contain nine sheets. The data relates to the European project 'Protecting the health of Europeans by improving methods for the detection of pathogens in drinking water and water used in food preparation', grant agreement ID: 311846. Further information about this project can be found at: https://cordis.europa.eu/project/id/311846.

Please find below a brief explanation of each data sheet.



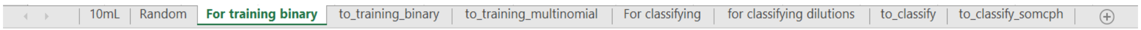
1) The sheet entitled "10mL" details the occurrence of several markers measured in 10 mL of water.

The markers are labelled as follows:

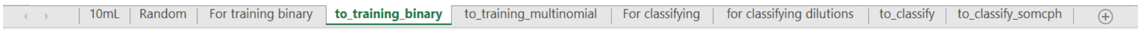
- ID_n: Identification number
- ID: Identification of the sample.
- Class_multinomial: Origin of pollution (Bovine, Equine, Human, Porcine and Poultry)
- Class_binomial: Origin of pollution (Human and Non-Human)
- EC: *Escherichia coli*
- FE: Faecal (intestinal) enterococci
- CP: Spores of *Clostridium perfringens*
- SomPhg: Somatic coliphages
- HMBactPhg: Human phage
- CWBactPhg: Bovine phage
- PGBactPhg: Porcine phage
- PLBactPhg: Poultry phage
- BifSorb: Bifidobacteria cultivated onto sorbitol
- BifTot: Total bifidobacteria 1
- HMBif: Human bifidobacteria
- CWBif: Bovine bifidobacteria
- PGNeo: Porcine Bifidobacteriaceae
- PLBif: Poultry bifidobacteria
- TLBif: Total bifidobacteria 2
- BacR: Marker of bovine contamination
- Pig2Bac: Marker of porcine contamination
- AllBac: Unspecific marker of faecal contamination
- HF183TaqMan: Marker of human contamination
- FEqPCR: Faecal enterococci by qPCR
- HMMit: Human mitochondrial marker
- CWMit: Bovine mitochondrial marker
- PGMit: Porcine mitochondrial marker
- PLMit: Poultry mitochondrial marker

- Acesulfame: Artificial sweetener E-950
- Cyclamate: Artificial sweetener E-952
- Saccharain: Artificial sweetener E-954
- Sucralose: Artificial sweetener E-955
- Adeno: Human adenovirus
- NoV: Norovirus, marker of human and porcine

2) The "Random" sheet contains two columns. The formula in the left column (labelled "Random selection") selects a random number from 1 to 113. The right column (labelled "Selected") contains the serie of random numbers that have been used to select samples with the same ID_n. These samples have been extracted from the complete dataset, and will be used in the classification process.

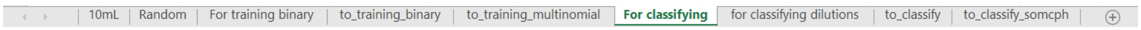


3) The sheet named "For training binomial" contains the data that will be used to define the training set. Please note that selected samples in the "Random" sheet have now been extracted.

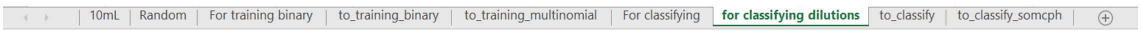


4) The "to_training_binomial" sheet contains the same data as the "For training binomial" sheet, but with additional metadata. Please note that the ID_n, ID and Class_multinomial have been removed, and the column named Class_binomial has been labelled as Class. It will be saved as a separate file with the same name of the data sheet.

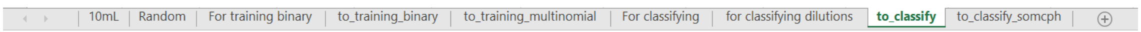
5) The sheet named "to_training_multinomial" contains the same data as the "For training binomial" sheet. In addition, the "to_training_multinomial" sheet includes metadata. Please note that the ID_n, ID and Class_binomial have been removed, and the column Class_multinomial has been renamed as Class. It will be saved as a separate file with the same name of the data sheet.



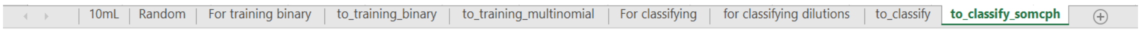
6) The "For classifying" sheet contains the data that has been extracted from the "10mL" sheet. This data will be used latter in the classification process.



7) The sheet named "for classifying dilutions" contains the same data and four decimal dilutions have been added (1/10; 1/100, 1/1000 and 1/10000). As per the established protocol, the occurrence of these diluted samples has been amended in accordance with their detection limits (DL). In the event of an occurrence < DL, the occurrence has been considered to be 0.



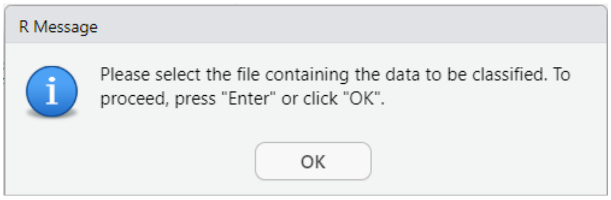
8) The sheet named "to_classify" contains the same data as the sheet named "for classifying dilutions", with the exception that non-MST markers have been removed. It will be saved as a separate file with the same name of the data sheet.



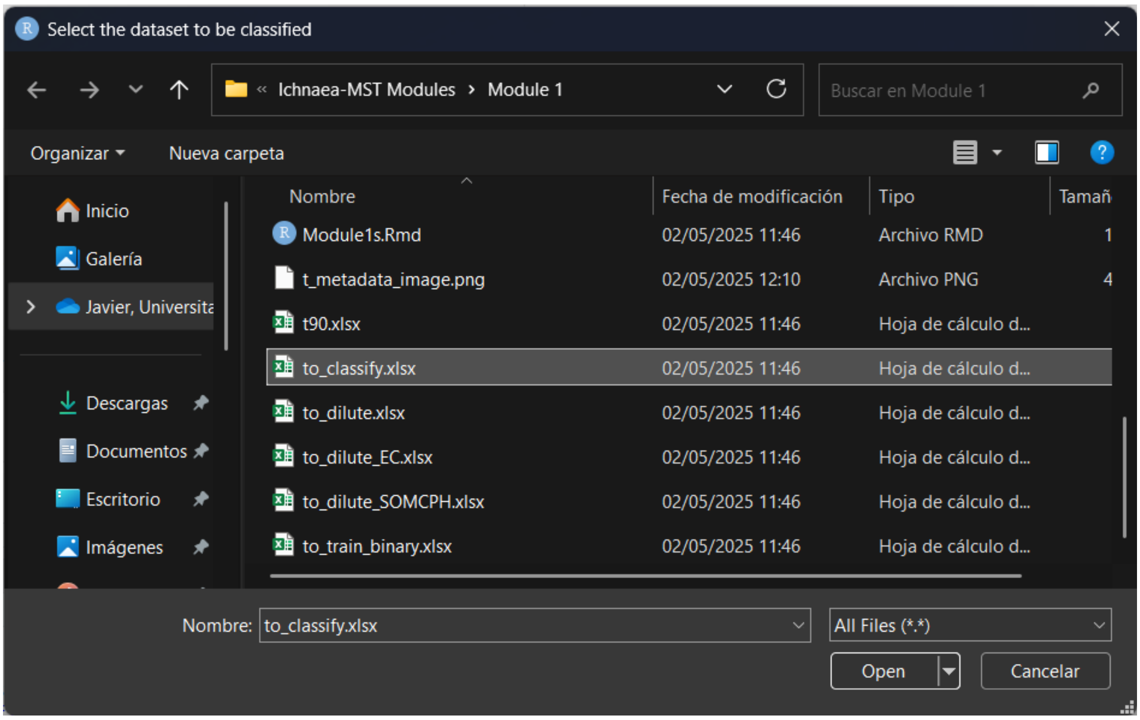
9) The sheet named "to_classify_somcph" contains the same information as the "for classifying dilutions" sheet. Please note that non-MST markers (with the exception of the "SomPhg" column) have been removed. This dataset will serve later as an illustrative example of how to modify a dataset. It will be saved as a separate file with the same name of the data sheet.

This manual will utilise the provided data for its intended purpose.

Once the chunk is executed the following pop-up window will appear.

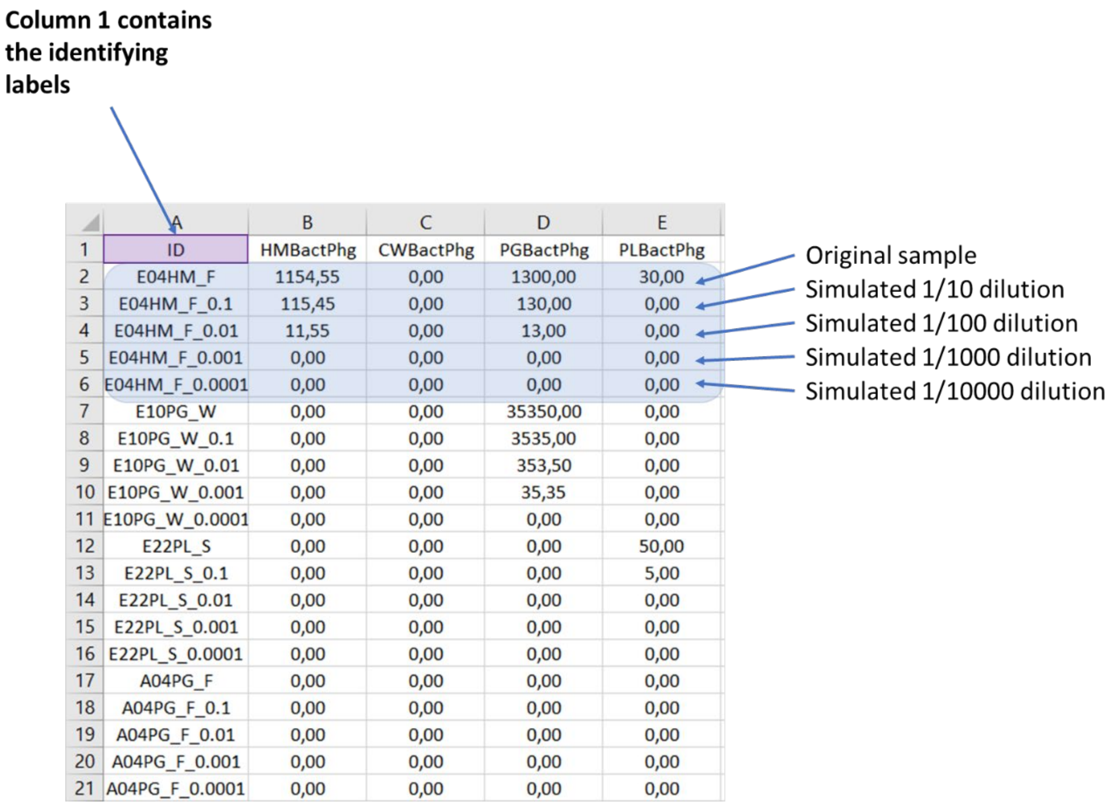


After clicking on the ‘OK’ button, a window will open where the user can select the file.



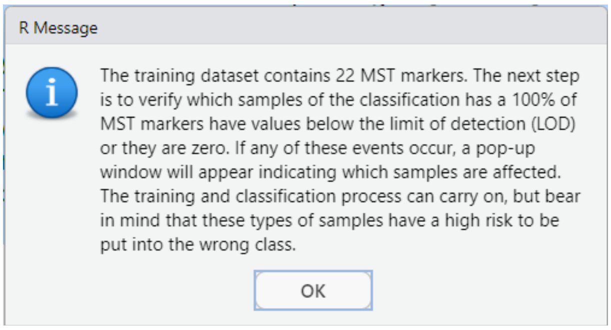
The only precaution to be exercised when organising the data is to place the labels to be used for identifying the samples in column 1 and to use the same labels as those used for identifying the MST markers in the dataset for training. When feasible, use of the same MST markers that are to be employed in the training process.

For instance, the Excel file named 'to_classify.xlsx' will be utilised as a case study. A mere segment of the file is displayed.



	A	B	C	D	E
1	ID	HMBactPhg	CWBactPhg	PGBactPhg	PLBactPhg
2	E04HM_F	1154,55	0,00	1300,00	30,00
3	E04HM_F_0.1	115,45	0,00	130,00	0,00
4	E04HM_F_0.01	11,55	0,00	13,00	0,00
5	E04HM_F_0.001	0,00	0,00	0,00	0,00
6	E04HM_F_0.0001	0,00	0,00	0,00	0,00
7	E10PG_W	0,00	0,00	35350,00	0,00
8	E10PG_W_0.1	0,00	0,00	3535,00	0,00
9	E10PG_W_0.01	0,00	0,00	353,50	0,00
10	E10PG_W_0.001	0,00	0,00	35,35	0,00
11	E10PG_W_0.0001	0,00	0,00	0,00	0,00
12	E22PL_S	0,00	0,00	0,00	50,00
13	E22PL_S_0.1	0,00	0,00	0,00	5,00
14	E22PL_S_0.01	0,00	0,00	0,00	0,00
15	E22PL_S_0.001	0,00	0,00	0,00	0,00
16	E22PL_S_0.0001	0,00	0,00	0,00	0,00
17	A04PG_F	0,00	0,00	0,00	0,00
18	A04PG_F_0.1	0,00	0,00	0,00	0,00
19	A04PG_F_0.01	0,00	0,00	0,00	0,00
20	A04PG_F_0.001	0,00	0,00	0,00	0,00
21	A04PG_F_0.0001	0,00	0,00	0,00	0,00

Following the selection of the file, the following window will pop-up.

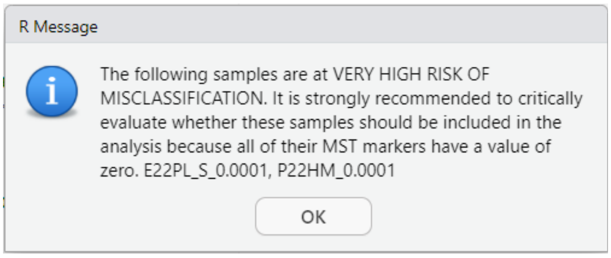


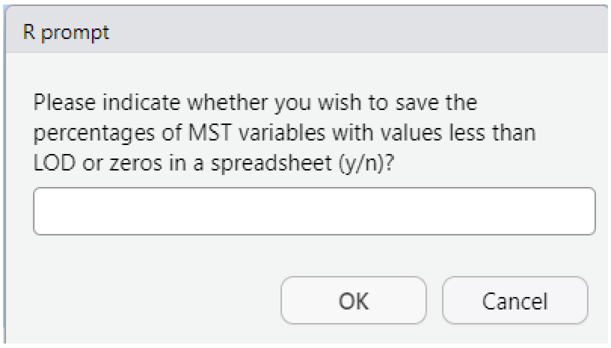
This step is a verification step. Please note that it is normal for classification samples to contain MST markers below their limit of detection (LOD), or even with values of zero. These markers can still retain discriminatory power, as their absence or low occurrence is often characteristic of samples originating from sources for which they are not indicative. In other words, the lack of signal can itself be informative in distinguishing between sources.

However, users should exercise caution when interpreting results from samples in which the overall presence of MST markers is extremely low or entirely absent. When all markers fall below the LOD or are zero, the classification model may lack sufficient information to make a reliable prediction. This can lead to increased uncertainty or even misclassification. In such cases, it is strongly recommended to critically evaluate whether these samples should be included in the analysis. Excluding them may help preserve the robustness and reliability of the classification outcomes.

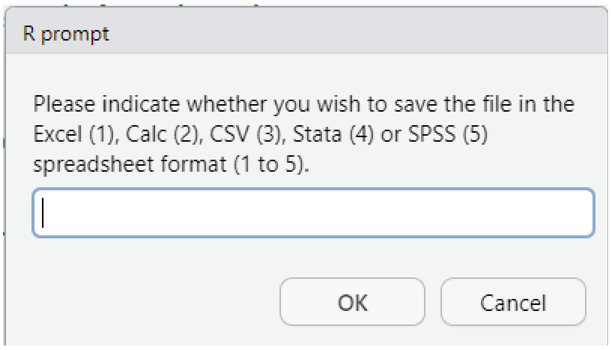
That said, the program is still capable of performing both training and classification even under these conditions. However, users must be aware that the results in such scenarios may be highly unreliable or entirely inaccurate.

After clicking on the 'OK' button, the verification is carried out. If any sample is detected a set of pop-ups will display.

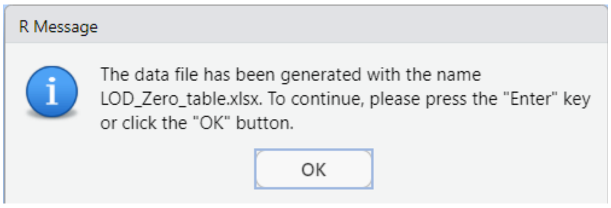




For instance, is selected the option to save the file. After indicating 'y' and clicking the 'OK' button the following pop-up will appear.



After selecting to save the file as an Excel spreadsheet. The following window will appear.



The following pages show information regarding the % of MST markers contained in each sample whose values are below the LOD or are zeros.

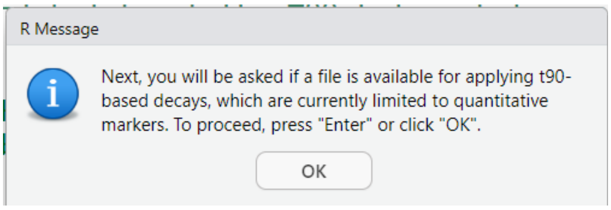
ID	LOD	Zeros
E04HM_F	45,45455	45,45455
E04HM_F_0.1	63,63636	63,63636
E04HM_F_0.01	72,72727	72,72727
E04HM_F_0.001	95,45455	95,45455
E04HM_F_0.0001	95,45455	95,45455
E10PG_W	59,09091	59,09091
E10PG_W_0.1	72,72727	72,72727
E10PG_W_0.01	72,72727	72,72727
E10PG_W_0.001	90,90909	90,90909
E10PG_W_0.0001	95,45455	95,45455
E22PL_S	72,72727	72,72727
E22PL_S_0.1	72,72727	72,72727
E22PL_S_0.01	81,81818	81,81818
E22PL_S_0.001	95,45455	95,45455
E22PL_S_0.0001	100	100
A04PG_F	68,18182	68,18182
A04PG_F_0.1	81,81818	81,81818
A04PG_F_0.01	86,36364	86,36364
A04PG_F_0.001	95,45455	95,45455
A04PG_F_0.0001	95,45455	95,45455
A08HM_W	54,54545	54,54545
A08HM_W_0.1	68,18182	68,18182
A08HM_W_0.01	81,81818	81,81818
A08HM_W_0.001	90,90909	90,90909
A08HM_W_0.0001	90,90909	90,90909
A22CW_S	77,27273	72,72727
A22CW_S_0.1	81,81818	81,81818
A22CW_S_0.01	86,36364	86,36364
A22CW_S_0.001	90,90909	90,90909
A22CW_S_0.0001	90,90909	90,90909
P06PG_W	63,63636	59,09091
P06PG_W_0.1	72,72727	72,72727
P06PG_W_0.01	90,90909	90,90909
P06PG_W_0.001	90,90909	90,90909
P06PG_W_0.0001	95,45455	95,45455

ID	LOD	Zeros
P16CW_SP	81,81818	81,81818
P16CW_SP_0.1	81,81818	81,81818
P16CW_SP_0.01	81,81818	81,81818
P16CW_SP_0.001	90,90909	90,90909
P16CW_SP_0.0001	90,90909	90,90909
P19PL_SP	77,27273	77,27273
P19PL_SP_0.1	77,27273	77,27273
P19PL_SP_0.01	86,36364	86,36364
P19PL_SP_0.001	90,90909	90,90909
P19PL_SP_0.0001	90,90909	90,90909
P22HM	50	45,45455
P22HM_0.1	72,72727	72,72727
P22HM_0.01	86,36364	86,36364
P22HM_0.001	90,90909	90,90909
P22HM_0.0001	100	100
D01HM_F	40,90909	40,90909
D01HM_F_0.1	68,18182	68,18182
D01HM_F_0.01	81,81818	81,81818
D01HM_F_0.001	90,90909	90,90909
D01HM_F_0.0001	90,90909	90,90909
F06HM_W	54,54545	54,54545
F06HM_W_0.1	68,18182	68,18182
F06HM_W_0.01	81,81818	81,81818
F06HM_W_0.001	90,90909	90,90909
F06HM_W_0.0001	90,90909	90,90909
F07PG_W	81,81818	81,81818
F07PG_W_0.1	86,36364	86,36364
F07PG_W_0.01	95,45455	95,45455
F07PG_W_0.001	95,45455	95,45455
F07PG_W_0.0001	95,45455	95,45455
F11HM_P	45,45455	45,45455
F11HM_P_0.1	63,63636	63,63636
F11HM_P_0.01	77,27273	77,27273
F11HM_P_0.001	86,36364	86,36364
F11HM_P_0.0001	90,90909	90,90909
F15HM_S	45,45455	45,45455
F15HM_S_0.1	59,09091	59,09091
F15HM_S_0.01	86,36364	86,36364
F15HM_S_0.001	86,36364	86,36364
F15HM_S_0.0001	90,90909	90,90909

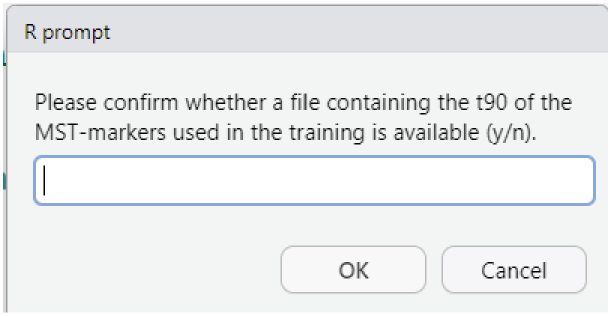
Chunk 8

This chunk contains the code responsible for importing the file (if it exists) that contains the t90 data for each of the MST markers. Ichanea-MST is able to take into account the natural decay by using a linear model based on t90 values. The utilisation of these values is optional. Regardless of whether or not these values are disposed of, please ensure that this chunk is executed, as it contains a prompt related to the application of natural decay.

Once is run the following informative pop-up appears.



After clicking the ‘OK’ button, a new pop-up window opens.



This window prompts the user to confirm the availability of a file containing the t90 values. If the file is found, the natural decay of each MST marker will be applied, along with any necessary amendments. When the user wishes to upload the file, the data must be organised such that the name of the markers is in row 1 and the t90 value is in row 2. It is imperative to ensure that the units of time employed for all the markers are consistent, for example, by using hours, days, or weeks, as appropriate.

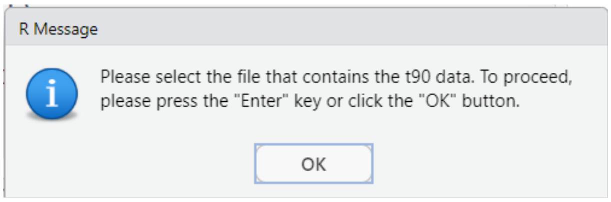
With the objective of providing a demonstrative illustration of the program's functionality, a t90 value of 48 hours is utilised for all the markers. Please refer to the bibliography to identify the most appropriate t90 value. A list of references is provided below.

Marker	**DOI**
AllBac	https://doi.org/10.1016/j.watres.2012.02.048
BacH	https://doi.org/10.1016/j.envpol.2021.116654
BACH	https://doi.org/10.1021/es2024498
BacH	https://doi.org/10.1016/j.watres.2012.02.048
BacH	https://doi.org/10.1038/srep35311
BacHum	https://doi.org/10.1016/j.envpol.2021.116654
BacR	https://doi.org/10.1021/es2024498
Bacteroidales	https://doi.org/10.1016/j.watres.2014.10.032
Bacteroidales	https://doi.org/10.1016/j.watres.2014.10.032
Bacteroidales	https://doi.org/10.1128/AEM.00444-15
BifCW	https://doi.org/10.1111/jam.14058
Bifidobacterum adolescentis	https://doi.org/10.1038/srep35311

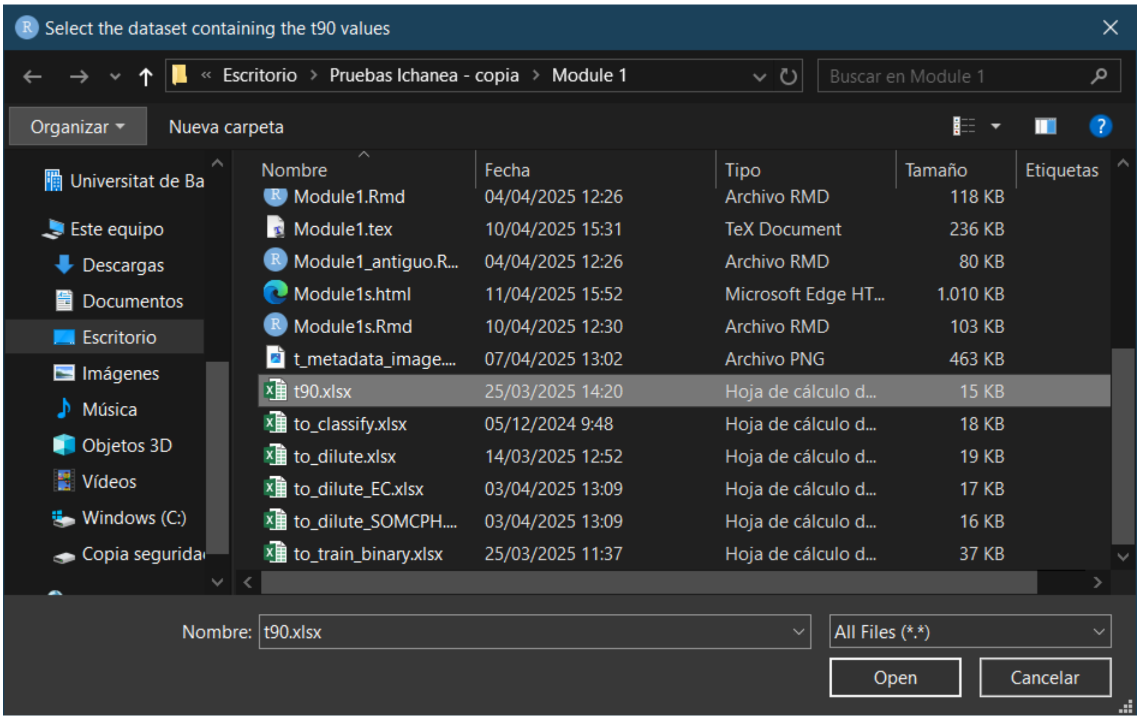
Marker	DOI
BifPL	https://doi.org/10.1111/jam.14058
BT-am	https://doi.org/10.1016/j.watres.2015.02.039
BT-am	https://doi.org/10.1007/s00253-018-9079-1
CF128	https://doi.org/10.1111/j.1462-2920.2009.01868.x
CF193	https://doi.org/10.1016/j.watres.2011.11.004
CF193	https://doi.org/10.1111/j.1462-2920.2009.01868.x
Chicken	https://doi.org/10.1038/srep35311
Chicken	https://doi.org/10.1038/srep35311
Cow_bacteroidales	https://doi.org/10.1016/j.watres.2014.10.032
CowM2	https://doi.org/10.1016/j.scitotenv.2018.09.108
CowM2	https://doi.org/10.1016/j.watres.2012.02.048
CowM3	https://doi.org/10.1016/j.scitotenv.2018.09.108
CPQ_056	https://doi.org/10.1016/j.scitotenv.2020.141475
CPQ_056	https://doi.org/10.1016/j.wroa.2020.100067
CPQ_056	https://doi.org/10.1016/j.wroa.2020.100067
CPQ_056	https://doi.org/10.1016/j.envpol.2021.116654
CPQ_064	https://doi.org/10.1016/j.envpol.2021.116654
CrAssphage	https://doi.org/10.1016/j.watres.2018.10.088
CrAssphage	https://doi.org/10.1016/j.watres.2019.02.042
Dog_bacteroidales	https://doi.org/10.1016/j.watres.2014.10.032
EC23S857	https://doi.org/10.1016/j.scitotenv.2018.09.108
ENT-23	https://doi.org/10.1016/j.watres.2015.02.039
ENT-23	https://doi.org/10.1007/s00253-018-9079-1
Entero1a	https://doi.org/10.1016/j.scitotenv.2018.09.108
Escherichia coli	https://doi.org/10.1016/j.scitotenv.2020.141475
Escherichia coli	https://doi.org/10.1016/j.scitotenv.2018.09.108
Escherichia coli	https://doi.org/10.1016/j.watres.2015.02.039
Escherichia coli	https://doi.org/10.1016/j.watres.2018.10.088
Escherichia coli	https://doi.org/10.1007/s00253-018-9079-1
Escherichia coli	https://doi.org/10.1016/j.watres.2019.02.042
Escherichia coli	https://doi.org/10.1021/es2024498
Escherichia coli	https://doi.org/10.1128/AEM.00444-15
Escherichia coli	https://doi.org/10.1111/lam.12285
Escherichia coli	https://doi.org/10.1016/j.watres.2012.02.048
Escherichia coli	https://doi.org/10.1111/jam.14058
GenBac3	https://doi.org/10.1016/j.scitotenv.2018.09.108
HadVs	https://doi.org/10.1016/j.watres.2018.10.088
HAV	https://doi.org/10.1111/lam.12285
HDN5	https://doi.org/10.1038/srep35311
HDN6	https://doi.org/10.1038/srep35311
HF134	https://doi.org/10.1111/j.1462-2920.2009.01868.x
HF183	https://doi.org/10.1016/j.scitotenv.2020.141475
HF183	https://doi.org/10.1016/j.watres.2011.11.004
HF183	https://doi.org/10.1016/j.scitotenv.2022.156025
HF183	https://doi.org/10.1016/j.wroa.2020.100067
HF183	https://doi.org/10.1016/j.envpol.2021.116654
HF183	https://doi.org/10.1016/j.watres.2018.10.088

Marker	DOI
HF183	https://doi.org/10.1016/j.watres.2019.02.042
HF183	https://doi.org/10.1111/j.1462-2920.2009.01868.x
HF183	https://doi.org/10.1111/lam.12285
HF183	https://doi.org/10.1038/srep35311
HF183	https://doi.org/10.1111/jam.14058
HMBif	https://doi.org/10.1016/j.watres.2019.02.042
HPyV	https://doi.org/10.1016/j.watres.2018.10.088
Hum_bacteroidales	https://doi.org/10.1016/j.watres.2014.10.032
Human	https://doi.org/10.1038/srep35311
Intestinal enterococci	https://doi.org/10.1016/j.scitotenv.2018.09.108
Intestinal enterococci	https://doi.org/10.1016/j.scitotenv.2022.156025
Intestinal enterococci	https://doi.org/10.1016/j.wroa.2020.100067
Intestinal enterococci	https://doi.org/10.1016/j.watres.2018.10.088
Intestinal enterococci	https://doi.org/10.1021/es2024498
Intestinal enterococci	https://doi.org/10.1128/AEM.00444-15
Intestinal enterococci	https://doi.org/10.1111/lam.12285
Intestinal enterococci	https://doi.org/10.1111/jam.14058
LA35	https://doi.org/10.1128/AEM.00444-15
Lactobacillus amylovorus	https://doi.org/10.1038/srep35311
PCytB	https://doi.org/10.1038/srep35311
PDN5	https://doi.org/10.1038/srep35311
Pig	https://doi.org/10.1038/srep35311
Pig	https://doi.org/10.1038/srep35311
Pig2Bac	https://doi.org/10.1111/jam.14058
Pig-2-Bac	https://doi.org/10.1038/srep35311
PMMov	https://doi.org/10.1016/j.wroa.2020.100067
PMMov	https://doi.org/10.1016/j.wroa.2020.100067
Rum2Bac	https://doi.org/10.1016/j.scitotenv.2018.09.108
Rum2Bac	https://doi.org/10.1111/jam.14058
Salmonella sp.	https://doi.org/10.1128/AEM.00444-15
Somatic coliphages	https://doi.org/10.1016/j.wroa.2020.100067
Somatic coliphages	https://doi.org/10.1016/j.watres.2019.02.042
Somatic coliphages	https://doi.org/10.1021/es2024498
Total coliforms	https://doi.org/10.1021/es2024498

If the answer is negative, the program will continue in chunk 9. If the answer is in the affirmative, the following pop-up appears.

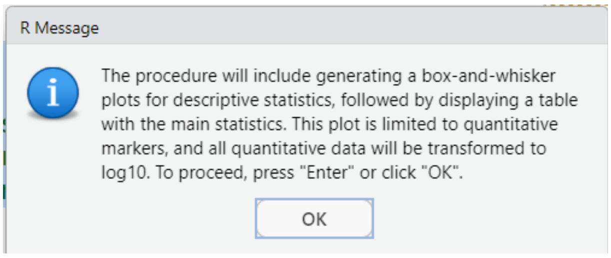


After that window the program will proceed to a file selection window, where the user can upload the data file.

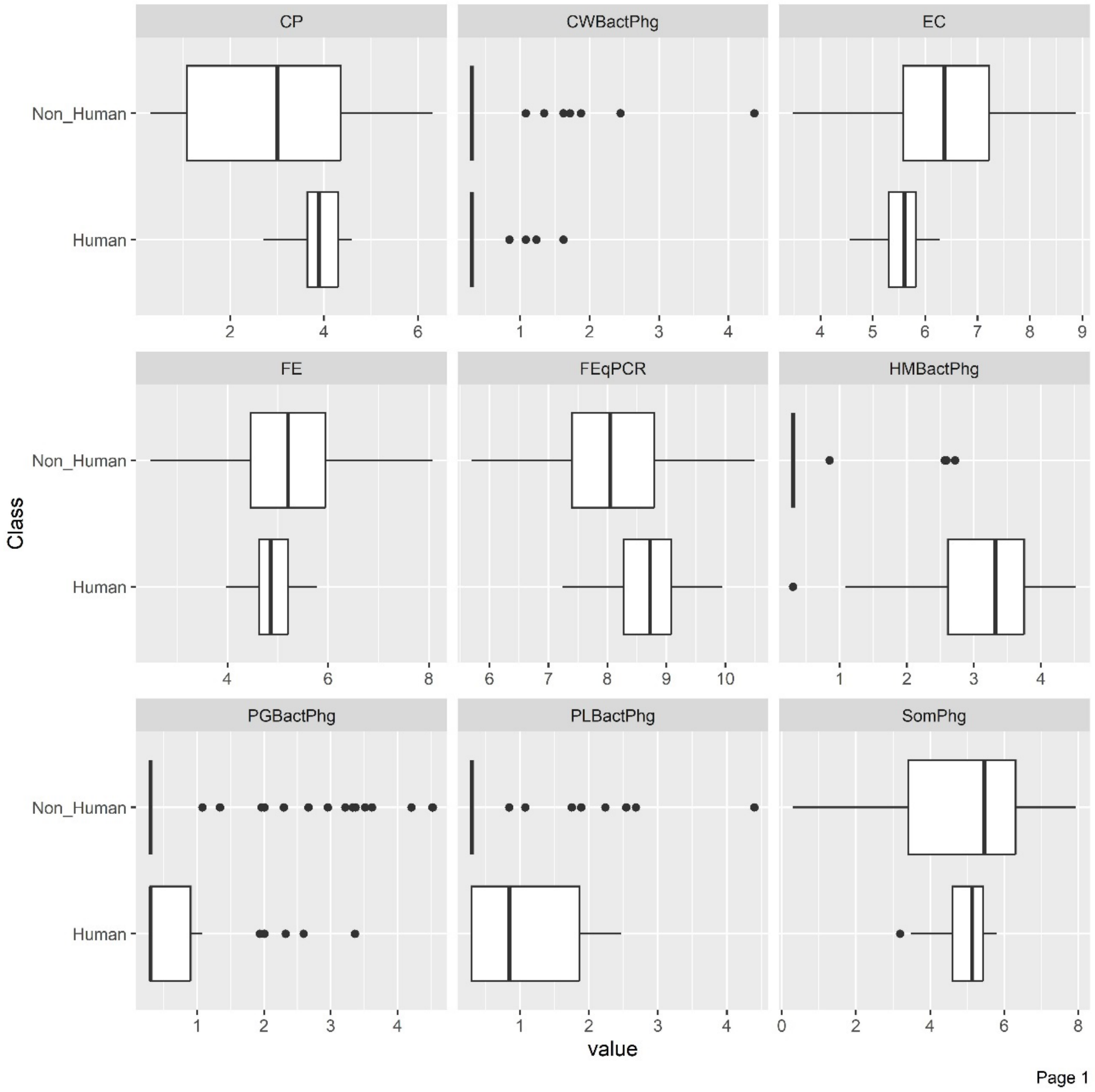


Chunk 9

This chunk contains the code that plots the log10 values of the parameters entered in the training file. It starts with an informative pop-up.



After clicking on the 'Ok' button, the program displays and saves as a JPEG file the boxplot graphs of each of the parameters, grouped into groups of 9 parameters per graph. For illustrative purposes, the following page contains a representation of one of the JPEG files containing the generated boxplot graphics. It is to be noted that by utilising the provided sample data, four graphic files will be obtained, which will be named Boxplot data to train_1.jpeg through Boxplot data to train_4.jpeg. It merits attention that the graphic sheets are numbered in the bottom right corner.



Furthermore, the results of the descriptive statistics can be viewed on the chunk's output screen. It is evident that the output screen displays multiple small preview windows, the content of which can be selected by clicking on the image.

Description: df [14 × 30]

	EC <dbl>	FE <dbl>	FEqPCR <dbl>	CP <dbl>	SomPhg <dbl>	HMBactPhg <dbl>	CWBactPhg <dbl>	PGBactPhg <dbl>	PLBactPhg <dbl>
nbr.val	98.000	98.000	98.000	98.000	98.000	98.000	98.000	98.000	98.000
nbr.null	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
nbr.na	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
min	3.477	2.467	5.696	0.301	0.301	0.301	0.301	0.301	0.301
max	8.870	8.079	10.492	6.316	7.925	4.516	4.370	4.527	4.398
range	5.392	5.612	4.796	6.015	7.624	4.215	4.069	4.226	4.097
sum	600.072	502.031	811.458	304.856	477.606	112.754	46.736	82.407	65.195
median	6.011	4.961	8.254	3.524	5.326	0.301	0.301	0.301	0.301
mean	6.123	5.123	8.280	3.111	4.874	1.151	0.477	0.841	0.665
SE.mean	0.111	0.097	0.104	0.162	0.188	0.138	0.057	0.113	0.078

1-10 of 14 rows | 1-10 of 30 columns Previous 1 2 Next

These descriptive statistics are calculated using the pastecs library. The individual statistics are explained below. The nbr.val indicates the number of samples of each parameter, nbr.null indicates the number of values that are 0 and nbr.na indicates the number of missing values. The min and max are the minimum and maximum values respectively, while the range is the difference between them. The sum is the sum of the parameter values, and the median is the median value. The mean is the average value and the SE.mean is the standard error of the mean. CI.mean is the confidence interval of the mean, var is the variance and std.dev is the standard deviation of the parameter values. Finally, coef.var is the coefficient of variation, which is the ratio of the standard deviation to the mean.

Chunk 10

This chunk comprises the code that performs the data augmentation process when the user elects to execute it based on a predefined threshold for the maximum dilution.

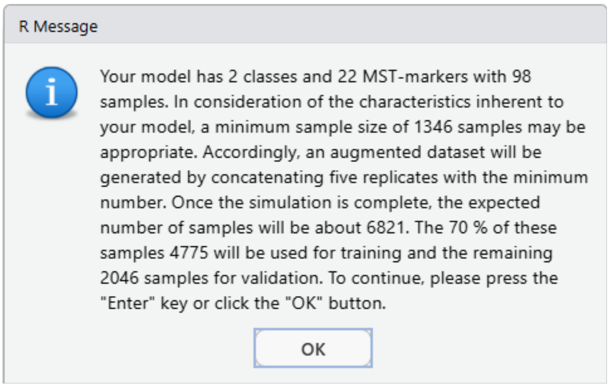
Chunk 11

This chunk comprises the code that performs the data augmentation process when the user elects to execute it based on the occurrence of a marker present in the pollution matrix and the matrix (or similar) to be classified.

In this context, it is crucial to highlight that the function's output is a list-type variable comprising the augmented dataset, and also encompasses the occurrence of the dilution marker measured in the matrix that is the origin of the pollution, as well as the occurrence of the dilution marker measured in the matrix that is similar to or equivalent to the matrix to be classified after applying the simulated dilution process. The latter dataset is not currently in use, but it has been included for informational purposes.

Chunk 12

This chunk contains the code that performs the power analysis to determine the minimum number of samples. After running the code, a pop-up window is displayed indicating the results.



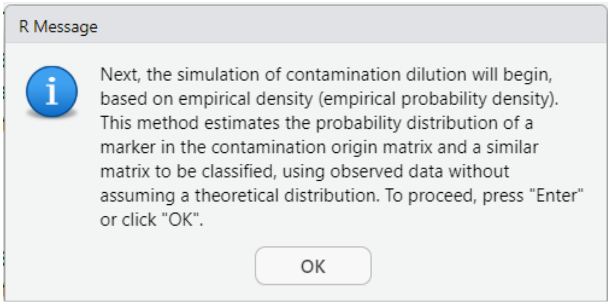
Please be advised that the pwr.f2.test function from the pwr library is utilized in this instance. This function has been specifically designed for power analysis in the context of general linear models (GLMs). GLMs represent the most straightforward algorithm that can be trained in H2O. It should be noted that other algorithms are more complex, and determining the appropriate sample size can also be more difficult due to the high capacity and flexibility of these models. As this test cannot be applied directly to those algorithms, using a sample size that is 5 times the minimum indicated by pwr.f2.test could be a reasonable approach for MST classification scenarios. Please verify that the performance metrics (e.g., accuracy, AUC) are satisfactory. Rajput et al. (2023) found that the sample size is considered suitable when it has appropriate effect sizes (≥ 0.5) and ML accuracy (≥ 80%). An increment in samples will not benefit as it will not significantly change the effect size and accuracy.

Rajput D, Wang WJ, Chen CC. Evaluation of a decided sample size in machine learning applications. BMC Bioinformatics. 2023 Feb 14;24(1):48. doi: 10.1186/s12859-023-05156-9. PMID: 36788550; PMCID: PMC9926644.

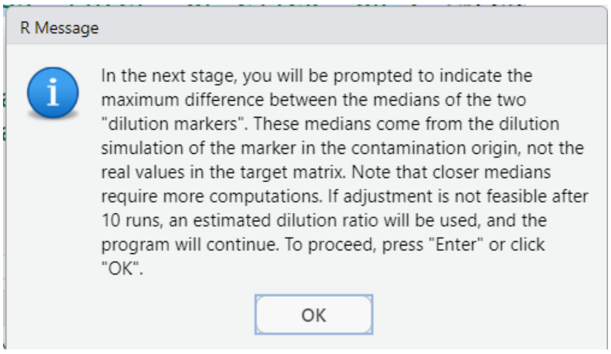
Chunk 13

This chunk contains the code to obtain the augmented dataset. There are two approaches. One involves using dilution markers, where the user only needs to indicate the degree of approximation that the occurrence in the sample to be classified and the simulated occurrence of the same marker in the original contamination matrix should have. The degree of concordance is measured as the log10 difference of the medians of both occurrences. A level of ±0.5 is generally more than adequate. Note that a smaller discrepancy between medians usually requires running more simulations. However, a very high discrepancy is counterproductive because it moves us further away from the 'most normal' values in the matrix to be classified. If after 10 simulations (in the console it is indicated how many simulations are running) it is not possible to meet the maximum discrepancy level between medians, the calculation continues, warning the user to choose a maximum dilution threshold. If you wish to skip this section, choose a discrepancy level of 0, as it is very unlikely that after the simulation the logarithmic values of the medians between the real values of the dilution marker in the matrix to be classified or similar and the log10 value of the simulated values will be exactly the same.

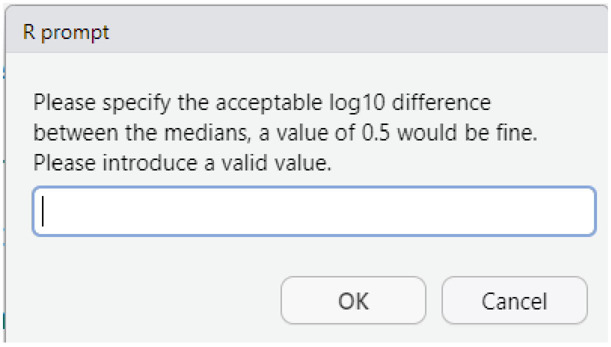
In the event that the user has selected the option to employ dilution markers, the following pop-up notification is displayed.



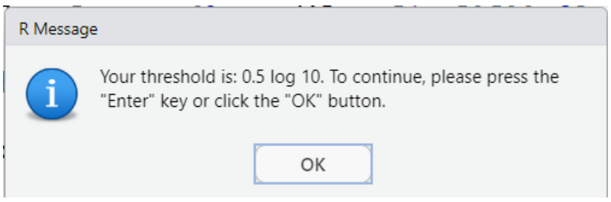
Following the selection of the 'OK' option, a new pop-up is displayed.



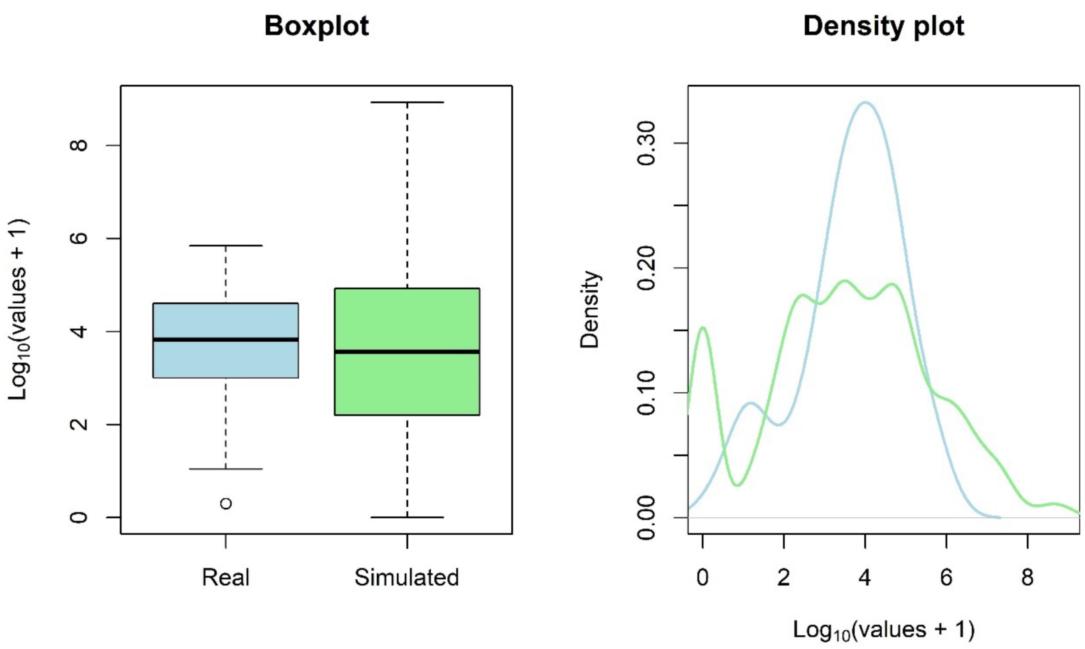
Upon selecting 'OK', the user is prompted to input the permissible range of variation between the log10-values of the medians. These correspond to the median occurrence of the marker used as a dilution control, measured in the target matrix, and the median occurrence of this marker measured in the source matrix of the contamination. The latter occurrence is subject to a simulated dilution process with resampling, and thus reflects the variation in occurrence that the rest of the markers will undergo. It should be noted that following the dilution simulation process, occurrences that have not been diluted are also included. This extends the range of occurrences.



For instance, a difference of 0.5 is indicated. Following the acceptance of the value, an informative pop-up is displayed.

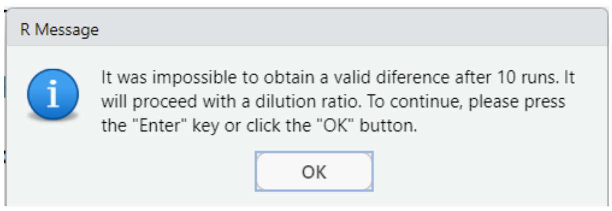


Following to the selection of 'OK', the augmentation of the data will commence. When the simulation is successfully concluded, the resultant data is presented as a boxplot and a density plot, in which the real values (light blue) of the occurrence of the selected marker in the matrix to be classified (or similar) are compared with the simulated values after applying the simulated dilutions to the real occurrence of the marker (light green). It should be noted that the marker selected in the matrix to be classified is used as a reference. Furthermore, it should be noted that the boxplot labelled 'Simulated' extends ‘upwards’ because it contains, in addition, the occurrences in the matrix origin of the contamination (the training dataset upload by the user).

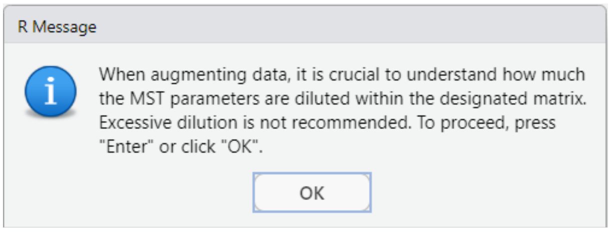


The range of the simulated data should be greater than the range of the real values. The density of the data depends on the degree of variability of the dilution ratio function and the degree of variability of the occurrence of the marker in the original matrix.

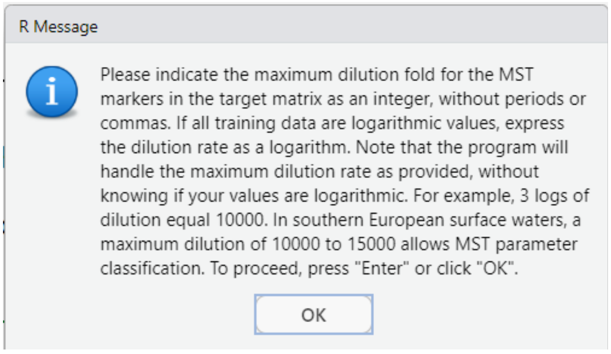
In the event of the difference being 0, the program continues trying to reach a no-difference. If no-difference is reached after 10 simulations (which is very likely), the data augmentation stops and the following pop-up will appear.



As null difference has not been reached, and consequently, no markers have been simulated. The code initiates an alternative process in which the user is prompted to specify the maximum dilution threshold. In the event of occurrence, the following warning is displayed.

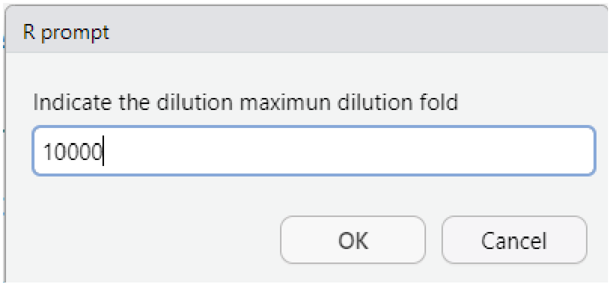


After clicking on the ‘OK’ button, a new pop-up will be displayed.

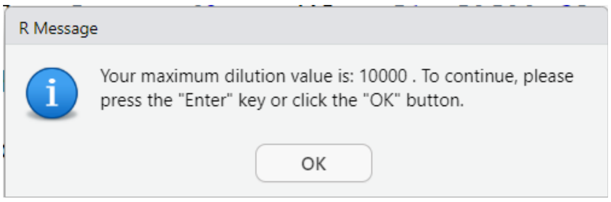


It is imperative to adjust to a reasonable degree of dilution. It should be noted that, despite the reluctance to employ logarithmic values, their utilisation remains a possibility in this context. It is evident that as the degree of dilution increases, there is a corresponding increase in the probability that the marker will be determined to be below the detection limit. Consequently, this would result in the marker being assigned a value of 0.

Subsequent to the selection of the 'OK' button, the user is prompted to input the desired value of dilution, for example 10000.



The value entered is displayed once more in the following pop-up window.

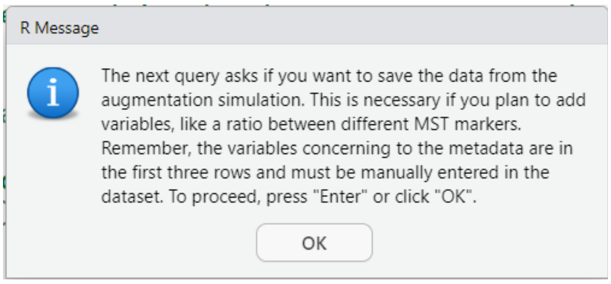


Following the selection of the 'OK' button, the process of data augmentation using the value indicated by the user the augmentation process is initiated.

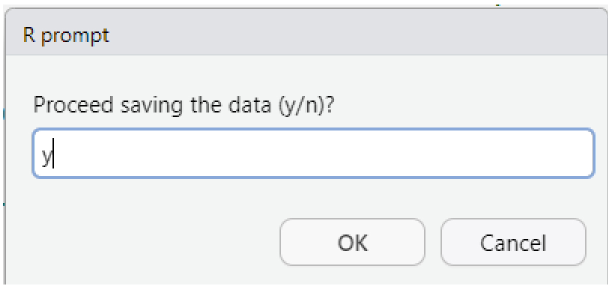
In case the user did not upload a file containing the marker in both matrices, the program will continue indicating

Chunk 14

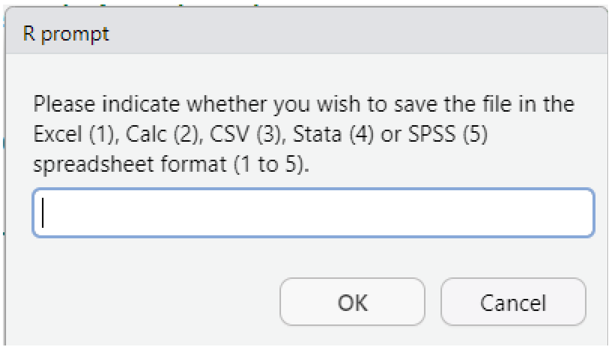
This section commences with a warning pop-up notification that an option to generate a file with the augmented dataset is available. This is particularly beneficial when dataset include a marker that is a covariate of another. One example of this is the use of the ratio between somatic coliphages (SOMCPH) and the GA17 phage of *Bacteroides thetaiotaomicron* (GA17PH) as MST marker. If this ratio – defined as log10(SOMCPH + 10)/log10(GA17PH + 10) – should be present in the dataset prior to the data augmentation process, it would also undergo a dilution and resampling simulation and therefore its value will become erroneous. This type of covariate or another one between one or several markers can be easily implemented manually in a data sheet. This part of the code has been designed to allow for this, allowing the user to modify the dataset. Just remember to add the metadata if you make modifications.



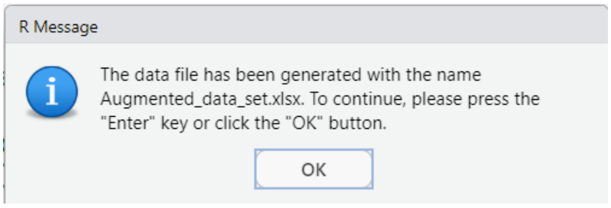
Once the 'OK' button has been selected, the user will be prompted to save the data in a new file. It is advisable to save the data, even if no changes are required.



If the answer is in the affirmative, the program will open a pop-up window prompting the user to enter the file format.



Once the format of the file has been selected, a new pop-up window will appear, displaying the name of the file. For instance, if option 1 is selected, the file will be saved as an Excel file named 'Augmented_data_set.xlsx'.

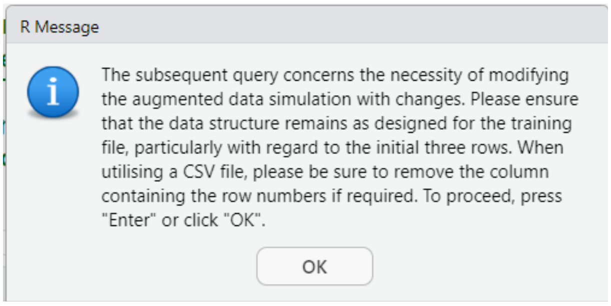


Chunk 15

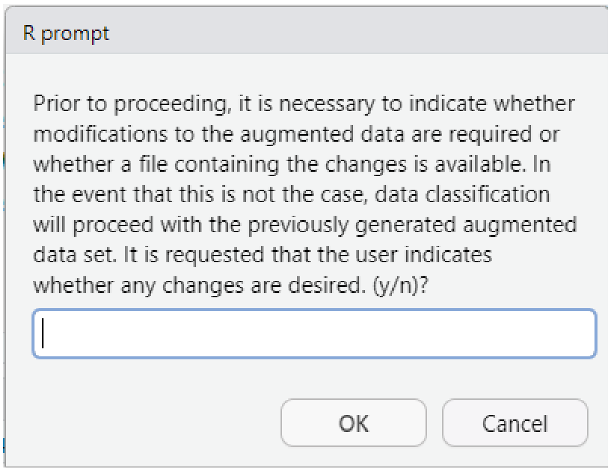
This component is responsible for loading any changes made to the augmented data. Please remember to indicate the metadata. R can perform this step of indicating the metadata, prompting the user to select this option.

Although unlikely, given that the training dataset requires hundreds to thousands of samples, it is possible that your dataset does not require data augmentation. Please execute the first eight chunks (including chunk 3) and reload the training data in chunk 15. This component has dual functionality, enabling the user to make the necessary amendments after data augmentation or to omit it when there are enough samples for machine learning.

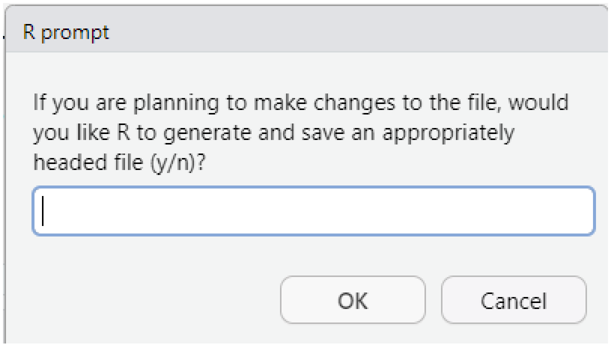
This chunk starts with a pop-up window.



Followed by a prompt, asking if you have a file with the modifications.

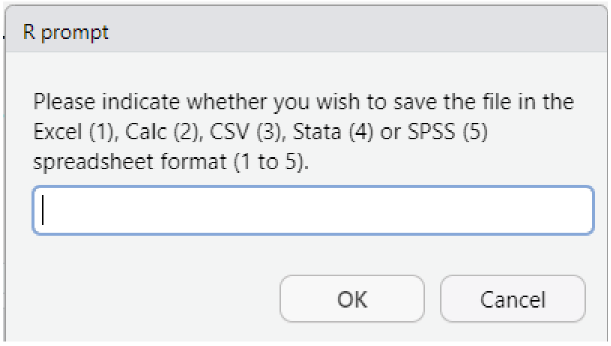


If the answer is affirmative, the programme will open a new pop-up window.

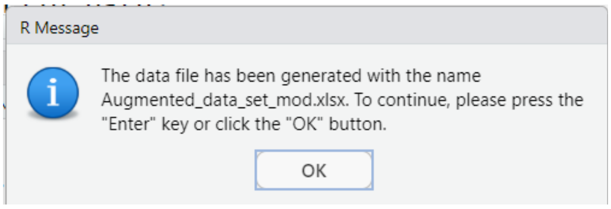


If the user answer affirmatively the program prompts the user to select the preferred option to save the file.

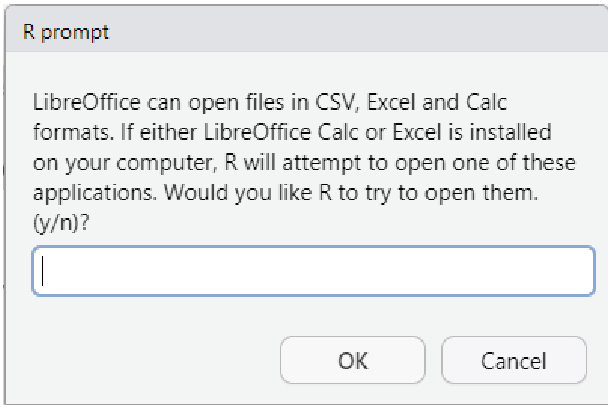
Important Note: This option is ideal if you intend to use the model currently being trained to classify a new dataset at a later stage (see the relevant section in Module II) and wish to verify that there is no significant divergence between the two datasets.



After the select the preferred option a file containing the metadata and the results of the augmentation is generated and saved in a new file named 'Augmented_data_set_mod'. For instance, is selected the first option.

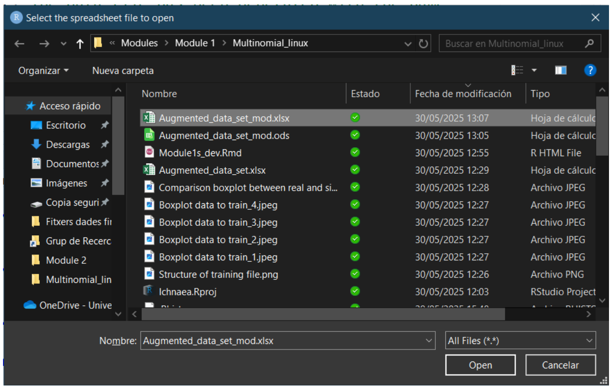


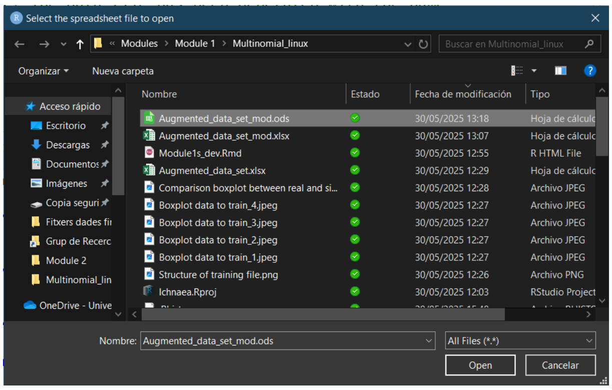
After that, the user is prompted to open the file containing the dataset to modify using LibreOffice Calc or Excel.



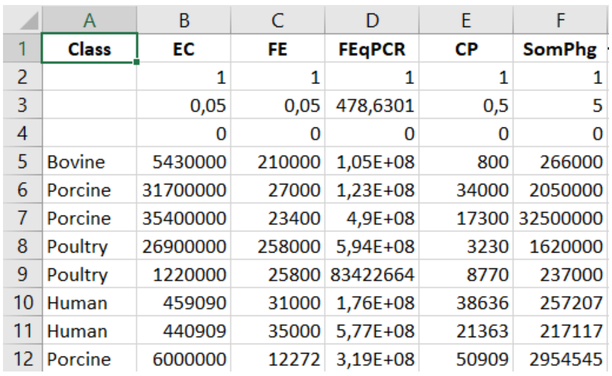
It should be noted that the selection or non-selection of any of these programs (LibreOffice Calc or Excel) is entirely at the discretion of the user; the spreadsheet containing the data with the decay modifications can be modified using the software with which the individual is most comfortable. Furthermore, the segment of the code that is responsible for initiating the application has been subjected to testing on the Windows 11 version 24h2 operating system and the Ubuntu versions 22.04 LTS (Jammy Jellyfish) and 24.04 LTS (Noble Numbat). It is important to note that, due to the inherent complexity of operating systems, there is a possibility that the application may not launch correctly. It is important to note that the R program has been designed to run on a continuous basis. This means that it will continue running whether or not the application is opened.

If the user answers affirmatively a window to upload the file will be displayed. Two screen captures follow this line: the left one shows the selection of an Excel file, and the right one shows the selection of a Calc file. After clicking the 'Open' button, R will try to launch Excel or Calc.

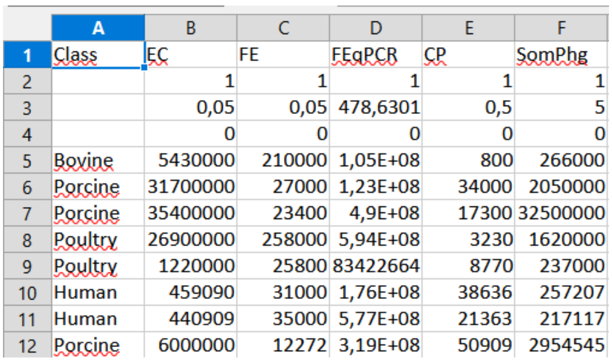




Two screen captures follow this line: the left one shows a capture of the file opened in Excel file, and the right one shows the file opened in Calc (this latter is the default option under Linux).

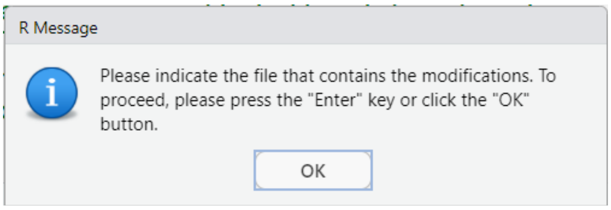


	A	B	C	D	E	F
1	**Class**	**EC**	**FE**	**FEqPCR**	**CP**	**SomPhg**
2		1	1	1	1	1
3		0,05	0,05	478,6301	0,5	5
4		0	0	0	0	0
5	Bovine	5430000	210000	1,05E+08	800	266000
6	Porcine	31700000	27000	1,23E+08	34000	2050000
7	Porcine	35400000	23400	4,9E+08	17300	32500000
8	Poultry	26900000	258000	5,94E+08	3230	1620000
9	Poultry	1220000	25800	83422664	8770	237000
10	Human	459090	31000	1,76E+08	38636	257207
11	Human	440909	35000	5,77E+08	21363	217117
12	Porcine	6000000	12272	3,19E+08	50909	2954545

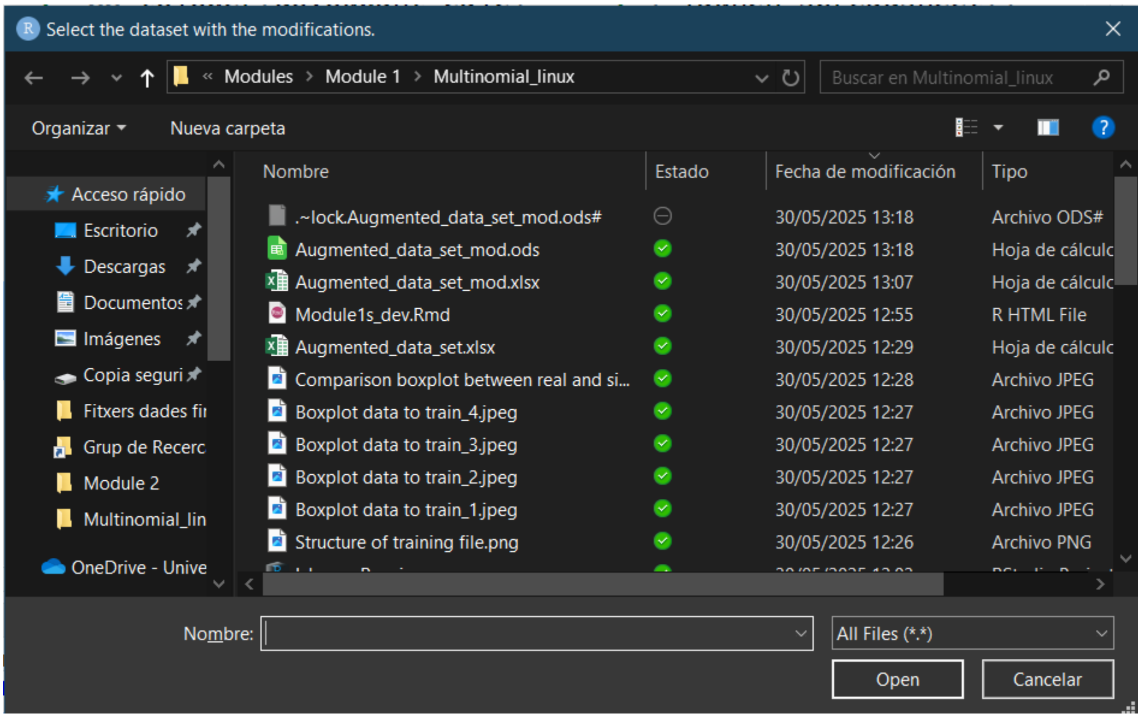


	A	B	C	D	E	F
1	Class	EC	FE	FEqPCR	CP	SomPhg
2		1	1	1	1	1
3		0,05	0,05	478,6301	0,5	5
4		0	0	0	0	0
5	Bovine	5430000	210000	1,05E+08	800	266000
6	Porcine	31700000	27000	1,23E+08	34000	2050000
7	Porcine	35400000	23400	4,9E+08	17300	32500000
8	Poultry	26900000	258000	5,94E+08	3230	1620000
9	Poultry	1220000	25800	83422664	8770	237000
10	Human	459090	31000	1,76E+08	38636	257207
11	Human	440909	35000	5,77E+08	21363	217117
12	Porcine	6000000	12272	3,19E+08	50909	2954545

After that or if the previous answer was negative or if you have edited the file manually, the program will then proceed with the following pop-up window.



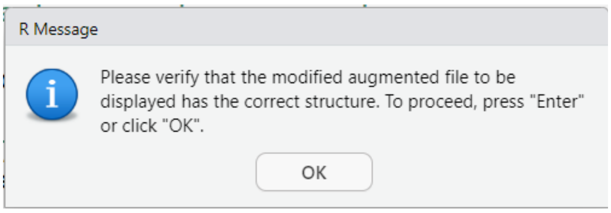
And after clicking the ‘OK’ button, a window to select the file to modify.



Chunk 16

This chunk shows the data structure of the augmented dataset if the user uploads a data file. For instance, a clone of the augmented dataset has been created, the only modification is to add the metadata.

When it is run the following window pops out.



After clicking the ‘OK’ button, it is shown the structure of the file.

R Console | data.frame

Description: df [30 x 4]

Parameter <chr>	Type <chr>	Det. Lim. <dbl>	ML_Predictor <chr>
EC	Quantitative	0.0500	No
FE	Quantitative	0.0500	No
FEqPCR	Quantitative	478.6301	No
CP	Quantitative	0.5000	No
SomPhg	Quantitative	5.0000	No
HMBactPhg	Quantitative	5.0000	Yes
CWBactPhg	Quantitative	5.0000	Yes
PGBactPhg	Quantitative	5.0000	Yes
PLBactPhg	Quantitative	5.0000	Yes
BifSorb	Quantitative	50.0000	Yes

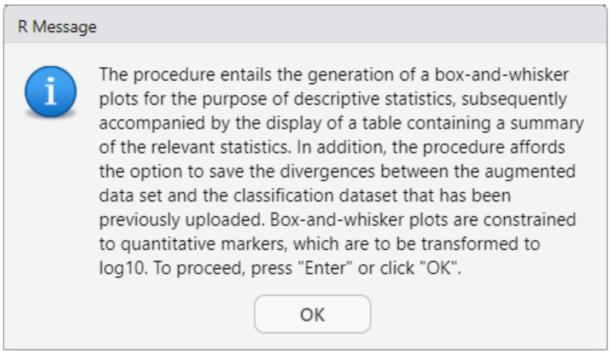
1-10 of 30 rows Previous 1 2 3 Next

In the current example, no changes have been made to the dataset. Therefore, the structure must be identical to the structure of the uploaded user training file (see Chunk 5). Furthermore, an image is generated, named 'modified_augmented_data.png'.

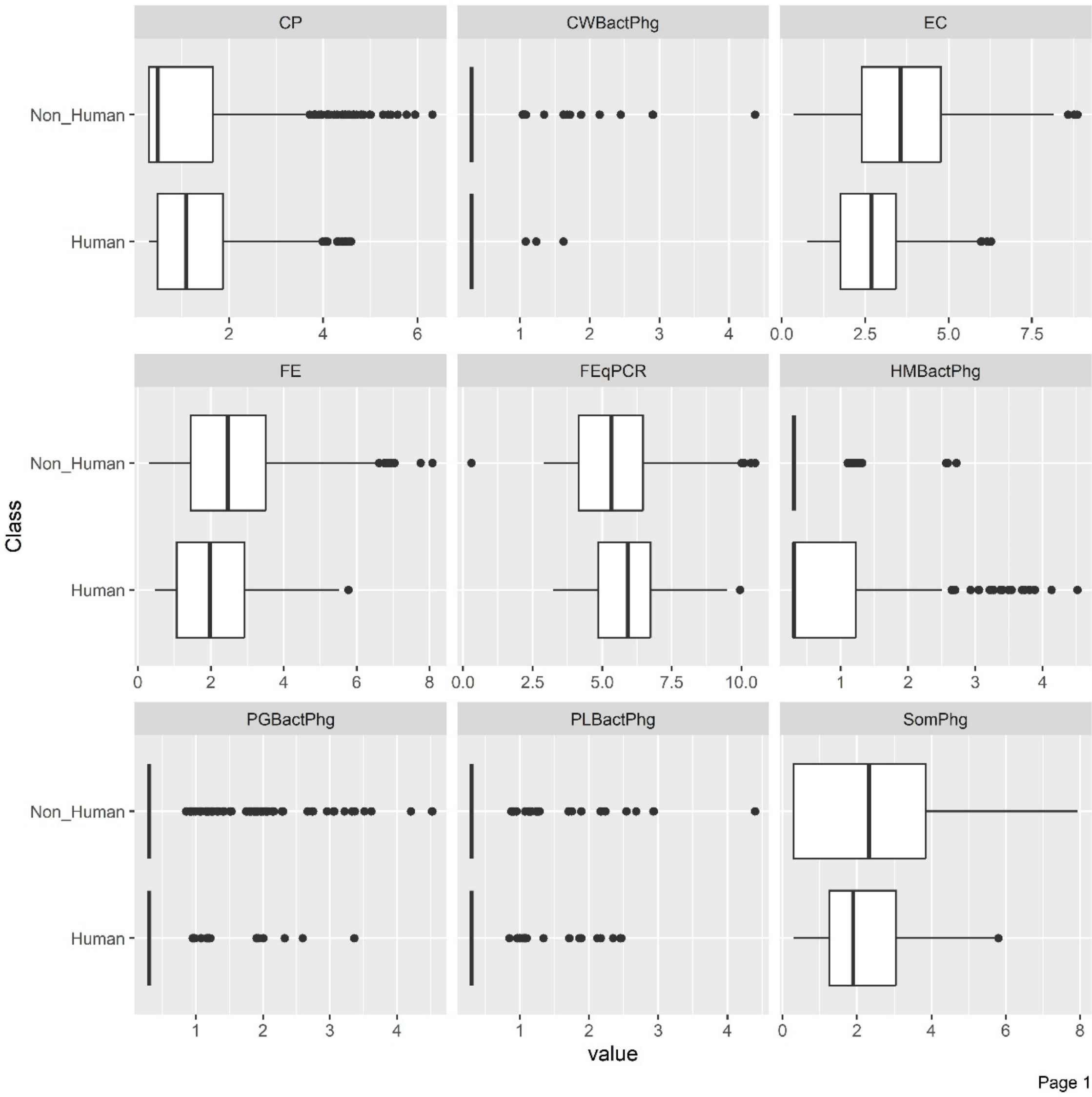
	Parameter	Type	Det. Lim.	ML_Predictor
1	EC	Quantitative	0.0500	No
2	FE	Quantitative	0.0500	No
3	FEqPCR	Quantitative	478.6301	No
4	CP	Quantitative	0.5000	No
5	SomPhg	Quantitative	5.0000	No
6	HMBactPhg	Quantitative	5.0000	Yes
7	CWBactPhg	Quantitative	5.0000	Yes
8	PGBactPhg	Quantitative	5.0000	Yes
9	PLBactPhg	Quantitative	5.0000	Yes
10	BifSorb	Quantitative	50.0000	Yes
11	BifTot	Quantitative	50.0000	No
12	HMBif	Quantitative	64000.0000	Yes
13	CWBif	Quantitative	30000.0000	Yes
14	PGNeo	Quantitative	36000.0000	Yes
15	PLBif	Quantitative	66000.0000	Yes
16	TLBif	Quantitative	64000.0000	No
17	BacR	Quantitative	478.6301	Yes
18	Pig2Bac	Quantitative	478.6301	Yes
19	AllBac	Quantitative	478.6301	No
20	HF183TaqMan	Quantitative	478.6301	Yes
21	HMMit	Quantitative	16000.0000	Yes
22	CWMit	Quantitative	10000.0000	Yes
23	PGMit	Quantitative	16000.0000	Yes
24	PLMit	Quantitative	10000.0000	Yes
25	Acesulfame	Quantitative	50.0000	Yes
26	Cyclamate	Quantitative	50.0000	Yes
27	Saccharain	Quantitative	50.0000	Yes
28	Sucralose	Quantitative	50.0000	Yes
29	Adeno	Quantitative	40000.0000	Yes
30	NoV	Quantitative	800.0000	Yes

Chunk 17

The chunk 17 has a similar function to the Chunk 9, i.e. it generates descriptive statistics and displays the box plots of the augmented dataset. Once the code has been executed, a pop-up window will appear.



Once the 'OK' button is clicked, the program continues showing the descriptive statistics and saving the boxplots as jpeg files.



Compare these boxplots with the calculated in chunk 9.

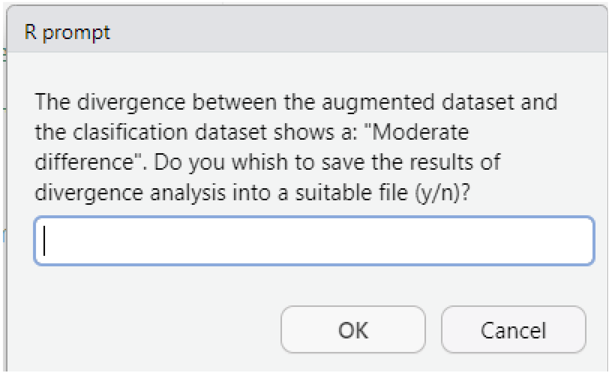
In the following page, a capture of the console output showing the descriptive statistics is shown. Remember that in the section named chunk 9 it is shown the definitions of the acronyms used in the descriptive statistics.

Description: df [14 x 30]

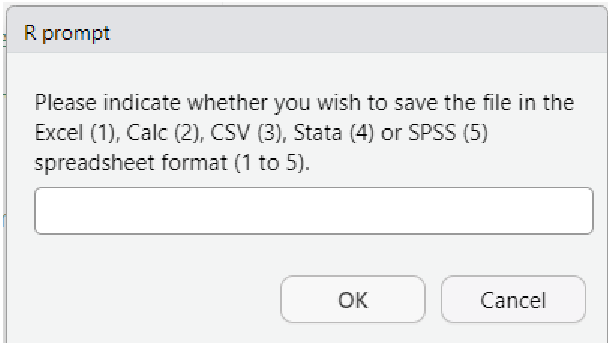
	EC <dbl>	FE <dbl>	FEqPCR <dbl>	CP <dbl>	SomPhg <dbl>	HMBactPhg <dbl>	CWBactPhg <dbl>	PGBactPhg <dbl>	PLBactPhg <dbl>
nbr.val	4243.000	4243.000	4243.000	4243.000	4243.000	4243.000	4243.000	4243.000	4243.000
nbr.null	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
nbr.na	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
min	0.342	0.301	0.301	0.301	0.301	0.301	0.301	0.301	0.301
max	8.870	8.079	10.492	6.316	7.925	4.516	4.370	4.527	4.398
range	8.527	7.778	10.191	6.015	7.624	4.215	4.069	4.226	4.097
sum	9409.601	5743.639	17823.290	2377.229	6121.129	1359.979	1293.418	1329.632	1310.788
median	2.036	1.056	4.249	0.301	1.320	0.301	0.301	0.301	0.301
mean	2.218	1.354	4.201	0.560	1.443	0.321	0.305	0.313	0.309
SE.mean	0.018	0.015	0.022	0.009	0.018	0.004	0.001	0.003	0.002

1-10 of 14 rows | 1-10 of 30 columns Previous 1 2 Next

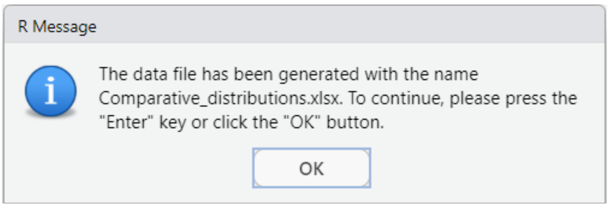
Concerning to divergences between the training and classification sets, the analysis indicates an overall moderate difference, which is indicated in the following pop-up.



For instance, is selected the option to save the results and the following window appears prompting to select a spreadsheet format.



Excel is selected, once selected a new window pop-up appears.

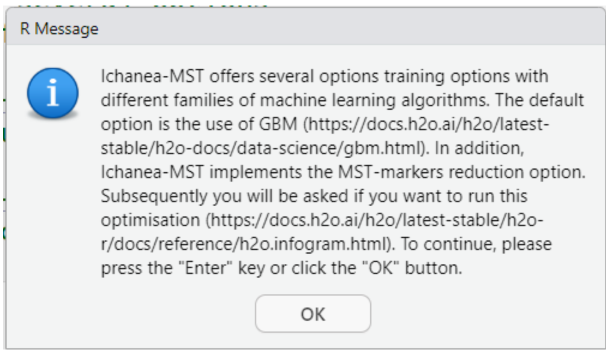


The file contains the comparative of each MST marker and the global results, which are presented in the following page.

Predictor	JSD	JSD_Interpretation	TVD	TVD_Interpretation	Overall_Interpretation
HMBactPhg	0.0989	Very similar	0.2533	Moderate difference	Moderate difference
CWBactPhg	0.0009	Very similar	0.0026	Very similar	Very similar
PGBactPhg	0.0553	Very similar	0.1467	Moderate difference	Moderate difference
PLBactPhg	0.0633	Very similar	0.1724	Moderate difference	Moderate difference
BifSorb	0.4270	Notable difference	0.7557	Very different	Very different
HMBif	0.1048	Very similar	0.2667	Moderate difference	Moderate difference
CWBif	0.0494	Very similar	0.1333	Moderate difference	Moderate difference
PGNeo	0.0145	Very similar	0.0400	Very similar	Very similar
PLBif	0.0399	Very similar	0.1067	Moderate difference	Moderate difference
BacR	0.1056	Very similar	0.1709	Moderate difference	Moderate difference
Pig2Bac	0.1966	Very similar	0.3200	Notable difference	Moderate difference
HF183TaqMan	0.3342	Moderate difference	0.6267	Very different	Notable difference
HMMit	0.0840	Very similar	0.2131	Moderate difference	Moderate difference
CWMit	0.0412	Very similar	0.0667	Very similar	Very similar
PGMit	0.0591	Very similar	0.1467	Moderate difference	Moderate difference
PLMit	0.0541	Very similar	0.1200	Moderate difference	Moderate difference
Acesulfame	0.0489	Very similar	0.1324	Moderate difference	Moderate difference
Cyclamate	0.0596	Very similar	0.1595	Moderate difference	Moderate difference
Saccharain	0.0048	Very similar	0.0133	Very similar	Very similar
Sucralose	0.0399	Very similar	0.1059	Moderate difference	Moderate difference
Adeno	0.0296	Very similar	0.0800	Very similar	Very similar
NoV	0.1343	Very similar	0.3331	Notable difference	Moderate difference
OVERALL	**0.0930**	**Very similar**	**0.1984**	**Moderate difference**	**Moderate difference**

Chunk 18

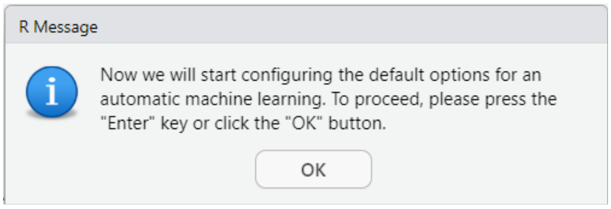
This chunk is just informative, from now on the code will execute the H2O machine learning framework. Once this chunk has been executed, the next pop-up will appear.



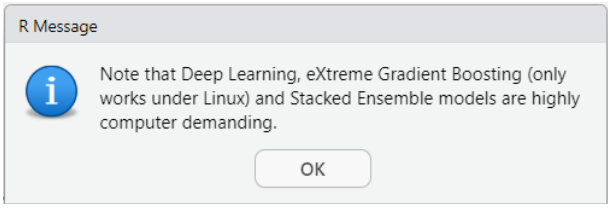
It is advisable to visit the H2O AutoML documentation website (https://docs.h2o.ai/h2o/latest-stable/h2o-docs/automl.html). As you will see, this R notebook is limited to the essential features, but the H2O framework contains much more. Do not hesitate to explore these features.

Chunk 19

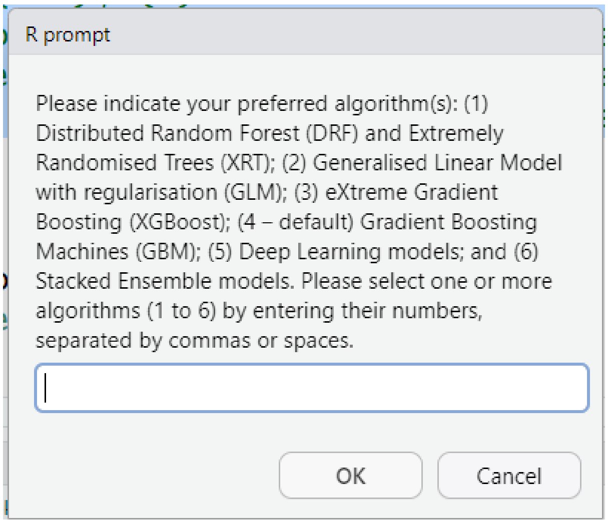
This chunk encodes the training process. Once the code starts to run the next pop-up appears.



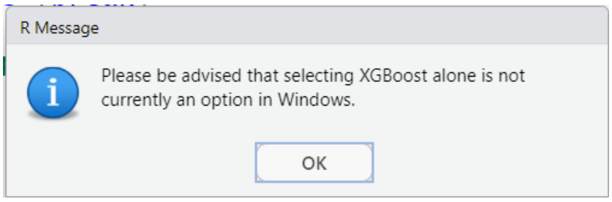
After clicking the 'OK' button, a new pop-up warns that Deep Learning, eXtreme Gradient Boosting and Stacked Ensemble models are computationally intensive.



Once the 'OK' button is pressed, the code prompts the user to select the machine learning algorithms. A short description is given in the introduction. It is recommended that readers familiarise themselves with the type of algorithms, if they have not already done so.

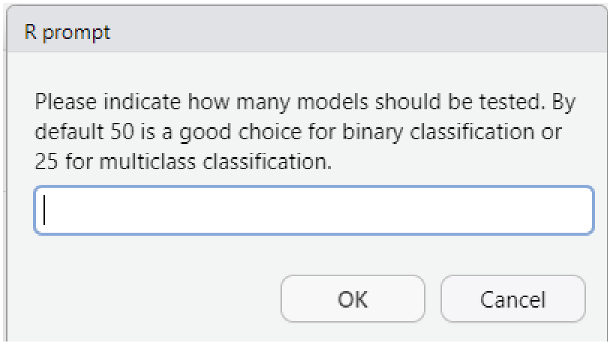


The default option is to include the GBM (4) algorithm. Please consider that multiple algorithms can be selected by indicating the number separated by commas or spaces. Please be advised that the XGBoost algorithm is not currently available for use with the H2O AutoML in Windows environments. In the event that this algorithm is the only selected option, the following pop-up warning will appear.



It is encouraged that users compare the results obtained from different trials that use the same or different algorithms.

In order to provide an illustrative example, option 4 was selected., the program will prompt the user to specify the number of models to be trained. As the number of models increases, the time required for training also increases. Although the number of models to be tested during training is indicated as a guide, users are at liberty to select other options. However, there is often only a slight improvement in evaluation metrics, such as accuracy, despite an increase in the number of models that are tested.



Following the selection of the required number of models, the program will initiate the H2O framework. It is imperative to verify that H2O is correctly installed and can be launched. In order to use this framework, it is necessary to install a Java runtime environment.

In https://docs.h2o.ai/h2o/latest-stable/h2o-docs/downloading.html is described how to download and install the latest stable version of H2O-3.

Below this line is shown the information of the version of H2O and the Java version that was used in this manual. H2O is launched with the default options.

```
java version "23.0.1" 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

Starting H2O JVM and connecting: . Connection successful!

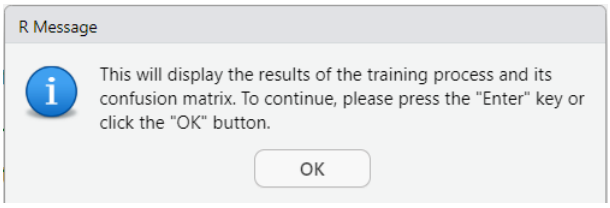
R is connected to the H2O cluster:
    H2O cluster uptime:         8 seconds 891 milliseconds
    H2O cluster timezone:       Europe/Madrid
    H2O data parsing timezone:  UTC
    H2O cluster version:        3.46.0.7
    H2O cluster version age:    1 month and 9 days
    H2O cluster name:           H2O_started_from_R_jmendez_xuc898
    H2O cluster total nodes:    1
    H2O cluster total memory:   7.96 GB
    H2O cluster total cores:    4
    H2O cluster allowed cores:  4
    H2O cluster healthy:        TRUE
    H2O Connection ip:          localhost
    H2O Connection port:        54321
    H2O Connection proxy:       NA
    H2O Internal Security:      FALSE
    R Version:                  R version 4.5.0 (2025-04-11 ucrt)
```

Please note that H2O opens port 54321. Please ensure that this port is accessible to H2O via your firewall. Once this port has been opened, users can try opening the following address in their web browser: http://localhost:54321, which provides access to H2O Flow, a web-based user interface for the H2O machine learning platform. This platform will be discussed in more detail at the end of this manual.

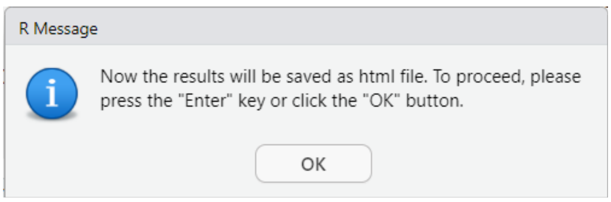
Please note that when training commences, it may take several minutes to complete the training process, particularly when using more computationally intensive algorithms or testing a greater number of models.

Chunk 20

This chunk contains the code responsible for displaying the metrics of the training. Please note that the augmented dataset is divided into a training set and a validation set. Chunks 20 and 21 present the metrics used to assess the performance of the best model from the range of tested models.



After a few moments, is displayed a pop-up indicating that the results will be saved to a HTML file.



Below this line is shown a screen capture of the results.

Description: df [6 x 7]

	mean	sd	cv_1_valid	cv_2_valid	cv_3_valid	cv_4_valid
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
pr_auc	0.999260	0.000247	0.999019	0.999111	0.999207	0.999660
precision	0.981097	0.003870	0.974719	0.983003	0.981366	0.985030
r2	0.933768	0.018149	0.902821	0.933445	0.948159	0.939952
recall	0.999394	0.001355	1.000000	1.000000	1.000000	0.996970
rmse	0.113948	0.011506	0.134193	0.110803	0.105447	0.110151
specificity	0.948821	0.016885	0.920354	0.946429	0.958042	0.961240

6 rows | 1-7 of 7 columns

The generated output is stored as an HTML file, which, when opened in a browser, displays its content on this page as well as subsequent pages. Note that it is possible that when the HTML file is generated this will appear truncated, this may occur due to limitations in the R notebook. The following image shows the R code responsible to save the results in an HTML file.

```
# Load the htmltools library
library(htmltools)

# Capture the output of the best model details
model_details <- capture.output(print(results_best_model))

# Ensure all lines are captured by concatenating the outputs
full_output <- c(model_details)

# Create HTML content with both outputs
html_content <- tags$html(
  tags$head(tags$title("Best Model Details and Confusion Matrix with training
dataset")),
  tags$body(
    tags$h1("Summary of the Best Model with the Training Dataset"),
    tags$pre(paste(full_output, collapse = "\n"))
  )
)

# Save the HTML content to a file
save_html(html_content, file =
"Summary_of_the_best_trained_model_and_confusion_matrix_training_dataset.html")
```

When this occurs, the code should be copied and run in the console.

```
> # Load the htmltools library
  library(htmltools)

  # Capture the output of the best model details
  model_details <- capture.output(print(results_best_model))

  # Ensure all lines are captured by concatenating the outputs
  full_output <- c(model_details)

  # Create HTML content with both outputs
  html_content <- tags$html(
      tags$head(tags$title("Best Model Details and Confusion Matrix with training dataset")),
      tags$body(
          tags$h1("Summary of the Best Model with the Training Dataset"),
          tags$pre(paste(full_output, collapse = "\n"))
      )
  )

  # Save the HTML content to a file
  save_html(html_content, file = "Summary_of_the_best_trained_model_and_confusion_matrix_training_dataset
  .html")
```

This approach may be employed to ensure that the generated HTML is not truncated.

The results of the HTML file are displayed on this page and the ensuing page. It should be noted that even when working with the same dataset and algorithm, there may be slight variations in the results.
The following is a concise elucidation of the results obtained.

Summary of the Best Model with the Training Dataset

```
Model Details:
==============

H2OBinomialModel: gbm
Model ID:  GBM_grid_1_AutoML_1_20250507_110124_model_40
Model Summary:
  number_of_trees number_of_internal_trees model_size_in_bytes min_depth max_depth
1              92                       92              164492         0        15
  mean_depth min_leaves max_leaves mean_leaves
1   12.47826          1        295   136.93478


H2OBinomialMetrics: gbm
** Reported on training data. **

MSE:  0.01164491
RMSE:  0.1079116
LogLoss:  0.04570297
Mean Per-Class Error:  0.0234375
AUC:  0.9985139
AUCPR:  0.9994202
Gini:  0.9970279
R^2:  0.9420785
AIC:  NaN

Confusion Matrix (vertical: actual; across: predicted) for F1-optimal threshold:
          Human Non_Human    Error      Rate
Human       610        30 0.046875   =30/640
Non_Human     0      1656 0.000000   =0/1656
Totals      610      1686 0.013066  =30/2296

Maximum Metrics: Maximum metrics at their respective thresholds
```

```
                        metric threshold       value idx
1                       max f1  0.884263    0.991023 279
2                       max f2  0.884263    0.996390 279
3                 max f0point5  0.885768    0.986641 278
4                 max accuracy  0.884263    0.986934 279
5                max precision  0.999847    1.000000   0
6                   max recall  0.884263    1.000000 279
7              max specificity  0.999847    1.000000   0
8             max absolute_mcc  0.884263    0.967556 279
9   max min_per_class_accuracy  0.884263    0.953125 279
10 max mean_per_class_accuracy  0.884263    0.976562 279
11                     max tns  0.999847  640.000000   0
12                     max fns  0.999847 1654.000000   0
13                     max fps  0.000438  640.000000 399
14                     max tps  0.884263 1656.000000 279
15                     max tnr  0.999847    1.000000   0
16                     max fnr  0.999847    0.998792   0
17                     max fpr  0.000438    1.000000 399
18                     max tpr  0.884263    1.000000 279

Gains/Lift Table: Extract with `h2o.gainsLift(<model>, <data>)` or
`h2o.gainsLift(<model>, valid=<T/F>, xval=<T/F>)`

H2OBinomialMetrics: gbm
** Reported on cross-validation data. **
** 5-fold cross-validation on training data (Metrics computed for combined holdout
predictions) **

MSE:  0.01308044
RMSE:  0.1143697
LogLoss:  0.04852276
Mean Per-Class Error:  0.02734375
AUC:  0.9978898
AUCPR:  0.9991744
Gini:  0.9957796
R^2:  0.9349382
AIC:  NaN
Confusion Matrix (vertical: actual; across: predicted) for F1-optimal threshold:
          Human Non_Human    Error      Rate
Human       605        35 0.054688   =35/640
Non_Human     0      1656 0.000000   =0/1656
Totals      605      1691 0.015244  =35/2296

Maximum Metrics: Maximum metrics at their respective thresholds
                        metric threshold       value idx
1                       max f1  0.459546    0.989543 261
2                       max f2  0.459546    0.995791 261
3                 max f0point5  0.891748    0.984751 243
4                 max accuracy  0.459546    0.984756 261
5                max precision  0.999780    1.000000   0
6                   max recall  0.459546    1.000000 261
7              max specificity  0.999780    1.000000   0
8             max absolute_mcc  0.459546    0.962157 261
9   max min_per_class_accuracy  0.860460    0.963768 252
10 max mean_per_class_accuracy  0.854927    0.973792 255
11                     max tns  0.999780  640.000000   0
12                     max fns  0.999780 1652.000000   0
13                     max fps  0.000423  640.000000 399
14                     max tps  0.459546 1656.000000 261
15                     max tnr  0.999780    1.000000   0
16                     max fnr  0.999780    0.997585   0
17                     max fpr  0.000423    1.000000 399
18                     max tpr  0.459546    1.000000 261

Gains/Lift Table: Extract with `h2o.gainsLift(<model>, <data>)` or
`h2o.gainsLift(<model>, valid=<T/F>, xval=<T/F>)`
Cross-Validation Metrics Summary:
              mean       sd cv_1_valid cv_2_valid cv_3_valid cv_4_valid cv_5_valid
accuracy  0.985629 0.002904   0.980435   0.986928   0.986928   0.986928   0.986928
aic             NA 0.000000         NA         NA         NA         NA         NA
auc       0.998036 0.000874   0.997003   0.997298   0.998274   0.999131   0.998473
err       0.014371 0.002904   0.019565   0.013072   0.013072   0.013072   0.013072
err_count 6.600000 1.341641   9.000000   6.000000   6.000000   6.000000   6.000000

---
                mean       sd cv_1_valid cv_2_valid cv_3_valid cv_4_valid cv_5_valid
pr_auc      0.999260 0.000247   0.999019   0.999111   0.999207   0.999660   0.999302
```

```
precision   0.981097 0.003870   0.974719   0.983003   0.981366   0.985030   0.981366
r2          0.933768 0.018149   0.902821   0.933445   0.948159   0.939952   0.944461
recall      0.999394 0.001355   1.000000   1.000000   1.000000   0.996970   1.000000
rmse        0.113948 0.011506   0.134193   0.110803   0.105447   0.110151   0.109144
specificity 0.948821 0.016885   0.920354   0.946429   0.958042   0.961240   0.958042
```

The file is initiated with the following segment, which pertains to the particulars of the model.

```
Model Details:
==============

H2OBinomialModel: gbm
Model ID:  GBM_grid_1_AutoML_1_20250507_110124_model_40
Model Summary:
  number_of_trees number_of_internal_trees model_size_in_bytes min_depth max_depth
1              92                       92              164492         0        15
  mean_depth min_leaves max_leaves mean_leaves
1   12.47826          1        295   136.93478
```

This section provides a concise clarification that the training was conducted utilising a GBM model and a dataset comprising solely two classes (a binomial model). The model that has been trained to demonstrate the most optimal performance metrics is entitled 'GBM_grid_1_AutoML_1_20250507_110124_model_40'. GBM is a tree-based algorithm that utilises 92 trees, with a mean number of leaves of 136.93. In addition, the nomenclature assigned to the model can be categorised as autodescriptive. The term 'grid' refers to a grid search process employed by H2O AutoML to optimise hyperparameters for the GBM algorithm. In the course of this process, the system undertakes a systematic testing of multiple combinations of hyperparameter values with a view to identifying the model that achieves the optimum level of performance. The model in question (model_40) is one of the candidates that were generated within the initial grid search (grid_1) of the AutoML run. This run was identified by the timestamp 20250507_110124, which refers to a training performance on the 7th of May 2025 at 11:01:24 hours.

The following section reports on the overall performance metrics in the training process:

```
H2OBinomialMetrics: gbm
** Reported on training data. **

MSE:  0.01164491
RMSE:  0.1079116
LogLoss:  0.04570297
Mean Per-Class Error:  0.0234375
AUC:  0.9985139
AUCPR:  0.9994202
Gini:  0.9970279
R^2:  0.9420785
AIC:  NaN
```

The **MSE** metric refers to the Mean Squared Error, which measures the average of the squares of the errors, that is, the differences between predicted and actual values. **RMSE** stands for Root Mean Squared Error, which is the square root of the MSE. **LogLoss**, also known as logistic loss or cross-entropy loss, is a metric used to evaluate the accuracy of a classification model by penalising false classifications. In the context of binomial classification, an elevated logarithmic loss is indicative of an increased probability of error on the part of the binary classifier. **Mean Per-Class Error** is a metric that calculates the average error rate across all classes in a classification problem. This provides insight into the model's performance in each class individually. The four metrics under consideration are associated with errors in classification. However, it is possible to interpret these metrics as a measure of how 'bad' the model is.

The subsequent set of three parameters, ranging from 0 to 1, has been demonstrated to be of considerable utility in the context of binomial classifications. The term Area Under the Curve (**AUC**) is employed to denote this area. This metric possesses a numerical value in binomial classification models; if not, it is displayed as 'NaN' (Not a Number), in which case it is not quantifiable. The metric functions as a tool to assess the model's capacity to differentiate between classes. In this particular context, the area under the Receiver Operating Characteristic (ROC) curve is of particular relevance. A slightly more sophisticated metric that is analogous to AUC is **AUCPR, which** stands for the Area Under the Precision-Recall Curve, which calculates the area under the precision-recall curve. This metric appraises the trade-off between precision and recall for varying threshold settings, a process that is of particular utility in the context of imbalanced datasets (for example, datasets in which one class is more prevalent than another). The Gini coefficient can be regarded as a standardised version of the AUC. For the three metrics higher values are indicative of enhanced model performance.

R-squared (**R^2**) or the coefficient of determination in the context of a classification model refers to how well the model explains the variance in the binary outcome. The utility of the metric is found in the context of regression problems, rather in that of classification problems.

Finally, **AIC** is the Akaike Information Criterion. This metric is used for model selection, balancing model fit and complexity. Lower AIC values indicate a better model, considering both the goodness of fitness and the number of parameters used.

In this link https://docs.h2o.ai/h2o/latest-stable/h2o-docs/performance-and-prediction.html#r2-r-squared the user would find additional information.

In the context of a binomial classification (the actual instance), the user must be centred on AUC, AUCPR and the Gini index. The closer these metrics are to 1, the better the classification.

The following section displays the confusion matrix:

```
Confusion Matrix (vertical: actual; across: predicted) for F1-optimal threshold:
          Human Non_Human    Error       Rate
Human       610        30 0.046875   =30/640
Non_Human     0      1656 0.000000   =0/1656
Totals      610      1686 0.013066  =30/2296
```

Briefly, the dataset that has been used in the training comprises 2296 samples: 640 are human polluted samples and 1656 are non-human polluted samples. 610 human samples were correctly classified as human polluted, but 30 samples were incorrectly classified as non-human polluted. On the other hand, all samples with non-human pollution have been correctly classified.

Finally, the following section provides a summary of the maximum values obtained for all the performance metrics. Please take care to note that the maximum values are indicated in the values column (please note that these should not be confused with the threshold column).

```
Maximum Metrics: Maximum metrics at their respective thresholds
                        metric threshold       value idx
1                       max f1  0.884263    0.991023 279
2                       max f2  0.884263    0.996390 279
3                 max f0point5  0.885768    0.986641 278
```

```
4                   max accuracy  0.884263    0.986934 279
5                  max precision  0.999847    1.000000   0
6                     max recall  0.884263    1.000000 279
7                max specificity  0.999847    1.000000   0
8               max absolute_mcc  0.884263    0.967556 279
9     max min_per_class_accuracy  0.884263    0.953125 279
10 max mean_per_class_accuracy  0.884263    0.976562 279
11                       max tns  0.999847  640.000000   0
12                       max fns  0.999847 1654.000000   0
13                       max fps  0.000438  640.000000 399
14                       max tps  0.884263 1656.000000 279
15                       max tnr  0.999847    1.000000   0
16                       max fnr  0.999847    0.998792   0
17                       max fpr  0.000438    1.000000 399
18                       max tpr  0.884263    1.000000 279
```

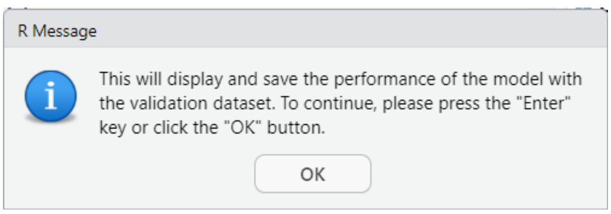
The **metric** column contains the specific metric being optimised, such as F1, F2, F0.5 (f0point5), etc. The column threshold refers to the probability cutoff that is used to maximise the different metrics. The **value** column contains the highest value achieved by the corresponding metric at the specified threshold. Finally, the **idx** column contains the index or position in the list of predictions, sorted by probability, where the optimal metric value is found. This index is used for plotting ROC or precision-recall curves. The **F1** score is the harmonic mean of precision and recall, indicating a balanced approach between the two. The **F2** score is a variation of the F1 score that prioritises recall over precision, while the **F0.5** score prioritises precision over recall. The **accuracy** rate is calculated by dividing the number of true results (both true positives and true negatives) by the total number of cases examined. **Precision** is defined as the proportion of true positive results among all positive results. The **recall** or sensitivity is defined as the proportion of true positive results among all actual positive cases. The **specificity** is the proportion of true negative results among all actual negative cases. The *MCC* or the Matthews correlation coefficient is a measure of the quality of binary classifications. The **'min_per_class_accuracy**' and **'mean_per_class_accuracy**' functions calculate, respectively, the minimum and average accuracy across all classes. The **tns**, **fns**, **fps** and **tps** states refer to true negatives, false negatives, false positives and true positives, respectively. The **tnr**, **fnr**, **fpr** and **tpr** states indicate the rate of true negatives, false negatives, false positives and true positives, respectively.

The rest of the file displays the results of the 5-fold cross-validation on training data.

Chunk 21

The function of this section is to encode and display the results when the validation data is tested. It should be recalled that the validation data constitutes the part of the augmented data reserved for the validation of the training process.

Once the chunk begins to run, a pop-up window is displayed



As the before chunk, a HTML file with the metrics can be saved. The results are presented below.

Summary of the Best Model with the Validation Dataset

```
H2OBinomialMetrics: gbm

MSE:  0.008666439
RMSE:  0.09309371
LogLoss:  0.03878032
Mean Per-Class Error:  0.01465201
AUC:  0.9987601
AUCPR:  0.9995128
Gini:  0.9975201
R^2:  0.9568228
AIC:  NaN

Confusion Matrix (vertical: actual; across: predicted) for F1-optimal threshold:
          Human Non_Human    Error    Rate
Human       265         8 0.029304  =8/273
Non_Human     0       709 0.000000  =0/709
Totals      265       717 0.008147  =8/982

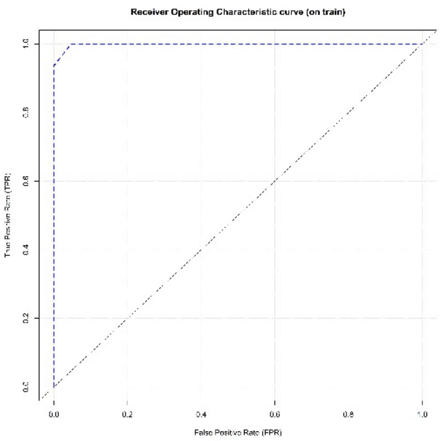
Maximum Metrics: Maximum metrics at their respective thresholds
                        metric threshold      value idx
1                       max f1  0.657475   0.994390 273
2                       max f2  0.657475   0.997748 273
3                 max f0point5  0.657475   0.991054 273
4                 max accuracy  0.657475   0.991853 273
5                max precision  0.999834   1.000000   0
6                   max recall  0.657475   1.000000 273
7              max specificity  0.999834   1.000000   0
8             max absolute_mcc  0.657475   0.979727 273
9   max min_per_class_accuracy  0.884263   0.970696 271
10 max mean_per_class_accuracy  0.657475   0.985348 273
11                     max tns  0.999834 273.000000   0
12                     max fns  0.999834 708.000000   0
13                     max fps  0.000438 273.000000 399
14                     max tps  0.657475 709.000000 273
15                     max tnr  0.999834   1.000000   0
16                     max fnr  0.999834   0.998590   0
17                     max fpr  0.000438   1.000000 399
18                     max tpr  0.657475   1.000000 273

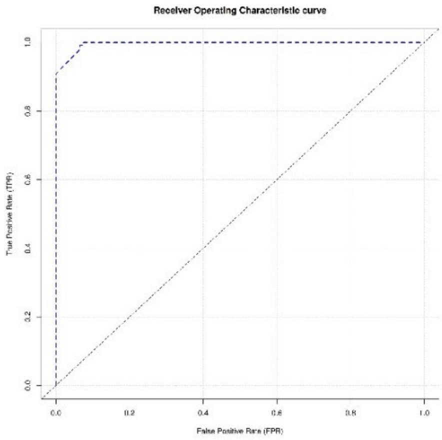
Gains/Lift Table: Extract with `h2o.gainsLift(<model>, <data>)` or
`h2o.gainsLift(<model>, valid=<T/F>, xval=<T/F>)`
```

The results of AUC: 0.9987601, AUCPR: 0.9995128, and Gini: 0.9975201 indicate a very good classification.

Along with this metric, the ROC curve is also plotted. The plot differs between binomial or multinomial models. For binomial models is plotted using the own utility provide for H2O AutomL as is shown in the illustration below these lines. The ROC curve can be obtained using the performance metrics of the training and validation datasets. It is imperative to prioritise the performance metrics derived from the validation dataset, which will be stored in the R notebook.

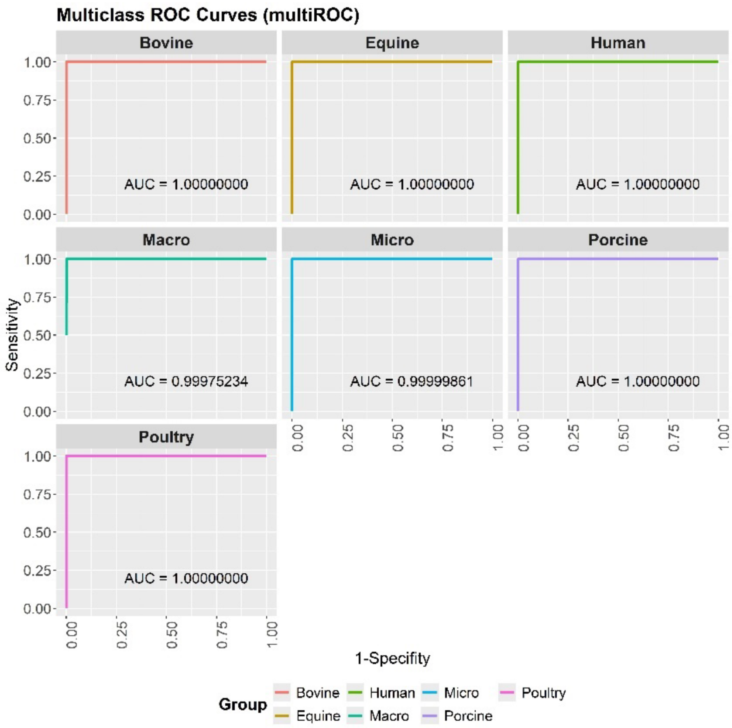
The ROC curves are displayed below. On the left, using the training dataset, and on the right, using the validation dataset.





In the context of multinomial models, the direct calculation of the ROC curve is not feasible. However, the evaluation of this metric is possible through the calculation of both micro-average and macro-average ROC curves, in conjunction with their respective AUC values. The micro-average ROC/AUC is derived by aggregating the contributions of all classes into a single binary classification task. This objective is realised through the aggregation of all true positive, false positive, true negative, and false negative outcomes across all classes, and the subsequent treatment of these values as a large binary classification. Conversely, the macro-average ROC/AUC is calculated by first computing individual ROC curves for each class using a one-vs-rest approach, where each class is designated as the positive class while the others are designated as the negative class. The results from all classes are then averaged to produce a single summary curve. In order to ensure smoothness and comparability across classes, linear interpolation was applied between the ROC points prior to the implementation of the averaging process.

The following illustration is presented as an example of the ROC curve plot with their respective AUC values. It should be noted that the illustration corresponds to another training set of samples.



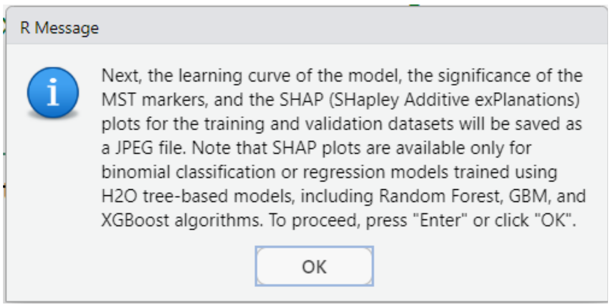
Chunk 22

This chunk provides an explanation of which MST markers contribute the most and how they contribute towards the prediction for a specific class. By default, the plots concerning the learning curve and the importance of each MST marker (with the exception of stacked models) are displayed. For binomial classification it is possible to obtain the SHAP (SHapley Additive exPlanations) plots. SHAP plots are a powerful tool for interpreting machine learning models. They do this by visualising the contribution of each feature to the model's predictions. Shapley values from cooperative game theory form the basis of SHAP plots, which ensure a fair distribution of the "payout" (prediction) among features. In this particular instance, it is interesting to observe the alteration in the disposition of the blue and red dots within human and non-human MST markers in the SHAP plots.

In contradistinction to binomial classifications, in which a single set of SHAP values is generated per sample, multinomial classifications necessitate the computation of a distinct set of SHAP values for each class per sample. This requirement significantly increases both computational complexity and processing time, particularly as the number of classes grows. It is important to highlight that, due to the inherent intricacy of this approach, SHAP visualisations in multinomial classification are generally restricted to a single class. This class is selected by the user based on the specific analytical focus or interest. The SHAP analysis is then performed using the validation dataset, ensuring that the insights derived are based on previously unseen data and thus provide a reliable indication of the model’s generalisation performance.

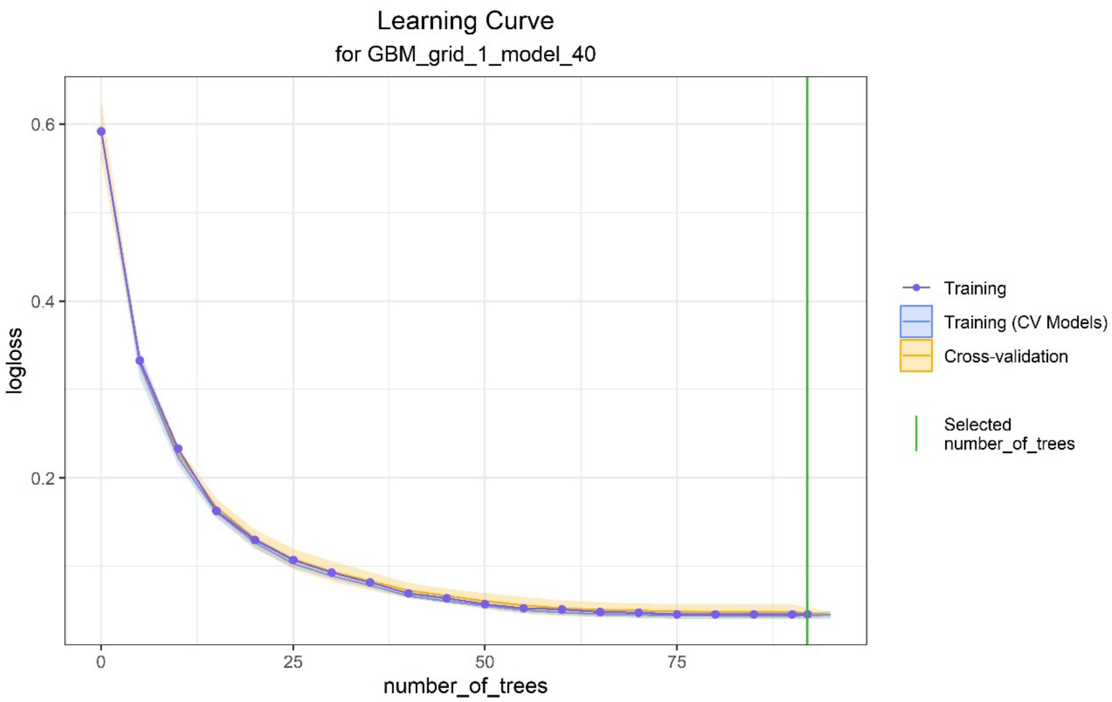
This targeted strategy enables a more focused interpretation of feature importance for the chosen class, while maintaining computational efficiency and avoiding unnecessary overhead associated with generating SHAP values for all classes simultaneously.

It should be noted that by default, all the MST markers are shown in the plots.



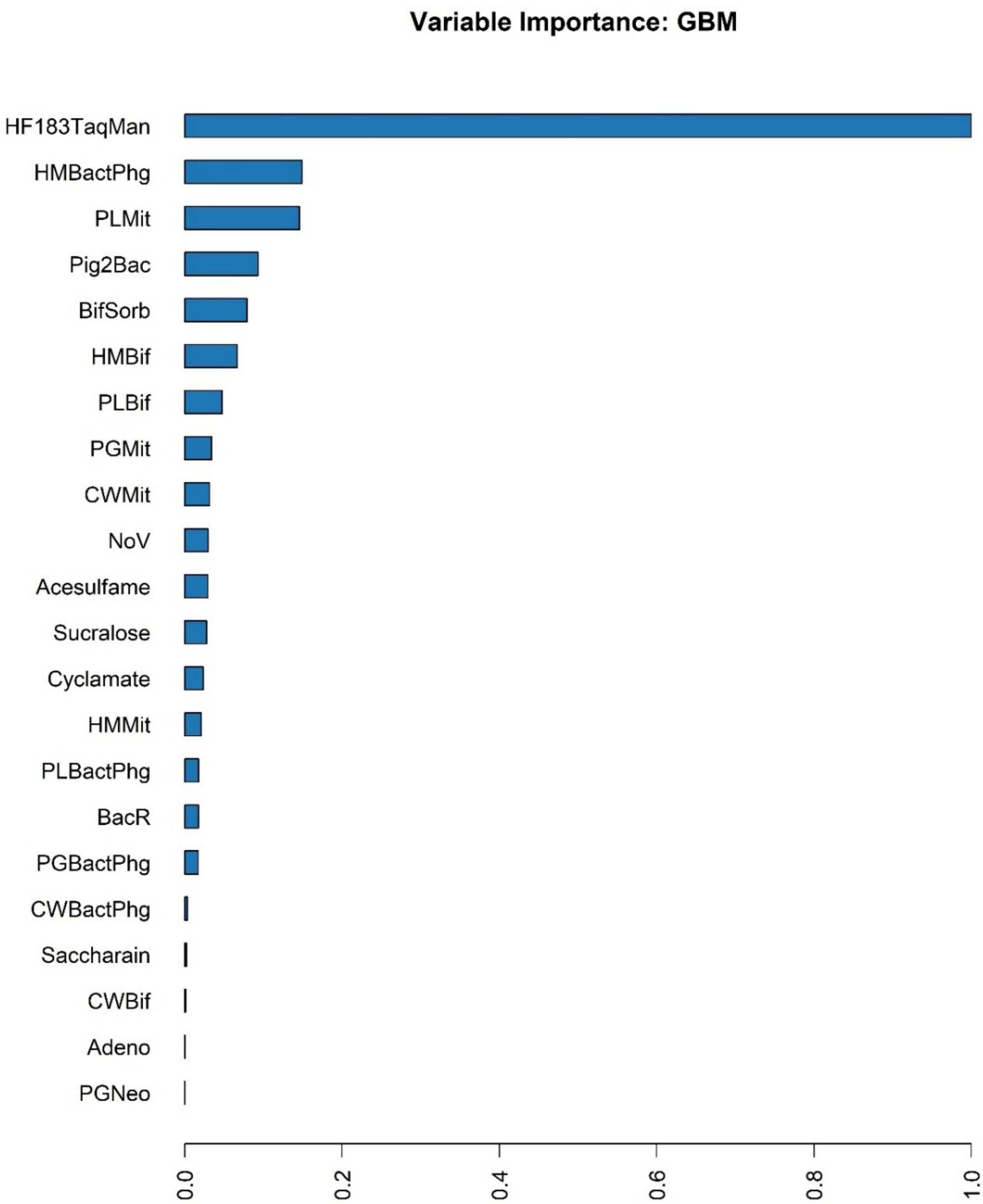
After clicking the 'OK' button, the program generates the following images, which are saved as JPEGs.

- Learning curve.jpeg



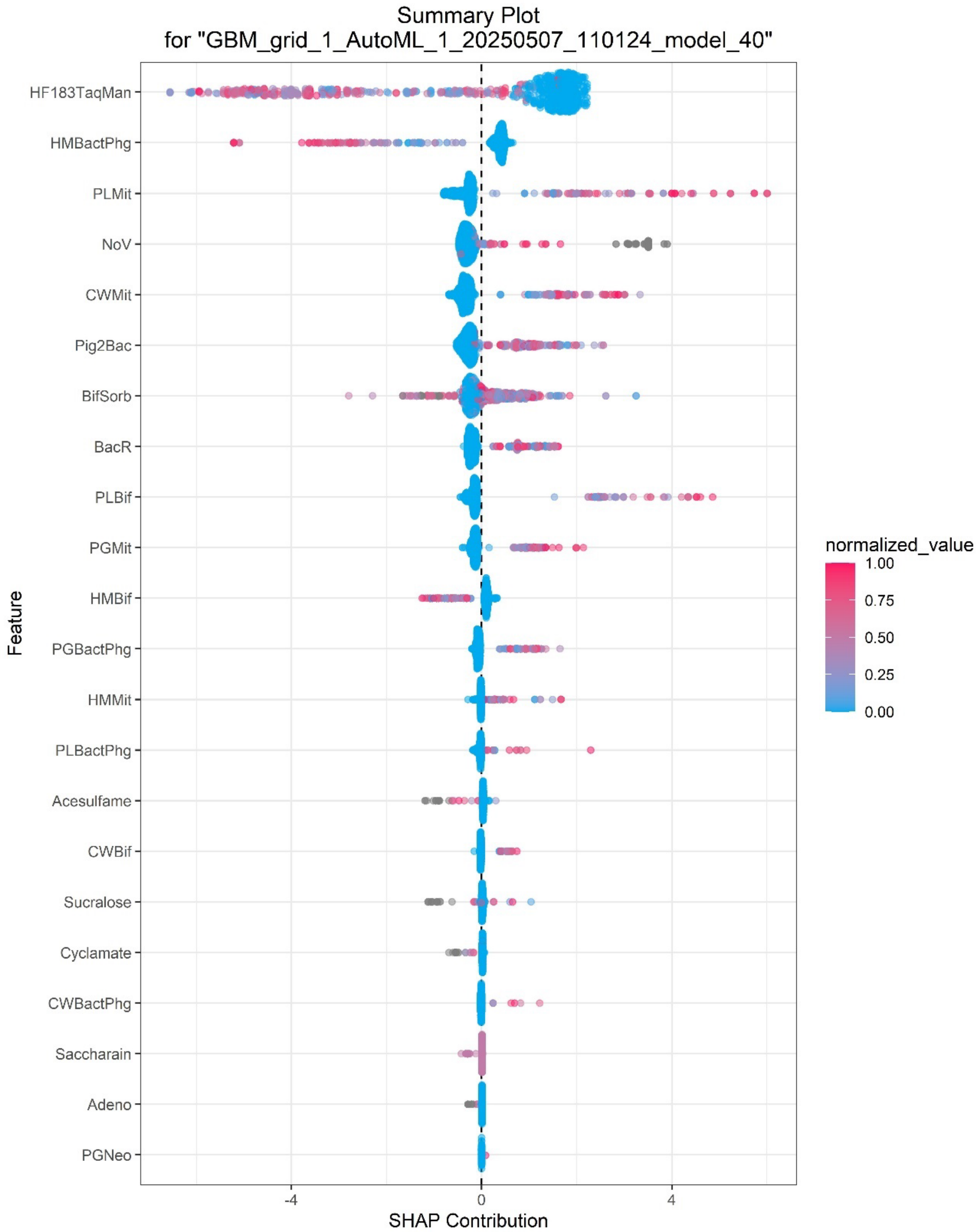
In the context of a learning curve, a decreasing logloss (a measure of prediction error for classification tasks) is indicative of improving performance. The proximity of the training and cross-validation curves to each other is indicative of the model's capacity for generalisation denoting low variance and low bias. Conversely, a significant disparity between the training score and the validation score, particularly if the training logloss is much lower than the validation logloss, it suggests overfitting. If the logloss remains high in both, the model may be underfitting. The number of trees refers to the number of trees been added to the model while the green vertical line indicates the number of trees used in the model.

- Importance.jpeg



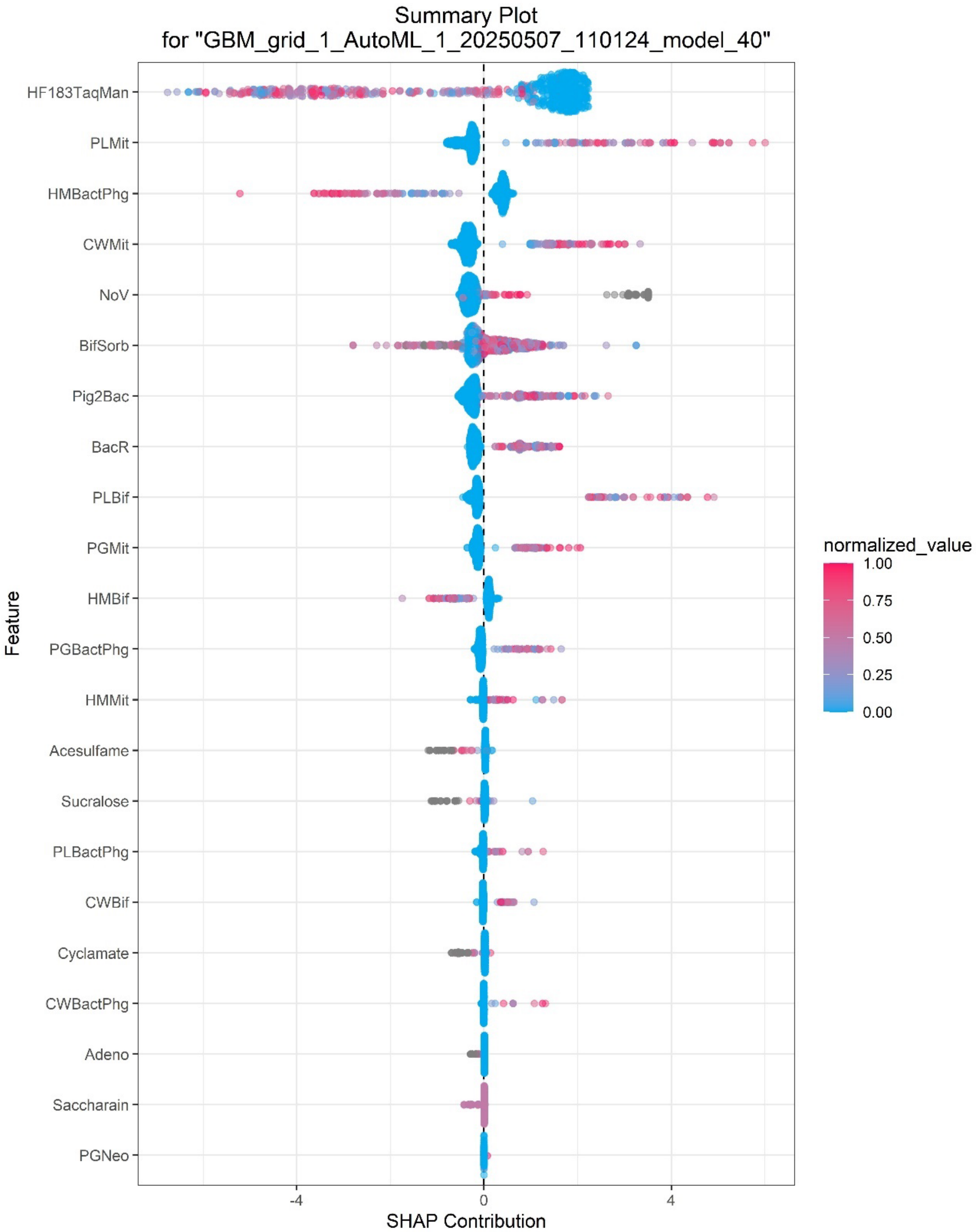
An importance plot is a visual representation of the contribution of each MST marker to the model's predictions. The x-axis of the graph represents the importance score, which has been scaled to range from 0 to 1. The y-axis of the graph lists the features that have been used in the model. In the classification model, the greater the size of the bar, the more significant the contribution of the MST marker. As the non-human class is pool of a number of different contamination origins, including bovines, porcines, equines and poultry. It is therefore not surprising that a human MST marker is of great significance. This is especially true if the marker is not detected in non-human samples.

- SHAP of training.jpeg

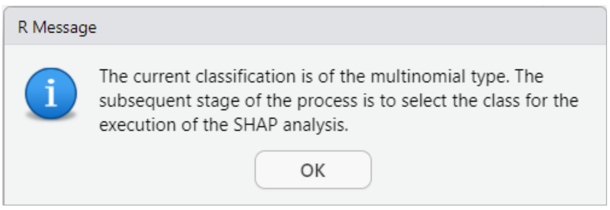


A SHAP plot is a graph where the MST markers are ranked by their overall impact on the model's predictions. The x-axis contains the SHAP values, that shows how much each MST marker contributes to increasing or decreasing the prediction. Each dot represents a single prediction (that corresponds to a row in the dataset), and colour varies according their normalized value. Note that human markers exhibit a tendency to cluster on the right side of the graph, while markers for other types of pollution (non-human) tend to cluster on the left side. Observe that the most significant markers tend to create a greater separation between these clusters and the centre line, while the least significant markers appear to be centred on the middle line.

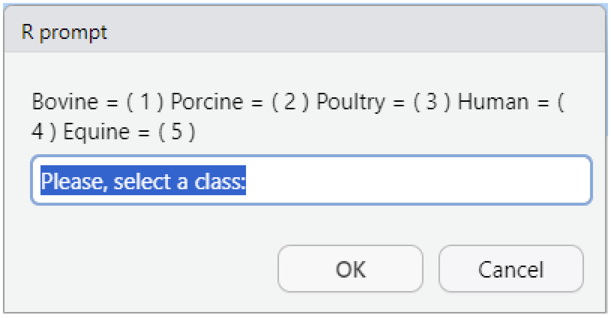
- SHAP of validation.jpeg



In the event of a multinomial classification, the user will be prompted to select the class to be used in the analysis. The subsequent pop-up notification is displayed, instructing the user to specify the class with which the analysis should be conducted.



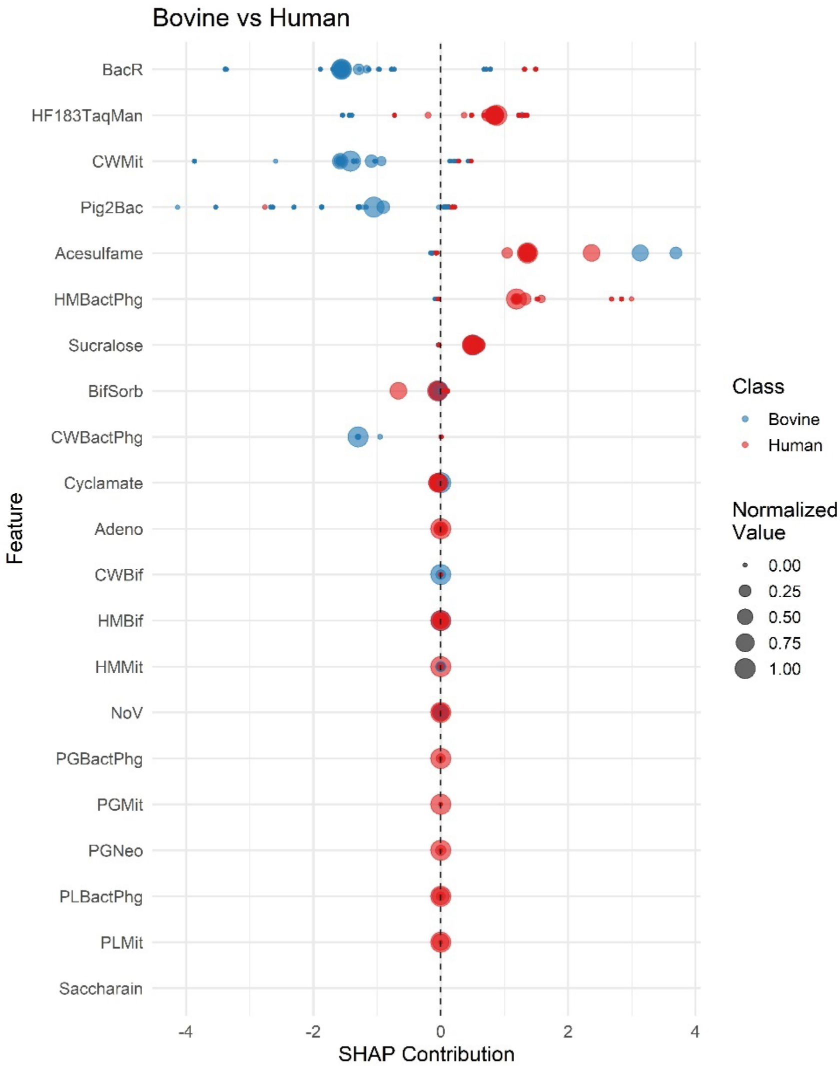
After clicking the “OK” button, the program continues prompting the user to select the class.



The user is required to select a number. For example, the number 4 (human) has been introduced.

Once the class has been selected, the analysis is commenced. The SHAP plots are then saved as JPEG files, with the number of files corresponding to the number of classes utilised in the training.

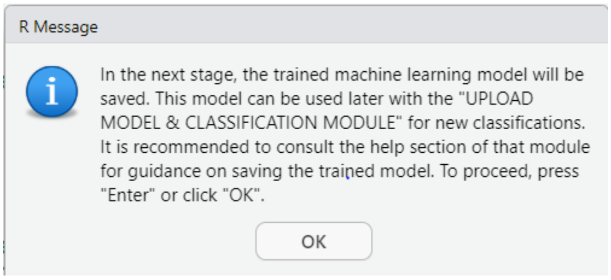
For instance, the following plot corresponds to a multiclass classification, where is represented the SHAP values of the comparison between the Human and the Bovine classes.



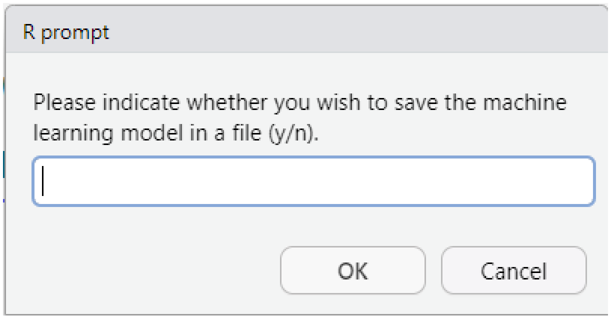
Chunk 23

This chunk encodes for saving the trained model. It is recommended that the model be saved for two key reasons. Firstly, it can be used to efficiently classify new samples. Secondly, it can be employed to make meaningful comparisons between models trained using different algorithms.

Once the chunk is run, a pop-up window appears.



After clicking the 'OK' button, the program will prompt the user to indicate whether they wish to save the model.



If the selection is affirmative, the model is saved onto the directory, with a name containing the identification of the model (Model ID: GBM_grid_1_AutoML_1_20250507_110124_model_40).

```
+ }
[1] "C:\\Users\\Javier\\Desktop\\Pruebas Ichanea - copia\\Module 1\\GBM_grid_1_AutoML_1_2025050
7_110124_model_40"
> |
```

This model can be latter upload using the Module II of Ichnaea-MST, which is a R notebook for uploading and performing the classification.

Chunk 24

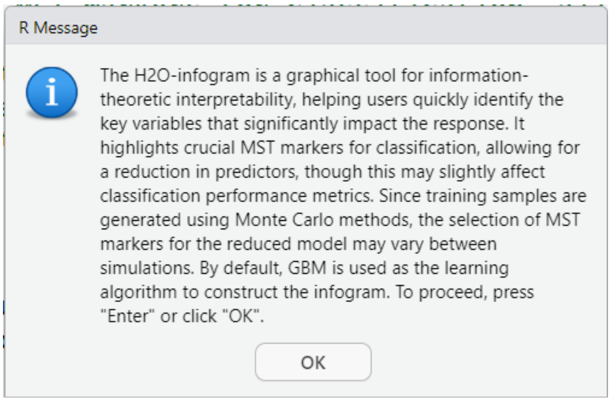
This chunk carries out the reduction of parameters using H2O-Infogram.

H2O Infogram is a sophisticated graphical tool designed to identify and highlight the most valuable and unique features that drive the response in supervised classification problems. This tool is especially beneficial for data scientists and analysts seeking to optimise the efficiency and performance of their predictive models. It provides a clear visual representation of the variables, highlighting their importance and uniqueness. By generating this graph, the H2O Infogram assists users in focusing on a streamlined set of Minimum Spanning Tree (MST) markers. These markers are essential for simplifying the model and improving its efficiency by reducing the number of necessary features without compromising predictive accuracy.

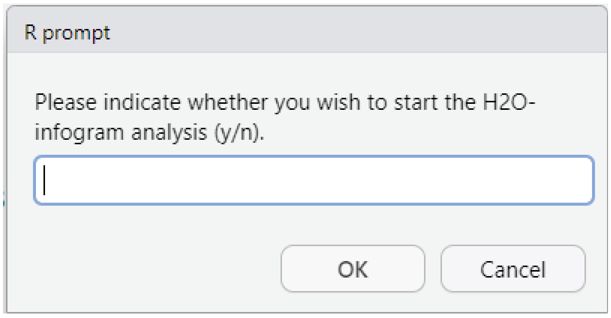
The script was written to employ three distinct datasets to ensure robust model development and evaluation:

1. Training Dataset (60%): This dataset comprises 60% of the samples of the augmented dataset and is used to calibrate the model's parameters. It is the primary dataset for model learning and fitting.
2. Validation Dataset (20%): This dataset, which constitutes 20% of the samples of the augmented dataset, is used to evaluate the model's performance during the training phase. It assists in fine-tuning hyperparameters and prevents overfitting by providing an unbiased assessment of the model.
3. Testing Dataset (20%): The remaining 20% of the samples of the augmented dataset are reserved for the final evaluation of the model. This dataset is used to compute performance metrics, generate confusion matrices, and create associated plots to assess the model's effectiveness in real-world scenarios.

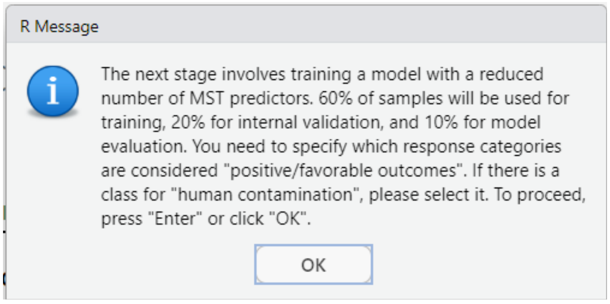
The chunk starts with a pop-up window.



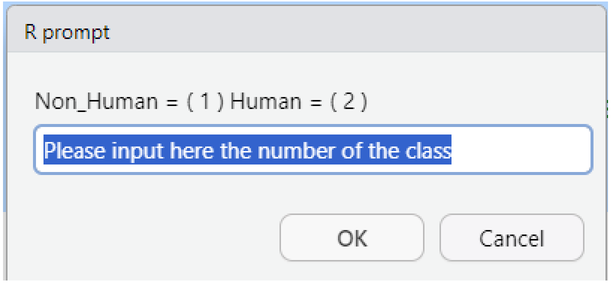
After clicking the ‘OK’ button, a new pop-up window will appear, prompting confirmation of the reduction.



If the answer is negative the program jumps to the next chunk. If the response is affirmative, the program will initiate the Infogram analysis. After a few minutes, a pop-up window will appear to inform you once again about the reduction process.



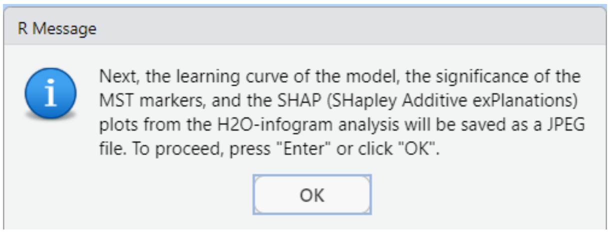
Following the selection of 'OK', a new pop-up will appear prompting the user to indicate which class they would like to prioritise.



In this instance, the human class (2) is selected.

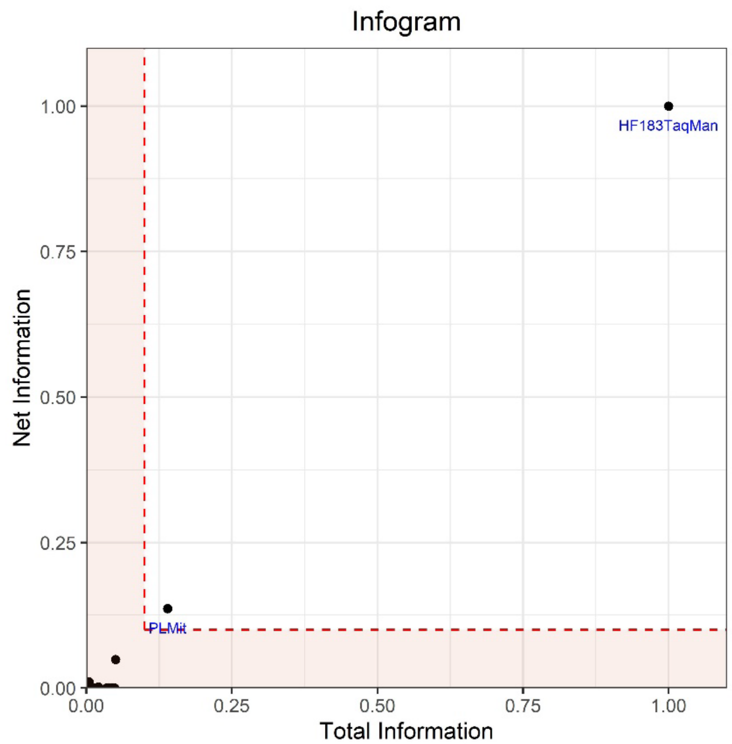
The selection of the 'favourable class' is of pivotal significance in this context of training models with H2O. In this regard, the 'favourable class' plays a crucial role in interpreting classification models, especially for binary or multiclass problems. The term is employed to denote the target class that should be regarded as the "positive" or desired outcome, with implications for evaluation metrics (e.g. AUC, precision, recall, or F1-score), model interpretation, and predictor selection. In this particular instance, the model will concentrate on accurately identifying human samples and prioritise predictors that best distinguish this class. In the absence of this option, H2O may be required to default to the first class alphabetically, or to optimise for no specific class at all. The latter option may potentially reduce the model's practical effectiveness.

Please be advised that the process may take a few minutes, depending on the complexity of the dataset and the computer's capacity. A window pop-up will appear to inform you that the analysis has been completed and the program is ready to save the results.

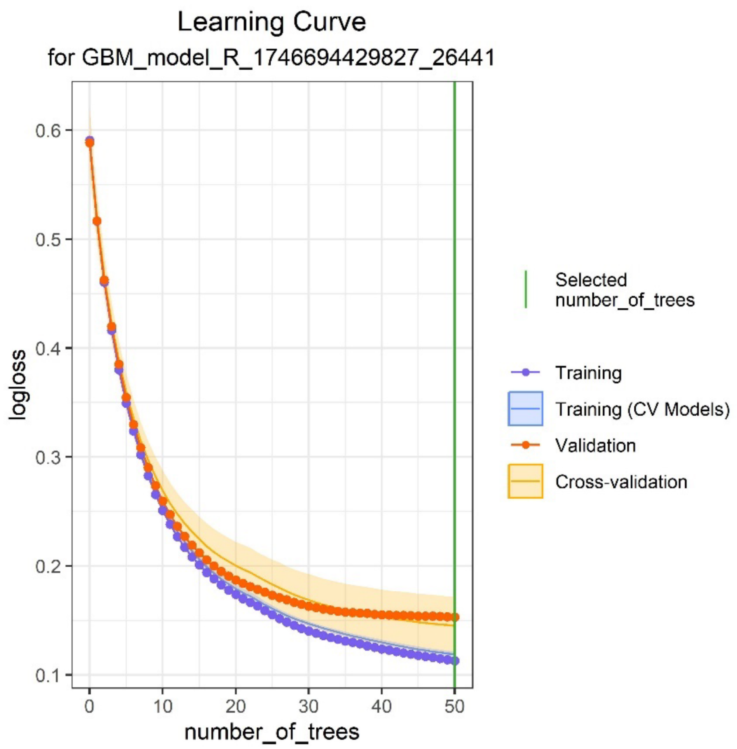


After clicking the 'OK' button, several JPEG files are generated.

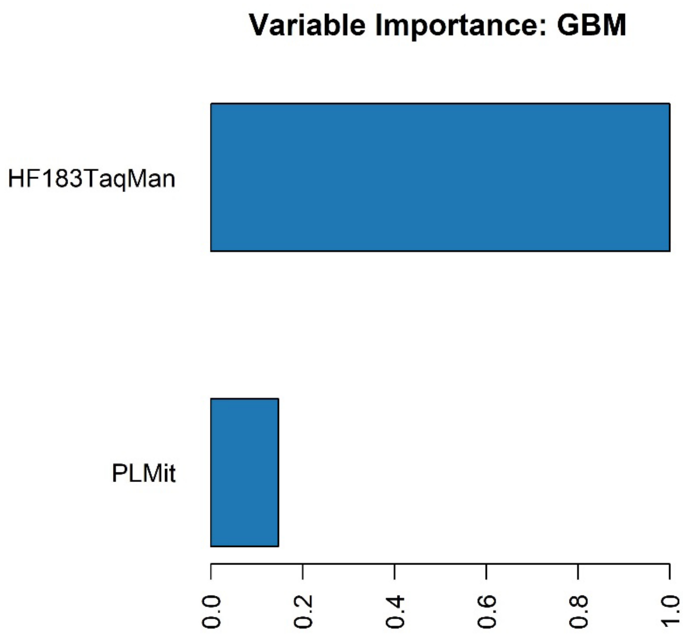
- Infogram plot.jpeg



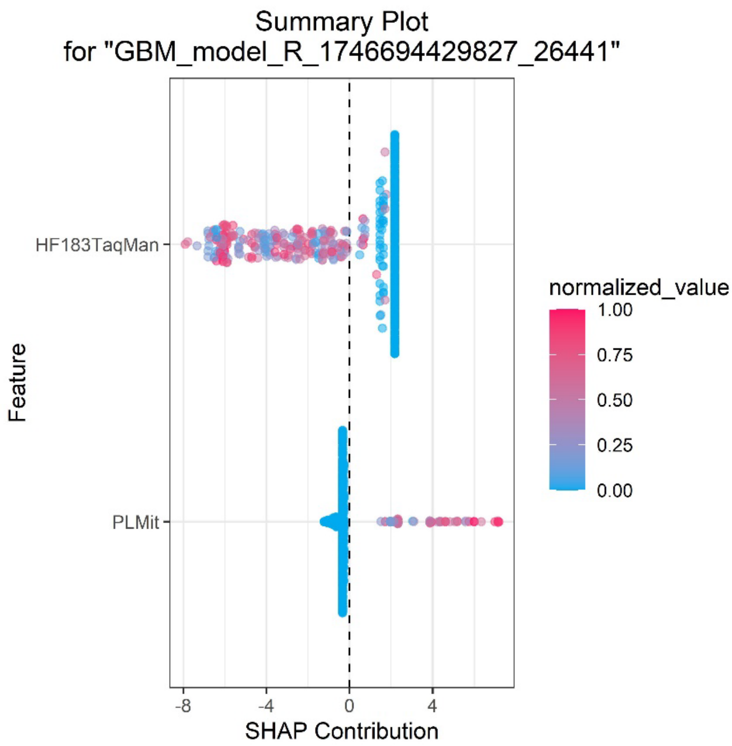
- Reduced_set_Learning curve.jpeg



- Reduced_set_Importance.jpeg

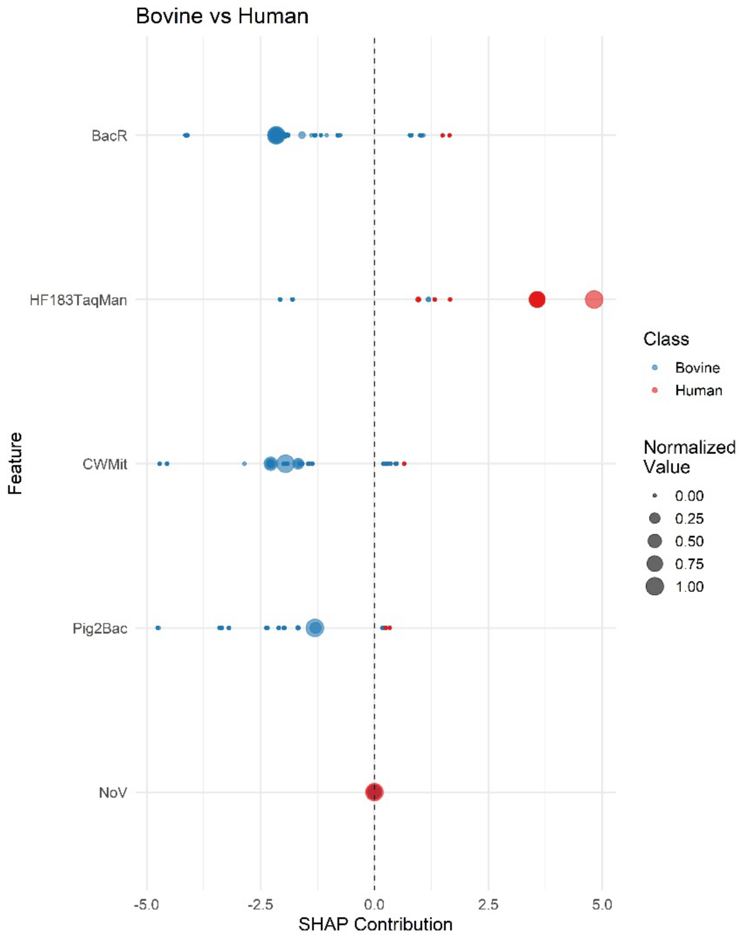


- Reduced_set_SHAP.jpeg



As indicated previously, in the event of a multimodal classification, a number of plots are generated using the positive class that has been selected for the infogram analysis.

For instance, the following graph presents a SHAP plot of a multinomial classification model following MST reduction parameters.



Here, the Infogram plot indicates that the MST marker based on HF183TaqMan is a strong candidate for selection, and the PLMit marker (a mitochondrial marker for poultry and the most important non-human marker) may also aid in classification.

The learning curve indicates that a potential overfitting begins to occur and adding > 50 trees no longer improves generalization even it may harm the classification.

In the variable importance and SHAP plots again the HF183 MST marker has the most importance in the contribution of the classification.

As illustrated on the subsequent page, the numerical output comprises the Infogram results and the confusion matrix.

R Console | data.frame | data.frame

Description: df [6 × 6]

	column <chr>	admissible <dbl>	admissible_index <dbl>	total_information <dbl>	net_information <dbl>
1	HF183TaqMan	1	1.00000000	1.00000000	1.00000000
2	PLMit	1	0.13779506	0.13934081	0.13623178
3	CWMit	0	0.04960044	0.05043723	0.04874928
4	Pig2Bac	0	0.03445361	0.04872476	0.00000000
5	HMBactPhg	0	0.02969466	0.04199459	0.00000000
6	PLBif	0	0.02817953	0.03985188	0.00000000

6 rows | 1-6 of 6 columns

These results are saved in a HTML file named ‘Infogram_results.html’.

Infogram results

```
        column  admissible  admissible_index  total_information  net_information    cmi_raw
1  HF183TaqMan           1        1.00000000         1.00000000       1.00000000 0.22229552
2        PLMit           1        0.13779506         0.13934081       0.13623178 0.03028371
3        CWMit           0        0.04960044         0.05043723       0.04874928 0.01083675
4      Pig2Bac           0        0.03445361         0.04872476       0.00000000 0.00000000
5    HMBactPhg           0        0.02969466         0.04199459       0.00000000 0.00000000
6        PLBif           0        0.02817953         0.03985188       0.00000000 0.00000000

[22 rows x 6 columns]
```

The confusion matrix, which is displayed below.

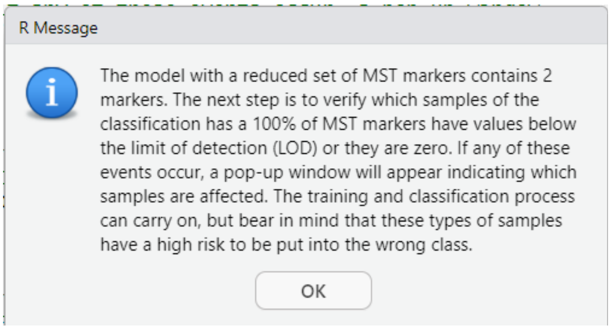
R Console | data.frame | data.frame

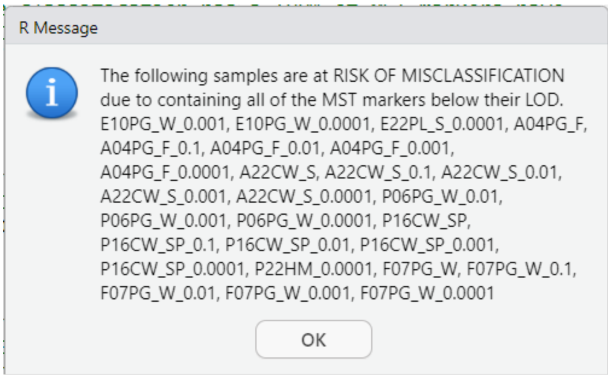
Description: df [3 × 4]

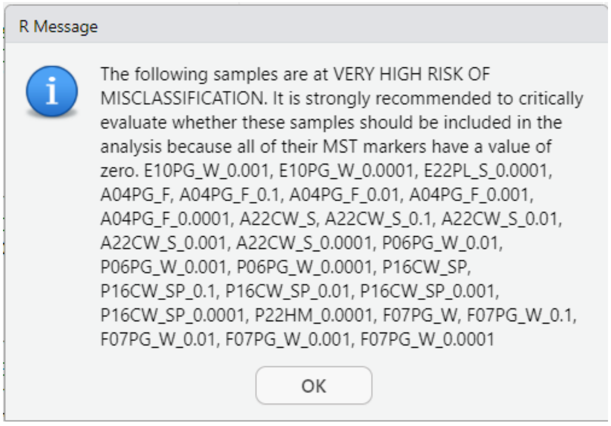
	Human <chr>	Non_Human <chr>	Error <chr>	Rate <chr>
Human	167	14	0.077348	=14/181
Non_Human	6	450	0.013158	=6/456
Totals	173	464	0.031397	=20/637

3 rows

After that the analysis of how many samples in the classification set that, according to the reduced set of MST markers, are likely to be misclassified due to having all MST markers below their LOD or being zero will be display as is shown in the following illustrations.



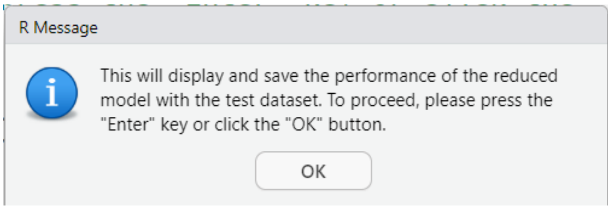




It is imperative to acknowledge that the reduction exhibits a propensity to augment the misclassification of the samples, albeit to a negligible extent when the MST markers demonstrate effective discriminatory capabilities. The rate of misclassification was found to be 1.31% and 0.81% for the complete set of MST markers when, respectively, the training and validation sets were used. Concerning to the reduced set of markers, the rate of misclassification increases to 3.14%. The user must evaluate the potential benefits of using a reduced set of MST markers against the potential drawbacks.

Chunk 25

This chunk displays the metrics of performance of this MST markers reduced model, starts by displaying a window pop-up.



And results are saved in a HTML file named:

'Summary_of_the_best_trained_model_and_confusion_matrix_reduced_set.html'.

Summary of the Best Model

```
H2OBinomialMetrics: gbm

MSE:  0.03564198
RMSE:  0.1887908
LogLoss:  0.1464374
Mean Per-Class Error:  0.04525298
AUC:  0.9725756
AUCPR:  0.9846425
Gini:  0.9451512
R^2:  0.8247745
AIC:  NaN

Confusion Matrix (vertical: actual; across: predicted) for F1-optimal threshold:
          Human Non_Human    Error     Rate
Human       167        14 0.077348  =14/181
Non_Human     6       450 0.013158   =6/456
Totals      173       464 0.031397  =20/637

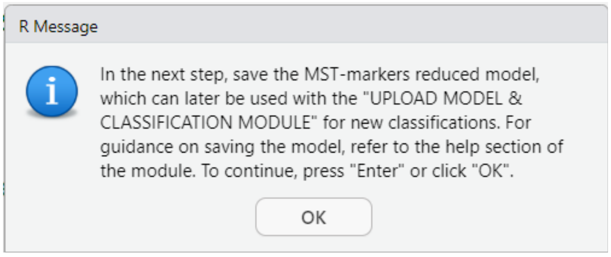
Maximum Metrics: Maximum metrics at their respective thresholds
                        metric threshold      value idx
1                       max f1  0.451923   0.978261  57
2                       max f2  0.451923   0.983392  57
3                 max f0point5  0.554359   0.976148  47
4                 max accuracy  0.451923   0.968603  57
5                max precision  0.999630   1.000000   0
6                   max recall  0.009197   1.000000 118
7              max specificity  0.999630   1.000000   0
8             max absolute_mcc  0.451923   0.922231  57
9   max min_per_class_accuracy  0.706504   0.950276  40
10 max mean_per_class_accuracy  0.554359   0.957025  47
11                     max tns  0.999630 181.000000   0
12                     max fns  0.999630 454.000000   0
13                     max fps  0.003586 181.000000 144
14                     max tps  0.009197 456.000000 118
15                     max tnr  0.999630   1.000000   0
16                     max fnr  0.999630   0.995614   0
17                     max fpr  0.003586   1.000000 144
18                     max tpr  0.009197   1.000000 118

Gains/Lift Table: Extract with `h2o.gainsLift(<model>, <data>)` or
`h2o.gainsLift(<model>, valid=<T/F>, xval=<T/F>)`
```

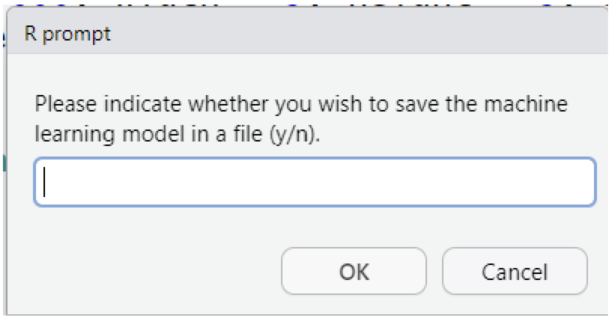
Please note that the AUC, AUCPR and the Gini index are all slightly lower than the metrics obtained in the validation with the complete set of MST markers.

Chunk 26

This chunk encodes to save the model in a file. The code starts displaying a pop-up window.



After clicking the 'OK' button, the user is prompted to save or not the model in a file.



This model is saved as: "GBM_model_R_1746694429827_55308" in the working directory.

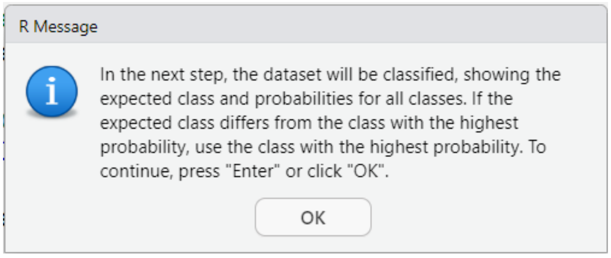
Chunk 27

This chunk encodes for the classification step for the model using the complete dataset of MST markers. The previously upload datafile will be evaluated. Following the classification of the samples, the subsequent information presented includes the class label, in this case 'Human' or 'Non-Human', and the probability of assignment to one of the two aforementioned classes. As stated in the introduction, this probability is known as the assignment probability and is independent of the percentage of mixture. Nevertheless, it is imperative for the user to evaluate each of the probabilities. Consequently, samples exhibiting a comparable assignment probability across both classes are those that, due to, but not limited to, their inherent complexity of the sample, high dilution or merely by the occurrence of MST markers, can present a challenge in terms of classification. Therefore, the user is encouraged to perform multiple repetitions of the machine learning process and compare the classification results obtained with different algorithms.

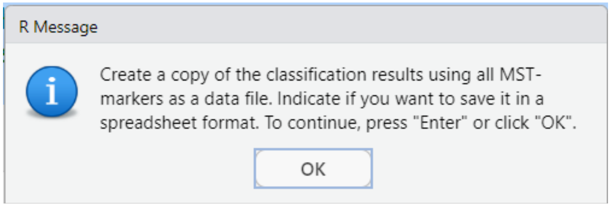
The dataset employed for the classification process consists of 15 water matrices derived from recognised contamination sources. It is evident that these samples have not been employed during the training process; consequently, they are considered novel and have not been presented to the model.

Furthermore, for each of them, an in-silico dilution of 1/10, 1/100, 1/1,000, and 1/10,000 was performed, with the occurrence of MST markers being adjusted according to their detection limit. It is important to note that the occurrence of these markers was evaluated in several EU laboratories, and minor variations in the detection limit were observed. Consequently, the first occurrence on the list was adopted as the reference point for all samples. It is intriguing for the user to observe the effect of the various dilutions on the classification of these parameters.

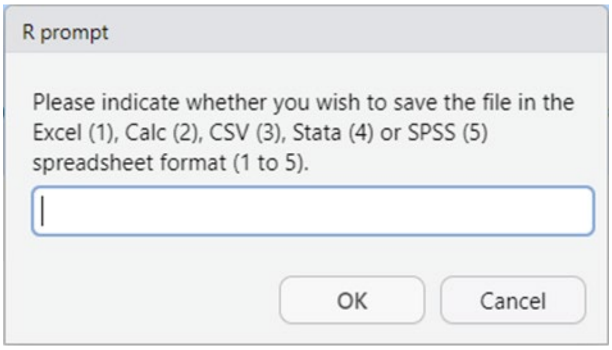
When the code is executed, a window pop-up appears, indicating that the classification process will commence.



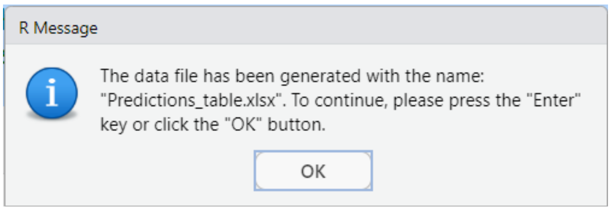
After clicking the 'OK' button, the classification is carried out. After that a new pop-up is displayed.



Once 'OK' button is clicked, the user will be prompted to choose the file type in which to save the classifications.



For instance, it is select 1.



The results of the classifications are displayed on this page and the following pages.

ID	predict	Human	Non_Human
E04HM_F	Human	0.998930473	0.001069527
E04HM_F_0.1	Human	0.999289940	0.000710060
E04HM_F_0.01	Human	0.996870973	0.003129027

ID	predict	Human	Non_Human
E04HM_F_0.001	Human	0.723937536	0.276062464
E04HM_F_0.0001	Human	0.130645200	0.869354800
E10PG_W	Non_Human	0.016335627	0.983664373
E10PG_W_0.1	Non_Human	0.006575106	0.993424894
E10PG_W_0.01	Non_Human	0.001156867	0.998843133
E10PG_W_0.001	Non_Human	0.007436496	0.992563504
E10PG_W_0.0001	Non_Human	0.039396246	0.960603754
E22PL_S	Non_Human	0.000419786	0.999580214
E22PL_S_0.1	Non_Human	0.001438105	0.998561895
E22PL_S_0.01	Non_Human	0.003883449	0.996116551
E22PL_S_0.001	Human	0.985796298	0.014203702
E22PL_S_0.0001	Non_Human	0.115737432	0.884262568
A04PG_F	Non_Human	0.006535958	0.993464042
A04PG_F_0.1	Non_Human	0.001707681	0.998292319
A04PG_F_0.01	Non_Human	0.001990098	0.998009902
A04PG_F_0.001	Non_Human	0.021175451	0.978824549
A04PG_F_0.0001	Non_Human	0.030741625	0.969258375
A08HM_W	Human	0.969227429	0.030772571
A08HM_W_0.1	Human	0.990851671	0.009148329
A08HM_W_0.01	Human	0.997918692	0.002081308
A08HM_W_0.001	Human	0.997139784	0.002860216
A08HM_W_0.0001	Human	0.982830284	0.017169716
A22CW_S	Non_Human	0.001117484	0.998882516
A22CW_S_0.1	Non_Human	0.002426990	0.997573010
A22CW_S_0.01	Non_Human	0.002905257	0.997094743
A22CW_S_0.001	Non_Human	0.007028406	0.992971594
A22CW_S_0.0001	Non_Human	0.011865262	0.988134738
P06PG_W	Non_Human	0.001345855	0.998654145
P06PG_W_0.1	Non_Human	0.000653926	0.999346074
P06PG_W_0.01	Non_Human	0.011026401	0.988973599
P06PG_W_0.001	Non_Human	0.025828396	0.974171604
P06PG_W_0.0001	Non_Human	0.039396246	0.960603754
P16CW_SP	Non_Human	0.000573159	0.999426841
P16CW_SP_0.1	Non_Human	0.000653029	0.999346971
P16CW_SP_0.01	Non_Human	0.000839621	0.999160379
P16CW_SP_0.001	Non_Human	0.004045827	0.995954173
P16CW_SP_0.0001	Non_Human	0.007158804	0.992841196
P19PL_SP	Non_Human	0.000281218	0.999718782
P19PL_SP_0.1	Non_Human	0.000998892	0.999001108
P19PL_SP_0.01	Non_Human	0.023575870	0.976424130
P19PL_SP_0.001	Non_Human	0.098818284	0.901181716
P19PL_SP_0.0001	Human	0.951796211	0.048203789
P22HM	Human	0.171877927	0.828122073
P22HM_0.1	Human	0.981407481	0.018592519
P22HM_0.01	Human	0.980128089	0.019871911
P22HM_0.001	Human	0.997597667	0.002402333
P22HM_0.0001	Non_Human	0.115737432	0.884262568

ID	predict	Human	Non_Human
D01HM_F	Human	0.999031761	0.000968239
D01HM_F_0.1	Human	0.999073658	0.000926342
D01HM_F_0.01	Human	0.999239853	0.000760147
D01HM_F_0.001	Human	0.975887820	0.024112180
D01HM_F_0.0001	Human	0.985233365	0.014766635
F06HM_W	Human	0.996599652	0.003400348
F06HM_W_0.1	Human	0.995704609	0.004295391
F06HM_W_0.01	Human	0.998698218	0.001301782
F06HM_W_0.001	Human	0.997330064	0.002669936
F06HM_W_0.0001	Human	0.920429292	0.079570708
F07PG_W	Non_Human	0.001736218	0.998263782
F07PG_W_0.1	Non_Human	0.005215753	0.994784247
F07PG_W_0.01	Non_Human	0.021006469	0.978993531
F07PG_W_0.001	Non_Human	0.034352460	0.965647540
F07PG_W_0.0001	Non_Human	0.039350872	0.960649128
F11HM_P	Human	0.994355059	0.005644941
F11HM_P_0.1	Human	0.985215986	0.014784014
F11HM_P_0.01	Human	0.998677444	0.001322556
F11HM_P_0.001	Human	0.998843225	0.001156775
F11HM_P_0.0001	Human	0.966606935	0.033393065
F15HM_S	Human	0.999269452	0.000730548
F15HM_S_0.1	Human	0.997690927	0.002309073
F15HM_S_0.01	Human	0.998700625	0.001299375
F15HM_S_0.001	Human	0.997595848	0.002404152
F15HM_S_0.0001	Human	0.975532397	0.024467603

It is important to note that the origin of the faecal contamination in these samples is known, thereby enabling the evaluation of the efficacy of the classification performed by the machine learning algorithm.

It is also important to note that the dilutions of these samples were deliberately engineered to allow the user to discern the impact of these dilutions on the classification outcomes

The following table illustrates the provenance of the sample. It is imperative to note that the display is limited exclusively to the authentic sample. It is considered that simulated dilutions are of the same origin.

ID	Class	Class_binomial
E04HM_F	Human	Human
E10PG_W	Porcine	Non_Human
E22PL_S	Poultry	Non_Human
A04PG_F	Porcine	Non_Human
A08HM_W	Human	Human
A22CW_S	Bovine	Non_Human
P06PG_W	Porcine	Non_Human
P16CW_SP	Bovine	Non_Human
P19PL_SP	Poultry	Non_Human

ID	Class	Class_binomial
P22HM	Human	Human
D01HM_F	Human	Human
F06HM_W	Human	Human
F07PG_W	Porcine	Non_Human
F11HM_P	Human	Human
F15HM_S	Human	Human

Please refer to the table below, which details the misclassifications.

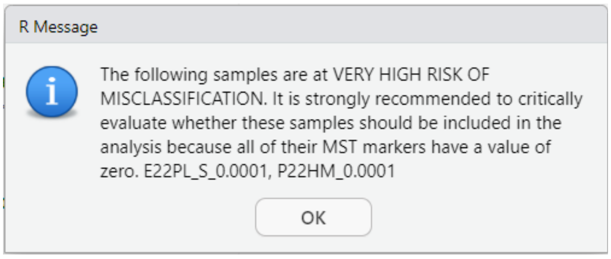
ID	predict	Human	Non_Human	Real	Mismatch
E04HM_F	Human	0.998930473	0.001069527	Human	
E04HM_F_0.1	Human	0.999289940	0.000710060	Human	
E04HM_F_0.01	Human	0.996870973	0.003129027	Human	
E04HM_F_0.001	Human	0.723937536	0.276062464	Human	
E04HM_F_0.0001	Human	0.130645200	0.869354800	Human	
E10PG_W	Non_Human	0.016335627	0.983664373	Non_Human	
E10PG_W_0.1	Non_Human	0.006575106	0.993424894	Non_Human	
E10PG_W_0.01	Non_Human	0.001156867	0.998843133	Non_Human	
E10PG_W_0.001	Non_Human	0.007436496	0.992563504	Non_Human	
E10PG_W_0.0001	Non_Human	0.039396246	0.960603754	Non_Human	
E22PL_S	Non_Human	0.000419786	0.999580214	Non_Human	
E22PL_S_0.1	Non_Human	0.001438105	0.998561895	Non_Human	
E22PL_S_0.01	Non_Human	0.003883449	0.996116551	Non_Human	
E22PL_S_0.001	Human	0.985796298	0.014203702	Non_Human	*
E22PL_S_0.0001	Non_Human	0.115737432	0.884262568	Non_Human	
A04PG_F	Non_Human	0.006535958	0.993464042	Non_Human	
A04PG_F_0.1	Non_Human	0.001707681	0.998292319	Non_Human	
A04PG_F_0.01	Non_Human	0.001990098	0.998009902	Non_Human	
A04PG_F_0.001	Non_Human	0.021175451	0.978824549	Non_Human	
A04PG_F_0.0001	Non_Human	0.030741625	0.969258375	Non_Human	
A08HM_W	Human	0.969227429	0.030772571	Human	
A08HM_W_0.1	Human	0.990851671	0.009148329	Human	
A08HM_W_0.01	Human	0.997918692	0.002081308	Human	
A08HM_W_0.001	Human	0.997139784	0.002860216	Human	
A08HM_W_0.0001	Human	0.982830284	0.017169716	Human	
A22CW_S	Non_Human	0.001117484	0.998882516	Non_Human	
A22CW_S_0.1	Non_Human	0.002426990	0.997573010	Non_Human	
A22CW_S_0.01	Non_Human	0.002905257	0.997094743	Non_Human	
A22CW_S_0.001	Non_Human	0.007028406	0.992971594	Non_Human	
A22CW_S_0.0001	Non_Human	0.011865262	0.988134738	Non_Human	
P06PG_W	Non_Human	0.001345855	0.998654145	Non_Human	
P06PG_W_0.1	Non_Human	0.000653926	0.999346074	Non_Human	
P06PG_W_0.01	Non_Human	0.011026401	0.988973599	Non_Human	
P06PG_W_0.001	Non_Human	0.025828396	0.974171604	Non_Human	
P06PG_W_0.0001	Non_Human	0.039396246	0.960603754	Non_Human	
P16CW_SP	Non_Human	0.000573159	0.999426841	Non_Human	
P16CW_SP_0.1	Non_Human	0.000653029	0.999346971	Non_Human	

ID	predict	Human	Non_Human	Real	Mismatch
P16CW_SP_0.01	Non_Human	0.000839621	0.999160379	Non_Human	
P16CW_SP_0.001	Non_Human	0.004045827	0.995954173	Non_Human	
P16CW_SP_0.0001	Non_Human	0.007158804	0.992841196	Non_Human	
P19PL_SP	Non_Human	0.000281218	0.999718782	Non_Human	
P19PL_SP_0.1	Non_Human	0.000998892	0.999001108	Non_Human	
P19PL_SP_0.01	Non_Human	0.023575870	0.976424130	Non_Human	
P19PL_SP_0.001	Non_Human	0.098818284	0.901181716	Non_Human	
P19PL_SP_0.0001	Human	0.951796211	0.048203789	Non_Human	*
P22HM	Human	0.171877927	0.828122073	Human	
P22HM_0.1	Human	0.981407481	0.018592519	Human	
P22HM_0.01	Human	0.980128089	0.019871911	Human	
P22HM_0.001	Human	0.997597667	0.002402333	Human	
P22HM_0.0001	Non_Human	0.115737432	0.884262568	Human	*
D01HM_F	Human	0.999031761	0.000968239	Human	
D01HM_F_0.1	Human	0.999073658	0.000926342	Human	
D01HM_F_0.01	Human	0.999239853	0.000760147	Human	
D01HM_F_0.001	Human	0.975887820	0.024112180	Human	
D01HM_F_0.0001	Human	0.985233365	0.014766635	Human	
F06HM_W	Human	0.996599652	0.003400348	Human	
F06HM_W_0.1	Human	0.995704609	0.004295391	Human	
F06HM_W_0.01	Human	0.998698218	0.001301782	Human	
F06HM_W_0.001	Human	0.997330064	0.002669936	Human	
F06HM_W_0.0001	Human	0.920429292	0.079570708	Human	
F07PG_W	Non_Human	0.001736218	0.998263782	Non_Human	
F07PG_W_0.1	Non_Human	0.005215753	0.994784247	Non_Human	
F07PG_W_0.01	Non_Human	0.021006469	0.978993531	Non_Human	
F07PG_W_0.001	Non_Human	0.034352460	0.965647540	Non_Human	
F07PG_W_0.0001	Non_Human	0.039350872	0.960649128	Non_Human	
F11HM_P	Human	0.994355059	0.005644941	Human	
F11HM_P_0.1	Human	0.985215986	0.014784014	Human	
F11HM_P_0.01	Human	0.998677444	0.001322556	Human	
F11HM_P_0.001	Human	0.998843225	0.001156775	Human	
F11HM_P_0.0001	Human	0.966606935	0.033393065	Human	
F15HM_S	Human	0.999269452	0.000730548	Human	
F15HM_S_0.1	Human	0.997690927	0.002309073	Human	
F15HM_S_0.01	Human	0.998700625	0.001299375	Human	
F15HM_S_0.001	Human	0.997595848	0.002404152	Human	
F15HM_S_0.0001	Human	0.975532397	0.024467603	Human	

Three samples are misclassified:

E22PL_S_0.001	Human	0.985796298	0.014203702	Non_Human	*
P19PL_SP_0.0001	Human	0.951796211	0.048203789	Non_Human	*
P22HM_0.0001	Non_Human	0.115737432	0.884262568	Human	*

As is observed, only three out of the 75 samples were incorrectly classified, and these three samples corresponded to those which had been subjected to high dilution. Two of this three samples were previously detected as highly risky to be misclassified.

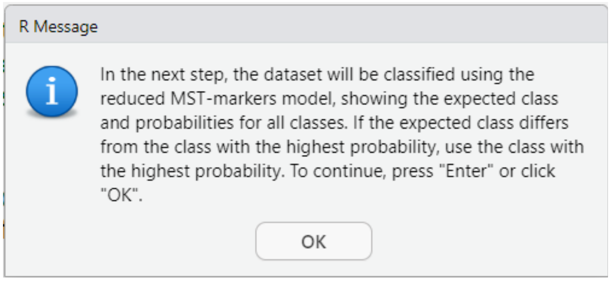


Therefore, as previously indicated, the user should anticipate this contingency in case they do not have an adequate MST marker to estimate the possible dilution. Furthermore, it is recommended that samples with known origins be incorporated into the classification dataset, whilst ensuring that these samples have not been utilised in the machine learning process. In the present example, a 96% accuracy rate was achieved in the classification of the samples.

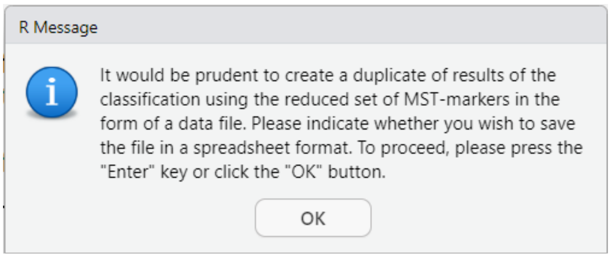
Chunk 28

This chunk is responsible for the classification using the model trained with a reduced set of MST markers. Please be aware that reducing the number of predictors may result in an increase in misclassification.

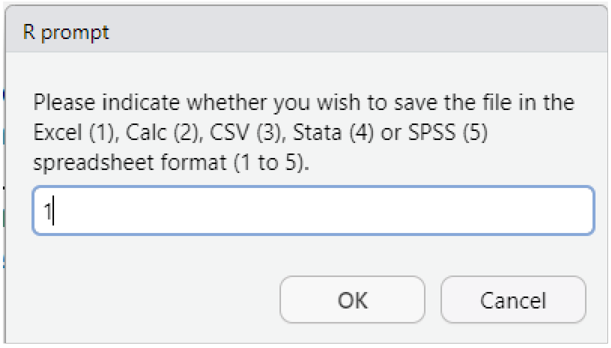
A pop-up information window will appear during the running process.



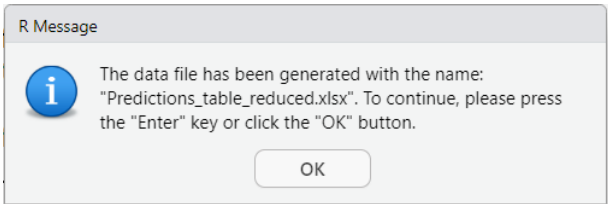
After clicking the 'OK' button, the classification is carried out and the user is prompted to save or not the file.



The selection of the option to save in the Excel file is to be made.



The file is saved as 'Predictions_table_reduced.xlsx'.



The results with the expected classification labels are shown in the following pages.

ID	predict	Human	Non_Human	Real	Mismatch
E04HM_F	Human	0.994065972	0.005934028	Human	
E04HM_F_0.1	Human	0.984311200	0.015688800	Human	
E04HM_F_0.01	Human	0.994885500	0.005114500	Human	
E04HM_F_0.001	Human	0.683692825	0.316307175	Human	
E04HM_F_0.0001	Non_Human	0.133630711	0.866369289	Human	*
E10PG_W	Human	0.994275170	0.005724830	Non_Human	*
E10PG_W_0.1	Human	0.683692825	0.316307175	Non_Human	*
E10PG_W_0.01	Non_Human	0.071594466	0.928405534	Non_Human	
E10PG_W_0.001	Non_Human	0.028661192	0.971338808	Non_Human	
E10PG_W_0.0001	Non_Human	0.028661192	0.971338808	Non_Human	
E22PL_S	Human	0.965813006	0.034186994	Non_Human	*
E22PL_S_0.1	Human	0.918615570	0.081384430	Non_Human	*
E22PL_S_0.01	Non_Human	0.424800443	0.575199557	Non_Human	
E22PL_S_0.001	Human	0.797006176	0.202993824	Non_Human	*
E22PL_S_0.0001	Non_Human	0.028661192	0.971338808	Non_Human	
A04PG_F	Non_Human	0.028661192	0.971338808	Non_Human	
A04PG_F_0.1	Non_Human	0.028661192	0.971338808	Non_Human	
A04PG_F_0.01	Non_Human	0.028661192	0.971338808	Non_Human	
A04PG_F_0.001	Non_Human	0.028661192	0.971338808	Non_Human	
A04PG_F_0.0001	Non_Human	0.028661192	0.971338808	Non_Human	
A08HM_W	Human	0.989096414	0.010903586	Human	
A08HM_W_0.1	Human	0.994065972	0.005934028	Human	
A08HM_W_0.01	Human	0.984311200	0.015688800	Human	
A08HM_W_0.001	Human	0.994887218	0.005112782	Human	
A08HM_W_0.0001	Non_Human	0.208418580	0.791581420	Human	*

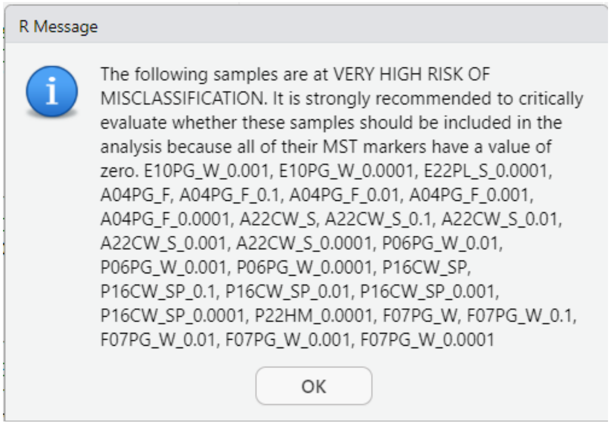
ID	predict	Human	Non_Human	Real	Mismatch
A22CW_S	Non_Human	0.028661192	0.971338808	Non_Human	
A22CW_S_0.1	Non_Human	0.028661192	0.971338808	Non_Human	
A22CW_S_0.01	Non_Human	0.028661192	0.971338808	Non_Human	
A22CW_S_0.001	Non_Human	0.028661192	0.971338808	Non_Human	
A22CW_S_0.0001	Non_Human	0.028661192	0.971338808	Non_Human	
P06PG_W	Non_Human	0.014457743	0.985542257	Non_Human	
P06PG_W_0.1	Non_Human	0.004238431	0.995761569	Non_Human	
P06PG_W_0.01	Non_Human	0.028661192	0.971338808	Non_Human	
P06PG_W_0.001	Non_Human	0.028661192	0.971338808	Non_Human	
P06PG_W_0.0001	Non_Human	0.028661192	0.971338808	Non_Human	
P16CW_SP	Non_Human	0.028661192	0.971338808	Non_Human	
P16CW_SP_0.1	Non_Human	0.028661192	0.971338808	Non_Human	
P16CW_SP_0.01	Non_Human	0.028661192	0.971338808	Non_Human	
P16CW_SP_0.001	Non_Human	0.028661192	0.971338808	Non_Human	
P16CW_SP_0.0001	Non_Human	0.028661192	0.971338808	Non_Human	
P19PL_SP	Human	0.989127714	0.010872286	Non_Human	*
P19PL_SP_0.1	Human	0.988940524	0.011059476	Non_Human	*
P19PL_SP_0.01	Human	0.968680561	0.031319439	Non_Human	*
P19PL_SP_0.001	Non_Human	0.166080993	0.833919007	Non_Human	
P19PL_SP_0.0001	Human	0.794800075	0.205199925	Non_Human	*
P22HM	Human	0.993233639	0.006766361	Human	
P22HM_0.1	Human	0.968680561	0.031319439	Human	
P22HM_0.01	Non_Human	0.139376841	0.860623159	Human	*
P22HM_0.001	Human	0.993558812	0.006441188	Human	
P22HM_0.0001	Non_Human	0.028661192	0.971338808	Human	*
D01HM_F	Human	0.989914847	0.010085153	Human	
D01HM_F_0.1	Human	0.986411081	0.013588919	Human	
D01HM_F_0.01	Human	0.980327866	0.019672134	Human	
D01HM_F_0.001	Human	0.760400571	0.239599429	Human	
D01HM_F_0.0001	Non_Human	0.329257600	0.670742400	Human	*
F06HM_W	Human	0.994027662	0.005972338	Human	
F06HM_W_0.1	Human	0.994065972	0.005934028	Human	
F06HM_W_0.01	Human	0.992043633	0.007956367	Human	
F06HM_W_0.001	Human	0.985414602	0.014585398	Human	
F06HM_W_0.0001	Human	0.960708189	0.039291811	Human	
F07PG_W	Non_Human	0.028661192	0.971338808	Non_Human	
F07PG_W_0.1	Non_Human	0.028661192	0.971338808	Non_Human	
F07PG_W_0.01	Non_Human	0.028661192	0.971338808	Non_Human	
F07PG_W_0.001	Non_Human	0.028661192	0.971338808	Non_Human	
F07PG_W_0.0001	Non_Human	0.028661192	0.971338808	Non_Human	
F11HM_P	Human	0.994027662	0.005972338	Human	
F11HM_P_0.1	Human	0.994065972	0.005934028	Human	
F11HM_P_0.01	Human	0.994802668	0.005197332	Human	
F11HM_P_0.001	Human	0.995273669	0.004726331	Human	
F11HM_P_0.0001	Human	0.902901065	0.097098935	Human	
F15HM_S	Human	0.994027662	0.005972338	Human	
F15HM_S_0.1	Human	0.984515112	0.015484888	Human	

ID	predict	Human	Non_Human	Real	Mismatch
F15HM_S_0.01	Human	0.952022566	0.047977434	Human	
F15HM_S_0.001	Human	0.984515971	0.015484029	Human	
F15HM_S_0.0001	Human	0.820305029	0.179694971	Human	

The misclassified samples are shown below.

E04HM_F_0.0001	Non_Human	0,133630711	0,866369289	Human	*
E10PG_W	Human	0,994275170	0,005724830	Non_Human	*
E10PG_W_0.1	Human	0,683692825	0,316307175	Non_Human	*
E22PL_S	Human	0,965813006	0,034186994	Non_Human	*
E22PL_S_0.1	Human	0,918615570	0,081384430	Non_Human	*
E22PL_S_0.001	Human	0,797006176	0,202993824	Non_Human	*
A08HM_W_0.0001	Non_Human	0,208418580	0,791581420	Human	*
P19PL_SP	Human	0,989127714	0,010872286	Non_Human	*
P19PL_SP_0.1	Human	0,988940524	0,011059476	Non_Human	*
P19PL_SP_0.01	Human	0,968680561	0,031319439	Non_Human	*
P19PL_SP_0.0001	Human	0,794800075	0,205199925	Non_Human	*
P22HM_0.01	Non_Human	0,139376841	0,860623159	Human	*
P22HM_0.0001	Non_Human	0,028661192	0,971338808	Human	*
D01HM_F_0.0001	Non_Human	0,329257600	0,670742400	Human	*

When the potential samples at high risk of misclassification were analysed (listed below), all misclassified samples (listed above) were identified as likely to be misclassified, except for the samples identify as E10PG_W and E10PG_W_0.1.

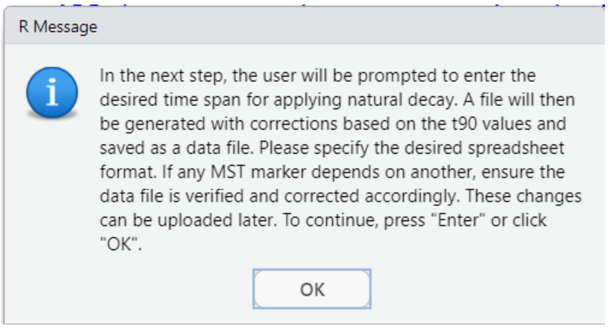


Following the classification process, it was found that 14 samples (18.7%) had been misclassified. Of these, 50% were samples containing the label PL. Notably, 50% of these misclassified samples carried the label 'PL', which denotes faecal pollution originating from poultry sources. As previously discussed, the decision to use a model based on a reduced set of MST markers, as well as whether to include or exclude samples with low marker abundance, ultimately rests with the user.

Chunk 29

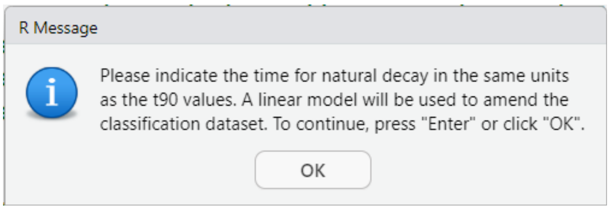
This chunk encodes for applying the natural decay. This option is only available if the user has previously indicated that a file containing the t90 for each MST markers is available.

Once it begins to run, a window pop-up is displayed.

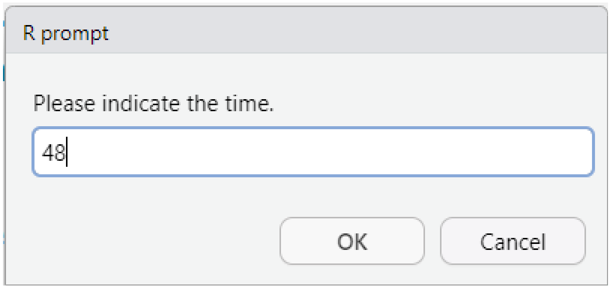


It is essential to incorporate the t90 values expressed at the same time units for all MST markers utilised in the training. In circumstances where t90 values are not available, a possible alternative may be to employ an analogous parameter. For instance, if the MST is regarded as a genetic marker, then use another genetic marker. However, it is imperative to acknowledge the limitations inherent in this approach. It is important to note that the values of t90 for the same marker can vary according to the natural conditions of the matrix.

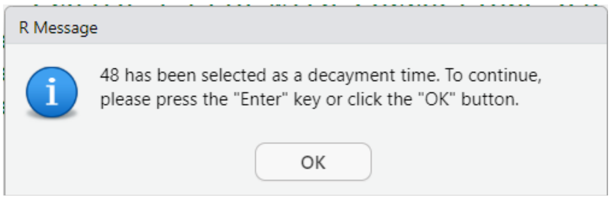
After clicking the 'OK' button, a new pop-up window will appear, informing the user that a linear decay model will be utilised to amend the occurrence of the samples contained within the classification file.



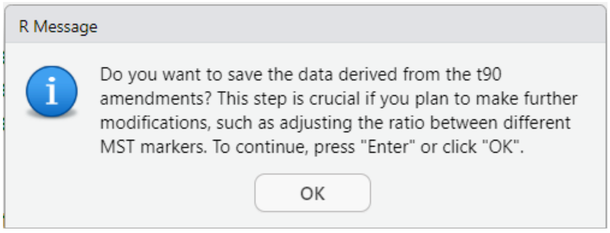
After clicking the 'OK' button, a new window opens prompting the user to indicate what time would apply. For demonstrative purposes 48 hours are indicated.



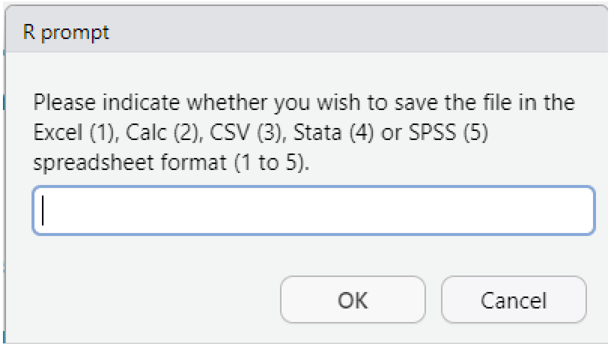
Once the time is introduced, a new pop-up will appear, displaying the input value (it should be noted that no time units are indicated).



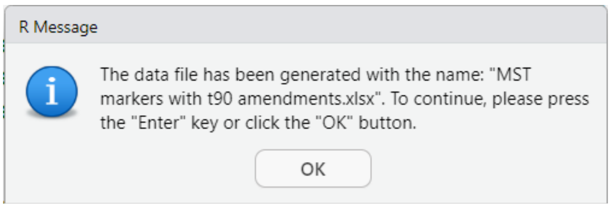
The program continues to prompt the user regarding the potential for saving the results of the amendments. It is recommended that the file be saved. In the event of any modifications to any marker being required (for example, for parameters that are covariates of MST markers), it is imperative that this file is available.



Once the ‘OK’ button is pressed, the following pop-ups display and the program prompts the user to specify the desired format for the file.



Upon selection of a file format in Excel, a pop-up notification is displayed, providing the file name.



Below is shown a part of the content of the saved file.

Classification samples			**Modified classification samples**			**With decayment**			
ID	**HMBactPhg**	**PGBactPhg**	**ID**	**HMBactPhg**	**PGBactPhg**	**t90**	**Time**	**HMBactPhg**	**PGBactPhg**
E04HM_F	1154.55	1300.00	E04HM_F	11545.4545	13000	48	48	1154.55	1300.00
E04HM_F_0.1	115.45	130.00	E04HM_F_0.1	1154.54545	1300	48	48	115.45	130.00
E04HM_F_0.01	11.55	13.00	E04HM_F_0.01	115.454545	130	48	48	11.55	13.00
E04HM_F_0.001	0.00	0.00	E04HM_F_0.001	0	0	48	48	0.00	0.00
E04HM_F_0.0001	0.00	0.00	E04HM_F_0.0001	0	0	48	48	0.00	0.00
E10PG_W	0.00	35350.00	E10PG_W	0	353500	48	48	0.00	35350.00
E10PG_W_0.1	0.00	3535.00	E10PG_W_0.1	0	35350	48	48	0.00	3535.00

E10PG_W_0.01	0.00	353.50	E10PG_W_0.01	0	3535	48	48	0.00	353.50
E10PG_W_0.001	0.00	35.35	E10PG_W_0.001	0	353.5	48	48	0.00	35.35
E10PG_W_0.0001	0.00	0.00	E10PG_W_0.0001	0	0	48	48	0.00	0.00
E22PL_S	0.00	0.00	E22PL_S	0	0	48	48	0.00	0.00
E22PL_S_0.1	0.00	0.00	E22PL_S_0.1	0	0	48	48	0.00	0.00
E22PL_S_0.01	0.00	0.00	E22PL_S_0.01	0	0	48	48	0.00	0.00
E22PL_S_0.001	0.00	0.00	E22PL_S_0.001	0	0	48	48	0.00	0.00
E22PL_S_0.0001	0.00	0.00	E22PL_S_0.0001	0	0	48	48	0.00	0.00
A04PG_F	0.00	0.00	A04PG_F	0	0	48	48	0.00	0.00
A04PG_F_0.1	0.00	0.00	A04PG_F_0.1	0	0	48	48	0.00	0.00
A04PG_F_0.01	0.00	0.00	A04PG_F_0.01	0	0	48	48	0.00	0.00
A04PG_F_0.001	0.00	0.00	A04PG_F_0.001	0	0	48	48	0.00	0.00
A04PG_F_0.0001	0.00	0.00	A04PG_F_0.0001	0	0	48	48	0.00	0.00

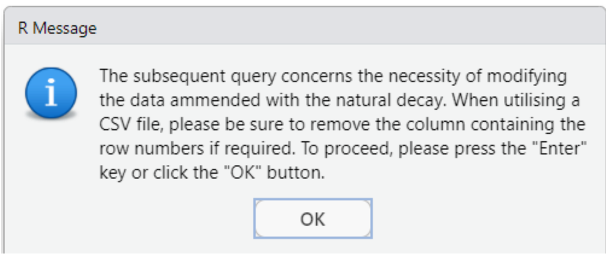
The following exposition shall elucidate the procedural framework. The t90 values are indicative of the time for which the matrix must be exposed in order to achieve a one-logarithmic-unit decrease in marker occurrence. In this instance, the calculation is performed in reverse order. That is to say, at a given time, $Time_t$, an occurrence is observed. However, the objective is to ascertain the occurrence before, at a specific point in time, designated as $Time_{(t-x)}$, where *x* denotes the time span selected by the user.

The data presented in the above table includes four samples, the first two columns of the table correspond to the occurrence in the classification file. The shaded columns illustrate the data for which the occurrence has been amended in accordance with the span time that has been designated by the user. For the sake of simplicity, a time equivalent to the t90 of the parameter (set at 48 hours) has been utilised.

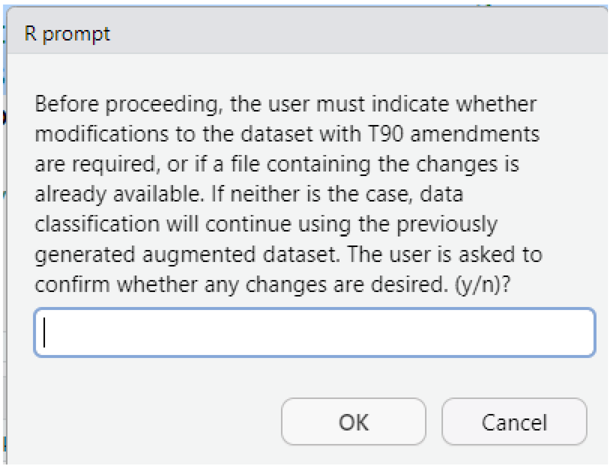
The right-hand side of the table presents the calculations applied to the decay, thus enabling verification of their congruence with the data in the classification table. However, it is imperative to acknowledge that the validity of this calculation is contingent upon the occurrence of events that surpass the detection limit. It is crucial to note that the events in question must not be categorised as zero in machine learning. Therefore, it is very important to exercise caution when employing a decay correction. It should be noted that markers with lower detection limits, which are less prone to having zero values, will likely be better adjusted.

This phenomenon must be given special consideration in the context of MST markers that play a pivotal role (higher importance) in the model's adjustment.

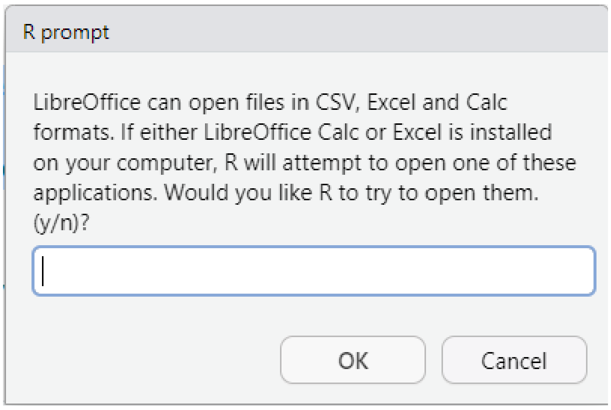
The program then displays the following pop-up window, which informs the user that the saved file may be modified. This is especially relevant when the file contains a covariable marker.



Once the 'OK' button is clicked, the program continues. A new pop-up windows prompting the user if any modifications have been made.

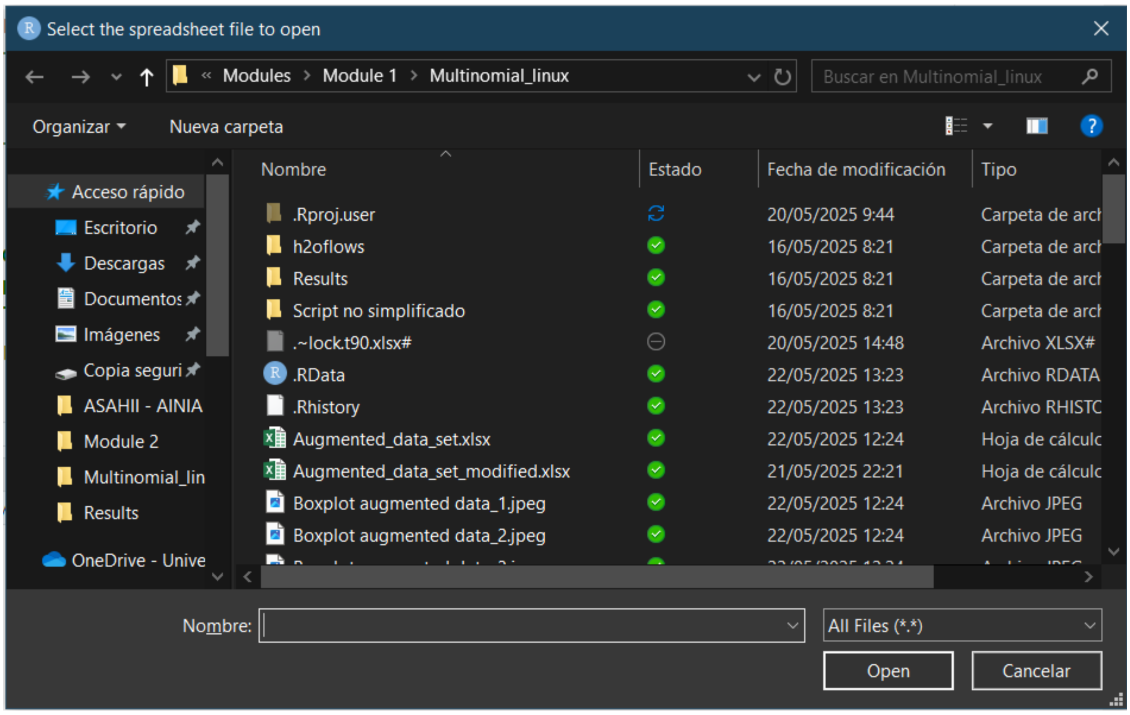


In case of affirmative response, the program displays the following pop-up.



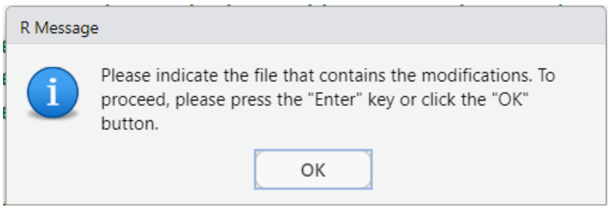
It should be noted that the selection or non-selection of any of these programs (LibreOffice Calc or Excel) is entirely at the discretion of the user; the spreadsheet containing the data with the decay modifications can be modified using the software with which the individual is most comfortable. Furthermore, the segment of the code that is responsible for initiating the application has been subjected to testing on the Windows 11 version 24h2 operating system and the Ubuntu versions 22.04 LTS (Jammy Jellyfish) and 24.04 LTS (Noble Numbat). It is important to note that, due to the inherent complexity of operating systems, there is a possibility that the application may not launch correctly. It is important to note that the R program has been designed to run on a continuous basis. This means that it will continue running whether or not any of these applications are opened.

If the user answers affirmatively a window to upload the file will be displayed.

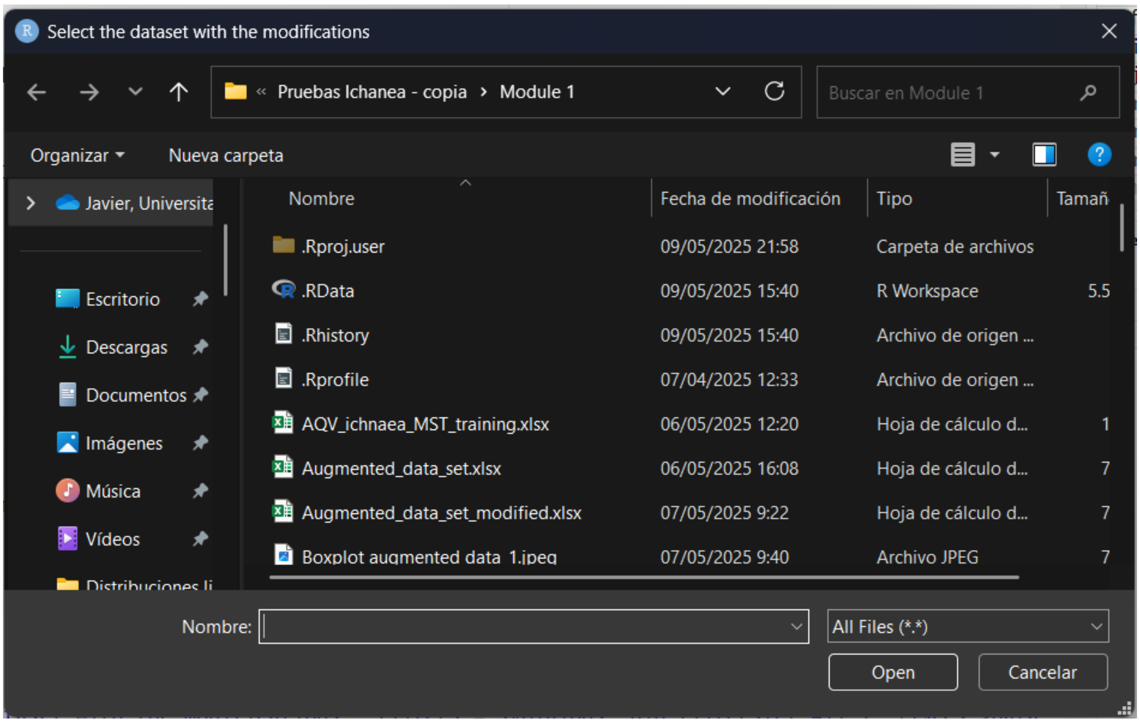


After that, Excel or LibreOffice Calc will be launched.

If the answer is negative or if you have edited the file manually, the program will then proceed with the following pop-up window.



And a window to select the file is displayed.

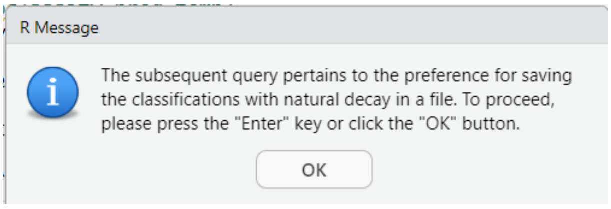


After selecting the file, the program continues to the following chunk.

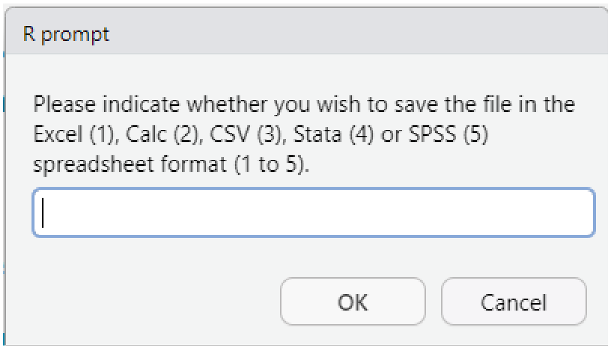
Chunk 30

This chunk is responsible for predicting using all the MST markers once the amendments have been carried out.

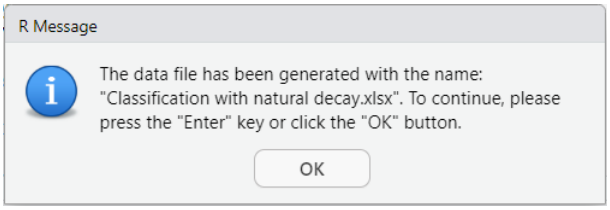
The chunk starts with the following pop-up window.



Upon clicking the 'OK' button, a prompt window will appear, prompting the user to specify the file format.



For instance, the Excel format is selected.



Upon successful completion of the file saving process, a pop-up notification is generated to inform the user that the data has been stored in an Excel file bearing the name 'Classification with natural decay.xlsx'.

In the following pages the results of the classification are shown. Samples misclassified are indicated with *'.

ID	predict	Human	Non_Human	Real	Mismatch
E04HM_F	Human	0.998930445	0.001069555	Human	
E04HM_F_0.1	Human	0.999126693	0.000873307	Human	
E04HM_F_0.01	Human	0.998567355	0.001432645	Human	
E04HM_F_0.001	Human	0.162986214	0.837013786	Human	
E04HM_F_0.0001	Human	0.987771753	0.012228247	Human	
E10PG_W	Non_Human	0.016327839	0.983672161	Non_Human	
E10PG_W_0.1	Non_Human	0.007995513	0.992004487	Non_Human	
E10PG_W_0.01	Non_Human	0.005524281	0.994475719	Non_Human	
E10PG_W_0.001	Non_Human	0.007442637	0.992557363	Non_Human	
E10PG_W_0.0001	Non_Human	0.039396246	0.960603754	Non_Human	
E22PL_S	Non_Human	0.000522925	0.999477075	Non_Human	
E22PL_S_0.1	Non_Human	0.001440538	0.998559462	Non_Human	
E22PL_S_0.01	Non_Human	0.005100189	0.994899811	Non_Human	
E22PL_S_0.001	Human	0.988018915	0.011981085	Non_Human	*
E22PL_S_0.0001	Non_Human	0.115737432	0.884262568	Non_Human	
A04PG_F	Non_Human	0.007897150	0.992102850	Non_Human	
A04PG_F_0.1	Non_Human	0.002803855	0.997196145	Non_Human	
A04PG_F_0.01	Non_Human	0.002008328	0.997991672	Non_Human	
A04PG_F_0.001	Non_Human	0.021281466	0.978718534	Non_Human	
A04PG_F_0.0001	Non_Human	0.030741625	0.969258375	Non_Human	
A08HM_W	Human	0.962496336	0.037503664	Human	
A08HM_W_0.1	Human	0.989906545	0.010093455	Human	
A08HM_W_0.01	Human	0.998046880	0.001953120	Human	
A08HM_W_0.001	Human	0.997607189	0.002392811	Human	
A08HM_W_0.0001	Human	0.480930251	0.519069749	Human	
A22CW_S	Non_Human	0.001126722	0.998873278	Non_Human	
A22CW_S_0.1	Non_Human	0.002422159	0.997577841	Non_Human	
A22CW_S_0.01	Non_Human	0.003261891	0.996738109	Non_Human	
A22CW_S_0.001	Non_Human	0.004726631	0.995273369	Non_Human	
A22CW_S_0.0001	Non_Human	0.011864441	0.988135559	Non_Human	
P06PG_W	Non_Human	0.001015362	0.998984638	Non_Human	
P06PG_W_0.1	Non_Human	0.000662132	0.999337868	Non_Human	
P06PG_W_0.01	Non_Human	0.010199610	0.989800390	Non_Human	

ID	predict	Human	Non_Human	Real	Mismatch
P06PG_W_0.001	Non_Human	0.025870541	0.974129459	Non_Human	
P06PG_W_0.0001	Non_Human	0.039396246	0.960603754	Non_Human	
P16CW_SP	Non_Human	0.000570780	0.999429220	Non_Human	
P16CW_SP_0.1	Non_Human	0.000655316	0.999344684	Non_Human	
P16CW_SP_0.01	Non_Human	0.000705173	0.999294827	Non_Human	
P16CW_SP_0.001	Non_Human	0.003570447	0.996429553	Non_Human	
P16CW_SP_0.0001	Non_Human	0.006784286	0.993215714	Non_Human	
P19PL_SP	Non_Human	0.000281218	0.999718782	Non_Human	
P19PL_SP_0.1	Non_Human	0.000996219	0.999003781	Non_Human	
P19PL_SP_0.01	Non_Human	0.004105714	0.995894286	Non_Human	
P19PL_SP_0.001	Human	0.846252198	0.153747802	Non_Human	*
P19PL_SP_0.0001	Human	0.982763692	0.017236308	Non_Human	*
P22HM	Non_Human	0.103290610	0.896709390	Human	*
P22HM_0.1	Human	0.958992250	0.041007750	Human	
P22HM_0.01	Human	0.995886272	0.004113728	Human	
P22HM_0.001	Human	0.99589474	0.00410526	Human	
P22HM_0.0001	Non_Human	0.115737432	0.884262568	Human	*
D01HM_F	Human	0.998816976	0.001183024	Human	
D01HM_F_0.1	Human	0.999074712	0.000925288	Human	
D01HM_F_0.01	Human	0.999237171	0.000762829	Human	
D01HM_F_0.001	Human	0.983894455	0.016105545	Human	
D01HM_F_0.0001	Human	0.990768198	0.009231802	Human	
F06HM_W	Human	0.997220609	0.002779391	Human	
F06HM_W_0.1	Human	0.990849213	0.009150787	Human	
F06HM_W_0.01	Human	0.998293952	0.001706048	Human	
F06HM_W_0.001	Human	0.997317922	0.002682078	Human	
F06HM_W_0.0001	Human	0.986836546	0.013163454	Human	
F07PG_W	Non_Human	0.001439765	0.998560235	Non_Human	
F07PG_W_0.1	Non_Human	0.005067708	0.994932292	Non_Human	
F07PG_W_0.01	Non_Human	0.021147672	0.978852328	Non_Human	
F07PG_W_0.001	Non_Human	0.030901186	0.969098814	Non_Human	
F07PG_W_0.0001	Non_Human	0.039339018	0.960660982	Non_Human	
F11HM_P	Human	0.96722387	0.03277613	Human	
F11HM_P_0.1	Human	0.988413581	0.011586419	Human	
F11HM_P_0.01	Human	0.998943505	0.001056495	Human	
F11HM_P_0.001	Human	0.998882042	0.001117958	Human	
F11HM_P_0.0001	Human	0.937556799	0.062443201	Human	
F15HM_S	Human	0.999269412	0.000730588	Human	
F15HM_S_0.1	Human	0.998519042	0.001480958	Human	
F15HM_S_0.01	Human	0.998700325	0.001299675	Human	
F15HM_S_0.001	Human	0.998811569	0.001188431	Human	
F15HM_S_0.0001	Human	0.966993568	0.033006432	Human	

Of the total samples, five samples exhibited misclassification, representing 6.67% of the misclassification cases.

The following samples were misclassified:

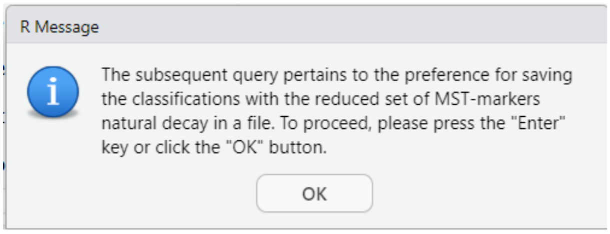
E22PL_S_0.001	Human	0.988018915	0.011981085	Non_Human	*
P19PL_SP_0.001	Human	0.846252198	0.153747802	Non_Human	*
P19PL_SP_0.0001	Human	0.982763692	0.017236308	Non_Human	*
P22HM	Non_Human	0.103290610	0.896709390	Human	*
P22HM_0.0001	Non_Human	0.115737432	0.884262568	Human	*

It was also found that samples E22PL_S_0.001, P19PL_SP_0.0001 and P22HM_0.0001 were misclassified when the decay amendment was not taken into account. Please note that the undiluted sample P22HM and the diluted (1/1000) sample P19PL_SP are now misclassified.

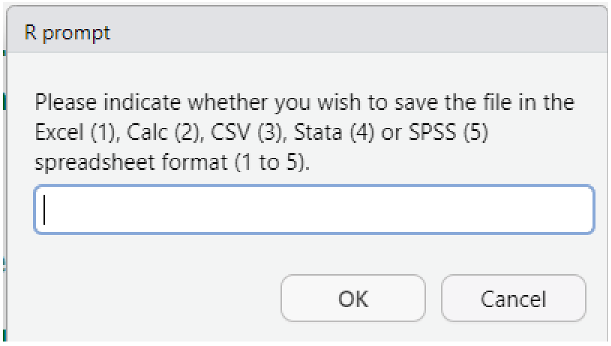
Chunk 31

This chunk is responsible for predicting using the reduced set MST markers once the amendments have been carried out.

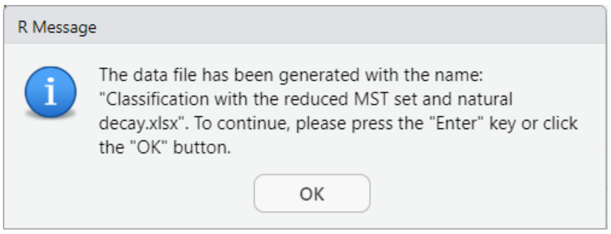
The chunk starts with the following pop-up window.



Upon clicking the 'OK' button, a prompt window will appear, prompting the user to specify the file format.



For instance, the Excel format has been selected.



Upon completion of the file save process, a pop-up notification is generated, confirming the successful storage of the data in an Excel file entitled 'Classification with the reduced MST set and natural decay.xlsx'.

In the ensuing pages, the outcomes of the classification process are presented. Samples that have been misclassified are indicated by an asterisk (*).

ID	predict	Human	Non_Human	Real	Mismatch
E04HM_F	Human	0.994028	0.005972338	Human	
E04HM_F_0.1	Human	0.992044	0.007956367	Human	
E04HM_F_0.01	Human	0.985724	0.014276052	Human	
E04HM_F_0.001	Non_Human	0.441312	0.558688462	Human	*
E04HM_F_0.0001	Human	0.720562	0.279437684	Human	
E10PG_W	Human	0.995402	0.004598479	Non_Human	*
E10PG_W_0.1	Human	0.955360	0.044639561	Non_Human	*
E10PG_W_0.01	Non_Human	0.143944	0.856055839	Non_Human	
E10PG_W_0.001	Non_Human	0.028661	0.971338808	Non_Human	
E10PG_W_0.0001	Non_Human	0.028661	0.971338808	Non_Human	
E22PL_S	Human	0.975280	0.024720100	Non_Human	*
E22PL_S_0.1	Human	0.969332	0.030667527	Non_Human	*
E22PL_S_0.01	Human	0.834972	0.165027881	Non_Human	*
E22PL_S_0.001	Human	0.984093	0.015906905	Non_Human	*
E22PL_S_0.0001	Non_Human	0.028661	0.971338808	Non_Human	
A04PG_F	Non_Human	0.028661	0.971338808	Non_Human	
A04PG_F_0.1	Non_Human	0.028661	0.971338808	Non_Human	
A04PG_F_0.01	Non_Human	0.028661	0.971338808	Non_Human	
A04PG_F_0.001	Non_Human	0.028661	0.971338808	Non_Human	
A04PG_F_0.0001	Non_Human	0.028661	0.971338808	Non_Human	
A08HM_W	Human	0.989096	0.010903586	Human	
A08HM_W_0.1	Human	0.994028	0.005972338	Human	
A08HM_W_0.01	Human	0.992044	0.007956367	Human	
A08HM_W_0.001	Human	0.984152	0.015847718	Human	
A08HM_W_0.0001	Non_Human	0.441312	0.558688462	Human	*
A22CW_S	Non_Human	0.028661	0.971338808	Non_Human	
A22CW_S_0.1	Non_Human	0.028661	0.971338808	Non_Human	
A22CW_S_0.01	Non_Human	0.028661	0.971338808	Non_Human	
A22CW_S_0.001	Non_Human	0.028661	0.971338808	Non_Human	
A22CW_S_0.0001	Non_Human	0.028661	0.971338808	Non_Human	
P06PG_W	Non_Human	0.015001	0.984999028	Non_Human	
P06PG_W_0.1	Non_Human	0.004236	0.995763627	Non_Human	
P06PG_W_0.01	Non_Human	0.028661	0.971338808	Non_Human	
P06PG_W_0.001	Non_Human	0.028661	0.971338808	Non_Human	
P06PG_W_0.0001	Non_Human	0.028661	0.971338808	Non_Human	
P16CW_SP	Non_Human	0.028661	0.971338808	Non_Human	
P16CW_SP_0.1	Non_Human	0.028661	0.971338808	Non_Human	
P16CW_SP_0.01	Non_Human	0.028661	0.971338808	Non_Human	
P16CW_SP_0.001	Non_Human	0.028661	0.971338808	Non_Human	
P16CW_SP_0.0001	Non_Human	0.028661	0.971338808	Non_Human	
P19PL_SP	Human	0.989128	0.010872286	Non_Human	*
P19PL_SP_0.1	Human	0.982734	0.017265560	Non_Human	*
P19PL_SP_0.01	Human	0.995373	0.004627329	Non_Human	*

ID	predict	Human	Non_Human	Real	Mismatch
P19PL_SP_0.001	Human	0.943120	0.056879793	Non_Human	*
P19PL_SP_0.0001	Non_Human	0.486630	0.513369871	Non_Human	
P22HM	Human	0.993945	0.006055276	Human	
P22HM_0.1	Human	0.968681	0.031319439	Human	
P22HM_0.01	Non_Human	0.132229	0.867770870	Human	*
P22HM_0.001	Human	0.901199	0.098801483	Human	
P22HM_0.0001	Non_Human	0.028661	0.971338808	Human	*
D01HM_F	Human	0.988936	0.011063613	Human	
D01HM_F_0.1	Human	0.984515	0.015484888	Human	
D01HM_F_0.01	Human	0.961399	0.038600989	Human	
D01HM_F_0.001	Human	0.924387	0.075612925	Human	
D01HM_F_0.0001	Human	0.995452	0.004548447	Human	
F06HM_W	Human	0.994028	0.005972338	Human	
F06HM_W_0.1	Human	0.994028	0.005972338	Human	
F06HM_W_0.01	Human	0.990803	0.009196976	Human	
F06HM_W_0.001	Human	0.908688	0.091312252	Human	
F06HM_W_0.0001	Human	0.935873	0.064126991	Human	
F07PG_W	Non_Human	0.028661	0.971338808	Non_Human	
F07PG_W_0.1	Non_Human	0.028661	0.971338808	Non_Human	
F07PG_W_0.01	Non_Human	0.028661	0.971338808	Non_Human	
F07PG_W_0.001	Non_Human	0.028661	0.971338808	Non_Human	
F07PG_W_0.0001	Non_Human	0.028661	0.971338808	Non_Human	
F11HM_P	Human	0.994028	0.005972338	Human	
F11HM_P_0.1	Human	0.994066	0.005934028	Human	
F11HM_P_0.01	Human	0.992082	0.007917591	Human	
F11HM_P_0.001	Human	0.994278	0.005721810	Human	
F11HM_P_0.0001	Human	0.683693	0.316307175	Human	
F15HM_S	Human	0.994028	0.005972338	Human	
F15HM_S_0.1	Human	0.989389	0.010611260	Human	
F15HM_S_0.01	Human	0.995014	0.004985580	Human	
F15HM_S_0.001	Human	0.995273	0.004726681	Human	
F15HM_S_0.0001	Human	0.897120	0.102880038	Human	

A total of 14 samples (18.67%) were misclassified. The samples are as follows: E04HM_F_0.001, E10PG_W, E10PG_W_0.1, E22PL_S, E22PL_S_0.1, E22PL_S_0.01, E22PL_S_0.001, A08HM_W_0.0001, P19PL_SP, P19PL_SP_0.1, P19PL_SP_0.01, P19PL_SP_0.001, P22HM_0.01 and P22HM_0.0001. The number of misclassifications remains consistent in instances where no natural decay has been applied, with the exception of the samples identified as D01HM_F_0.0001 and P22HM_0.0001.

R NOTEBOOK

MODULE II: MODEL UPLOAD AND CLASSIFICATION

R NOTEBOOK FOR UPLOADING A PREVIOUSLY TRAINED MODEL AND PERFORMING THE CLASSIFICATION

AIM OF THIS MANUAL

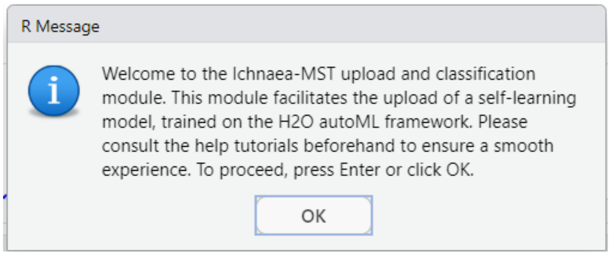
This R Notebook contains the code responsible for loading a previously trained H2O AutoML model, importing the dataset to be classified, and making the necessary modifications to this dataset if the effect of natural decay is to be considered.

It is imperative to acknowledge that this module has been meticulously crafted as a complementary component to Module I. It is presumed that the user has already accomplished the training process using the aforementioned module. It is also important to note that Module I incorporates a series of analytical tools that evaluate the extent of divergence (low, moderate, high) between the classification dataset and the training dataset. Furthermore, it draws attention to potential issues within the classification dataset, including the presence of very low occurrences of MST markers.

In the event that the user has access to both the augmented dataset and the original training dataset, they may undertake an analysis to verify the suitability of their values prior to proceeding with classification.

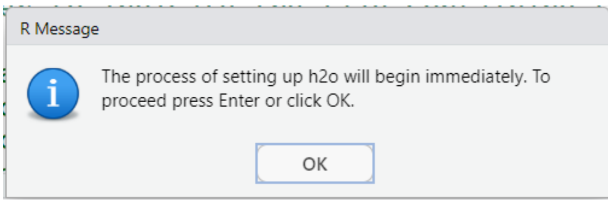
Chunk 1

This chunk encodes the following welcome screen.

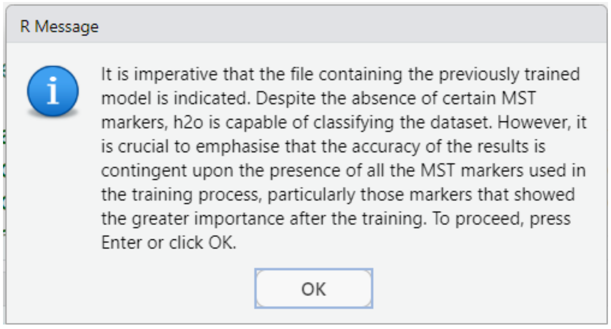


Chunk 2

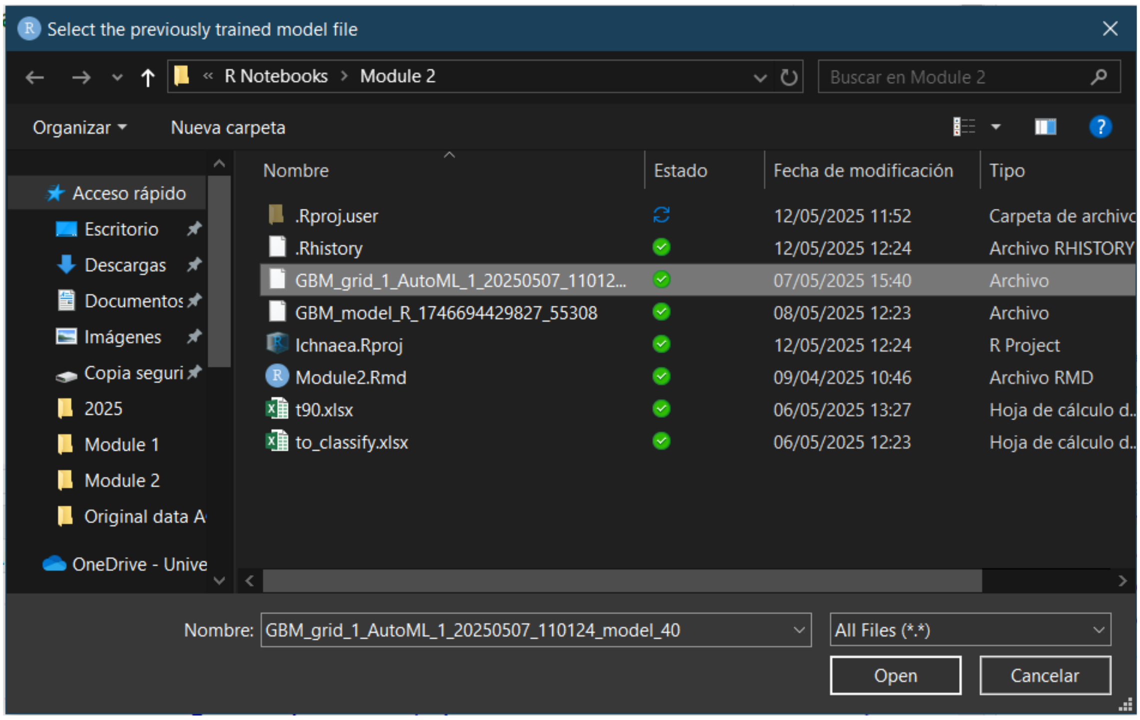
The second chunk of the program is responsible for initiating the H2O AutoML framework. After its initiation the following pop-up information window appears.



Following the selection of the "OK" button, the program will initiate the H2O framework. This process is a time-consuming one. Once the framework has been initiated, the code will launch a new pop-up window. This will indicate that the file containing the dataset to be classified should contain the same predictors used during the training process. For illustrative purposes, the same dataset used for the classification in Module I will be uploaded again.



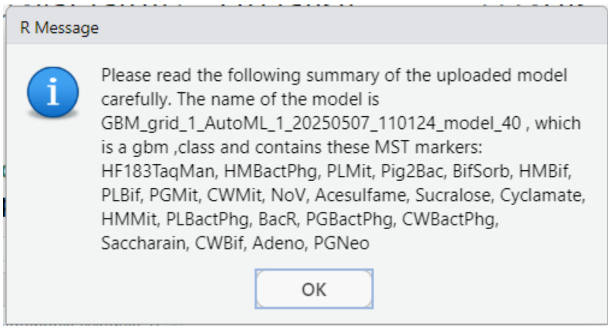
After clicking the “OK” button, a new window prompts the user to upload the trained model. The model contains the complete set of MST markers that have been trained in the example provided in Module I. This model has been named "GBM_grid_1_AutoML_1_20250507_110124_model_40".



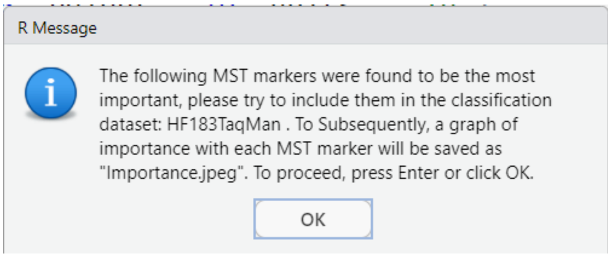
Once the H2O model has been uploaded, the process will be complete.

Chunk 3

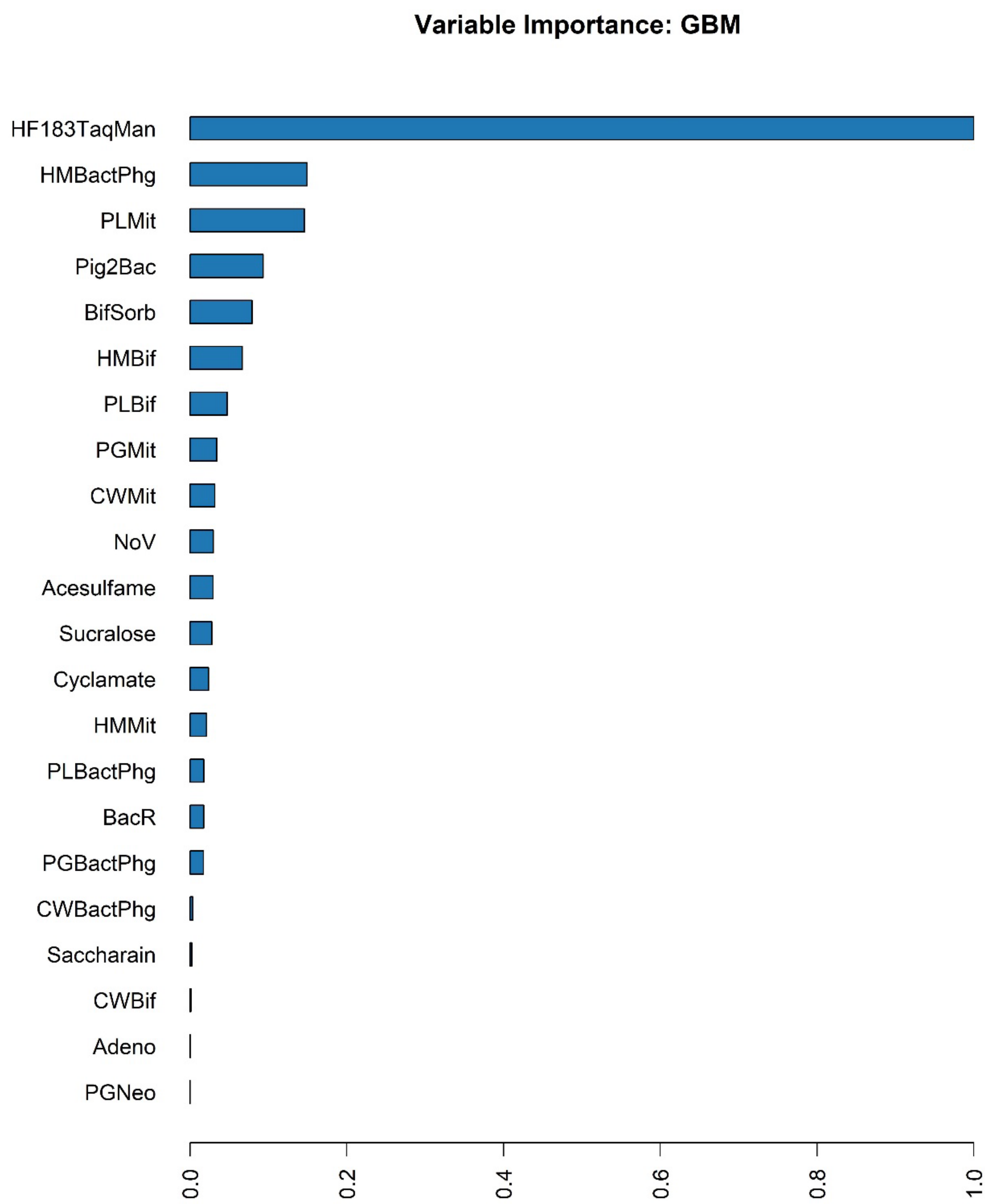
This chunk shows the main characteristics of the uploaded model. The program shows a pop-up windows with a summary of the information. Please ensure that you read this information carefully to confirm that the model specified is correct.



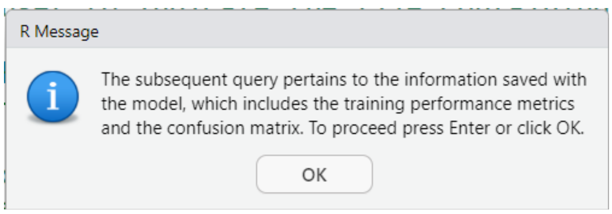
When the user clicks "OK", a new pop-up window will appear, displaying the MST markers that are significant in the training (with an importance rating of >0.5).



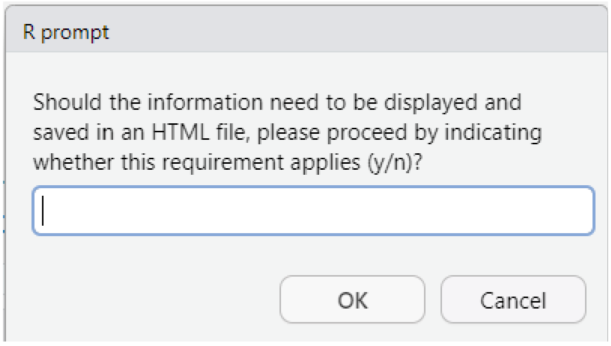
A plot with the importance of the MST markers is also saved as a JPEG file.



Following the selection of the "OK" option, the program will display a new pop-up.



After that, a new pop-up appears, prompting the user to save the information to an HTML file or not.



If the user responds affirmatively, the chunk saves the information in an HTML file named 'Previously trained model details.html', the contents of which are displayed on the following pages. Please refer to Chunk 20 of the Module I Help Section for a detailed description of each performance metric and how to deal with truncated HTML files.

Trained Model Summary

```
Model Details:
==============

H2OBinomialModel: gbm
Model ID:  GBM_grid_1_AutoML_1_20250507_110124_model_40
Model Summary:
  number_of_trees number_of_internal_trees model_size_in_bytes min_depth
1              92                       92              164492         0
  max_depth mean_depth min_leaves max_leaves mean_leaves
1        15   12.47826          1        295   136.93478

H2OBinomialMetrics: gbm
** Reported on training data. **

MSE:  0.01164491
RMSE:  0.1079116
LogLoss:  0.04570297
Mean Per-Class Error:  0.0234375
AUC:  0.9985139
AUCPR:  0.9994202
Gini:  0.9970279
R^2:  0.9420785
AIC:  NaN

Confusion Matrix (vertical: actual; across: predicted) for F1-optimal threshold:
          Human Non_Human    Error      Rate
Human       610        30 0.046875   =30/640
Non_Human     0      1656 0.000000    =0/1656
Totals      610      1686 0.013066  =30/2296

Maximum Metrics: Maximum metrics at their respective thresholds
                        metric threshold       value idx
1                       max f1  0.884263    0.991023 279
2                       max f2  0.884263    0.996390 279
3                 max f0point5  0.885768    0.986641 278
4                 max accuracy  0.884263    0.986934 279
5                max precision  0.999847    1.000000   0
6                   max recall  0.884263    1.000000 279
7              max specificity  0.999847    1.000000   0
8             max absolute_mcc  0.884263    0.967556 279
9   max min_per_class_accuracy  0.884263    0.953125 279
10 max mean_per_class_accuracy  0.884263    0.976562 279
11                     max tns  0.999847  640.000000   0
12                     max fns  0.999847 1654.000000   0
13                     max fps  0.000438  640.000000 399
14                     max tps  0.884263 1656.000000 279
15                     max tnr  0.999847    1.000000   0
16                     max fnr  0.999847    0.998792   0
17                     max fpr  0.000438    1.000000 399
18                     max tpr  0.884263    1.000000 279
```

```
Gains/Lift Table: Extract with `h2o.gainsLift(<model>, <data>)` or
`h2o.gainsLift(<model>, valid=<T/F>, xval=<T/F>)`

H2OBinomialMetrics: gbm
** Reported on cross-validation data. **
** 5-fold cross-validation on training data (Metrics computed for combined holdout
predictions) **

MSE:  0.01308044
RMSE:  0.1143697
LogLoss:  0.04852276
Mean Per-Class Error:  0.02734375
AUC:  0.9978898
AUCPR:  0.9991744
Gini:  0.9957796
R^2:  0.9349382
AIC:  NaN

Confusion Matrix (vertical: actual; across: predicted) for F1-optimal threshold:
          Human Non_Human    Error      Rate
Human       605        35 0.054688   =35/640
Non_Human     0      1656 0.000000   =0/1656
Totals      605      1691 0.015244  =35/2296

Maximum Metrics: Maximum metrics at their respective thresholds
                        metric threshold       value idx
1                       max f1  0.459546    0.989543 261
2                       max f2  0.459546    0.995791 261
3                 max f0point5  0.891748    0.984751 243
4                 max accuracy  0.459546    0.984756 261
5                max precision  0.999780    1.000000   0
6                   max recall  0.459546    1.000000 261
7              max specificity  0.999780    1.000000   0
8             max absolute_mcc  0.459546    0.962157 261
9   max min_per_class_accuracy  0.860460    0.963768 252
10 max mean_per_class_accuracy  0.854927    0.973792 255
11                     max tns  0.999780  640.000000   0
12                     max fns  0.999780 1652.000000   0
13                     max fps  0.000423  640.000000 399
14                     max tps  0.459546 1656.000000 261
15                     max tnr  0.999780    1.000000   0
16                     max fnr  0.999780    0.997585   0
17                     max fpr  0.000423    1.000000 399
18                     max tpr  0.459546    1.000000 261

Gains/Lift Table: Extract with `h2o.gainsLift(<model>, <data>)` or
`h2o.gainsLift(<model>, valid=<T/F>, xval=<T/F>)`
Cross-Validation Metrics Summary: 
              mean       sd cv_1_valid cv_2_valid cv_3_valid cv_4_valid
accuracy  0.985629 0.002904   0.980435   0.986928   0.986928   0.986928
aic             NA 0.000000         NA         NA         NA         NA
auc       0.998036 0.000874   0.997003   0.997298   0.998274   0.999131
err       0.014371 0.002904   0.019565   0.013072   0.013072   0.013072
err_count 6.600000 1.341641   9.000000   6.000000   6.000000   6.000000
          cv_5_valid
accuracy    0.986928
aic               NA
auc         0.998473
err         0.013072
err_count   6.000000

---
                mean       sd cv_1_valid cv_2_valid cv_3_valid cv_4_valid
pr_auc      0.999260 0.000247   0.999019   0.999111   0.999207   0.999660
precision   0.981097 0.003870   0.974719   0.983003   0.981366   0.985030
r2          0.933768 0.018149   0.902821   0.933445   0.948159   0.939952
recall      0.999394 0.001355   1.000000   1.000000   1.000000   0.996970
rmse        0.113948 0.011506   0.134193   0.110803   0.105447   0.110151
specificity 0.948821 0.016885   0.920354   0.946429   0.958042   0.961240
            cv_5_valid
pr_auc        0.999302
precision     0.981366
r2            0.944461
recall        1.000000
rmse          0.109144
specificity   0.958042
```

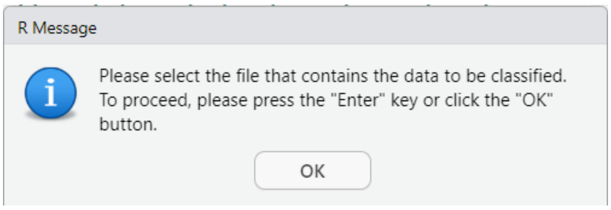
It should be noted that these results are identical to those achieved during the training process, as shown in Chunk 20 of Module I of the Help Section.

Chunk 4

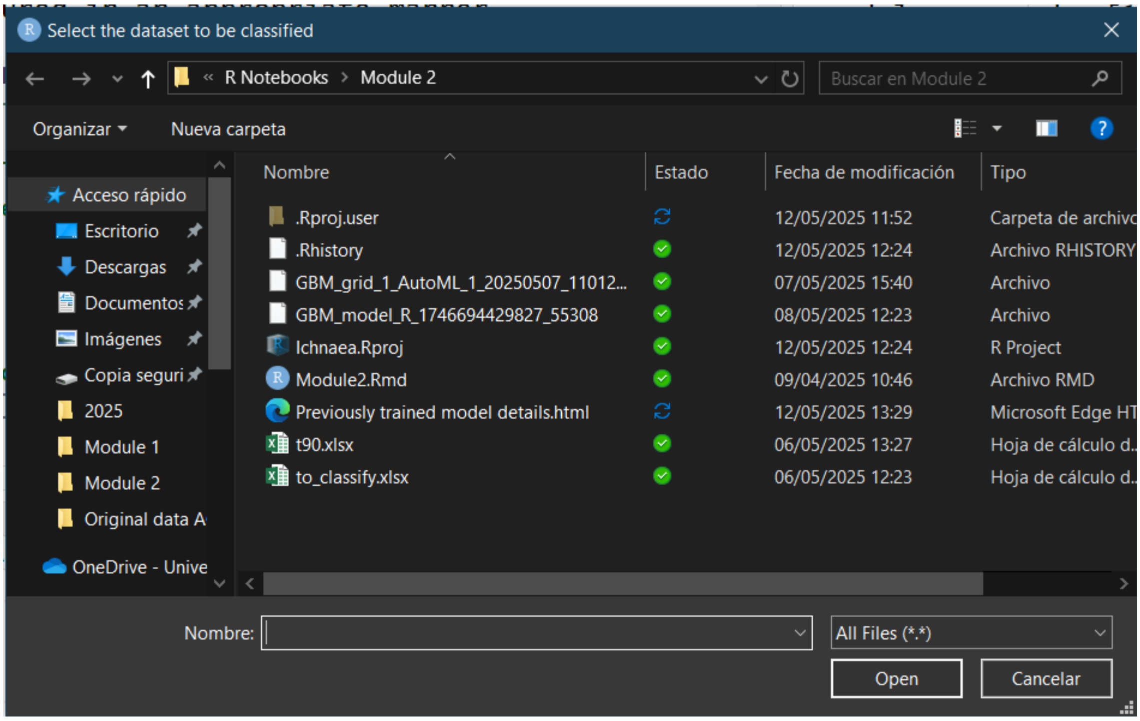
Chunk 4 is responsible for the encoding of data for opening and storage as datasheets.

Chunk 5

This chunk encodes the upload of the file containing the data to be classified. When executed, the following information pop-up window is displayed.



If the user responds in the affirmative, a window will appear prompting the user to specify the location of the file to be uploaded.

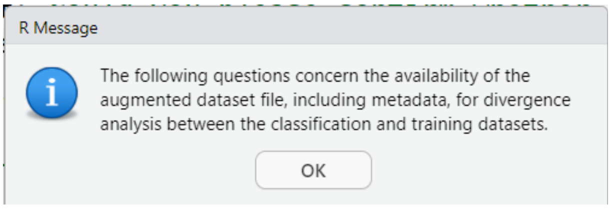


This chunk is terminated upon selection of the file.

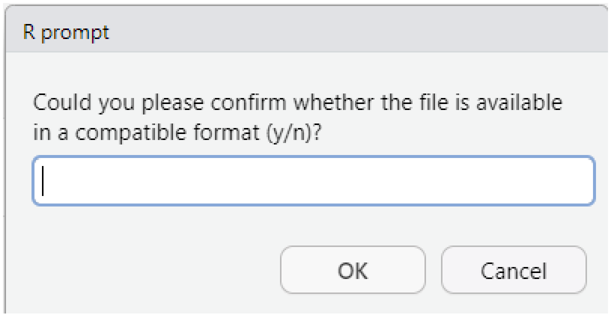
Chunk 6

This chunk is responsible for verifying the presence of any potential issues related to the divergence between the dataset utilised for training and the new classification dataset, or the low occurrence of MST markers in the latter. However, in our particular instance, the data provided are identical to those utilised in Module I.

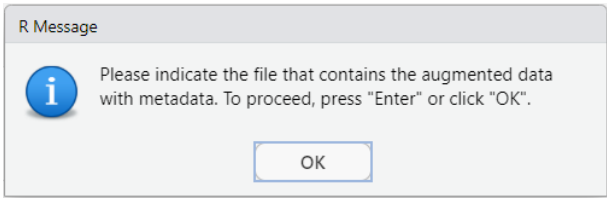
Upon initiation of the present chunk, the following pop-up will be displayed.



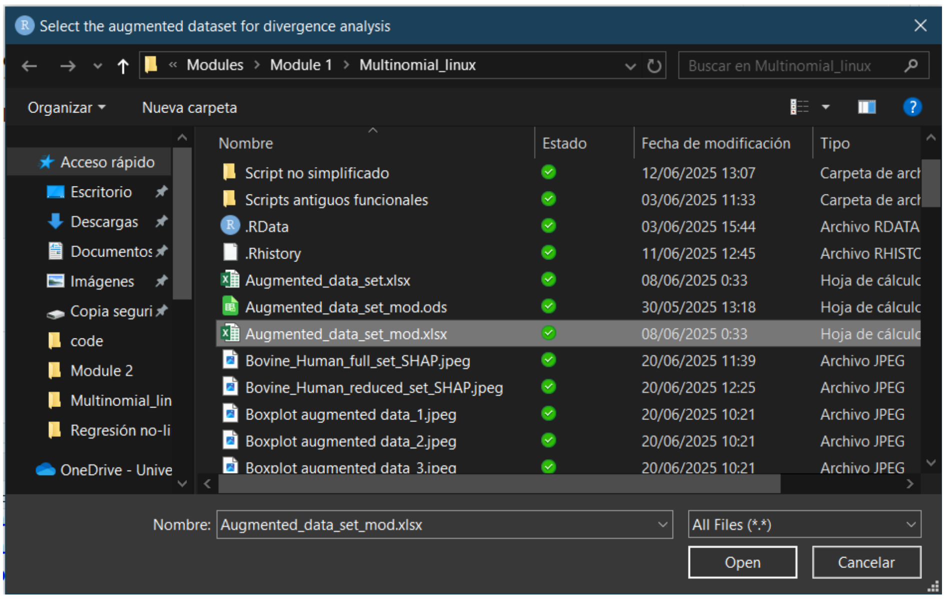
Subsequent to the pressing of the "OK" button, the user is prompted to indicate whether the file is available. Please be advised that this file is the only one that contains the information required for the divergence analysis.



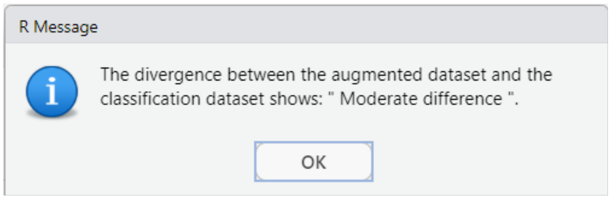
For instance, this chunk will be run in twice. The workflow should be displayed on two occasions: initially when the augmented file is present, and subsequently when only the original training file is available. Should any of the aforementioned items be unavailable, the process will be terminated at this point.



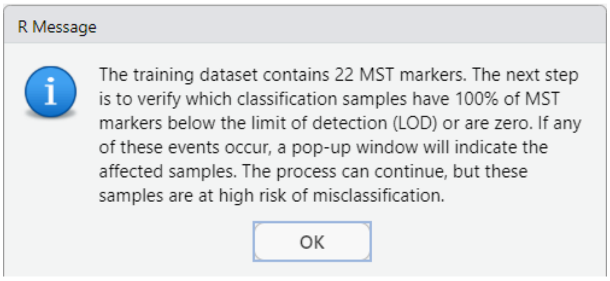
Following the selection of "OK", a window will appear in which the file can be selected.



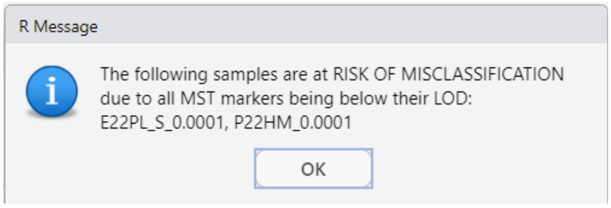
Following the selection of the appropriate file, the analysis will be conducted and the results will be displayed in the subsequent pop-up window.



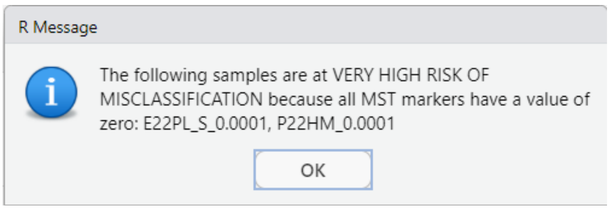
Subsequent to this, the analysis corresponding to the values below the limit of detection (LOD) and zero-values is presented in the following images. The first item to be displayed is a pop-up indicating the number of MST markers contained in the specified dataset.



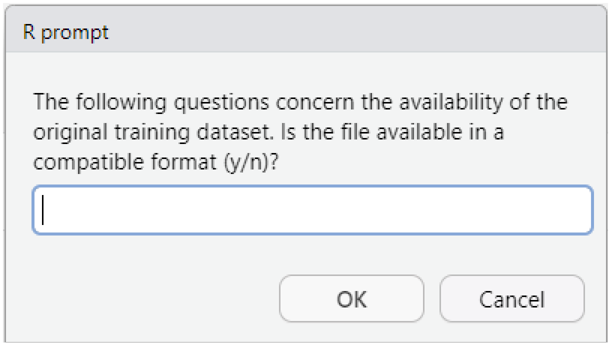
Subsequent to the selection of the "OK" option, the samples which classification that may be adversely affected by the low occurrence are displayed. Should no pop-up be visible, this is indicative of the absence of any issues detected in the specified dataset.



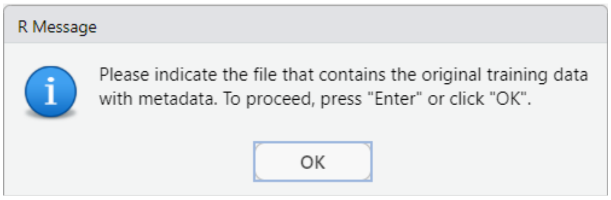
Followed by the samples with all MST-markers with a zero-value.



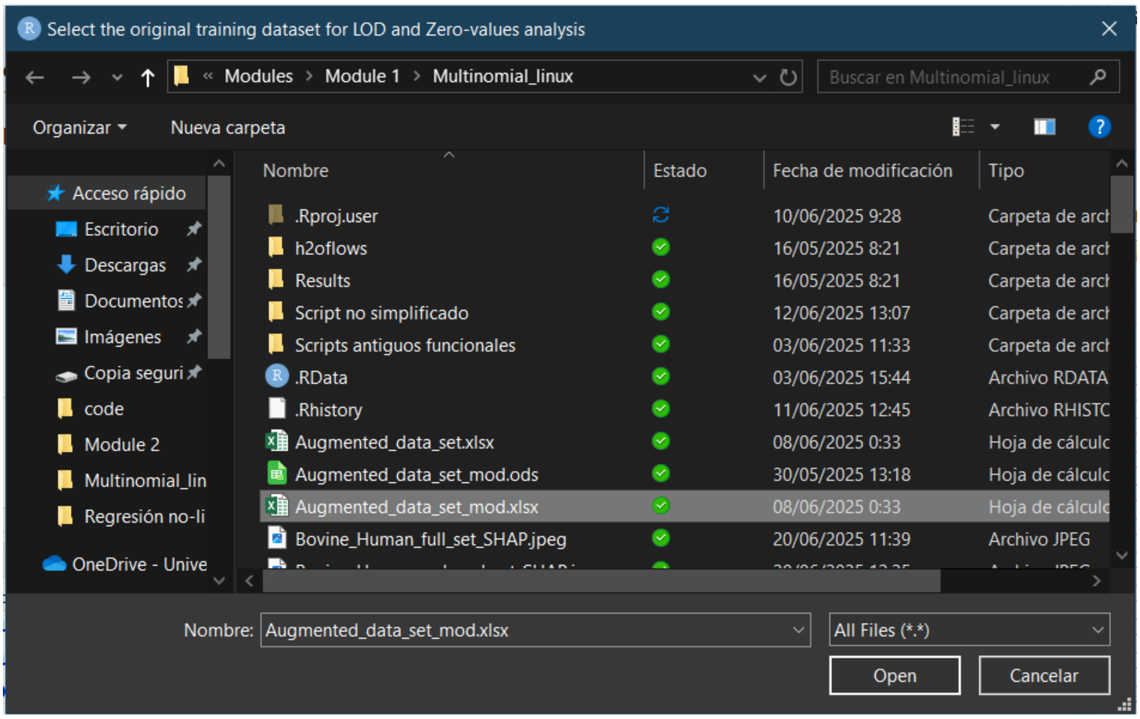
In the event that the file of the augmented dataset with the metadata is unavailable, the program omits the divergence analysis and proceeds to the detection of LOD and zero-values. At this point, the user is prompted to confirm the availability of the original training file.



In the event of a positive response from the user, the programme displays the following pop-up.



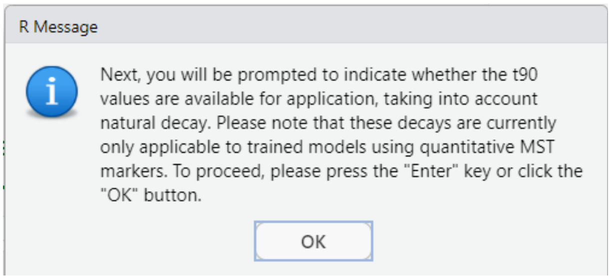
The program opens a window in which the user can select the required file.



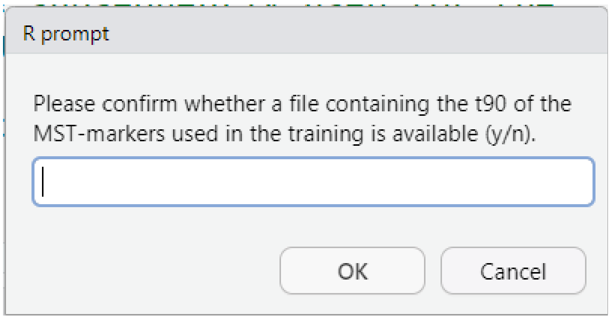
Subsequent to the selection of the file, the program will proceed with the analysis. As the metadata of both files is identical, given that they have been augmented with the original training file, it can be concluded that the results obtained are also identical. For further information, please refer to pages 14 and 15 of this manual.

Chunk 7

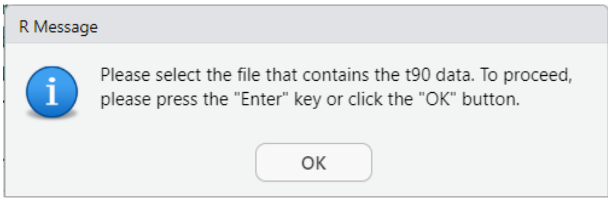
This particular chunk is responsible for the management of natural decay. The program commences with the appearance of the following information pop-up.



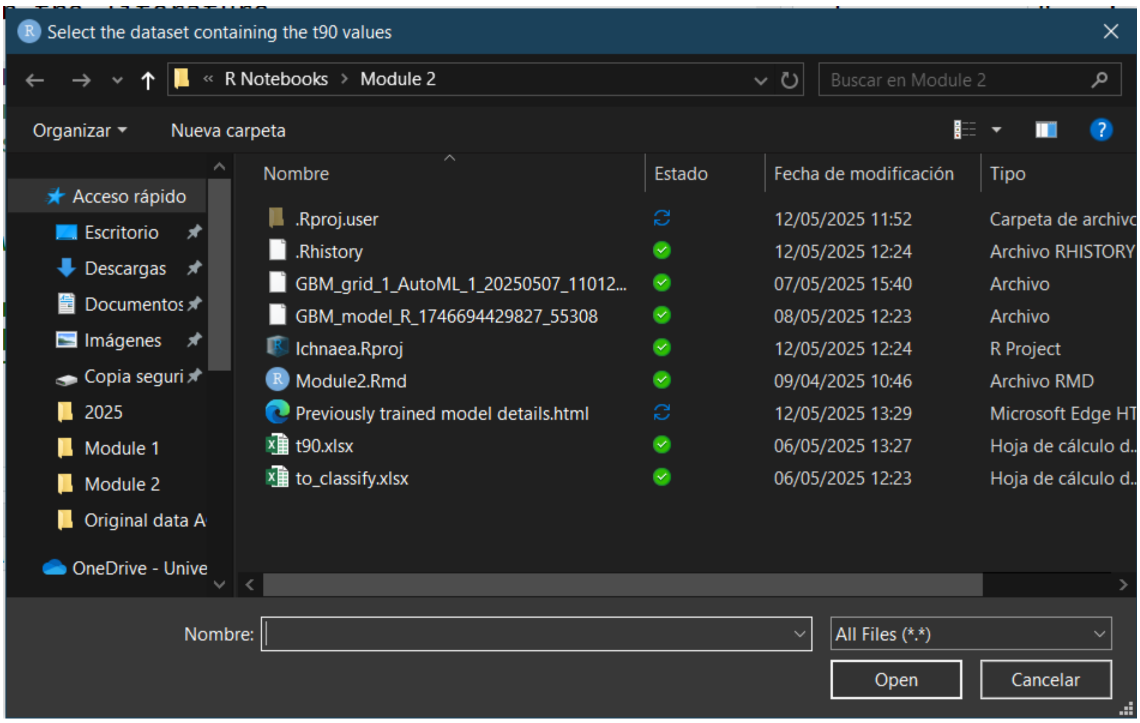
Upon clicking the "OK" button, a new window will appear, prompting the user to upload a file containing the t90-values of each marker, if deemed necessary.



In the event of a negative response, the chunk will be brought to a close at this point. Conversely, in the event of a positive response, the file will be requested.



Upon selecting the “OK” button, the following window will be initiated.

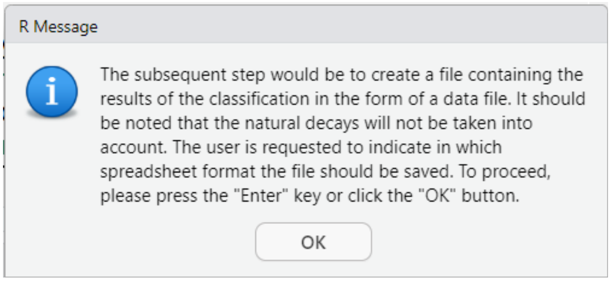


Once the file has been selected, the chunk terminates at this point.

Chunk 8

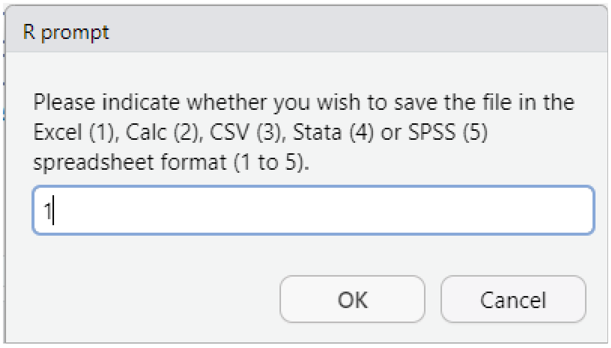
This chunk is responsible for the classification of the samples and the subsequent storage of the results of that classification in a datasheet file.

After the initiation of the chunk, the ensuing windows will be displayed.

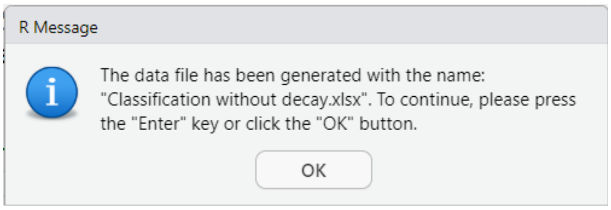


Upon clicking the "OK" button, the user is prompted to select the type of file in which the classification results should be saved.

For instance, the option of saving as an Excel file was selected.



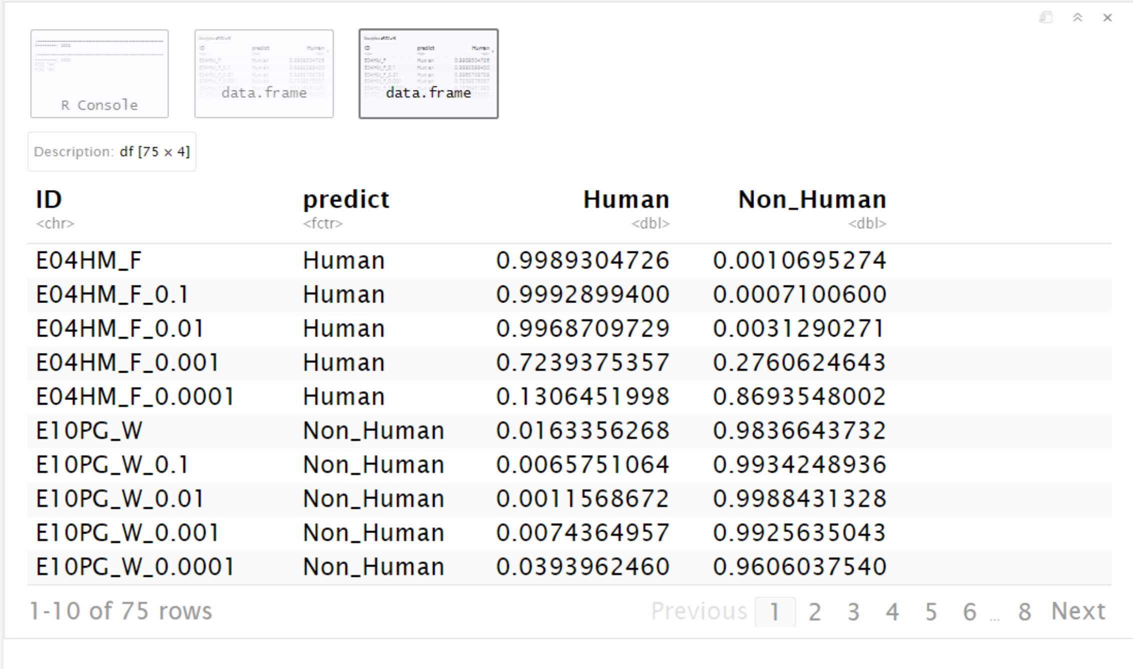
Upon selection of the aforementioned option, a pop-up notification is generated, indicating the creation of a file named “Classification without decay.xlsx”.



The present chunk concludes at the point at which the "OK" button is selected.

The results of the analysis are presented in the window located below the chunk, and these results are also saved to an Excel file.

As illustrated below, the screen capture of the results displayed in the window beneath the chunk is presented.



R Console | data.frame | data.frame

Description: df [75 x 4]

ID <chr>	predict <fctr>	Human <dbl>	Non_Human <dbl>
E04HM_F	Human	0.9989304726	0.0010695274
E04HM_F_0.1	Human	0.9992899400	0.0007100600
E04HM_F_0.01	Human	0.9968709729	0.0031290271
E04HM_F_0.001	Human	0.7239375357	0.2760624643
E04HM_F_0.0001	Human	0.1306451998	0.8693548002
E10PG_W	Non_Human	0.0163356268	0.9836643732
E10PG_W_0.1	Non_Human	0.0065751064	0.9934248936
E10PG_W_0.01	Non_Human	0.0011568672	0.9988431328
E10PG_W_0.001	Non_Human	0.0074364957	0.9925635043
E10PG_W_0.0001	Non_Human	0.0393962460	0.9606037540

1-10 of 75 rows Previous 1 2 3 4 5 6 ... 8 Next

On the following pages it is shown the table with the results that have been stored in the Excel file. A comparison of these results with those previously obtained during the classification with all MST markers in Module I (see Chunk 27) reveals that they are identical.

ID	predict	Human	Non_Human
E04HM_F	Human	0.998930473	0.001069527
E04HM_F_0.1	Human	0.999289940	0.000710060
E04HM_F_0.01	Human	0.996870973	0.003129027
E04HM_F_0.001	Human	0.723937536	0.276062464
E04HM_F_0.0001	Human	0.130645200	0.869354800
E10PG_W	Non_Human	0.016335627	0.983664373
E10PG_W_0.1	Non_Human	0.006575106	0.993424894
E10PG_W_0.01	Non_Human	0.001156867	0.998843133
E10PG_W_0.001	Non_Human	0.007436496	0.992563504
E10PG_W_0.0001	Non_Human	0.039396246	0.960603754
E22PL_S	Non_Human	0.000419786	0.999580214
E22PL_S_0.1	Non_Human	0.001438105	0.998561895
E22PL_S_0.01	Non_Human	0.003883449	0.996116551
E22PL_S_0.001	Human	0.985796298	0.014203702
E22PL_S_0.0001	Non_Human	0.115737432	0.884262568
A04PG_F	Non_Human	0.006535958	0.993464042
A04PG_F_0.1	Non_Human	0.001707681	0.998292319
A04PG_F_0.01	Non_Human	0.001990098	0.998009902
A04PG_F_0.001	Non_Human	0.021175451	0.978824549
A04PG_F_0.0001	Non_Human	0.030741625	0.969258375
A08HM_W	Human	0.969227429	0.030772571
A08HM_W_0.1	Human	0.990851671	0.009148329
A08HM_W_0.01	Human	0.997918692	0.002081308

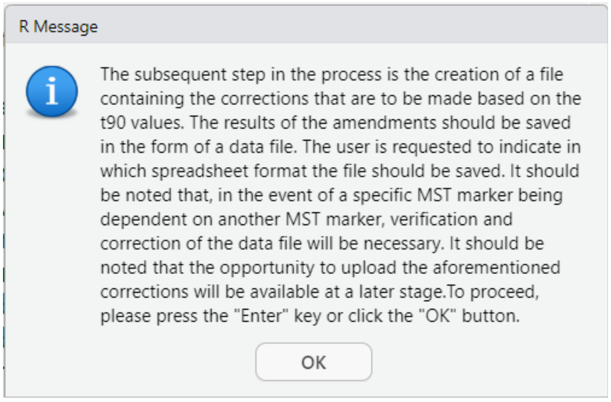
ID	predict	Human	Non_Human
A08HM_W_0.001	Human	0.997139784	0.002860216
A08HM_W_0.0001	Human	0.982830284	0.017169716
A22CW_S	Non_Human	0.001117484	0.998882516
A22CW_S_0.1	Non_Human	0.002426990	0.997573010
A22CW_S_0.01	Non_Human	0.002905257	0.997094743
A22CW_S_0.001	Non_Human	0.007028406	0.992971594
A22CW_S_0.0001	Non_Human	0.011865262	0.988134738
P06PG_W	Non_Human	0.001345855	0.998654145
P06PG_W_0.1	Non_Human	0.000653926	0.999346074
P06PG_W_0.01	Non_Human	0.011026401	0.988973599
P06PG_W_0.001	Non_Human	0.025828396	0.974171604
P06PG_W_0.0001	Non_Human	0.039396246	0.960603754
P16CW_SP	Non_Human	0.000573159	0.999426841
P16CW_SP_0.1	Non_Human	0.000653029	0.999346971
P16CW_SP_0.01	Non_Human	0.000839621	0.999160379
P16CW_SP_0.001	Non_Human	0.004045827	0.995954173
P16CW_SP_0.0001	Non_Human	0.007158804	0.992841196
P19PL_SP	Non_Human	0.000281218	0.999718782
P19PL_SP_0.1	Non_Human	0.000998892	0.999001108
P19PL_SP_0.01	Non_Human	0.023575870	0.976424130
P19PL_SP_0.001	Non_Human	0.098818284	0.901181716
P19PL_SP_0.0001	Human	0.951796211	0.048203789
P22HM	Human	0.171877927	0.828122073
P22HM_0.1	Human	0.981407481	0.018592519
P22HM_0.01	Human	0.980128089	0.019871911
P22HM_0.001	Human	0.997597667	0.002402333
P22HM_0.0001	Non_Human	0.115737432	0.884262568
D01HM_F	Human	0.999031761	0.000968239
D01HM_F_0.1	Human	0.999073658	0.000926342
D01HM_F_0.01	Human	0.999239853	0.000760147
D01HM_F_0.001	Human	0.975887820	0.024112180
D01HM_F_0.0001	Human	0.985233365	0.014766635
F06HM_W	Human	0.996599652	0.003400348
F06HM_W_0.1	Human	0.995704609	0.004295391
F06HM_W_0.01	Human	0.998698218	0.001301782
F06HM_W_0.001	Human	0.997330064	0.002669936
F06HM_W_0.0001	Human	0.920429292	0.079570708
F07PG_W	Non_Human	0.001736218	0.998263782
F07PG_W_0.1	Non_Human	0.005215753	0.994784247
F07PG_W_0.01	Non_Human	0.021006469	0.978993531
F07PG_W_0.001	Non_Human	0.034352460	0.965647540
F07PG_W_0.0001	Non_Human	0.039350872	0.960649128
F11HM_P	Human	0.994355059	0.005644941
F11HM_P_0.1	Human	0.985215986	0.014784014
F11HM_P_0.01	Human	0.998677444	0.001322556
F11HM_P_0.001	Human	0.998843225	0.001156775
F11HM_P_0.0001	Human	0.966606935	0.033393065

ID	predict	Human	Non_Human
F15HM_S	Human	0.999269452	0.000730548
F15HM_S_0.1	Human	0.997690927	0.002309073
F15HM_S_0.01	Human	0.998700625	0.001299375
F15HM_S_0.001	Human	0.997595848	0.002404152
F15HM_S_0.0001	Human	0.975532397	0.024467603

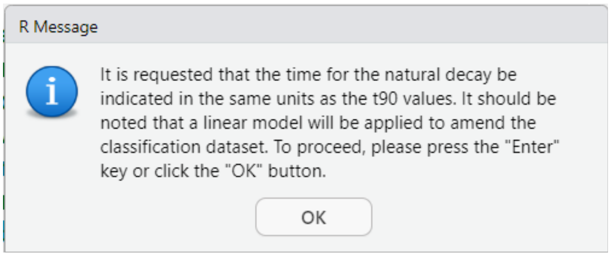
Chunk 9

This chunk is concerned with the management of amendments to the application of natural decay.

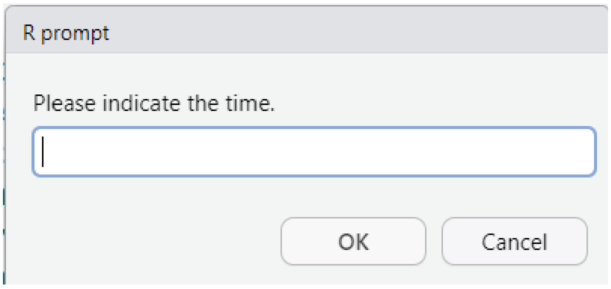
Upon execution of the code, the following information window will appear.



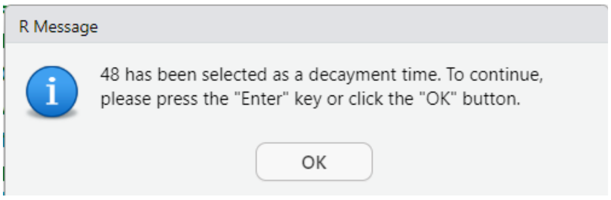
Following the selection of the "OK" button, the program proceeds by displaying the subsequent pop-up notification.



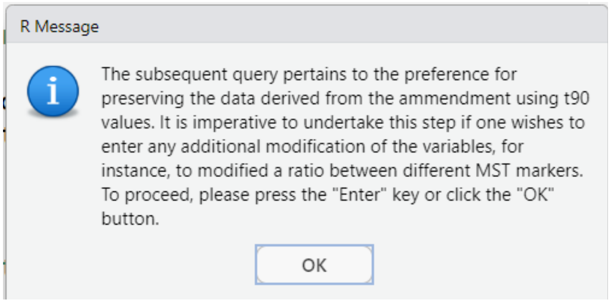
After clicking “OK” button, the program prompts the user to specify the span time that will be use for the purpose of data amendment.



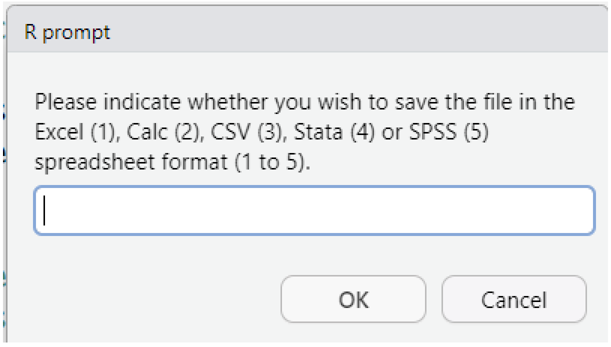
For instance, the same 48-hour time span is indicated as was previously carried out in the Module I. The program produces a pop-up window that displays the selected time.



Subsequent to the selection of the "OK" button, the program will display a pop-up notification indicating whether the result should be saved in a datasheet file. It is recommended that the results be stored, especially in the event that any modifications to these results are required.

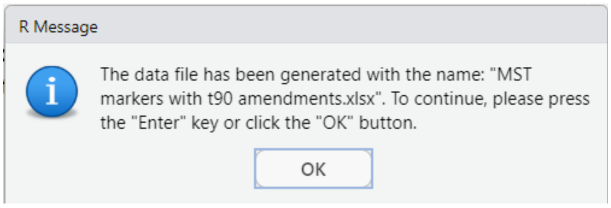


Upon selecting the "OK" button, the user is prompted to specify the desired file format.

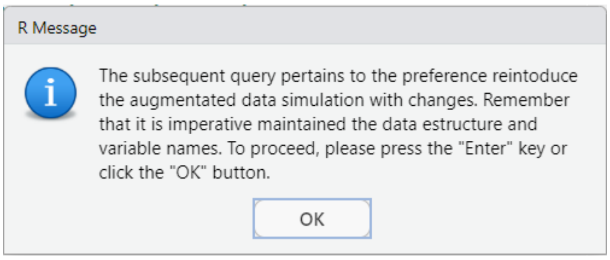


For instance, the first option (an Excel file) is selected.

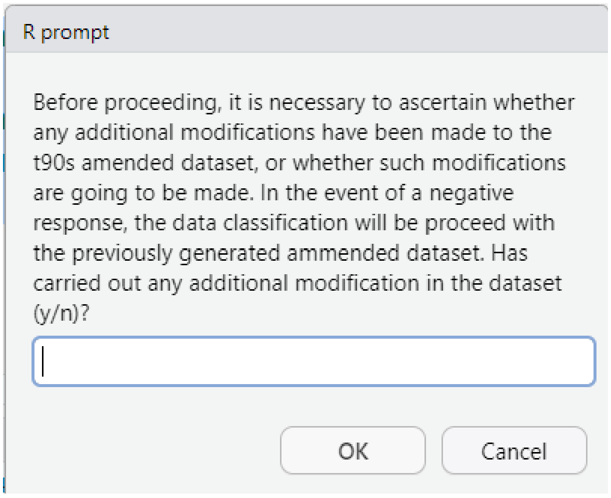
The generation of a file named "MST markers with t90 amendments.xlsx" is also initiated. The present file comprises the amendments pertaining to the t90 values.



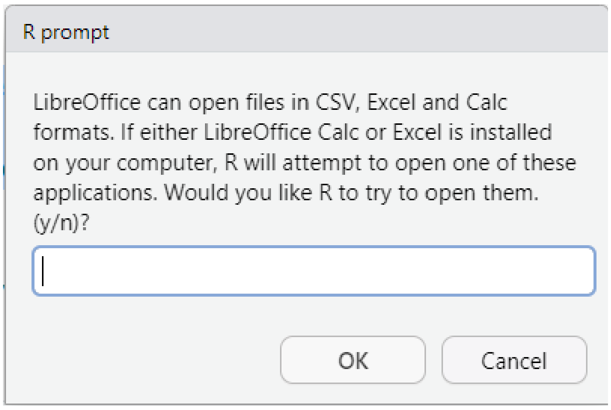
Upon pressing the "OK" button, a new pop-up will appear.



After clicking the “OK” button, a new pop-up appears.

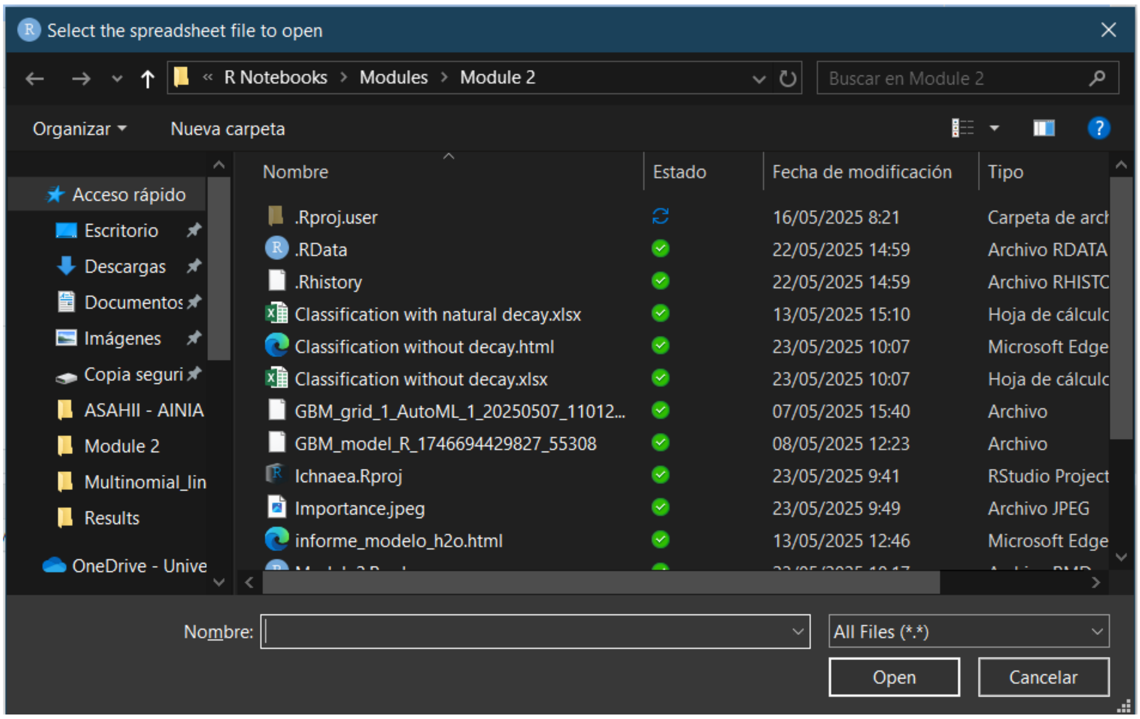


In the event of a positive response from the user, a new pop-up window will appear, enquiring whether the user wishes to implement the amendments using Excel or LibreOffice Calc.



It should be noted that the selection or non-selection of any of these programs (LibreOffice Calc or Excel) is entirely at the discretion of the user; the spreadsheet containing the data with the decay modifications can be modified using the software with which the individual is most comfortable. Furthermore, the segment of the code that is responsible for initiating the application has been subjected to testing on the Windows 11 version 24h2 operating system and the Ubuntu versions 22.04 LTS (Jammy Jellyfish) and 24.04 LTS (Noble Numbat). It is important to note that, due to the inherent complexity of operating systems, there is a possibility that the application may not launch correctly. It is important to note that the R program has been designed to run on a continuous basis. This means that it will continue running whether or not these applications are opened.

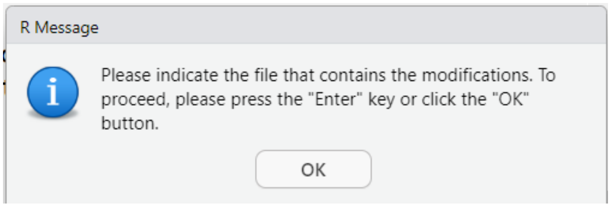
If the user responds affirmatively, the following windows to upload the file will be displayed.



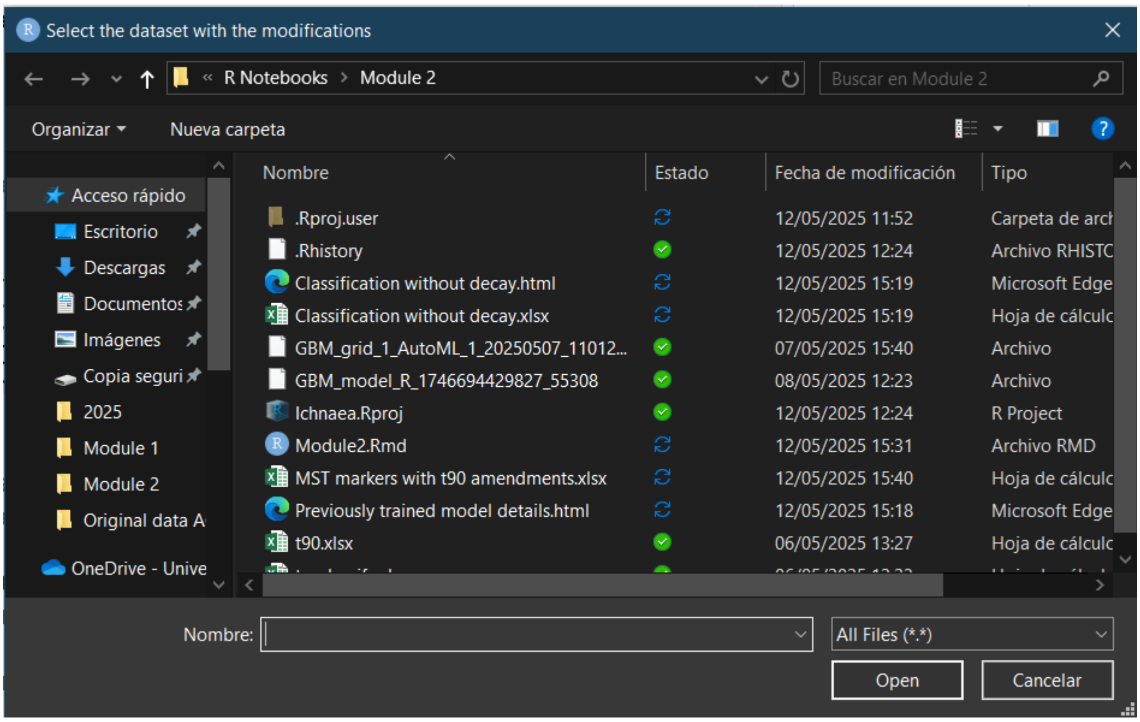
After selecting the file, Excel or LibreOffice Calc is launched.

In the event of a change to the file, it is imperative that the requisite alterations are made at this juncture. Subsequent to the selection of the OK option, the program prompts the user with a query regarding whether any alterations have been made to the file.

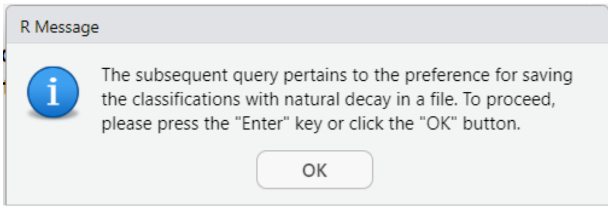
In the event of a negative user response, the program will proceed with the actual data. Conversely, if the response is positive, a pop-up window will be displayed, informing the user of the necessity of a fille.



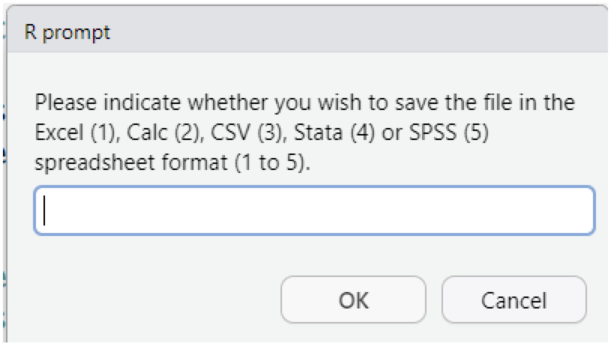
Upon selecting the "OK" button, a new window will be initiated.



For instance, the file named "MST markers with t90 amendments.xlsx" is utilised. It is evident that no alterations have been made. Upon completion of the upload process, the program proceeds to the subsequent pop-up notification.

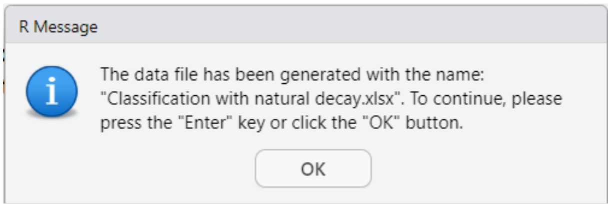


Subsequent to the selection of the "OK" option, the program will prompt the user to specify the preferred file format.



For instance, the initial option (an Excel file) is selected.

Upon selection, the programme prompts the user to confirm the creation of an Excel file bearing the name 'Classification with natural decay.xlsx'.



The termination of the chunk is initiated by the user's selection of the "OK" option.

The ensuing pages present the outcomes of the classification process with the aforementioned amendments.

ID	predict	Human	Non_Human
E04HM_F	Human	0.997871760	0.002128240
E04HM_F_0.1	Human	0.999076848	0.000923152
E04HM_F_0.01	Human	0.999289604	0.000710396
E04HM_F_0.001	Human	0.994996391	0.005003609
E04HM_F_0.0001	Human	0.723937536	0.276062464
E10PG_W	Non_Human	0.053567527	0.946432473
E10PG_W_0.1	Non_Human	0.024483506	0.975516494
E10PG_W_0.01	Non_Human	0.006575106	0.993424894
E10PG_W_0.001	Non_Human	0.003485542	0.996514458
E10PG_W_0.0001	Non_Human	0.038082545	0.961917455
E22PL_S	Non_Human	0.000307116	0.999692884
E22PL_S_0.1	Non_Human	0.000419786	0.999580214
E22PL_S_0.01	Non_Human	0.010209912	0.989790088
E22PL_S_0.001	Human	0.992461520	0.007538480
E22PL_S_0.0001	Non_Human	0.115737432	0.884262568
A04PG_F	Non_Human	0.005111386	0.994888614
A04PG_F_0.1	Non_Human	0.002828386	0.997171614
A04PG_F_0.01	Non_Human	0.001707681	0.998292319
A04PG_F_0.001	Non_Human	0.011252394	0.988747606
A04PG_F_0.0001	Non_Human	0.021175451	0.978824549
A08HM_W	Non_Human	0.092280989	0.907719011
A08HM_W_0.1	Human	0.997307262	0.002692738
A08HM_W_0.01	Human	0.988821958	0.011178042
A08HM_W_0.001	Human	0.993237396	0.006762604
A08HM_W_0.0001	Human	0.997139784	0.002860216
A22CW_S	Non_Human	0.000842079	0.999157921
A22CW_S_0.1	Non_Human	0.001125948	0.998874052
A22CW_S_0.01	Non_Human	0.003834695	0.996165305
A22CW_S_0.001	Non_Human	0.002959513	0.997040487
A22CW_S_0.0001	Non_Human	0.007028406	0.992971594
P06PG_W	Non_Human	0.000196124	0.999803876
P06PG_W_0.1	Non_Human	0.000444333	0.999555667
P06PG_W_0.01	Non_Human	0.006262760	0.993737240
P06PG_W_0.001	Non_Human	0.011026401	0.988973599
P06PG_W_0.0001	Non_Human	0.038082545	0.961917455
P16CW_SP	Non_Human	0.001360420	0.998639580
P16CW_SP_0.1	Non_Human	0.000573159	0.999426841

ID	predict	Human	Non_Human
P16CW_SP_0.01	Non_Human	0.000653029	0.999346971
P16CW_SP_0.001	Non_Human	0.002299690	0.997700310
P16CW_SP_0.0001	Non_Human	0.004045827	0.995954173
P19PL_SP	Non_Human	0.000169346	0.999830654
P19PL_SP_0.1	Non_Human	0.000281218	0.999718782
P19PL_SP_0.01	Non_Human	0.002584847	0.997415153
P19PL_SP_0.001	Human	0.996149263	0.003850737
P19PL_SP_0.0001	Non_Human	0.098818284	0.901181716
P22HM	Non_Human	0.025921175	0.974078825
P22HM_0.1	Non_Human	0.093292751	0.906707249
P22HM_0.01	Human	0.999280583	0.000719417
P22HM_0.001	Human	0.981041975	0.018958025
P22HM_0.0001	Non_Human	0.115737432	0.884262568
D01HM_F	Human	0.824070934	0.175929066
D01HM_F_0.1	Human	0.998821587	0.001178413
D01HM_F_0.01	Human	0.999073899	0.000926101
D01HM_F_0.001	Human	0.996511978	0.003488022
D01HM_F_0.0001	Human	0.975887820	0.024112180
F06HM_W	Human	0.889402141	0.110597859
F06HM_W_0.1	Human	0.996398242	0.003601758
F06HM_W_0.01	Human	0.994745170	0.005254830
F06HM_W_0.001	Human	0.994443075	0.005556925
F06HM_W_0.0001	Human	0.997330064	0.002669936
F07PG_W	Non_Human	0.000710881	0.999289119
F07PG_W_0.1	Non_Human	0.002453174	0.997546826
F07PG_W_0.01	Non_Human	0.016447928	0.983552072
F07PG_W_0.001	Non_Human	0.021006469	0.978993531
F07PG_W_0.0001	Non_Human	0.034352460	0.965647540
F11HM_P	Human	0.932100199	0.067899801
F11HM_P_0.1	Human	0.983928301	0.016071699
F11HM_P_0.01	Human	0.987288522	0.012711478
F11HM_P_0.001	Human	0.998669701	0.001330299
F11HM_P_0.0001	Human	0.996710023	0.003289977
F15HM_S	Human	0.997660772	0.002339228
F15HM_S_0.1	Human	0.999107822	0.000892178
F15HM_S_0.01	Human	0.989358864	0.010641136
F15HM_S_0.001	Human	0.998700625	0.001299375
F15HM_S_0.0001	Human	0.994436440	0.005563560

A comparison of the results obtained in 30 of the section Module I reveals no significant disparities in the classification outcomes. It is evident that both classifications are indistinguishable.

ICHANEA MST

WEB APPLICATION FOR TRAINING, CLASSIFICATION AND REDUCTION OF THE NUMBER OF MST PREDICTORS

WEB APPLICATION FOR TRAINING AND PERFORMING THE CLASSIFICATION

AIM OF THIS MANUAL

The present user manual has been designed with the intention of being utilised by users with limited or no prior knowledge of the R programming language and the RStudio software environment. For this purpose, a web application has been meticulously engineered to utilise Shiny as its interface. This web application is hosted on a remote server, which can be accessed via the provided link:

http://biost3.bio.ub.edu:3838/apps/Ichnaea_MST/

It is imperative that the tool is utilised in an appropriate manner. The classification issues associated with MST can be addressed with relative ease by employing algorithms that do not require significant computational capabilities.

However, it is advisable to refrain from using computationally intensive algorithms, as this may result in the computer being unavailable to other users. It is hoped that the utilisation of this instrument will engender a positive experience.

We have tried to make this web application as easy and accessible as possible. If you have any technical problems, we can help you. If you have any questions or need more information, please contact us. We would be very happy to help you and would be grateful if you could tell us what you think of the application. Please tell us if there is anything you think could be improved.

- Javier Méndez – jmendez@b.edu
- Alex Rodriguez – alejandrodriguez@ub.edu
- Antonio Monleón – amonleong@ub.edu
- Anicet Blanch – ablanch@ub.edu

Prior to the utilisation of Ichnaea-MST, it is advised that users consult the manual, specifically the sections ranging from pages 4 to 15, particularly for those with limited expertise in MST determination through machine learning methodologies.

ACCESS TO THE APPLICATION

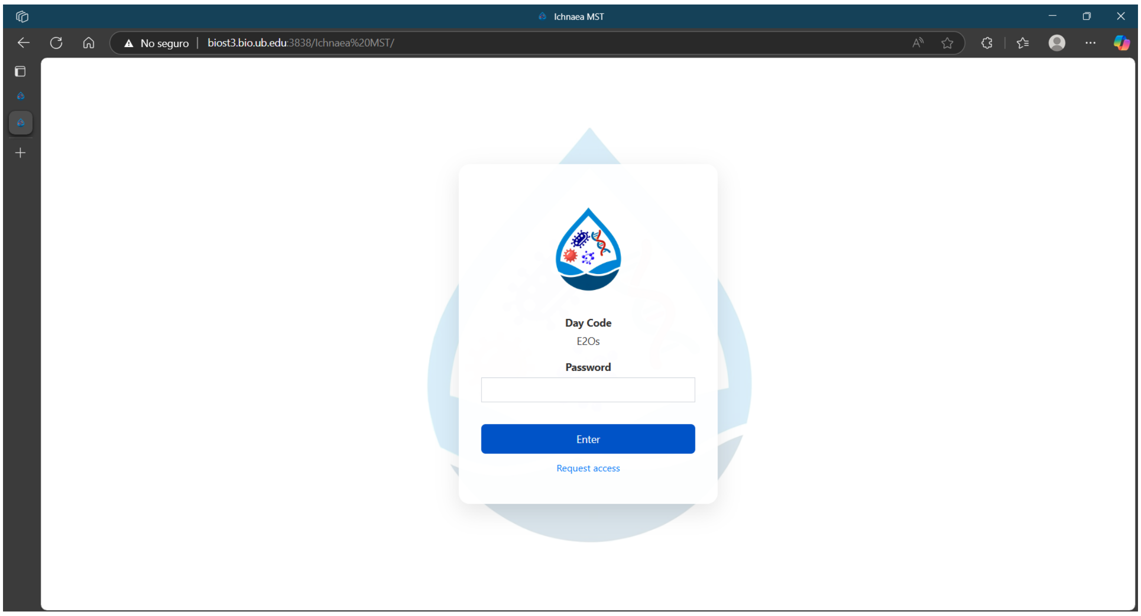
This web application can be accessed from any mobile device with an internet connection, but it is recommend using devices with a screen size of at least 10 inches. Once the URL is entered, the user is prompted to introduce a code to access.

This code is daily changed; in the following table it is shown the codes for access from 14th June to 5th of July.

Date	Day code	Password	Date	Day code	Password
2025-06-14	r39Y	xAgf	2025-06-25	7N1N	jSoz
2025-06-15	MQGP	TG87	2025-06-26	Yvaq	MVfv
2025-06-16	Rc8K	VP3U	2025-06-27	PhcU	KLBx
2025-06-17	QkZO	LhR1	2025-06-28	NjTc	wZMe
2025-06-18	Kh38	dUzD	2025-06-29	hYHD	ggd6
2025-06-19	mYX9	kJdN	2025-06-30	ks8w	X362
2025-06-20	dEgv	2wPt	2025-07-01	HarO	FxCC
2025-06-21	WC2O	FcIV	2025-07-02	QffZ	3qUn
2025-06-22	Hko3	cdb2	2025-07-03	AgOr	0gGq
2025-06-23	IBzX	TGzG	2025-07-04	ITDt	FPQq
2025-06-24	FQXQ	ovx8	2025-07-05	mP1G	5i3O

Once the link is accessed, the user will be prompted to enter a password to access the application. This screen will display a little window containing a Day Code and a field for entering the password. For instance, the illustration below these lines is a screen capture taken on 10 June. For this particular day, the Day Code is E20s. Please note that this code changes daily and this particular code is associated with a corresponding password (4wxC in this case) that grants access to the web application.

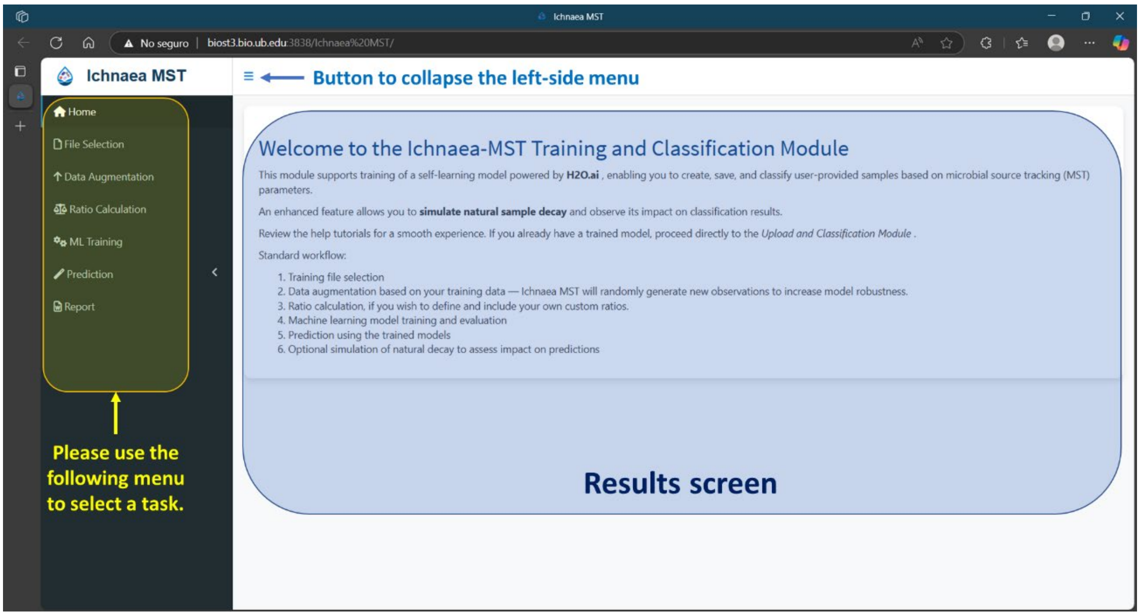
For instance, if a user accesses the system on 18 June, the screen will show this Day Code: Kh38, In order to gain access to the application, it is necessary to enter dUzD as a password.



Please be aware that beneath the 'Enter' button, there is a link to enable the sending of an email to our team.

How is Ichnaea-MST organised?

In essence, Ichnaea-MST has been conceived with the objective of ensuring maximum ease of use. It is comprised of two principal sections. The menu, which is accessible via the left-side panel (which displays a dark background), enables the user to perform a variety of tasks. The right screen characterised by a light grey and white background, serves to facilitate supplementary functionalities and the presentation of analytical outcomes.



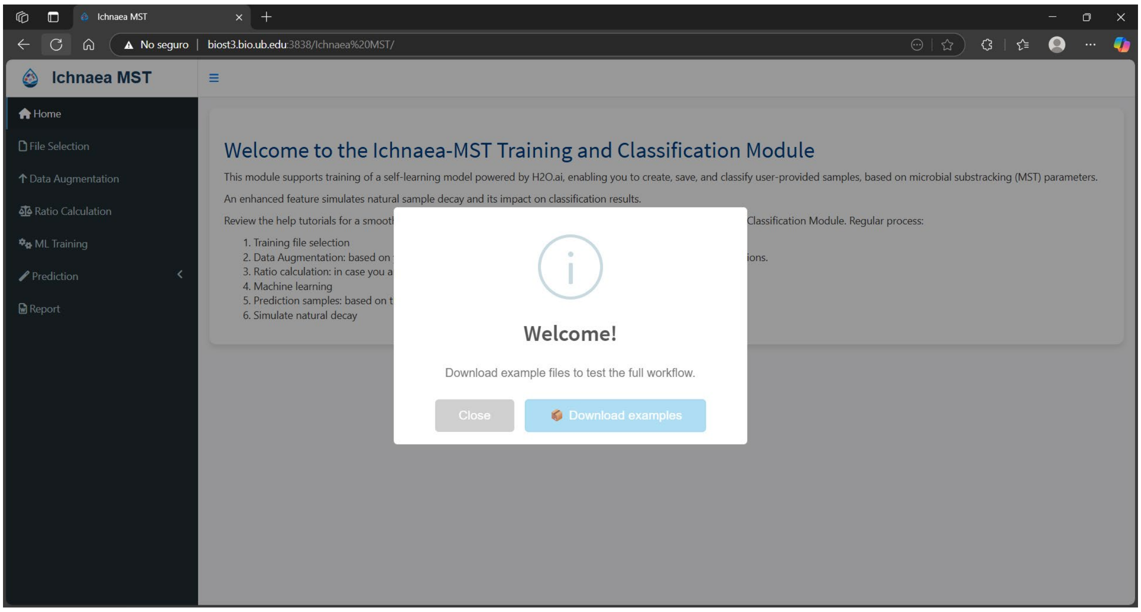
The selection of a task results in the task being highlighted in the menu. The accessible tasks are as follows:

- **Home**: this menu functions as a welcome screen, facilitating the download of a set of Excel files containing various datasets. The utilisation of these datasets enables users to assess the functionality of the program and to acquire the necessary skills for its operation.
- **File Selection**: this menu is used to facilitate the upload of the user dataset, which comprises the data required for the training of the machine learning model. Following the upload of the data, a view of the uploaded data is made available. The parameters of the dataset are represented as boxplots graphics.
- **Data Augmentation**: the function of this menu is to process the dataset that has been uploaded by the user, with a view to increasing the number of samples that are to be used in the training process. Two augmentation processes are currently available. Moreover, this section facilitates the storage of augmented data and offers a comprehensive overview of the complete augmented dataset, which is further represented using boxplots.
- **Ratio Calculation**: this menu has been developed for the specific purpose of facilitating the creation of covariables as ratios of log10-values between MST markers. In the event that there is no requirement for the user to create such types of variables, this menu should be ommited.
- **ML Training**: this menu proffers a plethora of functions, encompassing the utilisation of a pre-trained model and the capacity to train a novel model employing augmented data. It is possible to select one of six different algorithms.
- **Prediction**: this the menu is responsible for uploading the classification dataset. The menu has been divided into two separate branches. One branch is dedicated to the provision of classification that does not take into account natural decay. The other branch facilitates the modification of the classification dataset through the incorporation of natural decay.
- **Report**: this menu offers the option to generate a report containing the results.

For instance, it will present the workflow, including all the steps. The programme transitions to the 'File Selection Tasks' interface when the 'File Selection' option is selected. Subsequent to the act of clicking, a pop-up window will display. Note that when a section is selected (left-side menu) the option is highlighted.

Home menu

Upon successful authentication and subsequent authorisation, the end-user will be presented with the following screen.



This screen provides the option to download a zipped file named **samples_files.zip** which contains a set of Excel data files and two trained models with the objective of encouraging users to become familiar with the programme. The ZIP file comprises the following archives:

- The **t90.xlsx** file comprises an Excel spreadsheet containing the t90 values for utilisation in the context of natural decay. It should be noted that the t90 values in the file have been adjusted to 48 hours for the purpose of illustration.
- The **to_classify.xlsx** file comprises an Excel spreadsheet containing a set of real samples and simulated dilutions. This dataset pertains to the samples that are to be classified.
- The file entitled **to_train_binomial.xlsx** is an Excel file that has been structured for the purpose of training. The dataset under consideration can be represented as a binomial classification, characterised by the presence of two distinct classes. The initial column denotes the class, whilst the initial row indicates a marker. It is evident that the fourth row contains the metadata that serves to identify the type of marker. The second to fourth rows contain this same metadata.
- The file entitled **to_train_multinomial.xlsx** is an Excel file that has been structured for use in training. The dataset under consideration represents a multinomial classification, that is to say, it contains three or more classes. The initial column denotes the class, whilst the initial row indicates a marker. The subsequent rows, from 2 to 4, contain the metadata that identifies the type of marker.
- The **to_dilute.xlsx** file is an Excel spreadsheet that has been meticulously structured for the purpose of simulating the dilution process. This simulation is performed using two distinct occurrences: the first being the matrix that serves as the origin of contamination, and the second being the matrix that is intended to be classified. In this instance, the phenomenon under scrutiny is measured in wastewater and river water.
- A file entitled **Master data spreadsheet for Project 311846.xlsx**, which contains all the information from the files shown above. For more detailed information about this specific file, please refer to pages 35 to 37 of this manual.

With the exception of the file **t90.xlsx** whose data is entirely fictitious and intended solely for demonstration or testing purposes, all other files included in this dataset contain real-world data derived from actual field studies and laboratory analyses. These data collections were carried out within the framework of the European research project titled "Protecting the health of Europeans by improving methods for the detection of pathogens in drinking water and water used in food preparation", with grant agreement ID: 311846. These datasets reflect microbial counts and related parameters obtained from various sampling campaigns conducted across different European regions. Further information about this project can be found at: https://cordis.europa.eu/project/id/311846.

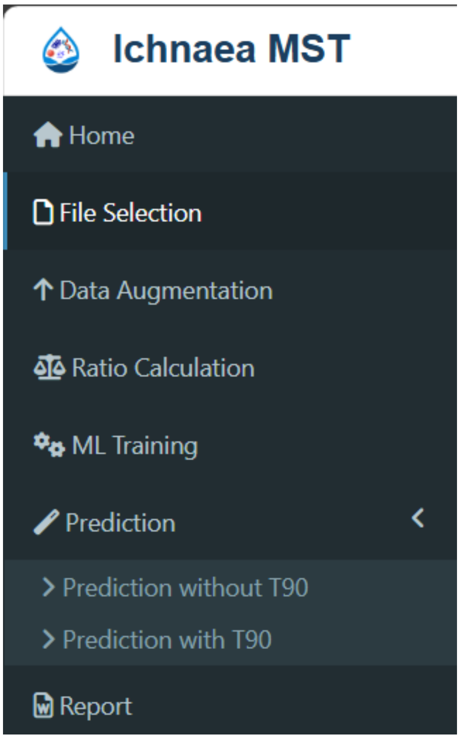
In the zipped file, you will find two additional files that corresponds to two trained models that were created with local execution of the Ichnaea-MST.

The first model (GBM_grid_1_AutoML_1_20250507_110124_model_40) was trained using all available MST markers.

The second model (GBM_model_R_1746694429827_55308) was trained using a reduced set of MST markers, selected automatically by the program based on their discriminative power.

These models can be uploaded to evaluate and test the functionality of Ichnaea-MST. The first model was trained using the full set of MST markers, while the second was generated using a reduced subset of markers selected automatically by the application.

File Selection

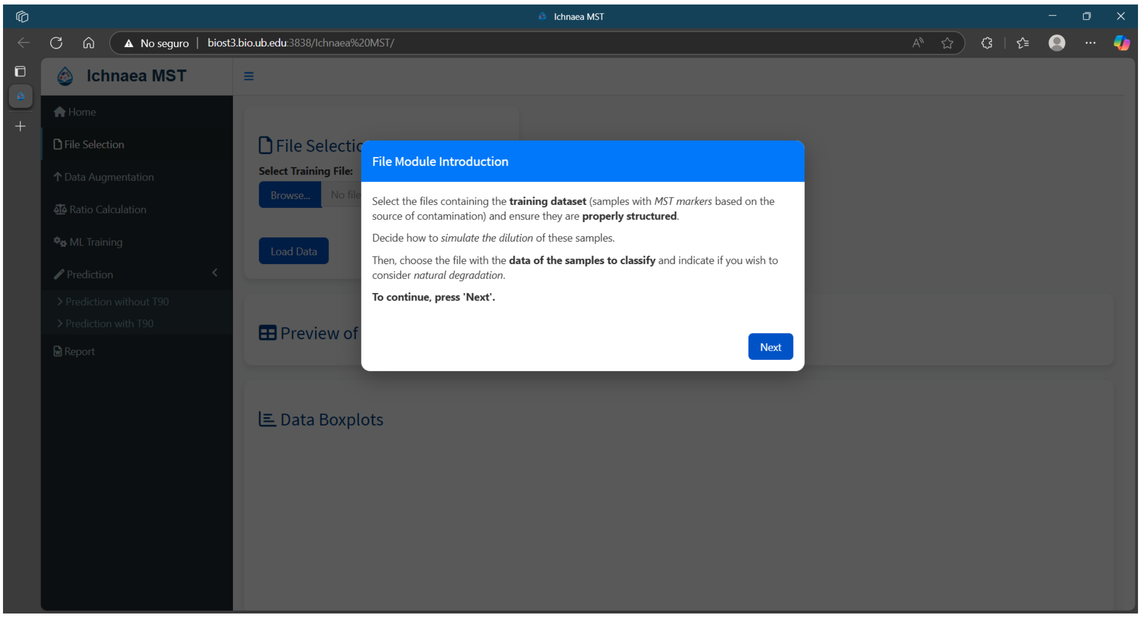


Kindly direct your attention to the following pages of the manual: 25 to 27, and 35 to 37.

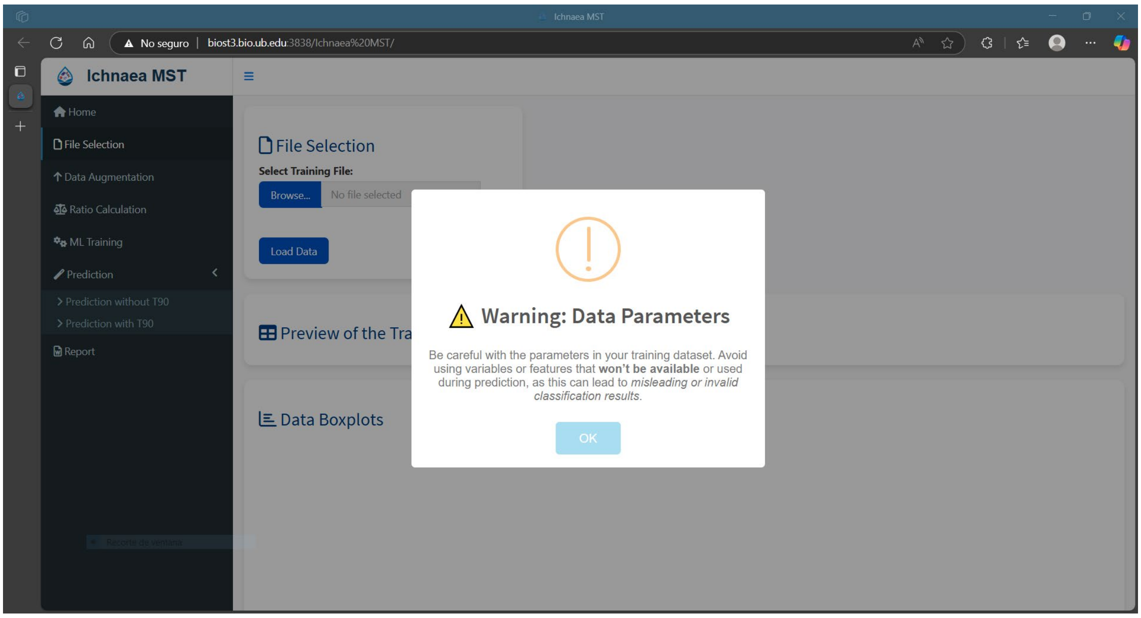
These pages contain comprehensive instructions on the organisation of data for the preparation of the training dataset and the dataset intended for utilisation as occurrences in the dilution process. The only concern regarding the dataset containing the T90 values is to use the same MST markers as during the training of the model. As mentioned above, the origin and structure of the datasets are explained on pages 35 to 37.

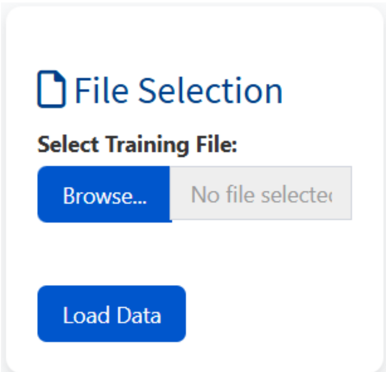
After that, the program proceeds with a series of pop-up notifications (following page), thereby providing a warning regarding the necessary structure of the data that will be utilised as a training dataset.

After closing or downloading the sample files, the program displays the following pop-up, indicating how the data must be organised.



After clicking the “next button’, Ichanea-MST also reminds the user that they need to use the same set of MST markers during training and classification.

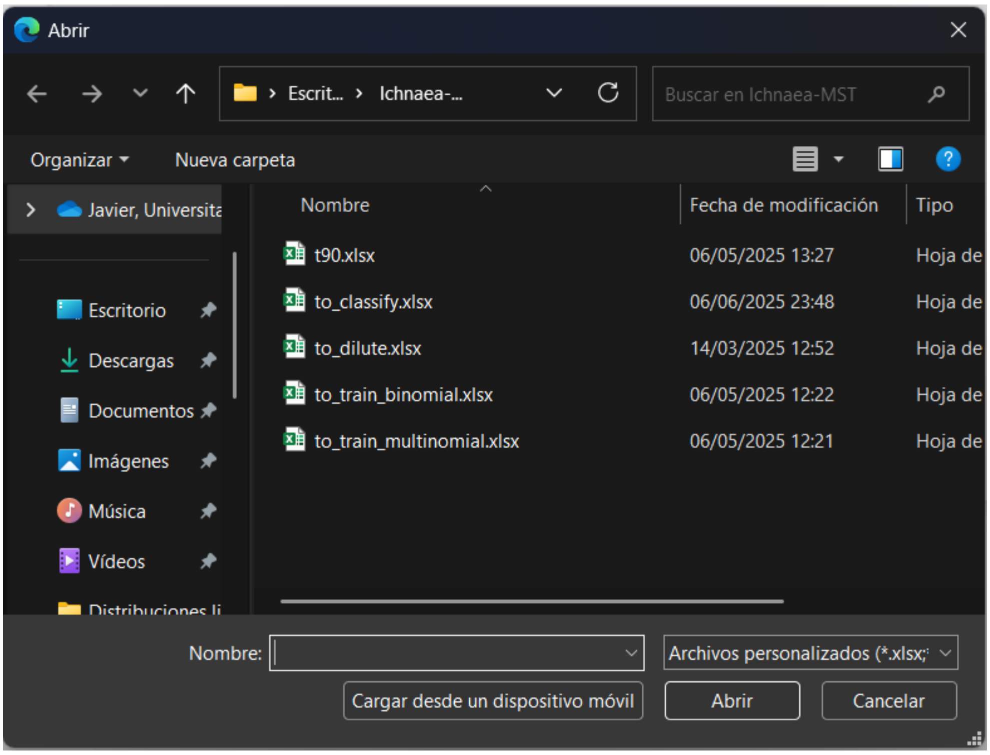




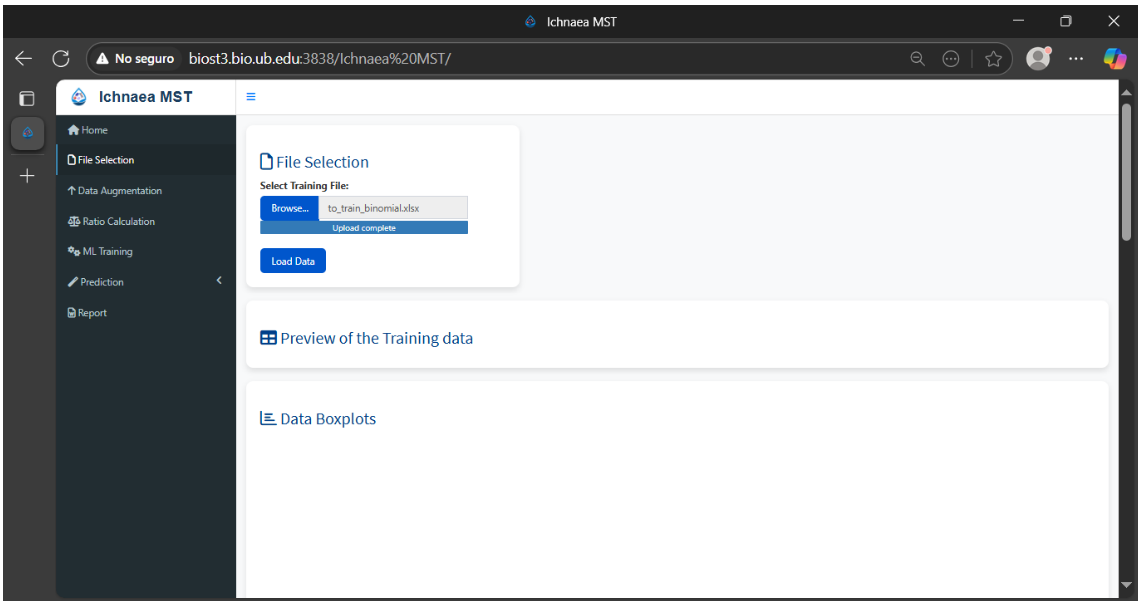
Now turn our attention to the function window named File Selection, this window allows the user to upload an appropriate formatted spreadsheet containing the data for the training process.

When the user clicks onto the button ‘Browse’ a window will appear allowing the user to select the adequate file.

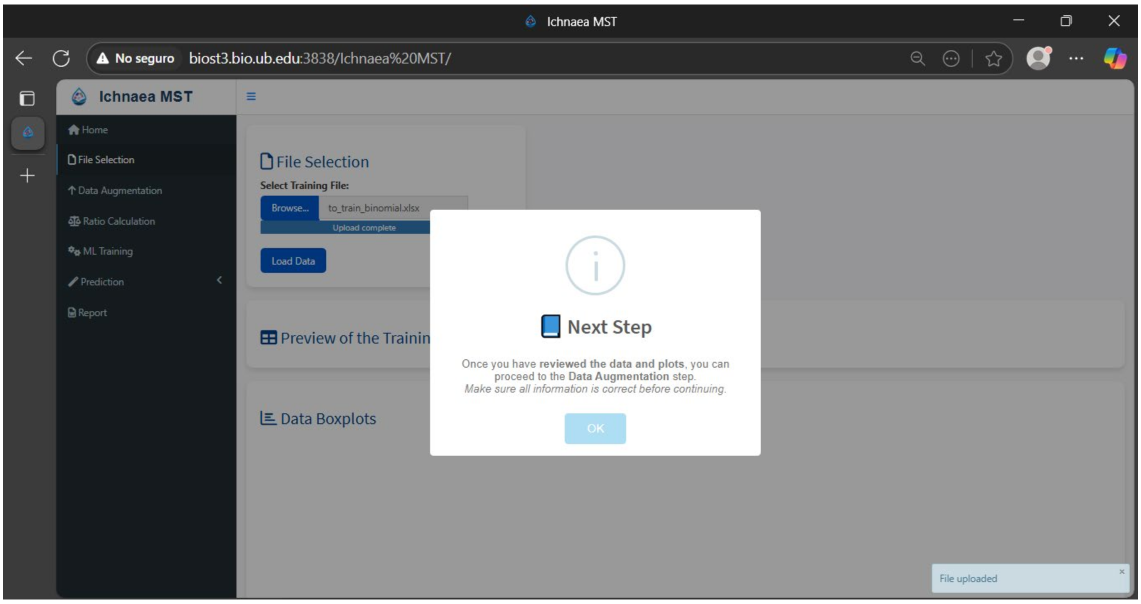
For instance, the file named **to_train_binomial.xlsx** will be uploaded.



Once the file has been selected, it will appear as uploaded.



In order to proceed, it is necessary to click on the button marked 'Load Data'. Upon clicking, Ichnaea-MST facilitates the preview of the uploaded file and provides a graphical representation of the uploaded dataset in the form of box plots. Furthermore, a pop-up notification is displayed, providing instructions for the subsequent step.



It should be noted that a pop-up notification will appear in the bottom right corner of the screen to indicate that the file has been successfully uploaded. This variety of miniature pop-up will materialise in other sections, thereby conveying information regarding the task that has been executed. Subsequent to the selection of the "OK" button, a preview of the data is displayed on the screen.

Preview of the Training data

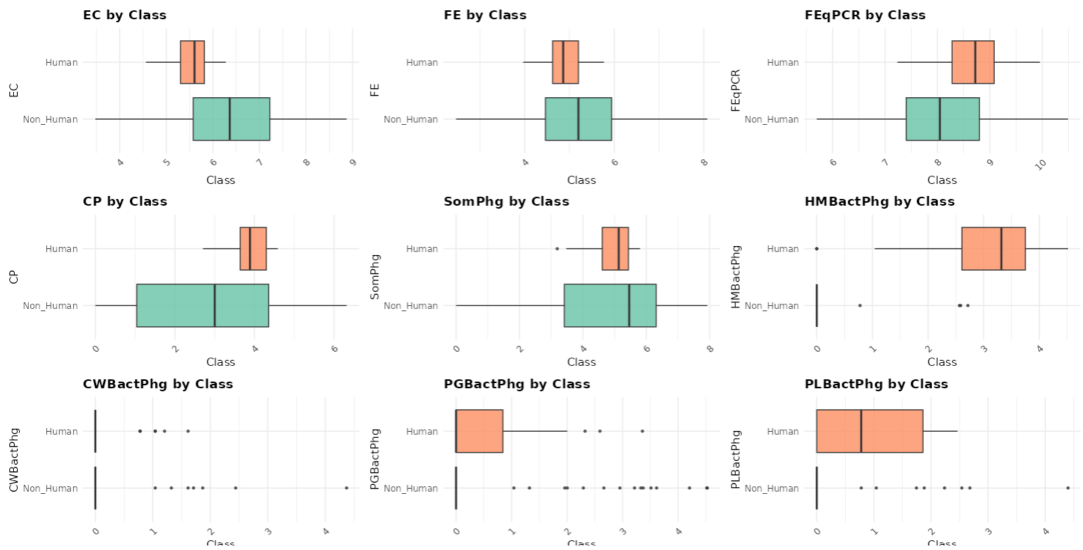
Show 10 entries Search:

Class	EC	FE	FEqPCR	CP	SomPhg	HMBactPhg	CWBactPhg	PGBactPhg	PLBactPhg	BifSorb	BifTot	HMBif	CWBif
Non_Human	6.73	5.32	8.02	2.9	5.42	0	1.71	0	0	0	5.92	0	
Non_Human	7.5	4.43	8.09	4.53	6.31	0	0	3.51	0	0	6.27	0	
Non_Human	7.55	4.37	8.69	4.24	7.51	0	0	4.53	0	0	6.25	0	
Non_Human	7.43	5.41	8.77	3.51	6.21	0	0	0	2.24	0	5.99	0	
Non_Human	6.09	4.41	7.92	3.94	5.37	0	0	1.32	2.54	3.81	5.29	0	
Human	5.66	4.49	8.25	4.59	5.41	3.69	0.78	2.32	0.78	3.7	4.85	0	
Human	5.64	4.54	8.76	4.33	5.34	3.8	0.78	0	0.78	3.3	4.88	6.63	
Non_Human	6.78	4.09	8.5	4.71	6.47	0	0	4.21	0.78	4.7	7.52	0	
Non_Human	6.29	5.02	8.42	4.81	6.31	0	0	4.52	0	3.74	6.74	0	
Non_Human	6.24	4.13	5.7	2.74	5.2	0	2.44	0	0	4.27	6.09	0	

Showing 1 to 10 of 98 entries Previous 1 2 3 4 5 ... 10 Next

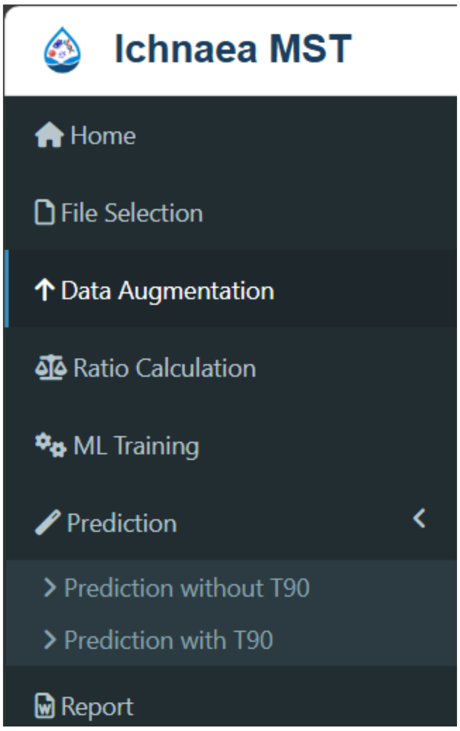
As illustrated in the following the boxplots are displayed beneath the preview of the dataset, located within the 'Data Boxplot' window.

Data Boxplots



It should be noted that the data presented in the boxplot has undergone log transformation. It is imperative to acknowledge that Ichnaea-MST does not differentiate between logarithmic values and non-transformed values. This is a salient factor that must be considered when interpreting the results.

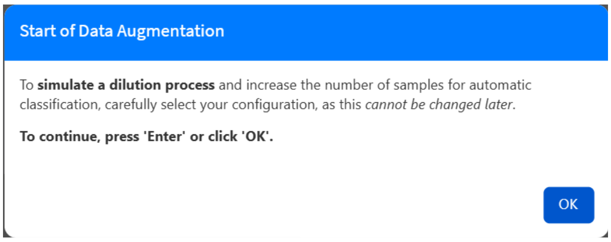
Data Augmentation



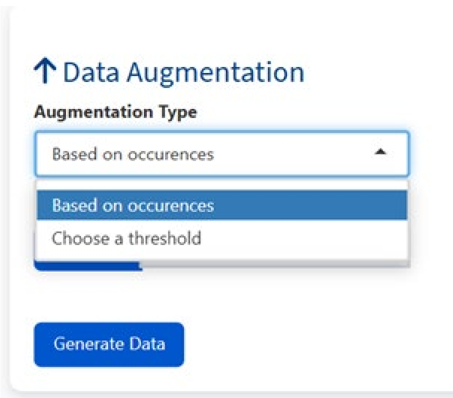
The subsequent step in the workflow pertains to the implementation of a data augmentation technique.

In the field of machine learning, the process of classification necessitates the utilisation of a substantial volume of data, frequently amounting to hundreds or thousands of examples, to ensure the effective training of models. In the absence of a sufficient quantity of training samples, the model may encounter difficulties in identifying patterns and making accurate predictions. The augmentation of data is a solution to this challenge. A more detailed piece of information can be found on page xxx of this manual

Upon selection of the aforementioned option by the user, the 'Data Augmentation' option is highlighted, and the following pop-up is displayed.



Once the user clicks on the “OK” button, the program allows to select the type of data augmentation available.

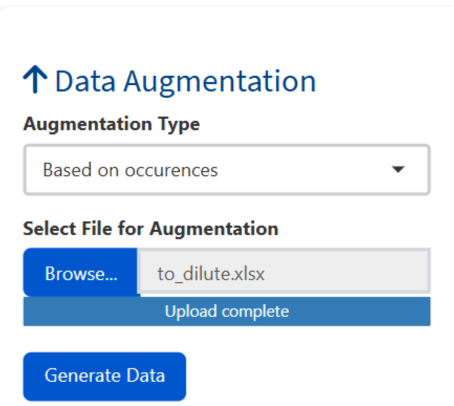


Currently, two distinct augmentation processes are available, each designed to enhance the accuracy and reliability of classification methods. The first approach, known as 'Based on occurrences', relies on identifying the same marker, whether Microbial Source Tracking (MST) or another type, within both the matrix used to determine the origin of pollution and the matrix set for classification. This technique strengthens the ability to trace contamination sources by leveraging the presence of consistent microbial indicators across datasets.

The fundamental premise behind this augmentation method is the hypothesis that a particular marker could serve as a universal indicator of faecal contamination. While this assumption holds significant value in environmental and water quality analysis, it is not entirely exclusive. In fact, several well-documented microbial groups, such as *Escherichia coli*, intestinal enterococci, and somatic coliphages, have been widely recognized as strong indicators of faecal contamination. These microorganisms are commonly used in pollution assessments due to their direct association with human and animal waste.

The second augmentation approach follows a random dilution process, which is carried out until a predetermined threshold is reached. This dilution strategy tries to ensure that the concentration of MST markers falls within the measurable range of the matrix to be classified. For example, in European surface waters, the estimated dilution threshold typically falls between 10,000 and 15,000, corresponding to a dilution factor of approximately 4 to 4.17 $\log_{10}$.

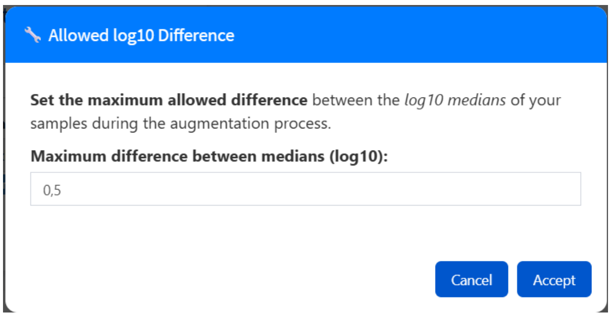
Please note that Ichanea-MST has been specifically designed with the assumption that raw microbial counts will be utilized in the training and classification process. Given this design principle, it is essential to exercise caution when working with dilution thresholds, particularly in instances where microbial concentrations are expressed as $\log_{10}$-values rather than absolute counts. Misinterpretation or incorrect application of dilution adjustments could lead to an inaccurate dataset to train the machine learning model.



For instance, an augmentation based on occurrences is selected. The file, entitled 'to_dilute.xlxs', contains the occurrence of somatic coliphages measured in the European project 311846 (the origin of contamination) and the occurrence of the same parameter measured in the Llobregat River (Catalonia, Spain).

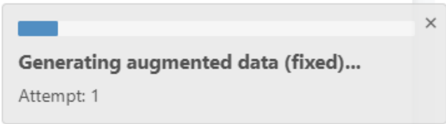
Once selected, the following step is clicking on the 'Generate Data' to start the augmentation process.

Upon clicking the button, a pop-up window will appear, prompting the user to select the maximum divergence between the medians of the simulated augmented marker used to determine occurrences and the real occurrences of this marker in the matrix to be classified.

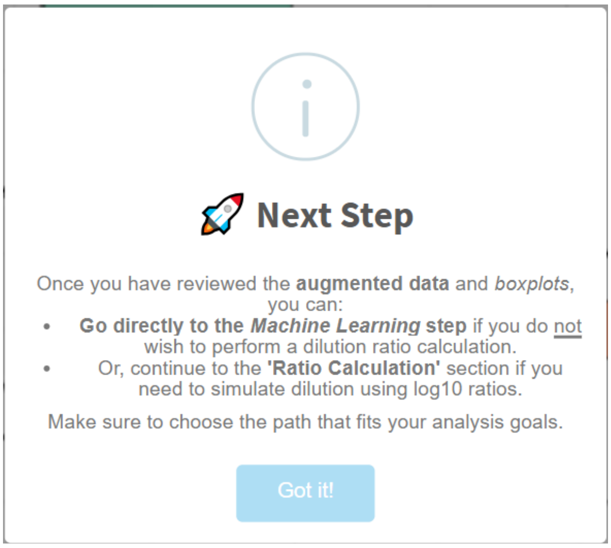


A value near 0.5 log10 difference between medians of the two occurrences should be sufficient. It is imperative to acknowledge that a value of 0 will be interpreted literally, and it is unlikely to achieve this null difference. In the event that the differences between medians were higher than the maximum permissible following ten simulations, the programme will proceed with the augmentation process based on a maximum dilution threshold.

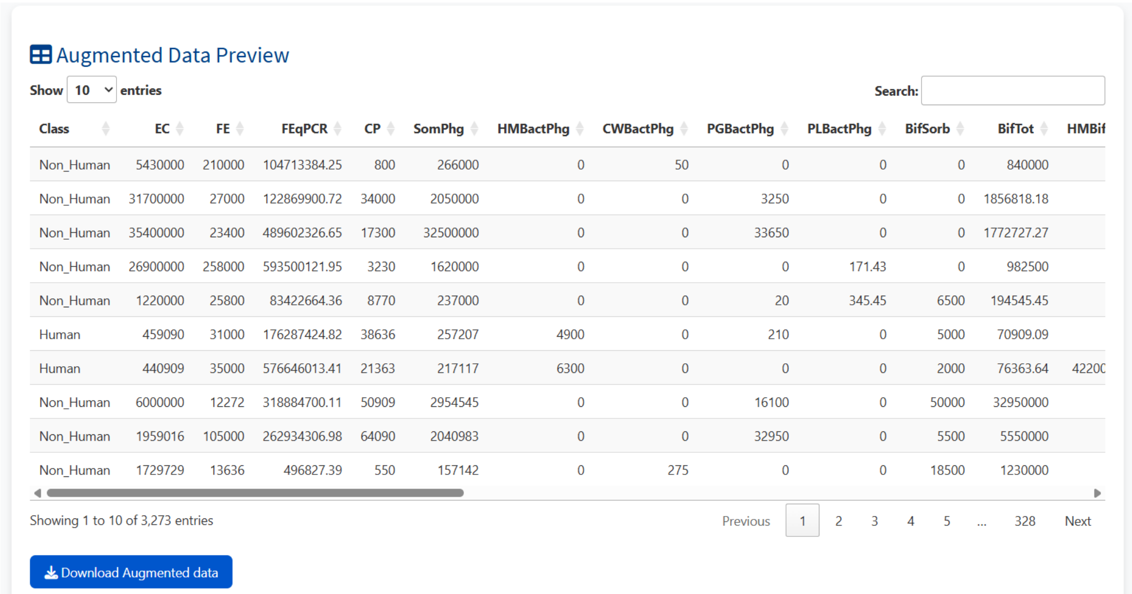
The number of attempts is indicated in the window that appears in the bottom right corner.



Subsequent to the augmentation process, a pop-up window will appear, indicating the next available step (see following illustration).



Upon clicking the "Got it!" button, the window will close and the results of the data augmentation process will be displayed, as it is shown in the following page.


Augmented Data Preview

Show 10 entries Search:

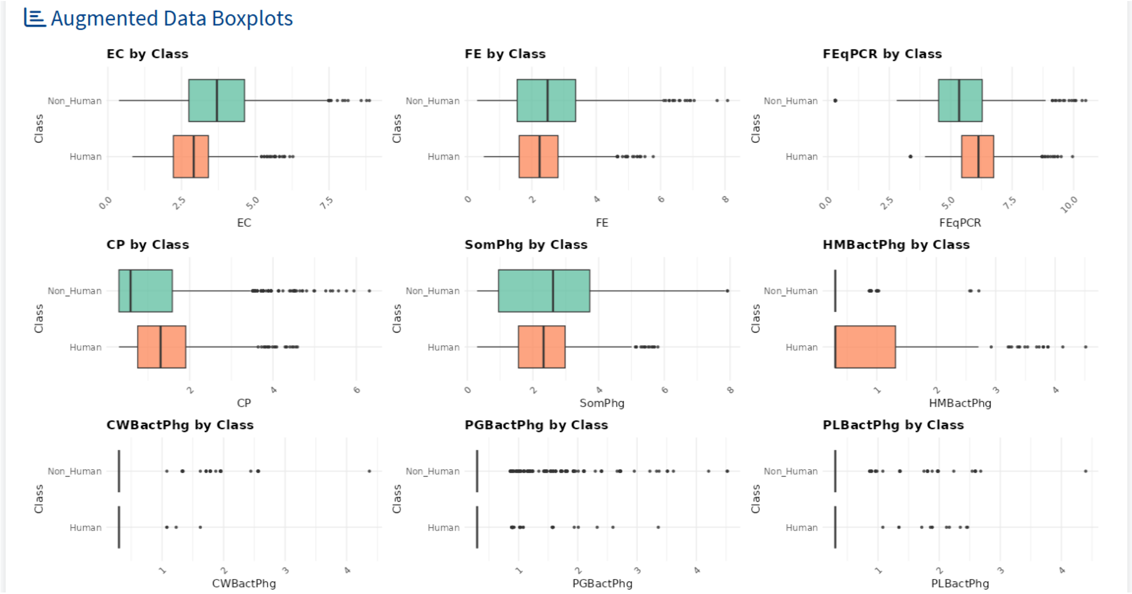
Class	EC	FE	FEqPCR	CP	SomPhg	HMBactPhg	CWBactPhg	PGBactPhg	PLBactPhg	BifSorb	BifTot	HMBif
Non_Human	5430000	210000	104713384.25	800	266000	0	50	0	0	0	840000	
Non_Human	31700000	27000	122869900.72	34000	2050000	0	0	3250	0	0	1856818.18	
Non_Human	35400000	23400	489602326.65	17300	32500000	0	0	33650	0	0	1772727.27	
Non_Human	26900000	258000	593500121.95	3230	1620000	0	0	0	171.43	0	982500	
Non_Human	1220000	25800	83422664.36	8770	237000	0	0	20	345.45	6500	194545.45	
Human	459090	31000	176287424.82	38636	257207	4900	0	210	0	5000	70909.09	
Human	440909	35000	576646013.41	21363	217117	6300	0	0	0	2000	76363.64	4220
Non_Human	6000000	12272	318884700.11	50909	2954545	0	0	16100	0	50000	32950000	
Non_Human	1959016	105000	262934306.98	64090	2040983	0	0	32950	0	5500	5550000	
Non_Human	1729729	13636	496827.39	550	157142	0	275	0	0	18500	1230000	

Showing 1 to 10 of 3,273 entries Previous 1 2 3 4 5 ... 328 Next

Download Augmented data

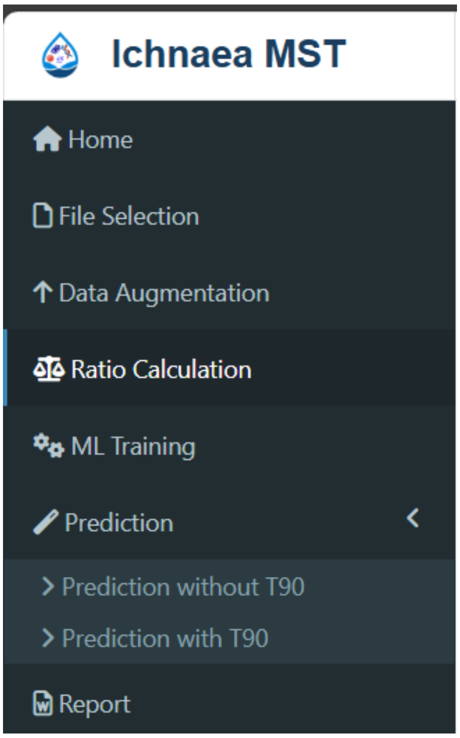
It should be noted that the data provided by the user is displayed at the commencement of the augmented dataset. The option to save this dataset as a CSV file is available by clicking the following button. 

Furthermore, the boxplots of the augmented dataset are presented in the Augmented Data Boxplots window.



The subsequent step in the sequence is the calculation of ratios. As previously indicated, this step can be omitted with confidence, unless there is a necessity to incorporate a log-10 ratio between the MST markers.

Ratio Calculation



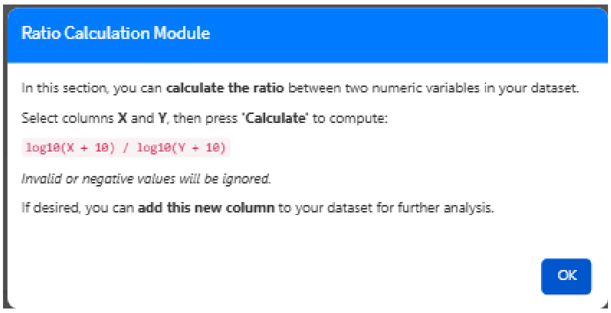
The subsequent step in the menu is the calculation of the ratio.

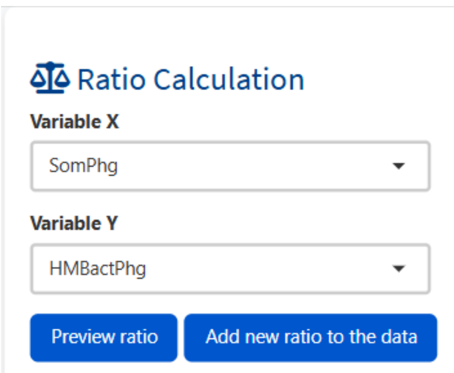
At is was indicated it is considered acceptable to omit this section, with the exception of instances in which a ratio between two MST markers is utilised as an MST predictor.

The ratio between the logarithmic values of somatic coliphages and bacteriophage infecting the strain GA17 of *Bacteroides tetraiotaomicron* has been proposed as a reliable Microbial Source Tracking (MST) marker, offering a valuable metric for distinguishing contamination sources. This ratio helps identify faecal pollution by assessing differences in the presence and concentration of these two viral indicators.

However, an important consideration arises when integrating this ratio into the training dataset. If the ratio is simply uploaded as part of the dataset file, it will be treated as a single marker rather than a covariate marker within the classification framework. To mitigate this issue, it is necessary to modify the dataset after the augmentation process to ensure that the ratio is appropriately calculated.

Upon initiation of this section, the subsequent windows will pop-up.

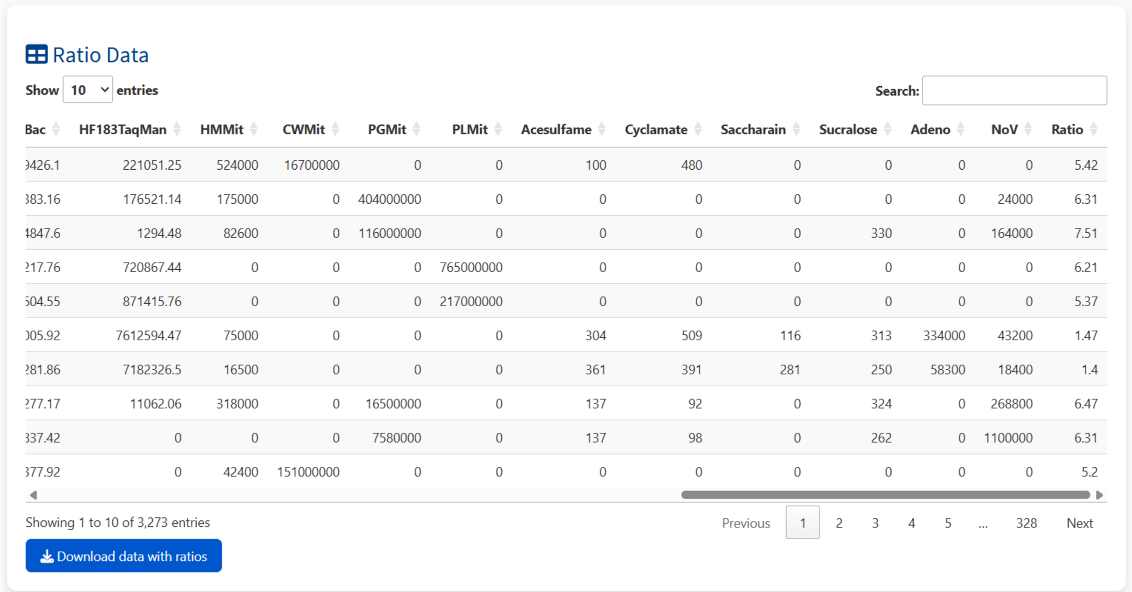




For instance, the variable SomPhg and HMBactPhg has been selected.

To preview the data without adding a new MST marker the user can use the “Preview ratio” button, only by pressing the button “Add new ratio to the data” the resultant ratio is added to the dataset for training.

In the following page is displayed a preview of this ratio.



Ratio Data

Show 10 entries   Search:

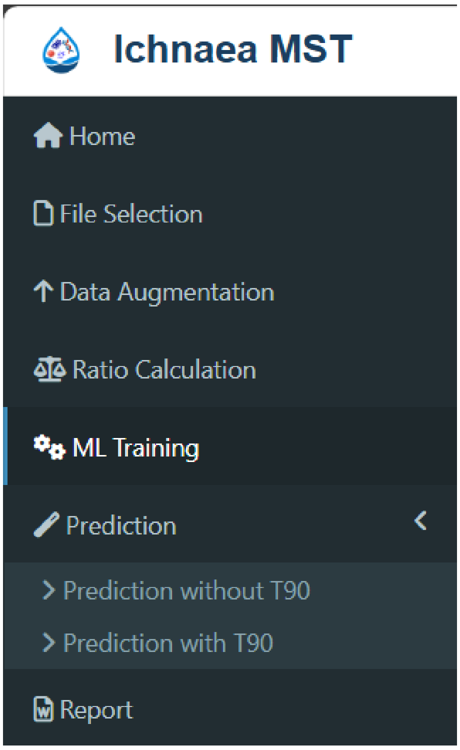
Bac	HF183TaqMan	HMMit	CWMit	PGMit	PLMit	Acesulfame	Cyclamate	Saccharain	Sucralose	Adeno	NoV	Ratio
426.1	221051.25	524000	16700000	0	0	100	480	0	0	0	0	5.42
83.16	176521.14	175000	0	404000000	0	0	0	0	0	0	24000	6.31
847.6	1294.48	82600	0	116000000	0	0	0	0	330	0	164000	7.51
17.76	720867.44	0	0	0	765000000	0	0	0	0	0	0	6.21
04.55	871415.76	0	0	0	217000000	0	0	0	0	0	0	5.37
05.92	7612594.47	75000	0	0	0	304	509	116	313	334000	43200	1.47
81.86	7182326.5	16500	0	0	0	361	391	281	250	58300	18400	1.4
77.17	11062.06	318000	0	16500000	0	137	92	0	324	0	268800	6.47
37.42	0	0	0	7580000	0	137	98	0	262	0	1100000	6.31
77.92	0	42400	151000000	0	0	0	0	0	0	0	0	5.2

Showing 1 to 10 of 3,273 entries   Previous 1 2 3 4 5 ... 328 Next

Download data with ratios

The new variable is displayed in the final column of the table that corresponds to the MST markers. It is possible to download the preview dataset as a CSV file without the requirement of additional data being added to the training set.

ML Training



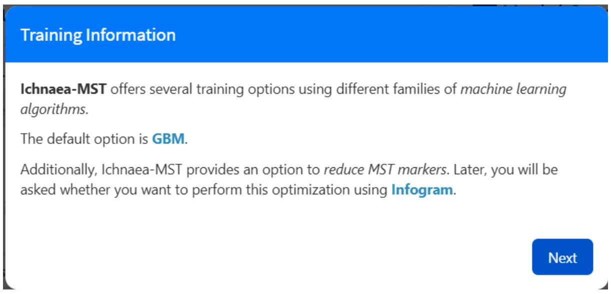
This section of the application can be accessed directly if the model has already been trained, or if data augmentation is not required. Nevertheless, it should be noted that, even in these cases, a submenu will appear with these two options for uploading your training or augmented dataset. In the ensuing pages, the process will be elucidated with these scenarios in mind.

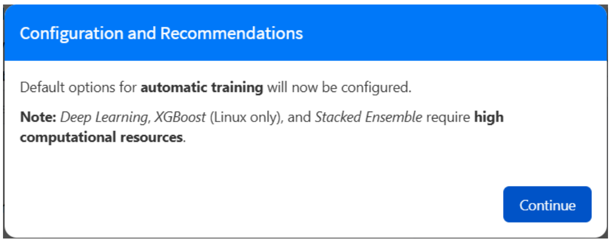
The training process is carried out using the H2O-3 framework provided by H2O.ai. If you are interested to learn more about why we chose H2O as the framework for the machine learning process, you will find a brief explanation on page 9.

Please considerer to visit the following links: https://h2o.ai/platform/ai-cloud/make/h2o/

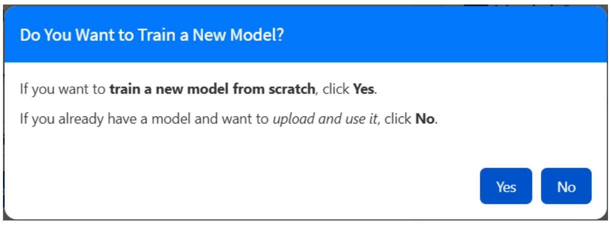
https://docs.h2o.ai/h2o/latest-stable/h2o-docs/automl.html

Once this section is accessed, the user is presented with the following pop-ups.

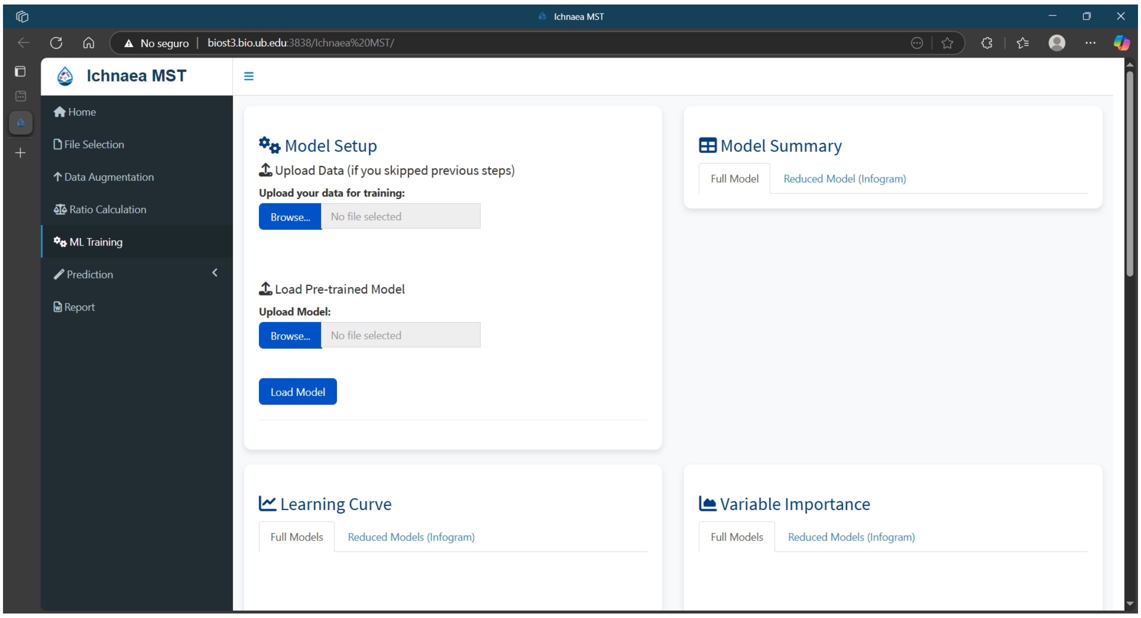




The preceding pop-ups serve exclusively an informational purpose; the subsequent pop-up, however, is of particular significance, as it determines whether the user is able to load a model that has been previously trained, thereby circumventing the process of creating a model from the beginning.

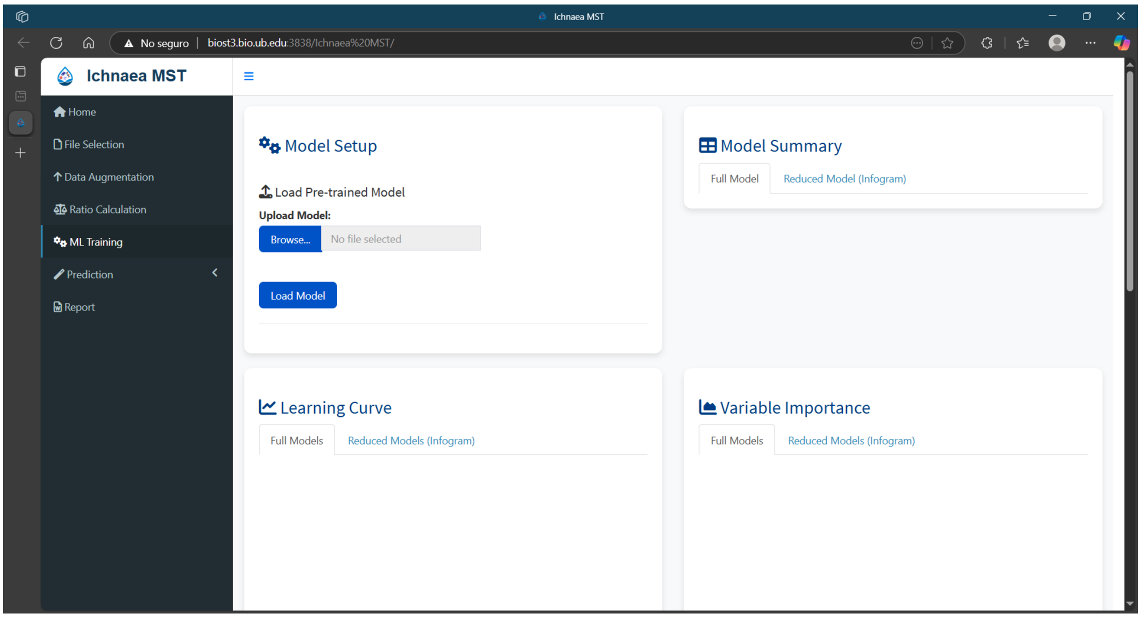


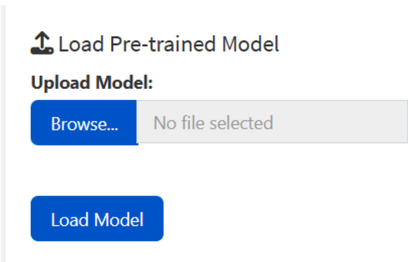
In the event of the user selecting to upload a trained model, the following screen will be displayed.

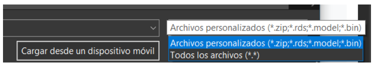


Although classification can be performed without loading additional data, the detection limits of the MST markers used for training may differ significantly from those calculated for classification purposes. For this reason, this app has been designed to oblige users to upload the original training data or an augmented dataset, in the correct format, and to include the necessary metadata (see pages 24 to 26). This enables the system to identify samples that may be prone to misclassification due to discrepancies in the detection limits or the presence of Zero-values across all MST markers.

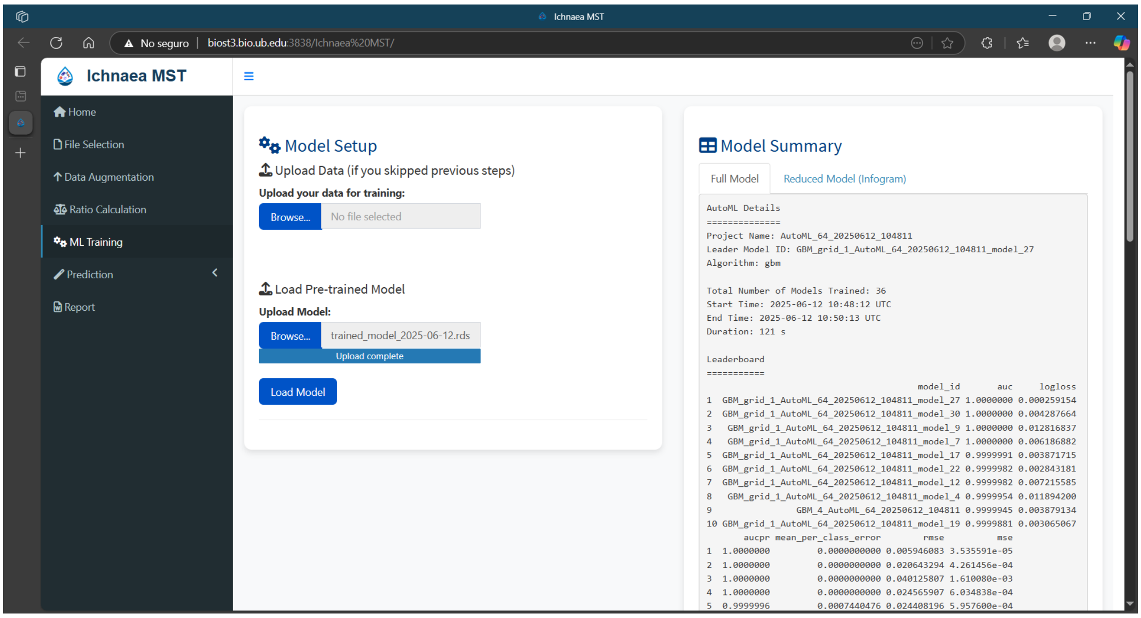
After the dataset has been uploaded, the only option that remains is to upload a trained model, as it is shown in following screen capture.





This menu allows the user to upload a previously trained model. To do so, click the Browse button to open a file selection window. Please ensure that the file type filter is set to All files (.), as the default setting may be configured to a custom filter that could prevent the model file from being displayed. 

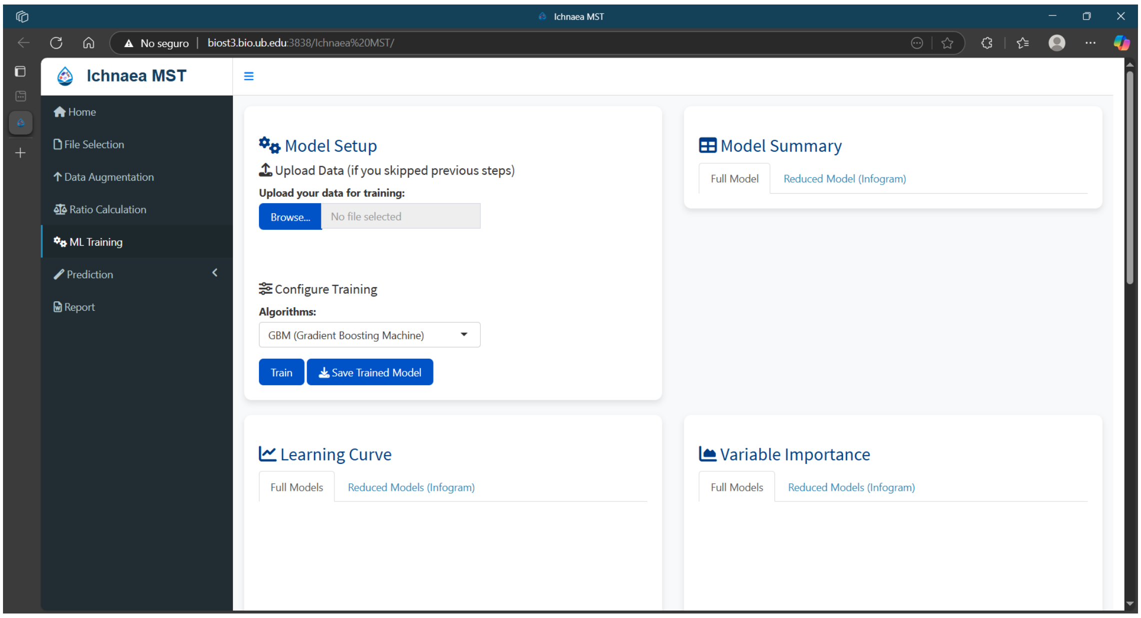
For instance, the following illustration shows a capture screen of a loaded trained model.



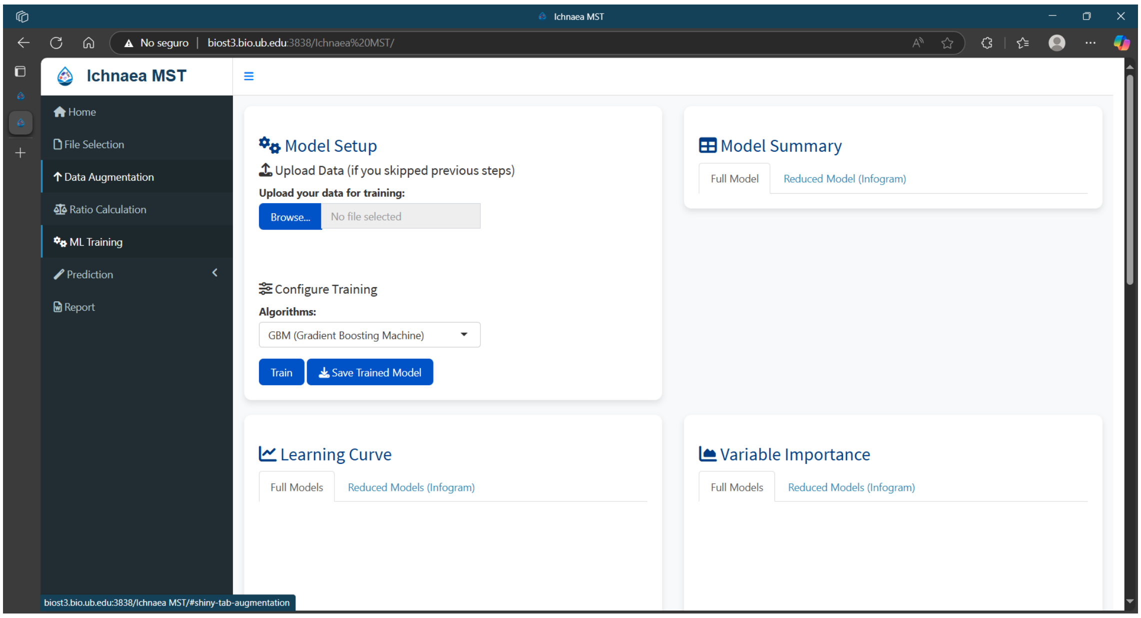
Note if a reduction of MST markers has also carried out both trained models are uploaded.

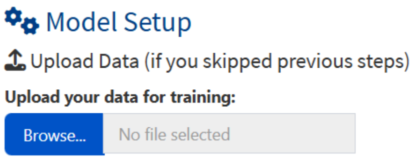
When the option to load a previously trained model is not selected, the corresponding menu is also omitted, can displayed one of the two following screen capture.

In the event that no model needs to be uploaded, the programme will proceed to the subsequent screen.



If the accession to this menu is carried out directly, the program will display two distinct menus to the user. The upper menu is intended for uploading a dataset for training purposes.



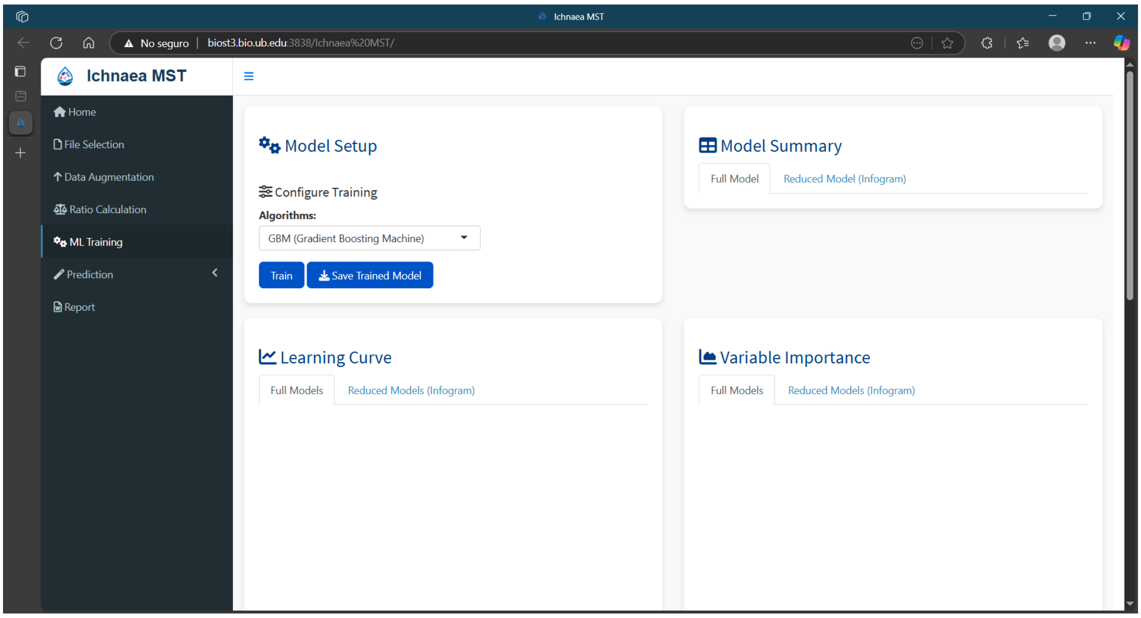


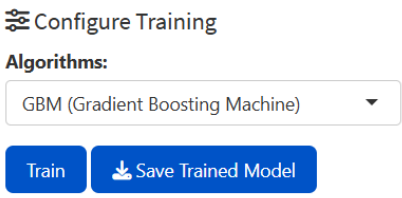
It is important to note that machine learning-based classification requires a sufficiently large and representative dataset to train a reliable model. If the user has such a dataset, this menu allows them to upload it directly to the application.

This menu option has also been included to accommodate scenarios in which the user has manually modified or augmented the dataset outside the Ichnaea-MST application, changes that cannot be made within the tool itself. This ensures flexibility and allows for the integration of externally curated data into the model training process.

Once the data has been uploaded, users can proceed with the training process.

If the user navigates to this menu following the completion of the previous steps using the application, only the menu of training will be available, as it is shown in following screen capture.



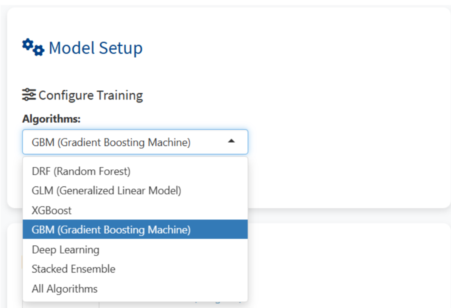


The menu for configuring the training process allows the user to select the machine learning algorithm. By default, the number of models to be tested is set to 25 for multinomial models and 50 for binomial models.

It is important to keep in mind that some algorithms are computationally intensive; therefore, the results may take some minutes to be calculated.

To facilitate the process and ensure usability, we have preconfigured default values for both the number of models and the algorithm type. These defaults have been carefully chosen based on the specific characteristics of the MST classification problem and are considered more than sufficient for achieving robust and reliable results in most scenarios. Nevertheless, advanced users may feel more comfortable using the local mode, which allows for finer control over these parameters according to their computational resources and specific analytical needs.

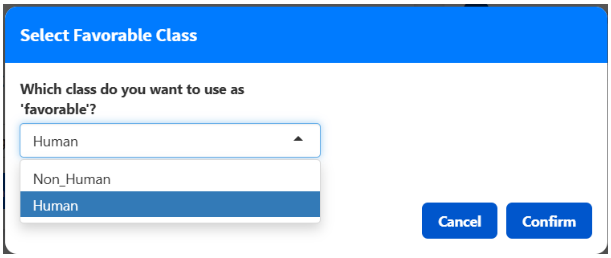
Users can select their preferred algorithm by navigating through the dropdown menu. The default algorithm is GBM.



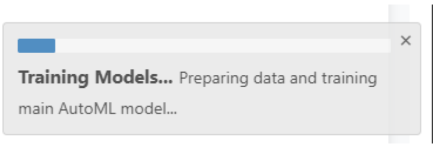
For more detailed information about the functionality of each algorithm, including their computationally advantages and disadvantages, please refer to pages 8 to 10 of this manual.

Once selected the kind of algorithm the user indicates which sample is the 'favourable class'. In this context of training models with H2O, the "favourable class" plays a crucial role in interpreting classification models, especially for binary or multiclass problems. It designates which target class should be treated as the "positive" or desired outcome, influencing evaluation metrics (like AUC, precision, recall, or F1-score), model interpretation, and predictor selection. For instance, if "human" is set as the favourable class in a faecal contamination classification task, the model will focus on accurately identifying human samples and prioritize predictors that best distinguish this class. without this option, H2O may default to the first class alphabetically or not optimize for any specific class, potentially reducing the model's practical effectiveness.

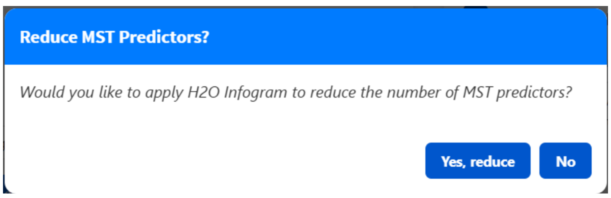
The use may select this class when the following window pops-up.



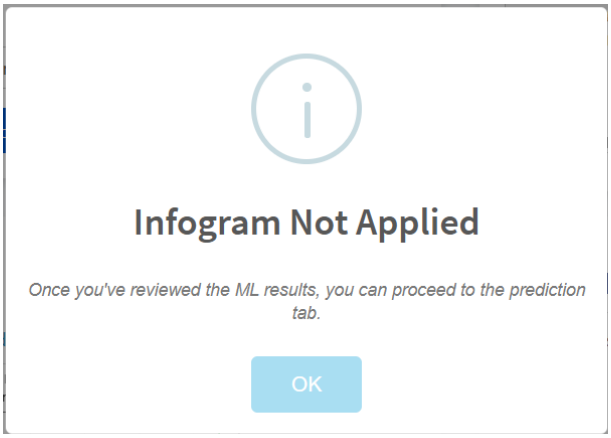
Once the model has been selected, the training process begins. This is indicated by the progress bar displayed in the bottom right corner.



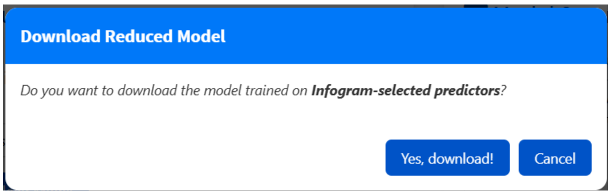
Once the training process is complete, a new pop-up window displays, prompting the user to indicate whether they want to run an analysis to reduce the MST markers. This analysis is performed using H2O Infogram, a tool for interpretable machine learning that identifies and ranks the most relevant, non-redundant features in a dataset. Using information theory, it measures how much each feature contributes to predicting a target variable while minimising overlap between features. This helps users to build simpler, more transparent models without sacrificing performance.



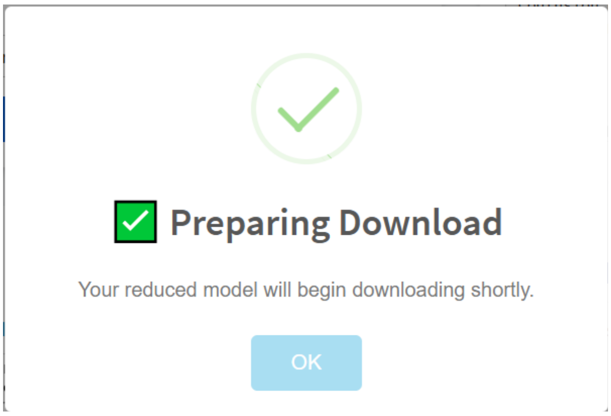
If the user selects to not performance the infogram analysis the following screen will appear.



In the event of opting for the Infogram analysis, the process will take several minutes to complete. Upon reaching the conclusion stage, a pop-up will be displayed indicating if the model should be saved.

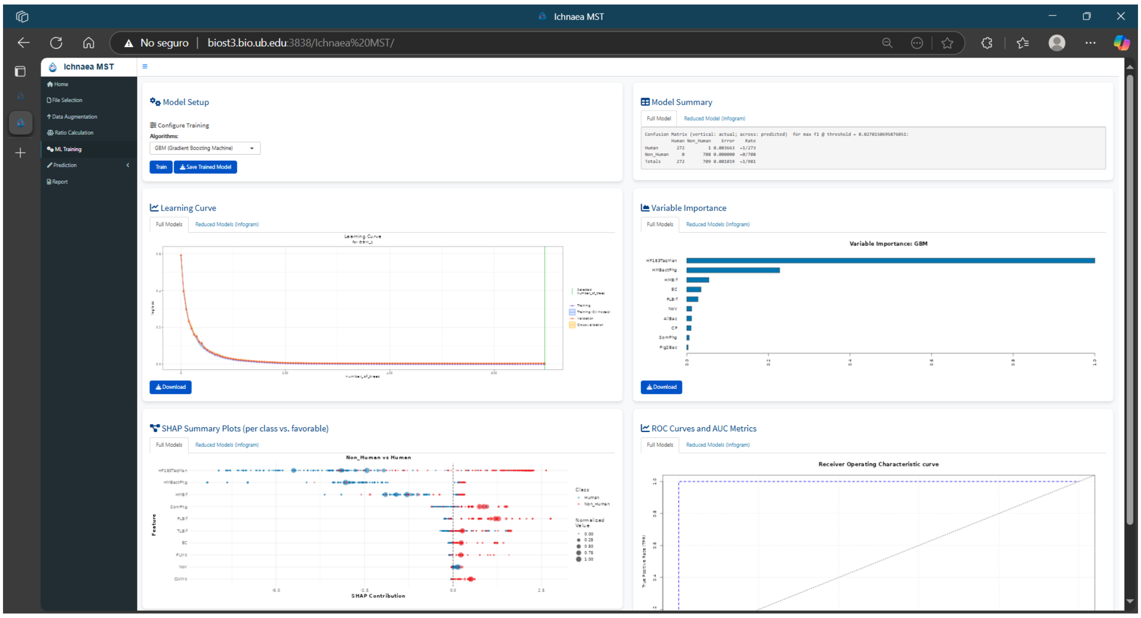


Following this, a new window will open to indicate that the infogram analysis is being downloaded.



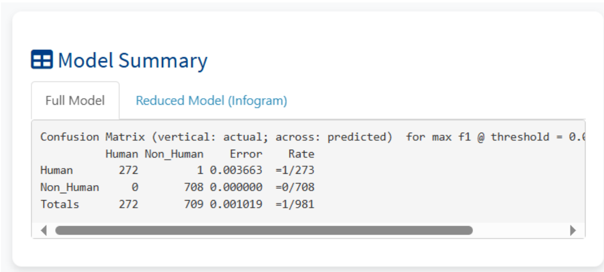
The program will display the training results, irrespective of whether or not parameter reduction analysis is performed. Should the infogram analysis not be performed, the results of said analysis will be displayed as blank.

The following screen capture shows the results of the training process.

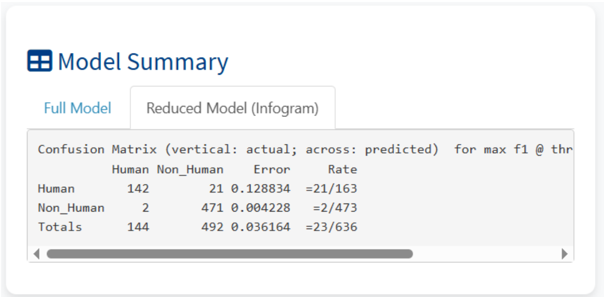


Let us now turn our attention to the results obtained.

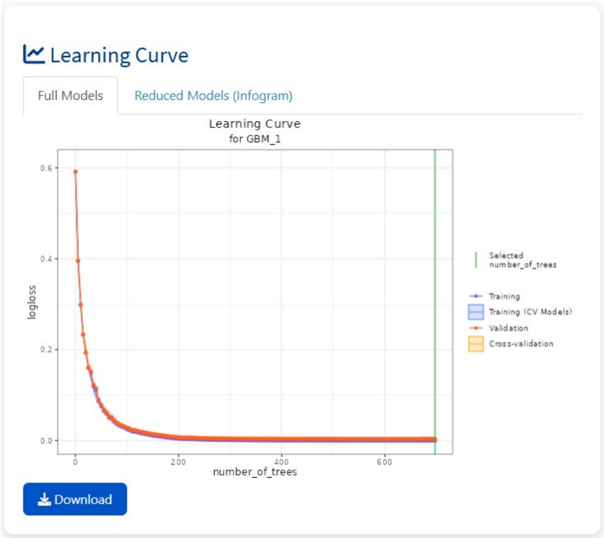
The model summary shows the confusion tables using the validation dataset. Both, the full model as the reduced model is presented.

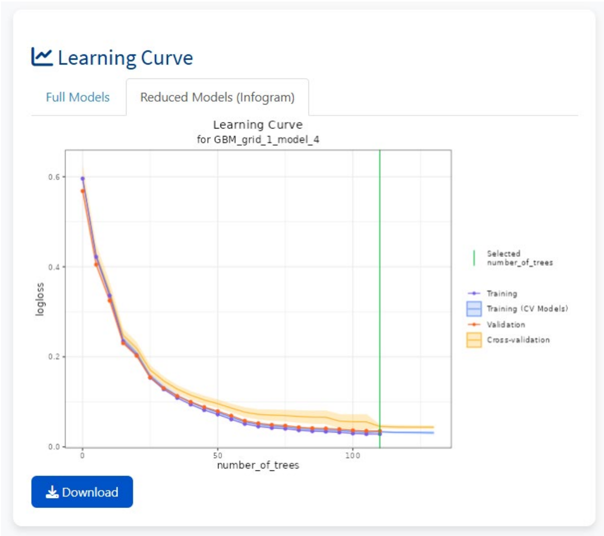



```
Confusion Matrix (vertical: actual; across: predicted)  for max f1 @ threshold = 0.(
          Human Non_Human    Error     Rate
Human       272         1 0.003663  =1/273
Non_Human     0       708 0.000000  =0/708
Totals      272       709 0.001019  =1/981
```

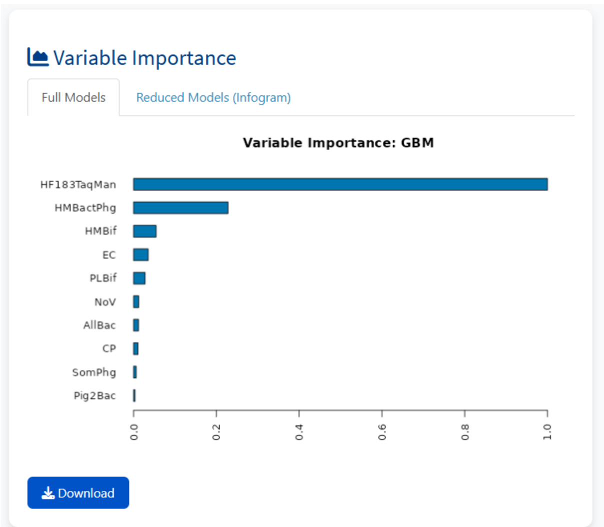


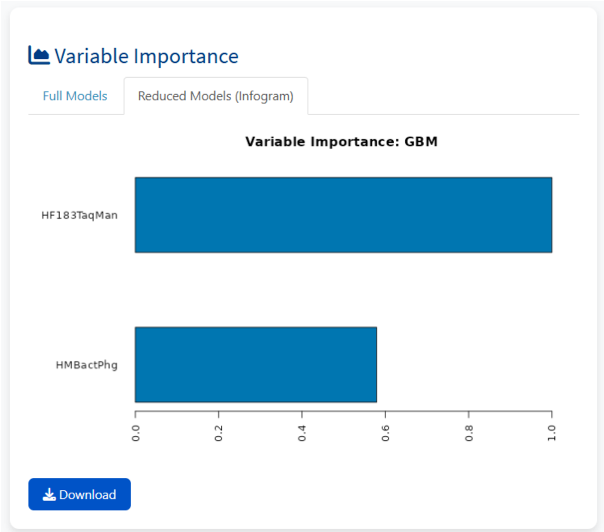

```
Confusion Matrix (vertical: actual; across: predicted)  for max f1 @ thr
          Human Non_Human    Error     Rate
Human       142        21 0.128834  =21/163
Non_Human     2       471 0.004228   =2/473
Totals      144       492 0.036164  =23/636
```



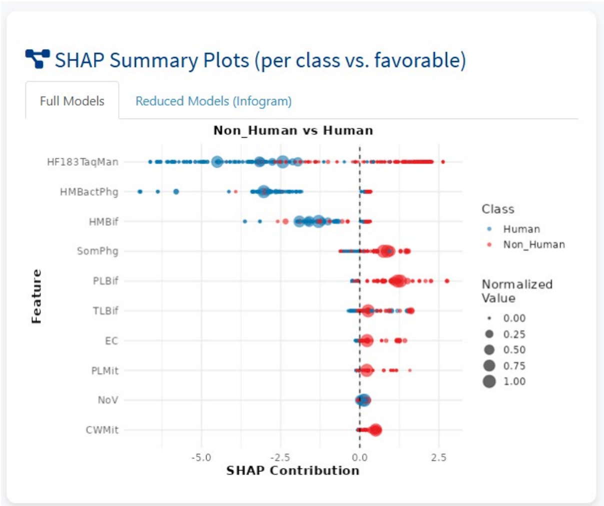


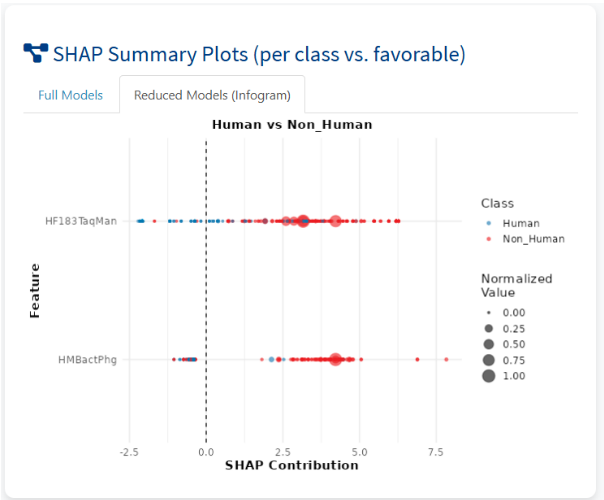
Kindly refer to page 72, where a concise exposition on the interpretation of a learning curve can be found.



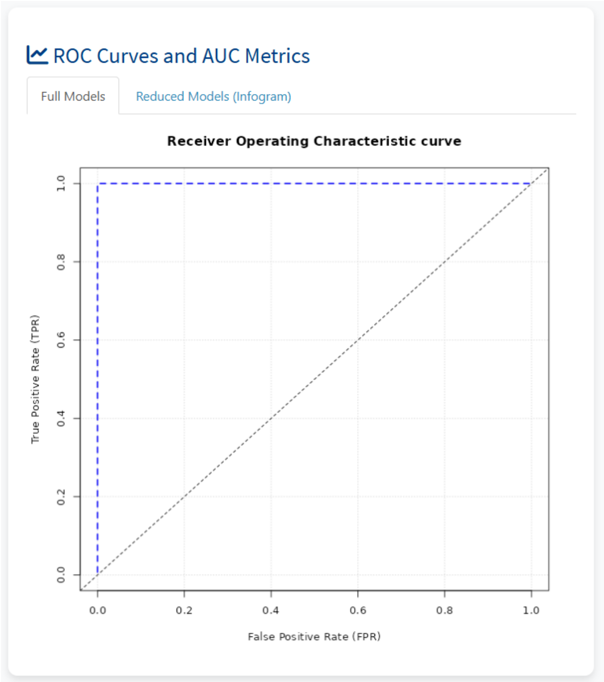


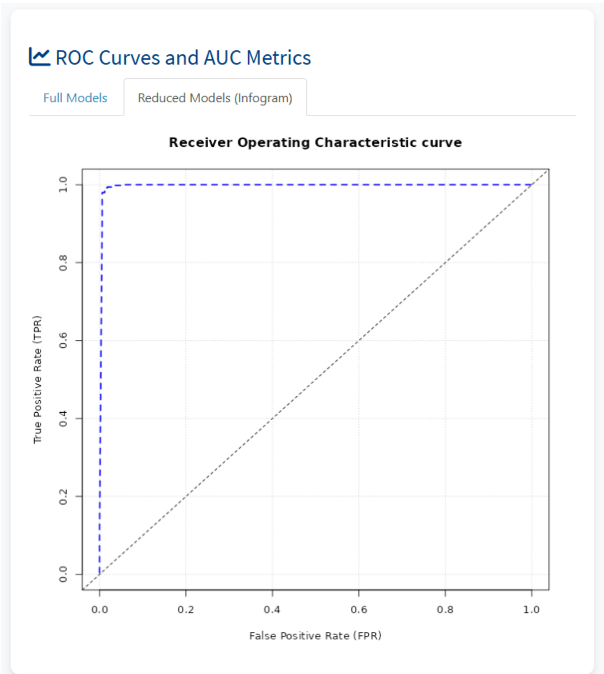
Please refer to page 73, where a concise explanation is provided to facilitate interpretation of a variable importance plot.





Please refer to page 74, where a concise explanation is provided to facilitate interpretation of a SHAP plot.

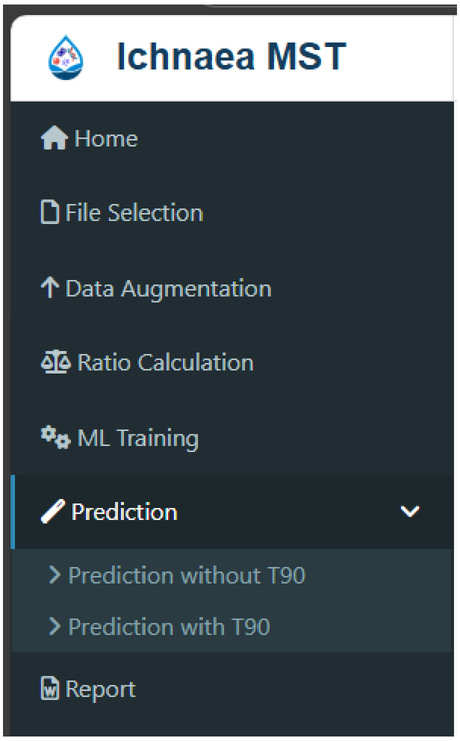




Kindly refer to page 70 to 71, where a concise exposition on the interpretation of a ROC curve plot can be found.

Upon successful completion of the training process, the user is permitted to save the model as an .rds file, which should be designated as 'compose trained_model_'+ data+.rds'. A .rds file is a file format used in R. In this particular instance, the file contains the model that has been trained with the full set of MST-markers, as well as the trained model with the reduced set (if the infogram analysis has been carried out). It is important to note that a different file type has been selected for the web application, as opposed to the R Notebook application, which saves and loads the native model file provided by H2O. This decision was made with the intention of facilitating the experience of users with limited knowledge of R. It is imperative that this is taken into consideration when working with both platforms.

Prediction

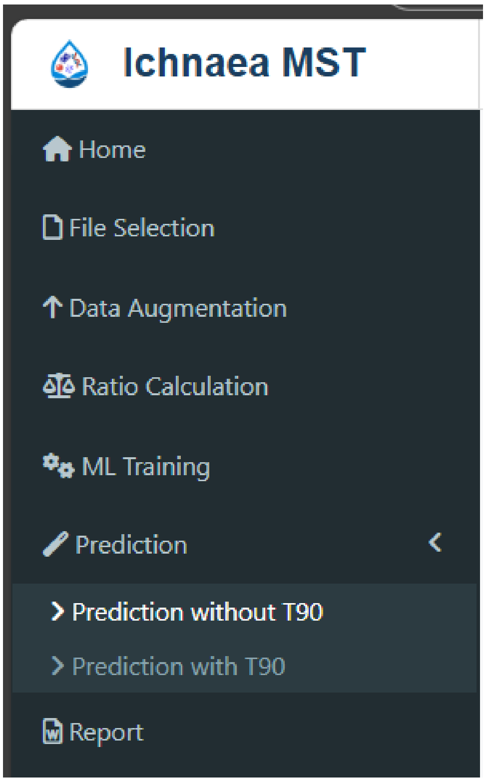


The Prediction menu offers two options: the first, labeled "Prediction without T90", and the second, "Prediction with T90". The first option allows for the prediction of the input classification data without applying any modifications. Conversely, the second option incorporates a linear model that accounts for the potential decay of various MST (Microbial Source Tracking) parameters over time.

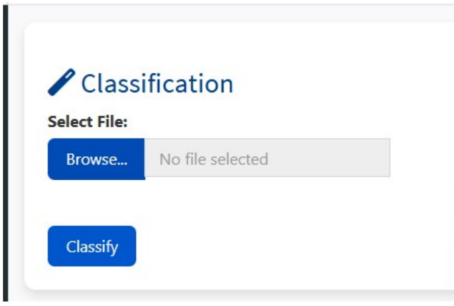
As previously mentioned for illustrative purposes, a dataset with artificially generated t90 values has been created.

These values are uniformly applicable across all parameters and are intended to simulate the natural decay process. In this scenario, the values of the markers are increased based on the estimated residence time to which the samples may have been exposed.

It is important to note that, in order to apply this decay adjustment accurately, the t90 values for all markers must be known. Furthermore, the process of increasing marker values based on residence time is only applicable to quantitative samples, as it relies on measurable concentrations.

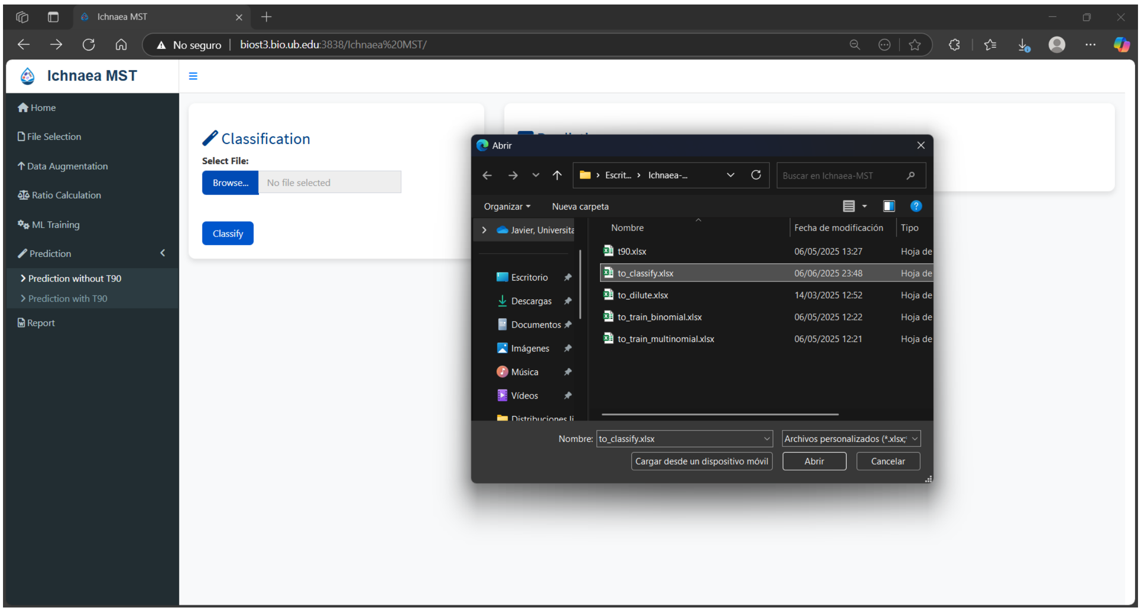


In the absence of t90, predictions are obtained without the application of natural decay. The predictions can be obtained from the model that has been trained with the full set of MST markers or from the reduced set (in the event that the infogram analysis was carried out).

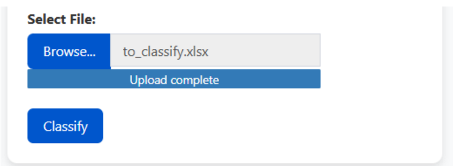


When this menu is selected the user is prompted to enter a file with the samples that it would be classified. Please refer to page 37 about the structure of this dataset.

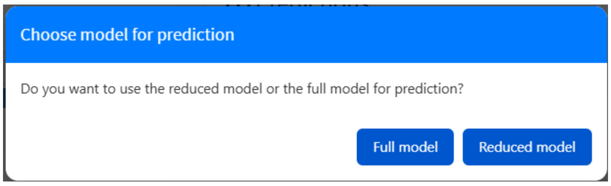
Please consider that it might be advisable to include some samples that have not been seen by the algorithm but whose origin is known, in order to verify that these samples are correctly classified. In the example used, samples with known origin were included, and a series of in silico dilutions were performed on them.



Once the file has been selected, the user can perform the classification by clicking on the 'Classify' button.



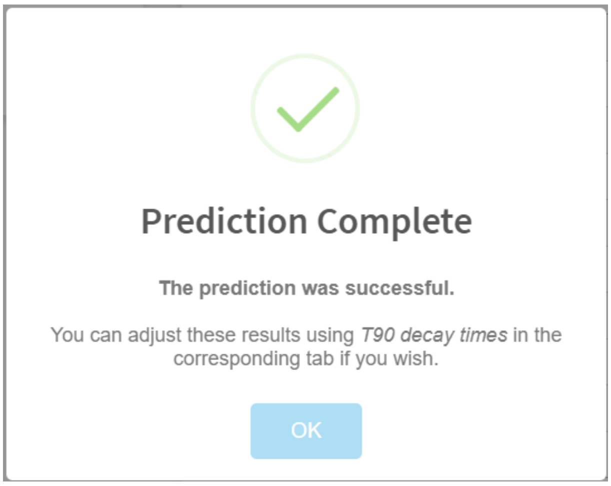
The user has the option of determining whether the classification should be carried out using the full and/or the reduced model.



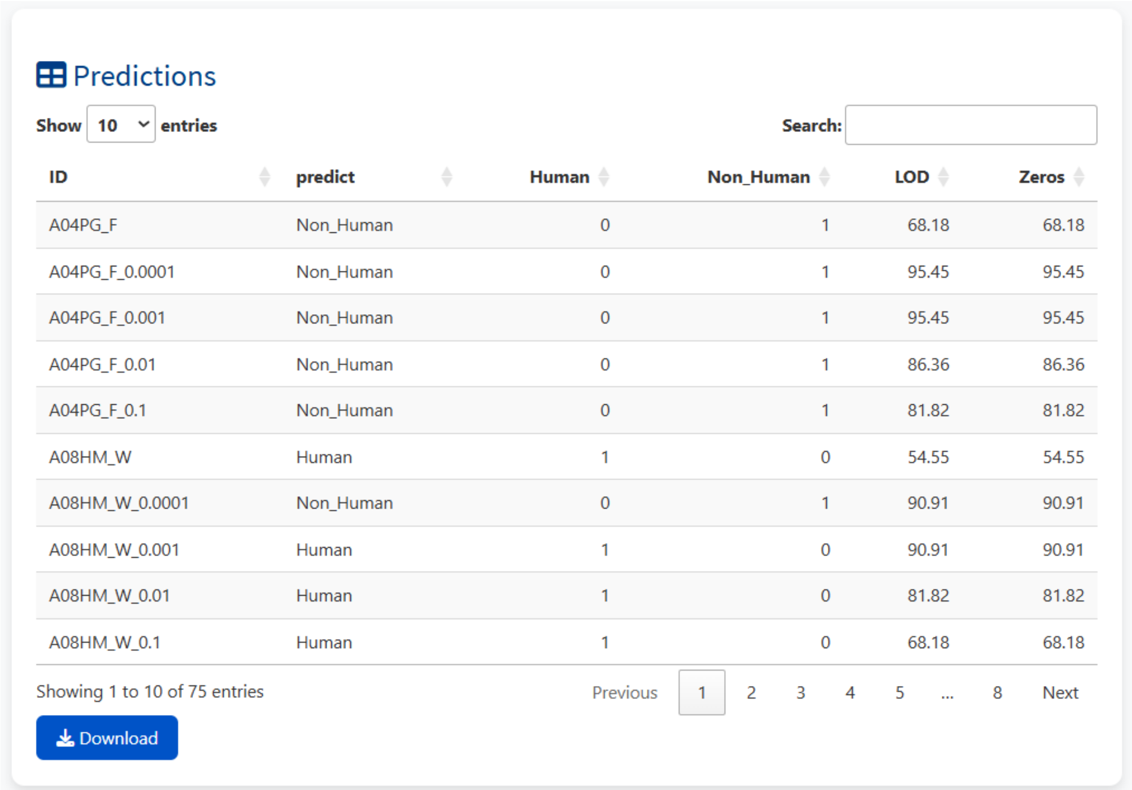
Furthermore, a test is conducted to identify the samples that present all the MST markers below the limit of detection (LOD) or have zero-values. This process helps to determine which samples are susceptible to misclassification. Please refer to page 37 for further details.

For instance, the results of this test are presented with the full set of MST markers.

Once these pop-ups have been closed, a notification will appear confirming that prediction process is complete.



After clicking th “OK” button the classification are displayed.



Predictions

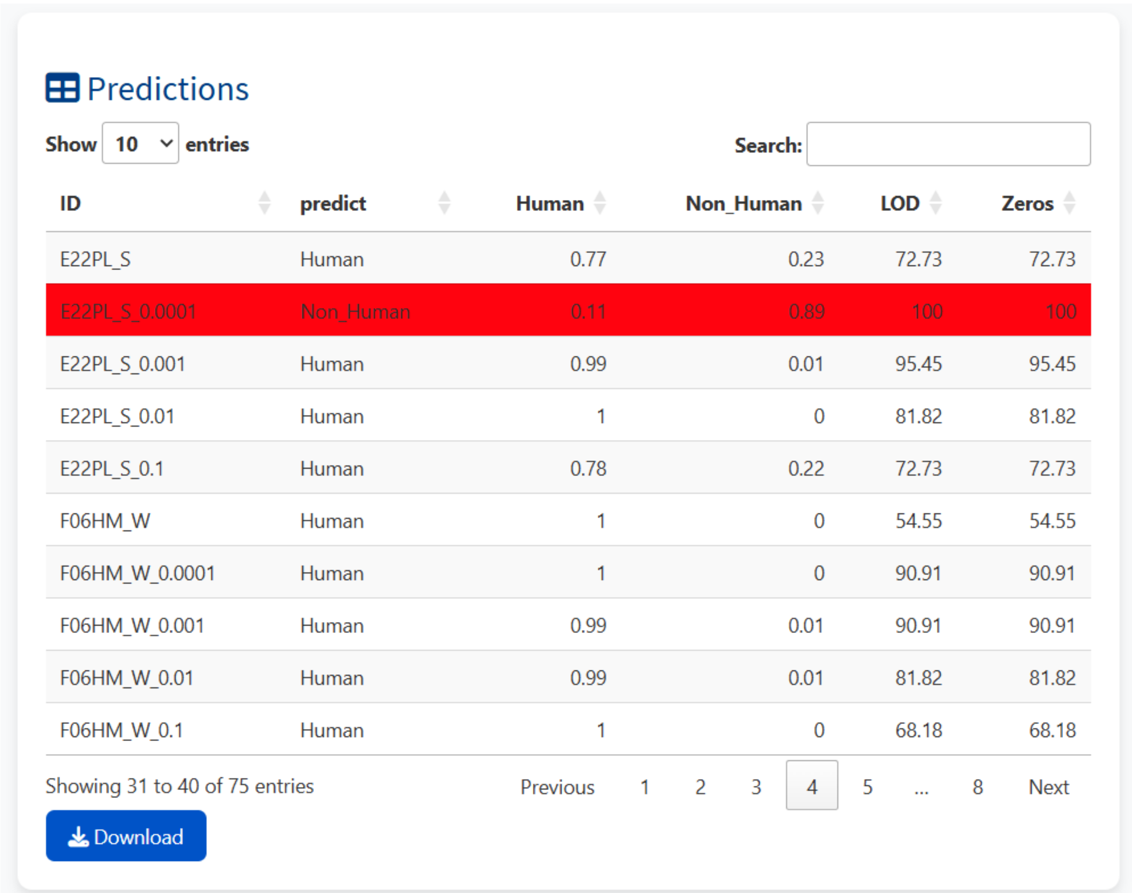
Show 10 entries Search:

ID	predict	Human	Non_Human	LOD	Zeros
A04PG_F	Non_Human	0	1	68.18	68.18
A04PG_F_0.0001	Non_Human	0	1	95.45	95.45
A04PG_F_0.001	Non_Human	0	1	95.45	95.45
A04PG_F_0.01	Non_Human	0	1	86.36	86.36
A04PG_F_0.1	Non_Human	0	1	81.82	81.82
A08HM_W	Human	1	0	54.55	54.55
A08HM_W_0.0001	Non_Human	0	1	90.91	90.91
A08HM_W_0.001	Human	1	0	90.91	90.91
A08HM_W_0.01	Human	1	0	81.82	81.82
A08HM_W_0.1	Human	1	0	68.18	68.18

Showing 1 to 10 of 75 entries Previous 1 2 3 4 5 … 8 Next

Download

The samples that have been marked as 'risk of misclassification' are highlighted in the table.



Predictions

Show 10 entries Search:

ID	predict	Human	Non_Human	LOD	Zeros
E22PL_S	Human	0.77	0.23	72.73	72.73
E22PL_S_0.0001	Non_Human	0.11	0.89	100	100
E22PL_S_0.001	Human	0.99	0.01	95.45	95.45
E22PL_S_0.01	Human	1	0	81.82	81.82
E22PL_S_0.1	Human	0.78	0.22	72.73	72.73
F06HM_W	Human	1	0	54.55	54.55
F06HM_W_0.0001	Human	1	0	90.91	90.91
F06HM_W_0.001	Human	0.99	0.01	90.91	90.91
F06HM_W_0.01	Human	0.99	0.01	81.82	81.82
F06HM_W_0.1	Human	1	0	68.18	68.18

Showing 31 to 40 of 75 entries Previous 1 2 3 4 5 … 8 Next

Download

As previously stated, when the reduced set of MST markers is applied to training, the probability of finding samples below the detection limit or with zero MST marker occurrences increases, as shown in the screenshot below.

Predictions

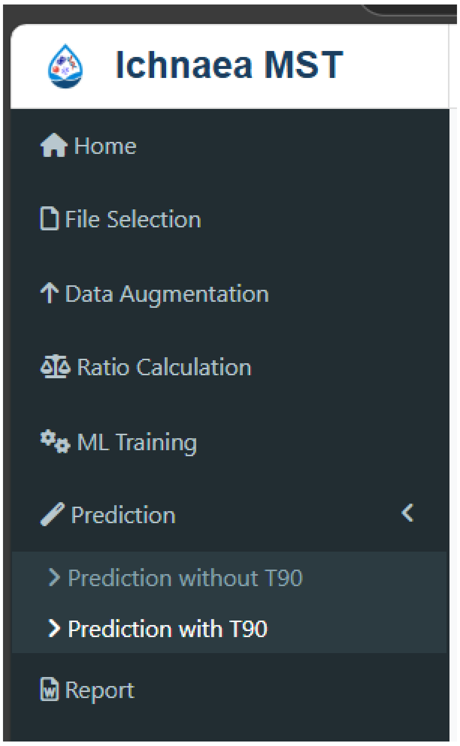
Show 10 entries Search:

ID	predict	Human	Non_Human	LOD	Zeros
A22CW_S	Non_Human	0.05	0.95	75	75
A22CW_S_0.0001	Non_Human	0.05	0.95	100	100
A22CW_S_0.001	Non_Human	0.05	0.95	100	100
A22CW_S_0.01	Non_Human	0.05	0.95	100	100
A22CW_S_0.1	Non_Human	0.05	0.95	100	100
D01HM_F	Human	0.98	0.02	25	25
D01HM_F_0.0001	Human	0.97	0.03	75	75
D01HM_F_0.001	Non_Human	0.17	0.83	75	75
D01HM_F_0.01	Non_Human	0.01	0.99	75	75
D01HM_F_0.1	Non_Human	0.89	0.11	25	25

Showing 11 to 20 of 75 entries Previous 1 2 3 4 5 … 8 Next

Download

For a comprehensive explanation of the interpretation of test results at the limit of detection and with zero-values, as well as the interpretation of prediction results, please refer to pages 84 to 93 of this manual.

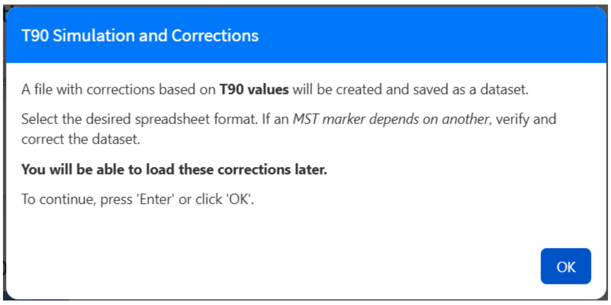


The other submenu in the prediction refers to the predictions considering the t90 values. As previously stated, the decay processes have the capacity to significantly alter the distribution of MST markers in the water matrix, especially if the decay affects the most important MST markers unevenly. In order to address this issue, Ichnaea-MST has introduced a small function that modifies the occurrence of the MST parameters according to a linear model based on t90.

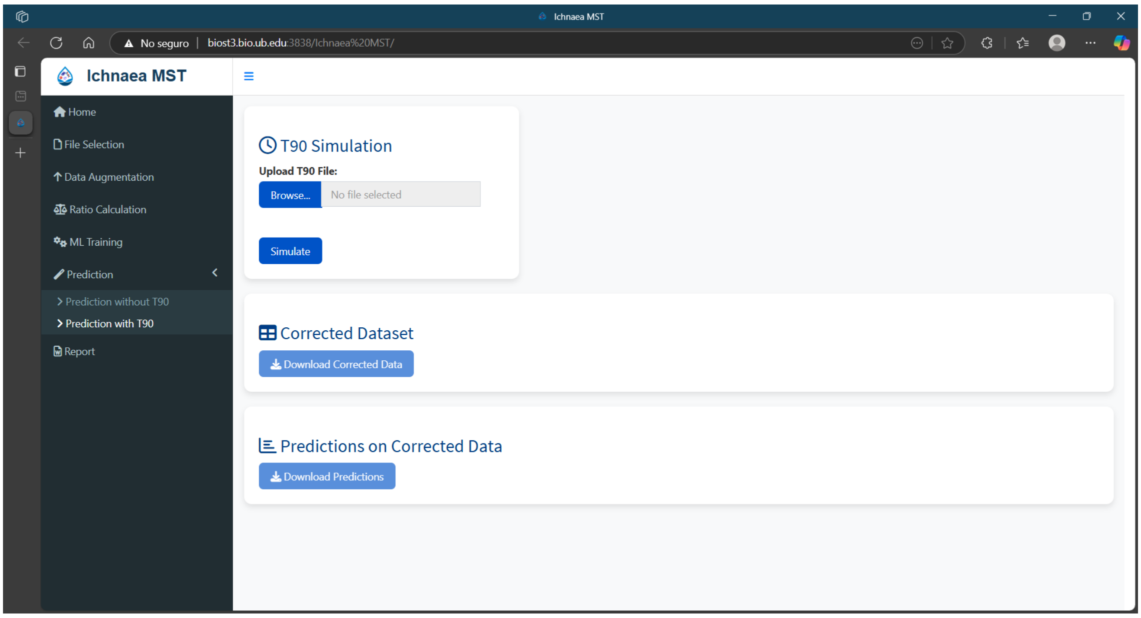
Remember that it is imperative to incorporate the t90 values expressed at the same time units for **all** MST markers utilised in the training. In circumstances where t90 values are not available, an analogous parameter may be employed as a possible alternative. For instance, if the MST is regarded as a genetic marker, then an alternative genetic marker should be used.

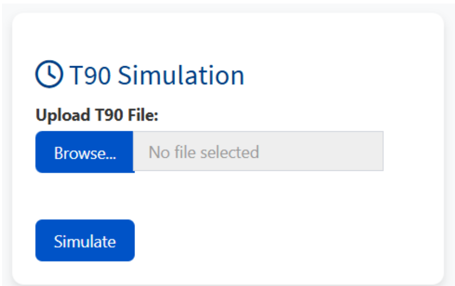
It is important to acknowledge the limitations of this approach, particularly in relation to the variability of the t90 values for a given marker, which is contingent on the inherent characteristics of the matrix.

Once the user has accessed this submenu, the following pop-up will be displayed.



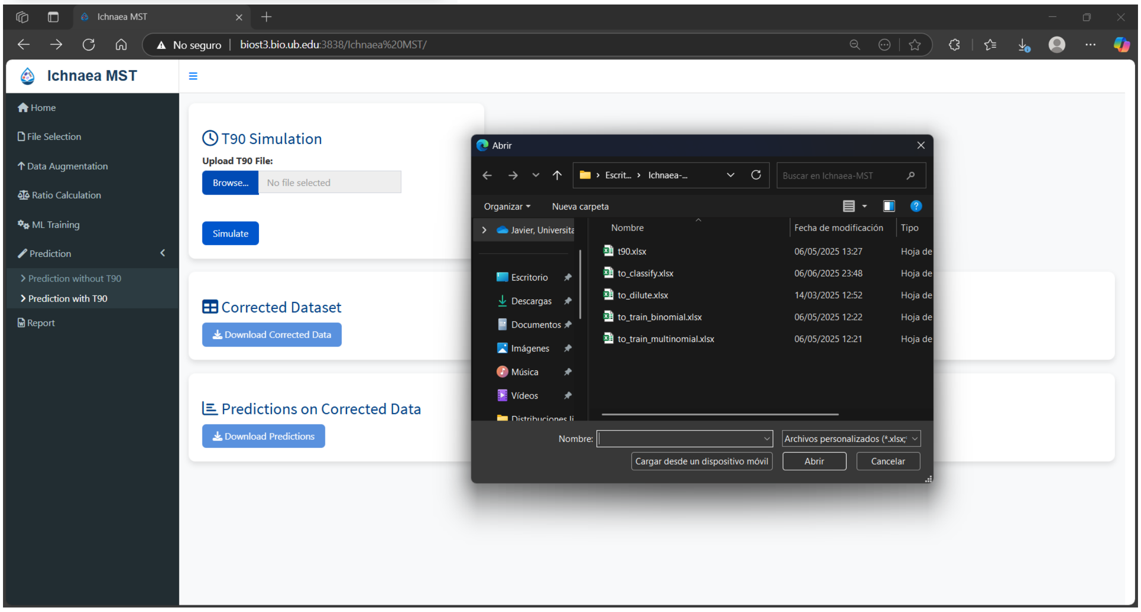
Once the user has confirmed, the following menu will be made available.

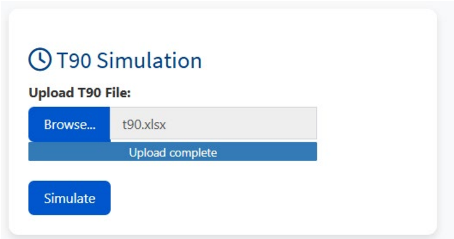




In this section, the available option prompts the user to upload the specified dataset containing the t90 values.

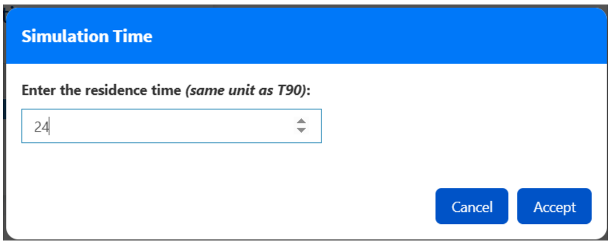
Once the user clicks on the "Browse" button, the following window will appear, prompting them to upload the file containing the t90 values, as it is shown in the following page.



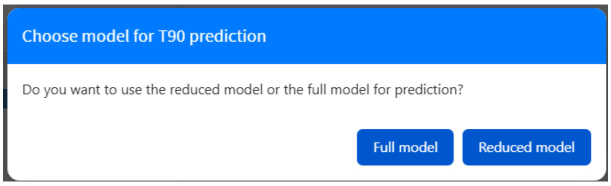


Once the file is uploaded, the application is ready to apply the t90 once the “Simulate button” is pressed.

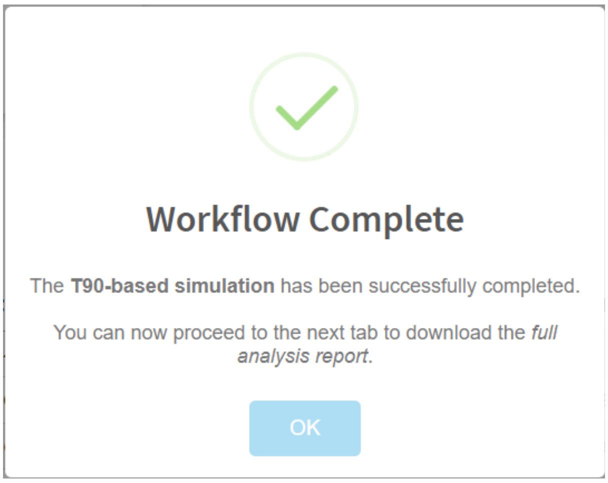
Following this, the application prompts the user to input the residence time that should be applied to amend the data with the t90. To select the desired time, please use the arrows to navigate upwards and downwards.



For instance, 24 hours has been introduced as a residence time.



For instance, the decision has been taken to show the full model prediction with the t90 amendments. Once selected, a pop-up window will appear to confirm that the process has been completed successfully.



The image below shows the values that have been adjusted in line with the t90 amendments.

Corrected Dataset

Show 10 entries Search:

Class	EC	FE	FEqPCR	CP	SomPhg	HMBactPhg	CWBactPhg	PGBactPhg	PLBactPhg	BifSorb	BifTot	HMBif	CWBif	PGNe
Non_Human	5430000	210000	104713384.25	800	266000	0	62.95	0	0	0	840000	0	0	
Non_Human	31700000	27000	122869900.72	34000	2050000	0	0	4091.51	0	0	1856818.18	0	0	
Non_Human	35400000	23400	489602326.65	17300	32500000	0	0	42362.84	0	0	1772727.27	0	0	
Non_Human	26900000	258000	593500121.95	3230	1620000	0	0	0	215.82	0	982500	0	0	
Non_Human	1220000	25800	83422664.36	8770	237000	0	0	25.18	434.9	8183.02	194545.45	0	0	
Human	459090	31000	176287424.82	38636	257207	6168.73	6.29	264.37	6.29	6294.63	70909.09	0	0	
Human	440909	35000	576646013.41	21363	217117	7931.23	6.29	0	6.29	2517.85	76363.64	5312665.24	0	
Non_Human	6000000	12272	318884700.11	50909	2954545	0	0	20268.7	6.29	62946.27	32950000	0	0	
Non_Human	1959016	105000	262934306.98	64090	2040983	0	0	41481.59	0	6924.09	5550000	0	0	
Non_Human	1729729	13636	496827.39	550	157142	0	346.2	0	0	23290.12	1230000	0	0	

Showing 1 to 10 of 98 entries Previous 1 2 3 4 5 ... 10 Next

Download Corrected Data

And below this line are the results of the classification.

Predictions on Corrected Data

Show 10 entries Search:

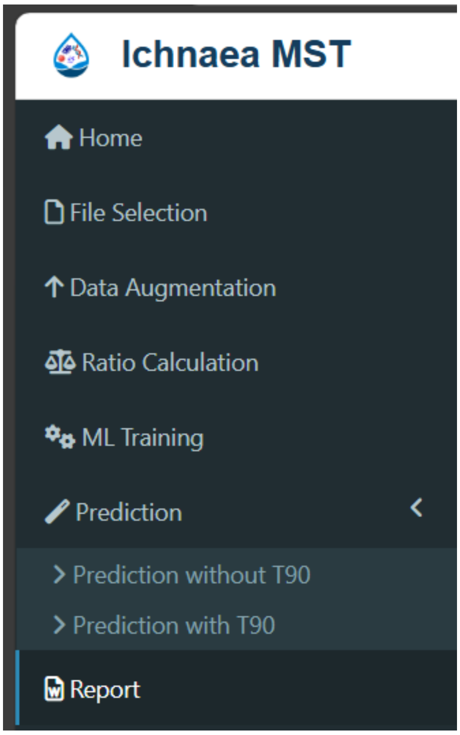
ID	predict	Human	Non_Human
Non_Human	Non_Human	0	1
Non_Human	Non_Human	0	1
Non_Human	Non_Human	0	1
Non_Human	Non_Human	0	1
Non_Human	Non_Human	0	1
Human	Human	1	0
Human	Human	1	0
Non_Human	Non_Human	0	1
Non_Human	Non_Human	0	1
Non_Human	Non_Human	0	1

Showing 1 to 10 of 98 entries Previous 1 2 3 4 5 ... 10 Next

Download Predictions

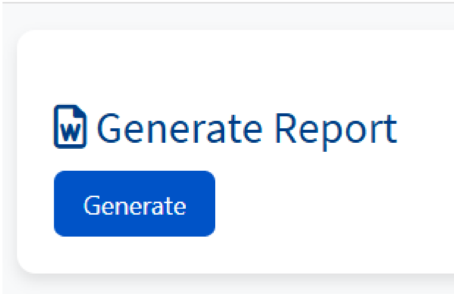
Please note that the predictions are adjusted on the results screen. They are shown with two decimal places, resulting in values of 0 and 1 where the probabilities are close to these values. However, the downloaded data shows more decimal places in the generated Excel file.

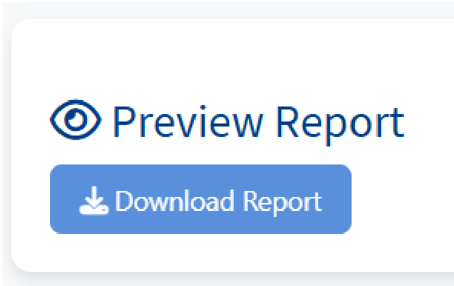
Report



This menu is responsible for displaying a report of the generated results and provides the option to save it in HTML format

Once accessed, the following submenus are available.





The initial menu is responsible for generating the report, for which a preview is subsequently displayed. In the 'Preview Report' section, the user can download the generated report as an HTML file, as illustrated below.